Can a single X-ray observation tell a black hole from a neutron star in a source never seen in training?

Leakage-safe, pre-registered tests with RXTE, MAXI and NICER
(the XRB-pilot project, versions v1–v14)

Technical report

TODO: author name

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Abstract

Can a single X-ray observation tell whether an accreting compact object is a black hole (BH) or a neutron star (NS), for a source that the classifier has never seen? The XRB-pilot project (versions v1–v14) answered this question with a strict design: dynamically confirmed BHs and burst- or pulse-confirmed NSs only; leave-one-source-out evaluation; source-level metrics with a class-stratified source bootstrap; one or a few pre-registered primary comparisons per version, judged by whether the 95% interval excludes zero; and frozen models, rules and pipelines for every test on new data.

Using RXTE/PCA spectra of 31 sources (7 BH, 24 NS), a single 5–25 keV spectrum classifies unseen sources well above chance (source balanced accuracy 0.80–0.87, source AUC 0.88–0.95), but essentially all of the information is in two count-space hardness ratios: the full spectral shape, intensity, 3–5 and 25–60 keV bands, flexible algorithms with nested selection, 17 times more observations per source and 15 additional sources add nothing detectable to the ranking of unseen sources. Variability-defined states show why: the colour classifier separates BHs from NSs in soft states (observation-level AUC 0.954) but not in hard states (0.646; difference +0.31 [+0.10, +0.51]). Inside the hard state, 0.016–4 Hz variability measured with cross-detector cospectra adds information: +0.26 [+0.05, +0.46] in AUC on the RXTE sample, consistently positive on NICER, +0.15 [+0.03, +0.26] when RXTE and NICER are pooled (86 sources, 16 dynamically confirmed BHs; not an independent confirmation), and +0.18 [+0.05, +0.34] in a prospective test with a frozen pipeline and a pre-chosen random forest on new NICER observations. At fixed colour the characteristic frequency of BH variability is lower by 0.3–0.6 dex in every data set but one, with pooled $p = 0.0012$ but non-significant single prospective tests. A burst catalogue, persistent BHs, event-mode frequencies up to 1 kHz and MAXI/GSC light curves (including an exact reproduction of a published classifier) qualify these results. All prospective confirmations are at the observation level on the same 16 BHs; a source-level test on new BH transients is the necessary next step. This report documents every phase, the full mathematics of features, models and statistics, and the mapping from the archived data to the machine-learning matrices; it re-runs no analysis and every number is read from files committed in the repository.

Executive summary

Question. Can one X-ray observation classify the compact object (BH or NS) of a source that was never used for training? **Design.** Dynamically confirmed BHs only; leave-one-source-out evaluation; source-level balanced accuracy and AUC with a 2000-draw class-stratified source bootstrap; a pre-registered primary comparison per sub-analysis, "detected" only if its 95% interval excludes 0 (or $p < 0.05$ for permutation tests); frozen models and rules for every test on new data. **Each pre-registration was written after the results of all earlier versions**; from v5 on they were designed and approved by the analyst under the user's standing instruction.

Supported by the data

1. Two count-space hardness ratios capture essentially all BH/NS information in a 5–25 keV RXTE spectrum: source BA 0.80, source AUC 0.92 for the two-colour logistic regression on 31 sources; nothing tested beyond them (full shape, intensity, 3–5 and 25–60 keV, eight algorithms with nested selection, all 7,949 observations, 15 more sources) improves the ranking of unseen sources (Chapters 6 and 7).
2. The colours work in soft states and fail in hard states: AUC 0.954 vs 0.646, difference +0.309 [+0.099, +0.510] (Chapter 9).
3. Inside the hard state, 0.016–4 Hz variability adds information: +0.264 [+0.048, +0.455] on the RXTE sample; +0.145 [+0.028, +0.257] pooled over RXTE and NICER; +0.184 [+0.049, +0.337] prospectively on new NICER observations with a frozen pipeline and a pre-chosen random forest (Chapters 9 to 11).
4. Type-I bursts do not affect the spectral conclusions (v4 detector, MINBAR catalogue).

Consistent in direction, marginal in significance

1. At fixed colour, BH variability has a lower characteristic frequency, by 0.3–0.6 dex in every data set but the four-BH external RXTE set; pooled $p = 0.0012$, but $p = 0.07$–0.21 in the background-corrected and prospective NICER tests; no Eddington-ratio control.
2. The timing gain is stable for random forests (+0.17 to +0.25) and unstable for logistic regression (+0.07 to +0.26).
3. On MAXI/GSC the 2–4 keV colour carries information beyond the ≥ 4 keV colour that survives a Galactic $N_{\rm H}$ correction.

Not supported, and the main caveat

1. Flexible models do not beat two colours; nested automatic selection is worse (inner AUC 0.96–1.00, outer 0.75).

2. A broad 512–1024 Hz "NS noise" criterion is not detectable at single-observation sensitivity (exploratory).
3. Linear colour models on MAXI with confirmed BHs have almost no discriminating power; a published high accuracy reproduces exactly but rests mainly on the NS classes.
4. **All confirmations are at the observation level on the same 16 dynamically confirmed BHs**; v10 pooling is overall evidence, not an independent confirmation; v11–v13 are new observations of the same sources, not new sources; RF was chosen post hoc in v12 and pre-specified in v13.

Table 1 summarises the six phases.

Table 1: Cross-phase summary. Primary results with 95% source-bootstrap intervals or permutation p.

Phase	Question	Data	Primary result(s)	Verdict
1 (v1–v3) Ch. 6	Does a spectrum classify unseen sources, and where is the information?	RXTE/PCA, 8 then 31 sources (7 BH), 456 obs.; 14 BH candidates	H+B − H: LR AUC −0.048 [−0.179, +0.048], RF −0.012 [−0.077, +0.048]; B−A RF source BA +0.235 [+0.042, +0.429]	better than chance; information in two colours
2 (v4) Ch. 7	Is the spectral information saturated?	v2 sample; HEXTE; 46 sources; 7,949 obs.	nested selection − H-LR: BA −0.062 [−0.146, +0.021], AUC −0.173 [−0.399, +0.012]; all other main comparisons n.d.	saturated at two colours
3 (v5–v6) Ch. 8	Does fast variability add information? Does it hold on unseen sources?	Standard-1 timing; 23 external sources (frozen)	v5 AUC +0.065 [0.000, +0.179]; v6a AUC 0.000 (saturated), BA +0.105 [0.000, +0.237]; log loss −0.127 [−0.217, −0.044]	n.d. (direction positive; calibration better)
4 (v7–v8) Ch. 9	Where do colours fail? Bursts, more BHs, kHz, MAXI	states; MINBAR; persistent BHs; event mode; MAXI	soft − hard AUC +0.309 [+0.099, +0.510]; hard-state timing +0.264 [+0.048, +0.455]; external +0.064 [−0.185, +0.367]; MAXI 2–4 keV +0.377 [+0.160, +0.595]	hard state is the problem; timing helps there
5 (v9–v10) Ch. 10	Another instrument; pooled evidence	NICER 368 obs. (9 BH sources); union 86 sources (16 BH)	NICER +0.112 [−0.004, +0.229], $D = -0.31$ ($p = 0.136$); pooled +0.145 [+0.028, +0.257], $D = -0.368$ ($p = 0.0012$)	replicated in size; pooled detected (not independent)
6 (v11–v13) Ch. 11	Prospective tests on new observations; background	318 + 238 new NICER obs.; 3C50	LR +0.066 [−0.095, +0.180]; $D = -0.663$ ($p = 0.023$); corrected RF +0.174 [+0.071, +0.300], $D = -0.514$ ($p = 0.073$); prospective RF +0.184 [+0.049, +0.337], $D = -0.427$ ($p = 0.21$)	RF timing gain confirmed prospectively (observation level)

Contents

Abstract i

Executive summary ii

Notation and abbreviations xviii

1 Introduction 1

1.1 The scientific question . . . 1
1.2 Why the test must be on unseen sources . . . 1
1.3 Data leakage and how large it is here . . . 2
1.4 Labels . . . 2
1.5 Design principles and their limits . . . 3
1.6 Reading guide . . . 3

2 Data sources and physical feature extraction 8

2.1 Instruments, data modes and catalogues . . . 8
2.2 Source labels . . . 8
2.3 Observation eligibility and time-stratified sampling . . . 11
2.4 RXTE/PCA Standard-2 spectra . . . 12
2.4.1 What the files contain . . . 12
2.4.2 Background subtraction . . . 13
2.4.3 Deadtime . . . 14
2.4.4 Common energy grid and rebinning . . . 14
2.4.5 Summary quantities, errors and quality rules . . . 15
2.4.6 Response check and response-normalised spectra . . . 15
2.4.7 Independent check with XSPEC . . . 15
2.5 Spectral representations . . . 16
2.6 HEXTE high-energy colours (v4b3) . . . 17
2.7 Type-I X-ray bursts . . . 18

2.8 Low-frequency timing from RXTE Standard-1 (v5–v10) . . . 19
2.8.1 Rows, detectors and background . . . 19
2.8.2 Fourier conventions and the binning correction . . . 20
2.8.3 Two-group cospectrum (v5) . . . 20
2.8.4 All-pairs estimator and single-PCU hybrid (v6) . . . 20
2.8.5 Characteristic frequency ν_c (v6) . . . 21
2.8.6 State index (v7a) . . . 22
2.9 High-frequency timing from RXTE event mode (v8c) . . . 22
2.10 NICER/XTI features (v9–v13) . . . 22
2.10.1 Reading the events . . . 22
2.10.2 Good intervals, segments and detectors . . . 23
2.10.3 Per-segment statistics and observation features . . . 23
2.10.4 Background proxy (v10) . . . 24
2.10.5 3C50 background correction and screening (v12, v13) . . . 24
2.10.6 Lorentzian fits (v11c) . . . 24
2.11 MAXI/GSC features (v7c, v8d, v9c) . . . 24
2.12 Feature summary . . . 25

3 Data model and machine-learning interface 27
3.1 Notation . . . 27
3.2 From files to matrices . . . 28
3.2.1 Missing values . . . 28
3.3 Catalogue of stored arrays . . . 29
3.3.1 Worked example: the first rows of the real v2 matrices . . . 30
3.4 The design matrix and the leave-one-source-out split . . . 32
3.5 End-to-end data flow . . . 32
3.6 Leakage controls implemented in the code . . . 32

4 Machine-learning methods 35
4.1 Shared components . . . 36
4.1.1 Standardisation . . . 36
4.1.2 Class-balanced weights . . . 36
4.1.3 Score conventions . . . 37
4.2 Dummy classifier . . . 37
4.3 Logistic regression . . . 37
4.4 Random forest . . . 38
4.5 Extremely randomised trees . . . 38

4.6 Histogram gradient boosting . . . 40
4.7 Support-vector machine with RBF kernel . . . 40
4.8 k-nearest-neighbour classifier . . . 41
4.9 Quadratic discriminant analysis . . . 42
4.10 Multi-layer perceptron (seed ensemble) . . . 42
4.11 Colour-conditional timing likelihood ratio (CCTLR, v6c) . . . 43
4.12 k-nearest-neighbour regression for colour residualisation . . . 44
4.13 Frozen characteristic-frequency rule (v11, v13) . . . 44
4.14 Models that were not used . . . 45

5 Evaluation and statistical inference 46
5.1 Leave-one-source-out cross-validation . . . 46
5.2 Source-level aggregation and thresholds . . . 48
5.3 Metrics . . . 48
5.4 Nested leave-one-source-out selection . . . 49
5.5 Source bootstrap . . . 50
5.6 Colour-conditional source permutation tests . . . 51
5.7 Source-level conformal prediction (v6e) . . . 52
5.8 Observation-budget curves (v6f) . . . 52
5.9 Agreement, bridging and reliability statistics . . . 53
5.10 Frozen and prospective protocol . . . 53
5.11 Implementation checks and multiple comparisons . . . 54

6 Phase 1 (v1–v3): spectral baselines 55
6.1 Motivation . . . 55
6.2 Pre-registration . . . 55
6.3 Data . . . 57
6.3.1 Instrument, mode, band and epoch . . . 57
6.3.2 Sources . . . 57
6.3.3 Data quality . . . 58
6.3.4 BH candidates (v3c) . . . 58
6.4 Data structure in this phase . . . 61
6.5 Models in this phase . . . 61
6.6 Results . . . 61
6.6.1 The pilot (v1, 8 sources) . . . 61
6.6.2 How much is in two colours? (v2, v3a, v3b) . . . 65
6.6.3 Label sensitivity: BH candidates (v3c) . . . 68

6.6.4 What the classifier looks like . . . 68
6.6.5 XSPEC cross-check . . . 73
6.7 Interpretation . . . 73
6.8 Deviations and caveats . . . 74
6.9 Takeaway . . . 74

7 Phase 2 (v4): is the spectral information saturated? 75
7.1 Motivation . . . 75
7.2 Pre-registration . . . 76
7.3 Data . . . 78
7.3.1 Type-I burst screening . . . 78
7.3.2 HEXTE cluster B (v4b3) . . . 78
7.3.3 Source-list verification and earlier gain epochs (v4b2) . . . 80
7.3.4 All observations of the same sources (v4b1) . . . 81
7.4 Data structure in this phase . . . 81
7.5 Models in this phase . . . 81
7.6 Results . . . 81
7.6.1 Flexible algorithms (v4a) . . . 83
7.6.2 Data quality: type-I bursts . . . 86
7.6.3 Wider band: HEXTE 25–60 keV (v4b3) . . . 86
7.6.4 More sources and earlier epochs (v4b2) . . . 89
7.6.5 All observations of each source (v4b1) . . . 91
7.6.6 Phase 1–2 comparisons at a glance . . . 93
7.7 Interpretation . . . 93
7.8 Deviations and caveats . . . 96
7.9 Takeaway . . . 97

8 Phase 3 (v5–v6): fast variability as a second axis 98
8.1 Motivation . . . 98
8.2 Pre-registration . . . 99
8.3 Data . . . 100
8.3.1 Timing inputs and coverage . . . 100
8.3.2 External sources (v6a) . . . 100
8.3.3 Implementation checks . . . 101
8.4 Data structure in this phase . . . 102
8.5 Models in this phase . . . 102
8.6 Results . . . 102

8.6.1 Timing as a new axis on the v2 sources (v5) . . . 102
8.6.2 Confirmation on unseen sources with frozen models (v6a) . . . 105
8.6.3 A structured classifier: CCTLR (v6c) . . . 107
8.6.4 Is the timing difference a class difference? (v6d) . . . 107
8.6.5 Usable outputs: conformal sets and observation budgets (v6e, v6f) . . . 110
8.6.6 Robustness: nuisance variables and frequency bands (v6h) . . . 111
8.7 Interpretation . . . 111
8.8 Deviations and caveats . . . 112
8.9 Takeaway . . . 113

9 Phase 4 (v7–v8): where the colours fail, and why 114
9.1 Motivation . . . 114
9.2 Pre-registration . . . 115
9.3 Data . . . 117
9.3.1 Variability states (v7a) . . . 117
9.3.2 Burst contamination from MINBAR (v7b1) . . . 117
9.3.3 Persistent and additional BHs (v7b2, v8b) . . . 118
9.3.4 Event-mode data (v8c) . . . 120
9.3.5 MAXI/GSC (v7c, v8d, v9c) . . . 120
9.4 Data structure in this phase . . . 121
9.5 Models in this phase . . . 121
9.6 Results . . . 122
9.6.1 Data quality: burst contamination from MINBAR (v7b1) . . . 122
9.6.2 States: soft-like separable, hard-like not (v7a) . . . 122
9.6.3 Timing inside the hard state (v8a, v8b) . . . 123
9.6.4 More BHs: the persistent systems (v7b2) . . . 125
9.6.5 Above 4 Hz: event-mode frequencies (v8c, exploratory) . . . 126
9.6.6 An independent instrument: MAXI/GSC (v7c, v8d, v9c) . . . 128
9.6.7 Phase summaries . . . 131
9.7 Interpretation . . . 134
9.8 Deviations and caveats . . . 135
9.9 Takeaway . . . 136

10 Phase 5 (v9–v10): a second instrument, and the pooled evidence 137
10.1 Motivation . . . 137
10.2 Pre-registration . . . 138
10.3 Data . . . 138

10.3.1 NICER sources and observations (v9) . . . 138
10.3.2 Full NICER observations (v10c) . . . 139
10.3.3 Source identity across instruments (v10) . . . 140
10.4 Data structure in this phase . . . 140
10.5 Models and statistics in this phase . . . 140
10.6 Results . . . 141
10.6.1 NICER alone: hard-state timing (v9a) . . . 141
10.6.2 NICER alone: the characteristic frequency (v9b) . . . 143
10.6.3 Pooled: the hard-state timing gain (v10a) . . . 143
10.6.4 Pooled: the characteristic frequency (v10b) . . . 143
10.7 Interpretation . . . 146
10.8 Deviations and caveats . . . 147
10.9 Takeaway . . . 148

11 Phase 6 (v11–v13): prospective tests on new observations 149
11.1 Motivation . . . 149
11.2 Pre-registration . . . 150
11.3 Data . . . 151
11.3.1 New NICER observations (v11, v13) . . . 151
11.3.2 Background: 3C50 (v12) . . . 151
11.4 Data structure in this phase . . . 152
11.5 Models in this phase . . . 152
11.6 Results . . . 153
11.6.1 Hard-state timing gain on new observations . . . 153
11.6.2 Characteristic frequency on new observations . . . 153
11.6.3 The frozen rule and never-used sources . . . 154
11.7 Interpretation . . . 156
11.8 Deviations and caveats . . . 156
11.9 Takeaway . . . 157

12 Synthesis across phases (v14) 158
12.1 Every primary comparison . . . 158
12.2 The two timing results across samples and instruments . . . 159
12.3 Conclusions in three tiers . . . 161
12.4 How independent is each piece of evidence? . . . 161
12.5 Why the series stopped . . . 162

13 Limitations **163**

13.1 Sample size and labels . . . 163

13.2 Independence and pre-registration . . . 163

13.3 Measurement . . . 164

13.4 Statistics . . . 164

13.5 Process and reproducibility . . . 164

14 Future work **166**

14.1 A source-level prospective test . . . 166

14.2 Physical normalisation . . . 166

14.3 Data improvements . . . 166

14.4 Methods . . . 167

References **168**

A Hyper-parameters and fixed settings **170**

B Reproduction **174**

B.1 Run scripts . . . 174

B.2 Scripts and their outputs . . . 174

B.3 Data locations . . . 176

B.4 Environment . . . 176

B.5 This report . . . 177

C Pre-registration summary **178**

D Decision-log summary **180**

E Glossary **182**

List of Figures

1.1 Leakage: LOSO versus observation-wise splits . . . 2
1.2 Independence of each version . . . 4
1.3 Sample sizes per version . . . 5
1.4 How a result is tagged and judged . . . 6
1.5 The chain of questions, v1–v14 . . . 7

2.1 Selection of sources and observations . . . 11
2.2 RXTE spectral pipeline and representations . . . 12
2.3 Two Standard-2 spectra used to validate the data flow . . . 13
2.4 Response variation between observations . . . 16
2.5 Count-space spectra of BHs and NSs with the colour bands . . . 18
2.6 Burst screening . . . 19
2.7 Timing-feature pipeline . . . 19
2.8 NICER feature pipeline . . . 23
2.9 MAXI pipeline . . . 25

3.1 The real v2 design matrix . . . 31
3.2 Design matrix and LOSO split . . . 33
3.3 End-to-end data flow . . . 34

5.1 Evaluation chain . . . 46
5.2 Nested leave-one-source-out . . . 49
5.3 Class-stratified source bootstrap . . . 50
5.4 Colour-conditional permutation test . . . 51
5.5 Source-level Mondrian conformal sets . . . 53
5.6 Frozen and prospective protocol . . . 53

6.1 Phase 1 analysis map . . . 56
6.2 Dataset diagnostics, v1 and v2 . . . 59
6.3 Net count-rate spectra per source, v1 and v2 . . . 61

6.4 Diagnostics of the v3c BH candidates . . . 63
6.5 v1 LOSO scores per source . . . 65
6.6 v3a and v3b . . . 67
6.7 v2 LOSO scores per source . . . 69
6.8 Hardness–intensity diagrams of the errors . . . 70
6.9 Per-source accuracy A vs B and score vs rate . . . 71
6.10 v3c candidate scores . . . 72
6.11 Colour–colour diagram with decision boundaries . . . 73

7.1 Phase 2 analysis map . . . 76
7.2 Burst-candidate light curves . . . 79
7.3 HEXTE colours . . . 80
7.4 v4a forest plot . . . 87
7.5 v4a decision boundaries . . . 88
7.6 Scores by gain epoch . . . 92
7.7 v4b1 source scores . . . 94
7.8 Phase 1–2 paired differences . . . 95

8.1 Phase 3 analysis map . . . 99
8.2 Timing vs hard colour . . . 105
8.3 External sources, frozen models . . . 108
8.4 CCTLR timing evidence map . . . 109
8.5 Observation-budget curves . . . 111

9.1 Phase 4 analysis map . . . 116
9.2 rms vs hard colour per source . . . 118
9.3 MINBAR vs Standard-1 bursts . . . 119
9.4 AUC by variability state . . . 123
9.5 Persistent BHs in the colour plane . . . 126
9.6 RXTE vs MAXI scores . . . 130
9.7 MAXI N_H control . . . 131
9.8 MAXI point-level AUC by C4 tertile . . . 132
9.9 v7 summary . . . 133
9.10 v8 summary . . . 133

10.1 Phase 5 analysis map . . . 138
10.2 NICER hard-like observations . . . 145
10.3 v9 summary . . . 145

10.4 Pooled tests 146

11.1 Phase 6 analysis map 150
11.2 v11 prospective test 155
11.3 v12 background correction 155
11.4 v13 prospective test 156

12.1 Timing results across versions 160

List of Tables

1 Cross-phase summary iii

1.1 Map of phases, versions and questions 5

2.1 Instruments, data modes and catalogues 9
2.2 Source labelling and eligibility rules by version 10
2.3 Sampling seeds and limits 12
2.4 The common energy grid 15
2.5 Colour bands 17
2.6 Timing frequency bands 21
2.7 Bins of the νP_ν centroid 21
2.8 Summary of all features 26

3.1 Catalogue of stored arrays 29
3.2 Example rows of the v2 design matrices 32
3.3 Example rows of the NICER feature table 32

4.1 Inventory of models 35
4.2 Logistic-regression hyper-parameters 38
4.3 Random-forest hyper-parameters 39
4.4 ExtraTrees hyper-parameters 39
4.5 HistGradientBoosting hyper-parameters 40
4.6 SVM hyper-parameters 41
4.7 kNN hyper-parameters 42
4.8 QDA hyper-parameters 42
4.9 MLP hyper-parameters 43

5.1 Summary of inferential procedures 47
5.2 What was frozen in the frozen and prospective tests 54

6.1 Phase 1 pre-registrations 56

6.2 v1 sources and attrition . . . 57
6.3 v2 sources and attrition . . . 57
6.4 Phase 1 arrays . . . 64
6.5 Phase 1 models . . . 64
6.6 v1 per-source results . . . 65
6.7 v2 representations and models . . . 66
6.8 Phase 1 paired comparisons . . . 67
6.9 v2 per-source scores . . . 68
6.10 v3c candidate predictions . . . 70
6.11 v3c: candidates added to training . . . 71

7.1 Phase 2 pre-registration . . . 77
7.2 Phase 2 samples . . . 78
7.3 Phase 2 arrays . . . 82
7.4 Phase 2 models . . . 82
7.5 Nested-selection grids . . . 83
7.6 v4a primary result . . . 83
7.7 v4a nested selection per fold . . . 84
7.8 v4a all combinations . . . 85
7.9 Burst exclusion . . . 88
7.10 HEXTE results . . . 89
7.11 v4b2 extended sample . . . 89
7.12 v4b2 paired differences . . . 90
7.13 v4b2 new sources . . . 91
7.14 v4b1 results . . . 92
7.16 v4b1 per-source scores . . . 92
7.15 v4b1 on the v2 observations . . . 94

8.1 Phase 3 pre-registrations . . . 100
8.2 Timing-feature coverage . . . 100
8.3 v6 external-source verification . . . 101
8.4 Split-half reliability . . . 102
8.5 Phase 3 arrays . . . 103
8.6 Phase 3 models . . . 103
8.7 v5 primary result . . . 104
8.8 v5 per-source scores . . . 104
8.9 v5 descriptive comparisons . . . 106

8.10 v6a external validation . . . 106
8.11 v6a per-source scores . . . 107
8.12 CCTLR results . . . 109
8.13 Colour-conditional permutation test . . . 110
8.14 Conformal prediction sets . . . 110
8.15 v6h robustness . . . 111

9.1 Phase 4 pre-registered comparisons . . . 115
9.2 State counts . . . 117
9.3 MINBAR vs Standard-1 bursts . . . 118
9.4 Persistent-BH attrition . . . 119
9.5 Phase 4 arrays . . . 121
9.6 Phase 4 models . . . 122
9.7 MINBAR exclusion . . . 122
9.8 AUC by state . . . 123
9.9 Timing in the hard-like state . . . 124
9.10 Hard-like scores per source . . . 124
9.11 External hard-like stratum . . . 125
9.12 External hard-like sources . . . 125
9.13 Adding persistent BHs . . . 126
9.14 Event-mode high frequencies . . . 127
9.15 SR2000-type rule per source . . . 127
9.16 de Beurs reproduction . . . 128
9.18 MAXI BH scores . . . 128
9.19 RXTE vs MAXI . . . 129
9.17 MAXI strict pipeline . . . 130
9.20 MAXI after N_H correction . . . 131
9.21 MAXI gain by C4 tertile . . . 132

10.1 Phase 5 pre-registrations . . . 139
10.2 NICER states . . . 140
10.3 NICER full-observation quality . . . 140
10.4 Phase 5 arrays . . . 141
10.5 NICER hard-state timing . . . 142
10.6 NICER hard-like scores per source . . . 142
10.7 NICER characteristic frequency test . . . 143
10.8 Pooled hard-state timing gain . . . 144

10.9 Pooled characteristic frequency 146

11.1 Phase 6 pre-registrations 151
11.2 Phase 6 arrays 152
11.3 Hard-state gain, prospective 153
11.4 Characteristic frequency, prospective 154
11.5 Frozen nu* rule 155

12.1 All primary comparisons 158
12.2 Hard-state timing gain across versions 160
12.3 Characteristic-frequency difference across versions 160
12.4 Independence ladder 162

A.1 Master table of settings 170

B.1 Script to output map 174

C.1 Pre-registrations 178

D.1 Decision-log summary 180

E.1 Glossary 182

Notation and abbreviations

Vectors and matrices are bold. Equation numbers refer to the defining equation.

Symbols

Symbol	Meaning	Def.
Data and evaluation		
n, p, S	number of observations (rows), features (columns), sources (groups)	§3.1
$\boldsymbol{X} \in \mathbb{R}^{n \times p}$	design matrix; row = one observation (RXTE/NICER ObsID or MAXI day)	§3.1
$\boldsymbol{y} \in \{0,1\}^n$	labels, $y_i = 1$ for BH	§3.1
$\boldsymbol{g}$, g_i	source (group) index of each row; n_k rows of source k	§3.1
S_1, S_0	numbers of BH and NS sources	§3.1
$\boldsymbol{w}$	sample weights (balanced or source-equal)	Eqs. (4.2) and (4.3)
ϕ	fold assignment (LOSO: $\phi(i) = g_i$)	§3.1
$\hat{s}_i$, $\hat{\boldsymbol{s}}$	out-of-fold BH score of observation i	Eq. (5.1)
$\bar{s}_k$, $\hat{Y}_k$	source score (mean OOF score) and source decision	Eq. (3.1)
t, t_k	decision threshold (0.5; fold-specific in nested v4a)	Eq. (5.2)
BA, AUC	balanced accuracy, area under the ROC curve (Mann–Whitney)	Eqs. (5.3) and (5.4)
Δ	paired difference of a metric between two models on the same resamples	§5.5
$B = 2000$	bootstrap resamples (class-stratified over sources, seed 42)	Eq. (5.6)
D, p	permutation statistic (BH − NS mean source residual) and p-value	Eqs. (5.7) and (5.8)
ε	conformal miscoverage level (0.1, 0.2)	Eq. (5.9)
RXTE spectral features		
x_{ij}	net (background-subtracted, deadtime-corrected) count rate of observation i in grid bin j, $\mathrm{c\,s^{-1}\,keV^{-1}\,PCU^{-1}}$	Eq. (2.2)
F_i	net 5–25 keV count rate (sum over the 45-bin grid)	Eq. (2.7)
A_{ij}, B_{ij}	intensity representation $\operatorname{asinh}(x_{ij}/0.1)$; shape representation x_{ij}/F_i	Eqs. (2.9) and (2.10)
h_1, h_2 (H)	soft colour $F(7\text{–}10)/F(5\text{–}7)$, hard colour $F(16\text{–}25)/F(10\text{–}16)$ (count space)	Eq. (2.11)
HI	$[h_1, h_2, \log_{10} F]$	§2.5
$\mathrm{pl2}_{ij}$; R, H_R	expected counts of a folded $\Gamma = 2$ power law; $x_{ij}/\mathrm{pl2}_{ij}$ and its colours	Eqs. (2.8) and (2.12)
X_1, X_2	HEXTE colours $F_{\mathrm{HEXTE}}(25\text{–}40)/F_{\mathrm{PCA}}(16\text{–}25)$, $F_{\mathrm{HEXTE}}(40\text{–}60)/F_{\mathrm{HEXTE}}(25\text{–}40)$	Eq. (2.13)
Timing features		

Symbol	Meaning	Def.
T_1, T_2, T_3	fractional rms (signed square root of the cospectral variance) in 0.016–0.094, 0.10–1.0, 1.0–4.0 Hz	Eqs. (2.15) and (2.16)
T_k^*	the same with the all-pairs estimator and single-PCU fallback (v6)	Eq. (2.17)
ν_c	νP_ν-weighted centroid frequency of the variability	Eq. (2.18)
ν_L	width of the lowest-frequency Lorentzian (v11c)	Eq. (2.22)
ν^*	frozen rule threshold $\sqrt{\tilde{\nu}_{\rm BH}\tilde{\nu}_{\rm NS}}$ (1.3007 and 1.3361 Hz)	Eq. (4.16)
rms (state)	0.1–4 Hz (RXTE) or 0.1–10 Hz (NICER) fractional rms; hard-like > 0.10, soft-like < 0.075	Eq. (2.19)
HF_1–HF_3	event-mode fractional rms in 4–64, 64–512, 512–1024 Hz (ratio of means)	Eq. (2.20)
c_1, c_2 (H_N)	NICER colours C(4–6)/C(2–4), C(6–10)/C(4–6)	§2.10
f, f_T	3C50 background fraction (2–10 keV)	Eq. (2.21)
MAXI features		
L, M, H	photon fluxes in 2–4, 4–10, 10–20 keV	Eq. (2.23)
SC, HC, C4, RelInt	$(M-L)/(M+L)$, $(H-L)/(H+L)$, $(H-M)/(H+M)$, total / 99.99th percentile	Eq. (2.23)
$N_{\rm H}$, t_b	Galactic hydrogen column (HI4PI) and band transmission of `tbabs`×power law	§2.11
Models		
$\boldsymbol{\beta}$, C	logistic-regression coefficients and inverse regularisation strength	Eqs. (4.4) and (4.5)
$\Lambda(\boldsymbol{T} \mid \boldsymbol{h})$	CCTLR timing log-likelihood ratio	Eq. (4.13)
k	neighbours (kNN classification, kNN residualisation with $k = 25$)	§4.8, §4.12

Abbreviations

Abbreviation	Meaning
BH, NS	black hole, neutron star
LMXB, HMXB, AMXP	low-/high-mass X-ray binary; accreting millisecond X-ray pulsar
NPNS, PSR	non-pulsing NS (burster without accretion-powered pulsations); pulsar (MAXI analyses)
RXTE, PCA, PCU, HEXTE	Rossi X-ray Timing Explorer; its Proportional Counter Array, Proportional Counter Unit, High Energy X-ray Timing Experiment
NICER, XTI, MPU, FPM	Neutron star Interior Composition Explorer; X-ray Timing Instrument; Measurement/Power Unit; Focal Plane Module
MAXI, GSC	Monitor of All-sky X-ray Image; Gas Slit Camera
GTI, ObsID	good time interval; observation identifier
LOSO, OOF	leave-one-source-out; out-of-fold
LR, RF, ET, HistGB	logistic regression, random forest, extremely randomised trees, histogram gradient boosting
SVM-RBF, kNN, QDA, MLP	support-vector machine with RBF kernel, k-nearest neighbours, quadratic discriminant analysis, multilayer perceptron
CCTLR	colour-conditional timing likelihood-ratio classifier (v6c)
BA, AUC, CI	balanced accuracy, area under the ROC curve, confidence (percentile) interval
srcw	source-equal sample weights
ICn	pre-declared implementation check (gate) n

Abbreviation	Meaning
RM06, SR2000	Remillard and McClintock (2006); Sunyaev and Revnivtsev (2000)
det., n.d.	detected (interval excludes 0 or $p < 0.05$); not detected under the pre-registered criterion

Chapter 1

Introduction

1.1 The scientific question

An accreting compact object in an X-ray binary is either a neutron star (NS) or a stellar-mass black hole (BH). The only direct way to tell them apart is a dynamical mass (from the companion's radial-velocity curve) above the maximum NS mass, or direct evidence of a solid surface (thermonuclear type-I X-ray bursts) or of a magnetosphere anchored to it (coherent pulsations). Most X-ray binaries – in particular new transients – have neither when they are discovered. The project asks whether a *single X-ray observation* carries enough information to say which kind of compact object it contains, *for a source that was never seen during training.*

Physical reasons to expect some information exist. In soft (thermal, disc-dominated) states, an NS has a boundary layer or surface emission in addition to the disc, while a BH shows a cooler disc plus a power-law tail, so their count-space colours occupy different regions (e.g. Done and Gierliński, 2003). In hard (Comptonisation-dominated) states the spectra are similar, but the variability may differ: the characteristic time-scales of the aperiodic variability are expected to scale with the size of the inner flow, hence with mass, and NSs show high-frequency noise above ∼500 Hz that BHs lack (Sunyaev and Revnivtsev, 2000). The variability amplitude also tracks the state (Muñoz-Darias et al., 2011; Remillard and McClintock, 2006). Whether these differences survive a strict test on unseen sources was the open question.

1.2 Why the test must be on unseen sources

Repeated observations of one source are strongly correlated: the same distance, absorption column, inclination, companion, outburst history and calibration epoch shape all of them. A classifier evaluated on a random split of observations can score well simply by recognising the source of an observation – through its brightness, its absorption or its particular track in a colour diagram – without having learned anything that transfers to a new object. The relevant population for the question is the population of *sources*; the effective sample size is the number of sources (here 7–16 BHs), not the number of observations (hundreds to thousands). Every primary result in this report is therefore obtained with leave-one-source-out (LOSO) evaluation, all choices that depend on data are made inside the training sources (Section 5.4), and uncertainty is quantified by resampling sources (Section 5.5).

1.3 Data leakage and how large it is here

Leakage is any path through which information about the test units enters training. In this problem the main path is source identity: if observations of the same source are in both training and test sets, the test score measures re-identification. The project quantified it directly: the same representations and models were evaluated once with LOSO and once with a stratified 5-fold split over observations (Figure 1.1). The leaky split overstates balanced accuracy by +0.04 to +0.20 (largest for the Random Forest on the intensity-keeping representation A, +0.16 in v2 and +0.20 in v1). Published machine-learning classifications of RXTE spectra used either LOSO with sigma-clipped source accuracies (87%; Pattnaik et al., 2021) or random 80:20 observation splits (90–94%; Garg et al., 2026); the latter cannot answer the question about new sources. A MAXI study (de Beurs et al., 2022) used LOSO but labelled seven BH candidates as BHs; it is reproduced and re-examined in Chapter 9.

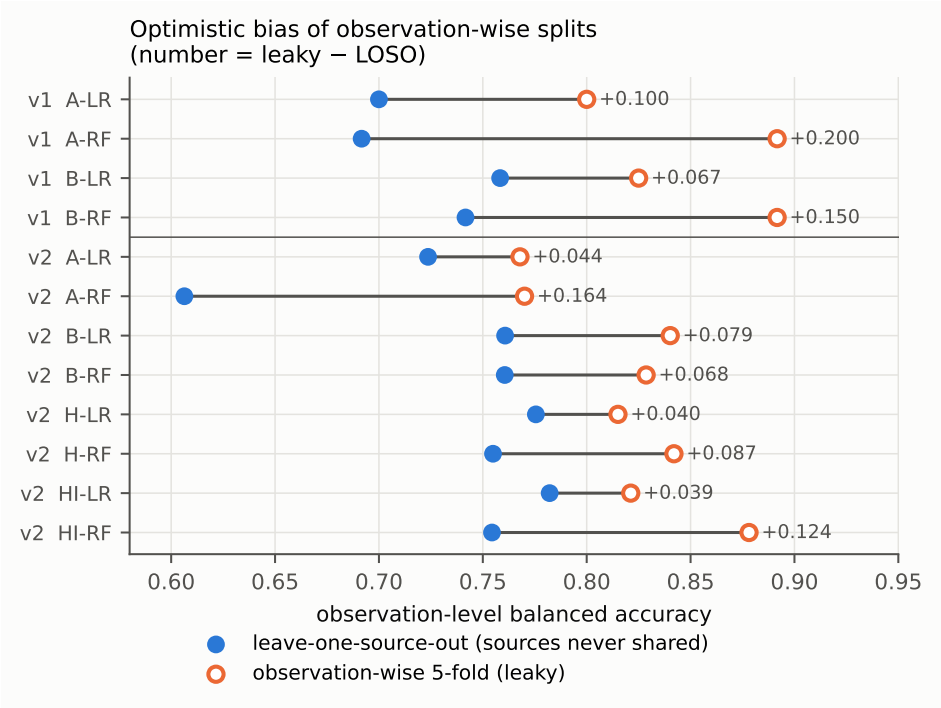


Figure 1.1: Optimism of observation-wise cross-validation. For every representation and model of v1 (8 sources) and v2 (31 sources), observation-level balanced accuracy under leave-one-source-out (filled blue) and under a stratified 5-fold split over observations in which sources appear in both training and test folds (open orange); the number is the difference. The leaky split is optimistic in every case (+0.04 to +0.20), most for models that can memorise intensity (A-RF). Generated for this report from `results/metrics_observation_level.csv`, `results/secondary_observation_split.csv` and their `results/v2/` counterparts (`report_en/figures/generated/make_figures.py`; values in `G4_leakage.csv`).

1.4 Labels

Only sources whose nature is established independently of their X-ray spectra and variability are used as labels: dynamically confirmed BHs (BlackCAT, Corral-Santana et al. 2016; Casares and Jonker 2014; from v7 the 25-system list of Marcel et al. 2026) and NSs with type-I bursts (Galloway et al., 2008, 2020) or accretion-powered pulsations (Patruno and Watts, 2021). BH *candidates* are never used as labels (they appear only as prediction targets and in a label-sensitivity analysis, v3c). This keeps the labels honest but limits the sample: across all instruments only 16

dynamically confirmed BHs have usable RXTE or NICER data, and only 7 of them are in the main RXTE sample.

1.5 Design principles and their limits

The project ran as 13 analysis versions (v1–v13) plus a wrap-up (v14), between 2026-09-29 and 2026-10-06, each with a written pre-registration committed before its data were downloaded or its models trained, and with a decision log (`reports/decision_log.md`) recording every change and whether it was made before or after seeing results. The rules (fixed from v4 on) are: LOSO; source score = mean OOF score, threshold 0.5; 2000 class-stratified source bootstrap replicates (seed 42); a difference is *detected* only if its 95% interval excludes 0 (or $p < 0.05$ for permutation tests); one (or very few) pre-declared primary comparison per sub-analysis; every selection nested inside the training sources.

Five facts that qualify every conclusion in this report

1. **Every pre-registration was written after the results of all earlier versions had been seen.** The pre-registrations prevent analysis choices from adapting to the data of their own version, but the hypotheses of later versions were suggested by earlier results (e.g. the hard-state timing hypothesis of v8 came from v5's descriptive results). They were approved by the analyst on the user's instruction, not by an external reviewer.
2. **The v10 pooled tests are overall evidence on all available data, not an independent confirmation**: the hypotheses and the pooling scheme were formulated after v5–v9.
3. **v11–v13 are new observations of the same sources**: the prospective confirmation (v13) is at the observation level; nothing has yet been confirmed on new sources.
4. **The Random Forest was chosen as primary post hoc in v12** (after seeing that its timing gain was stable in v9–v11 while LR's was not); **in v13 it was chosen before the data were downloaded.**
5. **Only 16 dynamically confirmed BH sources** have usable RXTE/NICER data; per instrument the hard-like BH sources number 5–9. A single BH source moves a source-level balanced accuracy by ≈ 0.07.

Figure 1.2 shows how much independence each version adds; Figure 1.3 shows the sample sizes.

1.6 Reading guide

Chapters 2 to 5 define all inputs, data structures, models and statistical procedures once. The six phase chapters then report the versions in groups of related analyses (Table 1.1); each starts with a "Phase at a glance" box and follows the same ten sections. Each result carries a tag: PRIMARY (the pre-declared primary comparison of its sub-analysis), SECONDARY (pre-declared, not primary), DESCRIPTIVE (pre-declared description without a decision rule), EXPLORATORY (pre-declared but downgraded, e.g. after a failed implementation check) or POST HOC (decided after seeing results); Figure 1.4 shows how a comparison receives its tag and its verdict. Chapter 12 brings the phases together; Chapters 13 and 14 discuss limitations and future work.

Figure 1.5 shows how each version followed from the results of the versions before it: the question it asked and what it found. The six phase chapters each open with an analysis map of the same

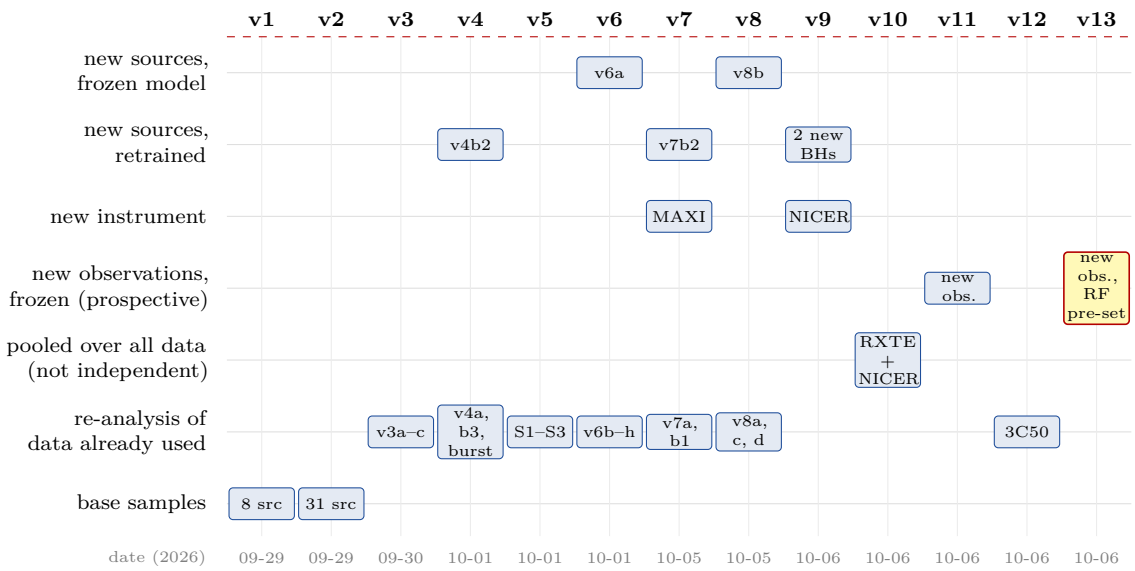


Figure 1.2: What each version adds in terms of independence from data already seen. Rows are ordered from most independent (top) to least; each box is a sub-analysis of the version in its column. External sources (v6a, v8b) were scored by frozen RXTE models, but few BHs were available (4–7). The NICER versions brought a new instrument and two BHs never used before; v10 pooled everything; v11 and v13 used never-read observations of the same sources with frozen pipelines, v13 with the Random Forest chosen before download (highlighted). Every version's pre-registration was written after all earlier results, so each column knew the results of every column to its left. Not reached by any version: new sources tested prospectively with a frozen model on new data. Drawn for this report from the pre-registrations and `reports/decision_log.md`.

kind at the level of sub-analyses (Figures 6.1, 7.1, 8.1, 9.1, 10.1 and 11.1), which shows what every version added – data, features, models, tests – and what it reused.

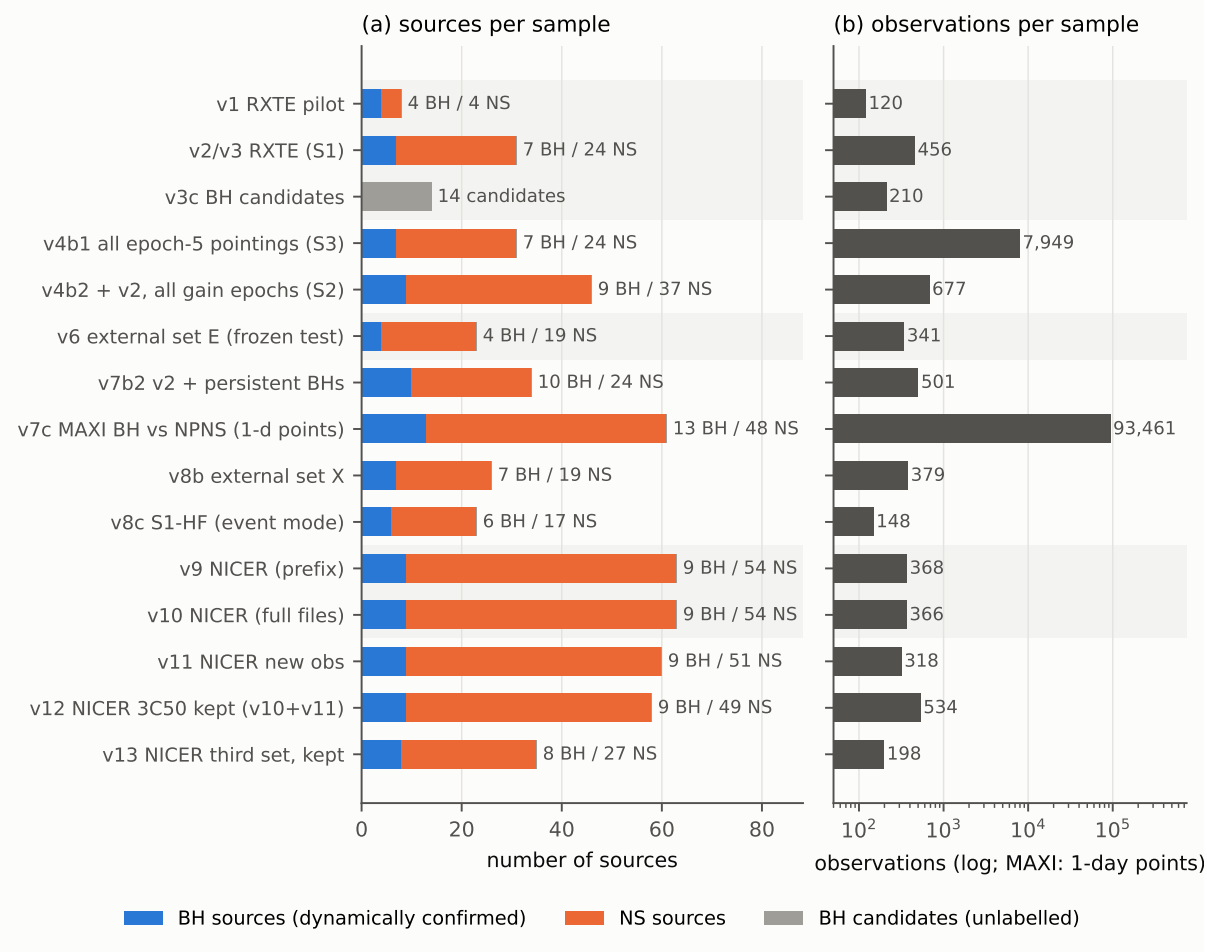


Figure 1.3: Sources (a) and observations (b) of every sample that enters a model. Bars in (a) are stacked: dynamically confirmed BH (blue), NS (orange), unlabelled BH candidates (grey, v3c only). Observations in (b) are on a log scale; for MAXI they are 1-day points. Shaded bands group the six phases. The number of BH sources never exceeds 13 in any single sample (16 over all instruments). Generated for this report from the `features.npz` files, `data/v6/sources.csv`, `data/v7c/maxi_points.csv`, `data/v8b/processed/external_states_timing.csv`, `data/v8c/processed/hf_features.csv`, the NICER feature tables and `results/v13/v13_frozen_predictions.csv` (values in `G1_sample_sizes.csv`).

Table 1.1: Map of phases, versions and questions.

Chapter	Versions	Question
6 Phase 1	v1–v3	Can 5–25 keV RXTE spectra classify unseen sources, and where is the information?
7 Phase 2	v4	Is the spectral information saturated (algorithms, more data, wider band, burst contamination)?
8 Phase 3	v5–v6	Does 0.016–4 Hz variability add information, and does it hold on external sources?
9 Phase 4	v7–v8 (+v9c)	Is the hard state the problem; does timing fix it; MINBAR, persistent BHs, high frequencies, MAXI?
10 Phase 5	v9–v10	Does an independent instrument (NICER) reproduce it; what does pooling all data say?
11 Phase 6	v11–v13	Does it hold prospectively on never-used observations with frozen pipelines and background correction?
12	v14	Synthesis

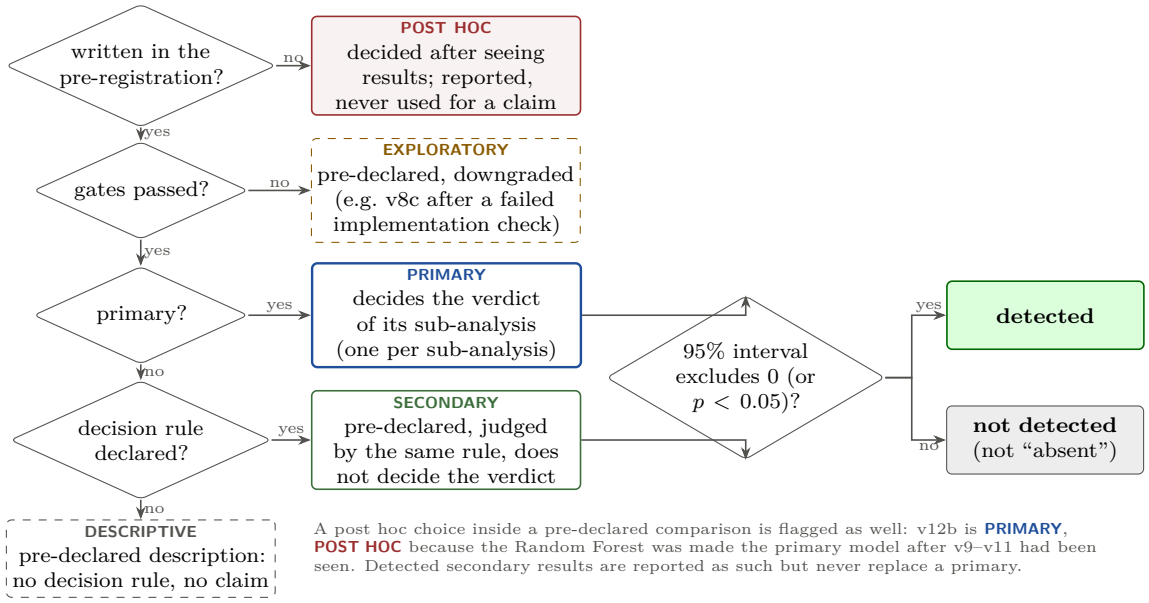


Figure 1.4: How a comparison receives its status tag and its verdict. Only primary and secondary comparisons are judged by the pre-registered rule; descriptive, exploratory and post hoc results are reported but never used to claim a difference. "Not detected" means that the interval includes 0 under the pre-registered criterion, not that the effect is absent. The interval is the paired 95% class-stratified source-bootstrap interval (Section 5.5). Drawn for this report from the pre-registrations and Section 5.11.



Figure 1.5: The chain of questions. Each version (white: its question) was designed after the results of all earlier versions (coloured: what it found, by the verdict of its pre-declared comparisons). Read down the left column (v1–v7), then the right column (v8–v14). The series moved from spectra (Phases 1–2) to timing (Phases 3–4), then to a second instrument and pooled evidence (Phase 5) and to prospective tests on new observations of the same sources (Phase 6). Detailed maps: Figures 6.1, 7.1, 8.1, 9.1, 10.1 and 11.1. Drawn for this report from the phase chapters.

Chapter 2

Data sources and physical feature extraction

This chapter defines, once, every input quantity used anywhere in the project: where the raw data come from, how a raw file becomes a number, in which unit, by which function of the code, and what happens to non-positive or missing values. The phase chapters (Chapters 6 to 11) refer back here and only state differences. When the description in a repository report and the code differ, the code is followed and the difference is flagged with a `% CHECK` comment in the LaTeX source.

2.1 Instruments, data modes and catalogues

Table 2.1 lists every data product that enters a model or a test. All raw files were obtained by scripts; every download is recorded with its URL, size in bytes and SHA256 in `logs/downloads.jsonl` (50 773 records, Appendix B). Raw data are not in git (`data/raw/` is ignored, large sets live outside the repository under `/xrb-pilot-data/`).

2.2 Source labels

The class label is a property of the *source*: $y = 1$ for a black hole (BH) and $y = 0$ for a neutron star (NS). Only labels with direct evidence are used for training and evaluation:

- **BH**: a dynamical mass function (radial-velocity curve of the companion). Versions v1–v4 used the dynamically confirmed systems of BlackCAT (Corral-Santana et al., 2016); v6 added high-mass X-ray binaries from Casares and Jonker (2014); from v7 on the list of Marcel et al. (2026) (25 systems in their Table 1) is used.
- **NS**: type-I X-ray bursts (thermonuclear burning on a solid surface) from Galloway et al. (2008) (v1–v6) or the MINBAR catalogue (Galloway et al., 2020) (v7c onwards), or coherent accretion-powered pulsations (Patruno and Watts, 2021) (v6 external sources, v7c pulsar class; Liu et al. 2006 for HMXB pulsars in v7c).
- **BH candidates** (X-ray properties only, no dynamical mass) are never used as labels. In v3c they appear only as prediction targets and in a label-sensitivity analysis.

The label rules of every version are summarised in Table 2.2; the realised source lists are given in the phase chapters.

Table 2.1: Instruments, data modes and catalogues used in the project. "Versions" gives where the product enters a model or test. Band and resolution are those actually used by the code (not the instrument limits). Code: the function that reads or processes the product.

Product	Versions	What is used	Code
RXTE/PCA Standard-2 Standard Products (`xp*_s2.pha`, `_b2.pha`, `.rsp`)	v1–v8, v10	129-channel total and `pcabackest` background spectra summed over the PCUs listed in the `ROWID` keywords; 5–25 keV (3–25 keV in v3b); response for the same PCU combination	`spectra.py`: `read_pha`, `read_rsp`, `net_spectrum`; `04_fetch_process.py`
RXTE/PCA Standard-1 (`pca/FS46_*`)	v2–v10	0.125 s light curves per PCU (`XeCntPcu0-4`), plus array rates `VpCnt`, `RemainingCnt`, `VLECnt`; 128 s rows of 1024 bins	`spectra.std1_rates`; `v5lib.py`; `v6lib.py`; `v7lib.py`
RXTE/PCA event mode (`pca/SE*.evt.gz`)	v8c	Events with `TIME`, `PCUID`; modes `E_{n}us_{m}M_0_{k}s`; binned at 2^{-12} s	`v8lib.py`: `read_events`, `hf_features`
RXTE/HEXTE cluster B Standard Products (`xh*_s1.pha`, `_b1.pha`)	v4b3	Net 25–40 and 40–60 keV count rates	`19_v4b3_hexte.py`
RXTE `MissionLongData` catalogues; HEASARC TAP `xtemaster`	v1–v8	Observation lists: `OBSID`, `TIME`, `EXPOSURE`, `STD1RATE`, `RA`, `DEC`, `NPCU`	`03_select_observations.py`, `20_v4b2_select.py`, `41_v8b_select.py`
MAXI/GSC standard products, 1-day light curves	v7c, v8d, v9c	Photon fluxes in 2–4, 4–10, 10–20 keV	`38_v7c2_select.py`
de Beurs et al. (2022) processed data (GitHub, commit `488b785`)	v7c1	Their 44 sources, ten 20% subsamples, features (RelInt, SC, HC)	`37_v7c1_repro.py`
NICER/XTI cleaned events (`ni*_0mpu7_cl.evt.gz`); HEASARC TAP `nicermastr`	v9–v13	Events with `TIME`, `PI`, `DET_ID`; 2–10 keV; 1/256 s bins, 128 s segments	`v9lib.py`, `v10lib.py`, `v11lib.py`
NICER unfiltered events (`_ufa`), NICER CALDB 3C50 library	v12, v13	Input of HEASoft `nibackgen3C50` (3C50 background model)	`v12_bkg3c50.sh`, `58_v12_bkg.py`, `62_v13_bkg.py`
MINBAR DR1 burst, observation and source tables	v7–v9	Burst start times and durations; burst-source positions	`v7lib.py`: `load_minbar`, `gti_utc`
HI4PI (via HEASoft `nh`); XSPEC `tbabs`	v8d	Galactic N_{H} and band transmissions	`v8d_heasoft.sh`, `45_v8d_nh.py`
CDS Sesame (SIMBAD) positions	v4b2–v9	Source positions for label and pointing matches	`20_v4b2_select.py`, `28_v6_select.py`, `38_v7c2_select.py`

Table 2.2: Source labelling and eligibility rules by version. "Pointing" is the maximum separation between an observation's pointing and the source position used. Eligibility always also requires catalogue exposure $\geq 1000\,\mathrm{s}$ (RXTE: and `STD1RATE` $\geq 5\,\mathrm{c\,s^{-1}\,PCU^{-1}}$; NICER: `nicermastr` exposure $\geq 1000\,\mathrm{s}$).

Version	BH rule	NS rule	Pointing / epoch / minimum
v1	4 BlackCAT dynamical BHs with ≥ 30 eligible pointings	4 Galloway+2008 bursters	0.1° from catalogue median; gain epoch 5, MJD 51677–54094
v2, v3	every BlackCAT Table 4 dynamical BH with a MissionLongData catalogue	every Galloway+2008 appendix-A source with a catalogue	0.1°; epoch 5 (MJD 51677–55931); ≥ 10 eligible
v3c	16 BlackCAT candidates (prediction only)	–	as v2
v4b2	BlackCAT dynamical BHs matched by SIMBAD position	Galloway+2008 Table 3 bursters matched by position	all gain epochs; ≥ 10
v6 (external)	Casares & Jonker (2014) HMXB BHs	Patruno & Watts (2021) AMXPs (+ 4 slow pulsars as stress set)	0.1° from SIMBAD; all epochs; ≥ 10
v7b2	Cyg X-1, LMC X-1, LMC X-3 (Marcel+2026)	–	0.1° from SIMBAD; epoch 5; ≥ 10
v7c	Marcel+2026 Table 1	MINBAR bursters without pulsations (NPNS); pulsars (PSR) secondary	MAXI source within 0.1° (Sesame; 0.3° fallback); ≥ 100 daily points
v8b	unused Marcel+2026 BHs with RXTE data	–	0.1° from SIMBAD; all epochs; ≥ 3
v9–v13	Marcel+2026 Table 1 (Sesame positions)	MINBAR bursters, excluding entries with another entry within 0.05°	target within 0.05°; ≥ 5

2.3 Observation eligibility and time-stratified sampling

For every source the eligible pointings are sorted by time and divided into K consecutive groups of (as nearly as possible) equal size, using `numpy.array_split` ($K = 15$ for RXTE, $K = 8$ for NICER). Inside each group the candidates are put in a random order generated by a seeded generator; the first candidate that passes the post-download quality rules (Section 2.4.5) is accepted. This keeps the sample spread over the source's history (several outbursts, several states) and makes the choice independent of the spectrum. If a source has fewer than K eligible pointings, some groups are empty and all pointings are used. Algorithm 1 gives the procedure; the seeds are listed in Table 2.3. Figure 2.1 shows the whole selection, from the label lists to the accepted observations.

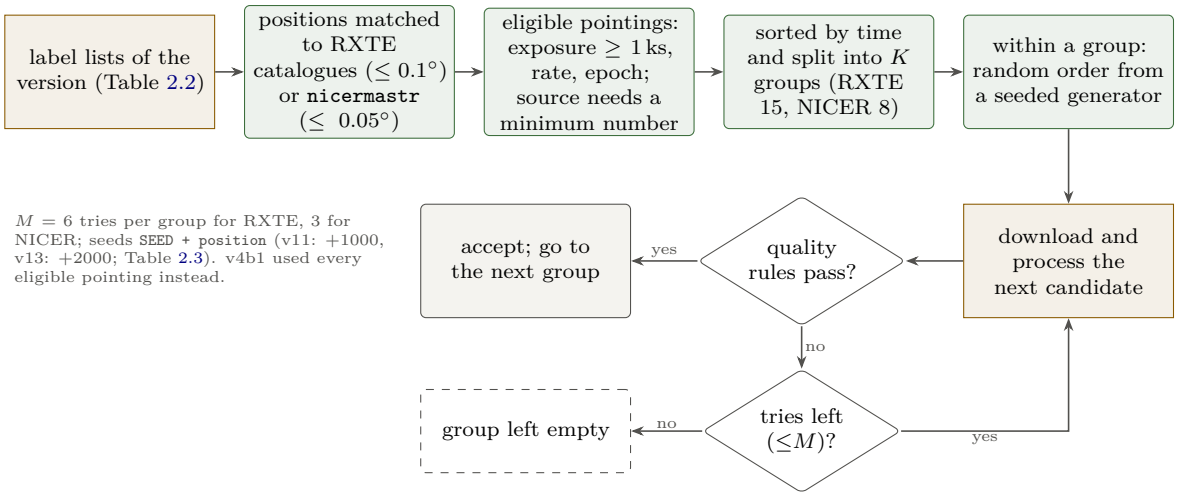


Figure 2.1: Selection of sources and observations (Algorithm 1). The choice of an observation does not depend on its spectral shape or its variability: apart from the quality rules of Section 2.4.5 (exposure, S/N, deadtime, position; NICER: segments and count rate, Section 2.10) it depends only on the label list, the catalogue position, the eligibility rules, the time ordering and a seeded random order. Drawn for this report.

Algorithm 1 Time-stratified candidate ranking and acceptance (`03_select_observations.py` + `04_fetch_process.py`; NICER: `47_v9_select.py`, `55_v11_select.py`, `60_v13_select.py`)

Input: eligible pointings $\{(o_i, t_i)\}$ of one source, number of groups K, seed σ, maximum tries M

sort pointings by time t_i; split the index list into K consecutive groups $G_1, \ldots, G_K$ (`array_split`)
rng $\leftarrow$ default_rng(σ)
for $b = 1, \ldots, K$ **do**
    $\pi \leftarrow$ rng.permutation(G_b) ▷ rank within the group
    for $r = 1, \ldots, \min(M, |G_b|)$ **do**
        download and process o_{π_r};
        if all quality rules pass **then** accept o_{π_r}; **break**
        end if
    end for
end for

Table 2.3: Sampling parameters. "position" is the 1-based position of the source in the source list processed by the script. The global seed is `SEED = 42` (`config.py`).

Sample	groups K	seed σ	max. tries M
RXTE v1, v2, v3c, v7b2, v8b	15	`SEED + position`	6
RXTE v4b2 new sources	15	`SEED + 100 + i`	6
RXTE v4b1	one group per pointing (all eligible)	–	1
NICER v9	8	`SEED + position`	3
NICER v11	8	`SEED + 1000 + position`	3
NICER v13	8	`SEED + 2000 + position`	3

2.4 RXTE/PCA Standard-2 spectra

Figure 2.2 summarises the processing chain from the archive files to the spectral representations, with the version that introduced each step or representation; the subsections below give the formulae.

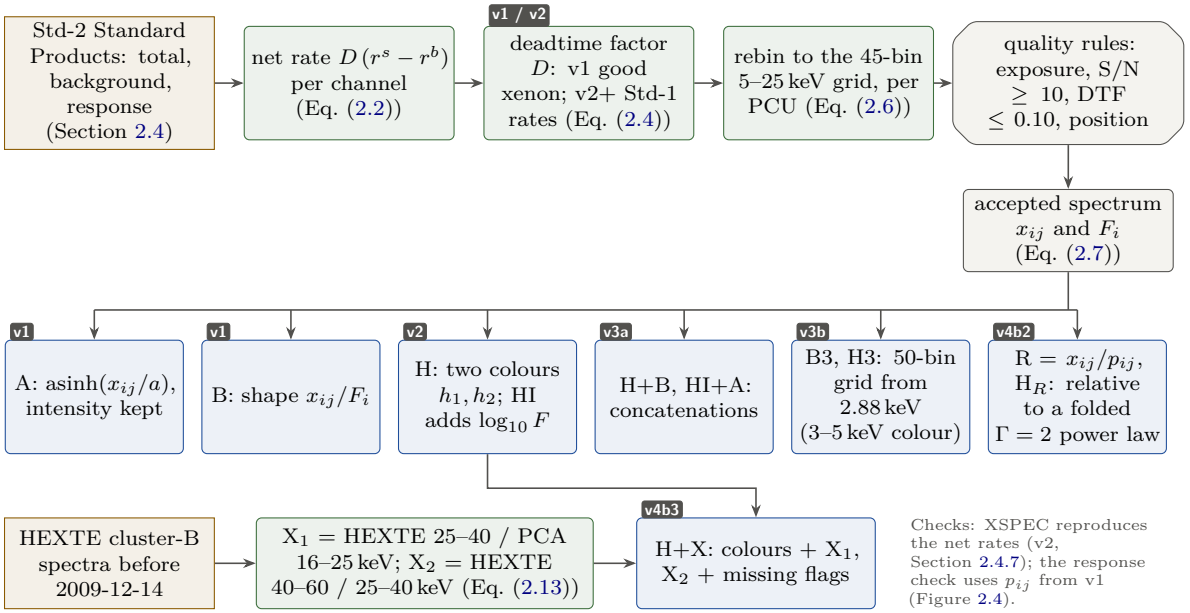


Figure 2.2: RXTE/PCA spectral pipeline. Top row: processing of every observation (the quality rules decide whether the next candidate of the same time group is tried, Algorithm 1). Middle: the representations built from the accepted spectrum, labelled with the version that introduced them; every model of Phases 1–4 uses one of them. Bottom: the HEXTE colours added in v4b3. Drawn for this report from Sections 2.4 to 2.6.

2.4.1 What the files contain

The Standard Products of every ObsID contain a total (source + background) spectrum `xp<obs>_s2.pha`, the `pcabackest` model background spectrum `xp<obs>_b2.pha` and the combined response `xp<obs>.rsp`. The header inspection of `05_two_source_check.py` (`reports/step2_fits_check.md`) established: `HDUCLAS2/3 = TOTAL/COUNT`, column `COUNTS` in counts, `POISSERR = F` with explicit `STAT_ERR`, `SYS_ERR = 0`, `BACKSCAL = AREASCAL = 1`, 129 channels, equal source and background exposures, and the PCUs summed listed in the `ROWID`n keywords.

The background statistical error is $\sqrt{\text{model counts}}$; the systematic uncertainty of the background model is *not* included anywhere. Standard Products are *not* deadtime corrected. Figure 2.3 shows the two spectra used for this check.

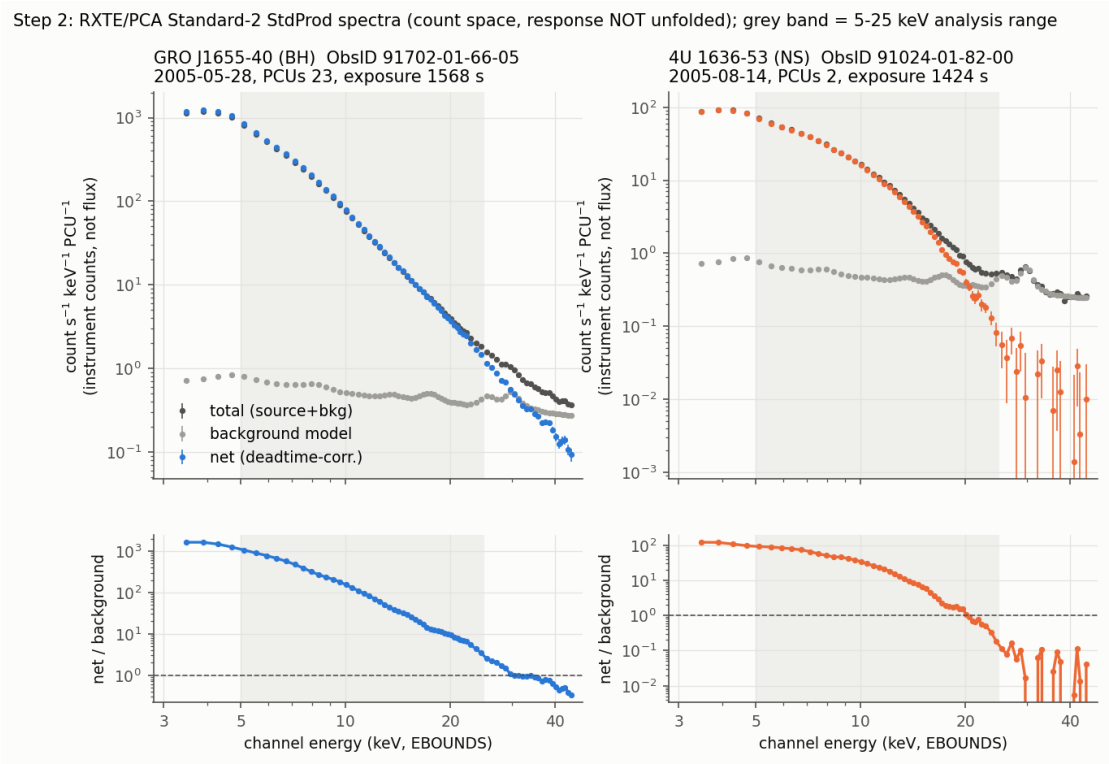


Figure 2.3: Data-flow validation on one BH (GRO J1655−40, ObsID 91702-01-66-05) and one NS (4U 1636−53, ObsID 91024-01-82-00). Top: total (dark grey), background model (light grey) and deadtime-corrected net (colour) count rates per keV per PCU against channel energy (EBOUNDS), log–log; the grey band is the 5–25 keV analysis range. Bottom: net/background ratio. The point is that the net spectrum is background-dominated only above ∼30 keV for these bright sources, and that everything is in instrument count space (no response unfolding). Repo: `figures/step2_two_sources.png`, script `scripts/05_two_source_check.py`.

2.4.2 Background subtraction

For channel c of observation i, with source counts C_c^s and errors σ_c^s, background counts C_c^b and errors σ_c^b (function `spectra.net_spectrum`):

$$t_s = \texttt{EXPOSURE}_s\,\texttt{AREASCAL}_s, \qquad t_b = \texttt{EXPOSURE}_b\,\texttt{AREASCAL}_b, \qquad \kappa = \texttt{BACKSCAL}_s/\texttt{BACKSCAL}_b,$$

$$r_c^s = C_c^s/t_s, \quad e_c^s = \sigma_c^s/t_s, \qquad r_c^b = \kappa\, C_c^b/t_b, \quad e_c^b = \kappa\, \sigma_c^b/t_b, \tag{2.1}$$

$$r_c^{\text{net}} = D\,(r_c^s - r_c^b), \qquad e_c^{\text{net}} = D\,\sqrt{(e_c^s)^2 + (e_c^b)^2}, \tag{2.2}$$

in count s^{-1} (summed over the PCUs of the spectrum), where $D = 1/(1 - \text{DTF})$ is the deadtime correction factor (Section 2.4.3). This is the OGIP/XSPEC convention: the same quantity that XSPEC uses after `data`, `backgrnd` and `response`. Non-positive net rates (possible where the source is faint, mainly at $E > 20$ keV in soft spectra) are kept unchanged: no absolute value, no clipping, no deletion.

2.4.3 Deadtime

v1 (lower bound). Only the good-xenon term was available from Standard-2:

$$\mathrm{DTF}^{(\mathrm{v1})} = 10^{-5}\,\mathrm{s} \times \frac{\sum_c C_c^s}{t_s\, N_{\mathrm{PCU}}}, \tag{2.3}$$

i.e. the total (source + background) count rate per PCU times 10 μs. Median 0.31% and maximum 6% in the v1 sample (`data/observations.csv`, column `dtf_xe`).

v2 onwards (RXTE GOF recipe from Standard-1). The Standard-1 files of the ObsID are read and the mean rates inside the spectrum's good time intervals (GTIs) are computed from all 0.125 s bins whose centre lies in a GTI (`spectra.std1_rates`): per-PCU good-xenon rates Xe_p ($p = 0, \ldots, 4$) and array rates Vp (propane), Rem (coincident events) and VLE (very large events). With N_{on} the number of PCUs with $\mathrm{Xe}_p > 1\,\mathrm{c\,s^{-1}}$ (`spectra.std1_dtf`):

$$\begin{aligned} \mathrm{DTF}_p &= 10^{-5}\left(\mathrm{Xe}_p + \frac{\mathrm{Vp} + \mathrm{Rem}}{N_{\mathrm{on}}}\right) + 6 \times 10^{-5}\,\frac{\mathrm{VLE}}{N_{\mathrm{on}}}, \\ \mathrm{DTF} &= \frac{1}{|\mathcal{P}|}\sum_{p \in \mathcal{P}} \mathrm{DTF}_p, \qquad D = \frac{1}{1 - \mathrm{DTF}}, \end{aligned} \tag{2.4}$$

with rates in count $\mathrm{s^{-1}}$ and $\mathcal{P}$ the PCUs of the spectrum. A version with 1.5×10^{-4} s per VLE event (quoted elsewhere on the same GOF page) was stored only as a sensitivity number (`dtf_std1_vle150us`). In the v2 sample the Standard-1 coverage of the spectrum GTIs has median 99% (minimum 88%), the DTF has median 1.38% and maximum 8.8%; no observation exceeded the 10% rule. Because D multiplies the whole spectrum, it changes intensities (representation A) by 1–9% and leaves shapes (B, H) unchanged.

2.4.4 Common energy grid and rebinning

Gain drifts change the channel-to-energy relation slightly between observations, so every spectrum is rebinned onto one fixed grid defined by a reference observation (`REFERENCE_OBS = 91702-01-66-05`). Let $\{E_k^{\mathrm{lo}}, E_k^{\mathrm{hi}}\}$ be the EBOUNDS of the reference response. The grid edges are the union of all EBOUNDS edges between the edge closest to 5 keV and the edge closest to 25 keV: $e_0 = 4.908$, …, $e_{45} = 25.008$ keV (45 bins, widths 0.41–0.86 keV; Table 2.4). For v3b the lower limit is the edge closest to 3 keV, giving 50 bins from 2.883 keV; the 45 bins above 4.908 keV are identical. For an observation with its own EBOUNDS, the overlap matrix (`spectra.overlap_matrix`)

$$W_{jk} = \frac{\max\!\left(0,\ \min(E_k^{\mathrm{hi}}, e_{j+1}) - \max(E_k^{\mathrm{lo}}, e_j)\right)}{E_k^{\mathrm{hi}} - E_k^{\mathrm{lo}}} \tag{2.5}$$

is the fraction of channel k that lies in grid bin j (counts assumed uniform within a channel). The rebinned net rate per keV per PCU and its error are

$$x_{ij} = \frac{\sum_k W_{jk}\, r_{ik}^{\mathrm{net}}}{N_{\mathrm{PCU},i}\,\Delta e_j}, \qquad \sigma_{ij} = \frac{\sqrt{\sum_k W_{jk}^2\,(e_{ik}^{\mathrm{net}})^2}}{N_{\mathrm{PCU},i}\,\Delta e_j}, \qquad \Delta e_j = e_{j+1} - e_j, \tag{2.6}$$

in count $\mathrm{s^{-1}\,keV^{-1}\,PCU^{-1}}$ (array `rate` and `err` of `features.npz`). Division by the number of PCUs is justified because the effective area grows linearly with the number of PCUs ($\approx 1070\,\mathrm{cm^2}$ per PCU at 6 keV). The error formula treats channels as independent; two grid bins that share one channel are therefore weakly correlated, which is ignored. The total (source + background) and background spectra are rebinned in the same way (`srate`, `brate`). The gain drift within epoch 5 is at most 0.85 channel (median 0.17) relative to the reference.

Table 2.4: The 45-bin common energy grid (keV), from the EBOUNDS of observation 91702-01-66-05 (`data/v2/processed/features.npz`, key `edges`; identical in every RXTE sample). Bin j spans $[e_j, e_{j+1})$. The v3b grid adds five bins below: 2.883, 3.287, 3.691, 4.096, 4.502, 4.908 keV.

j	e_j	j	e_j	j	e_j	j	e_j	j	e_j
0	4.908	10	9.001	20	13.149	30	17.346	40	21.588
1	5.315	11	9.414	21	13.566	31	17.768	41	22.014
2	5.722	12	9.827	22	13.985	32	18.191	42	22.441
3	6.130	13	10.240	23	14.403	33	18.614	43	23.295
4	6.538	14	10.654	24	14.822	34	19.038	44	24.151
5	6.948	15	11.069	25	15.242	35	19.462	45	25.008
6	7.357	16	11.484	26	15.662	36	19.886		
7	7.767	17	11.899	27	16.082	37	20.311		
8	8.178	18	12.315	28	16.503	38	20.736		
9	8.589	19	12.732	29	16.924	39	21.162		

2.4.5 Summary quantities, errors and quality rules

For each observation (`04_fetch_process.py`):

$$F_i = \sum_{j=0}^{44} x_{ij}\Delta e_j \quad [\mathrm{count\,s^{-1}\,PCU^{-1}}], \qquad \sigma_{F,i} = \Big(\sum_j (\sigma_{ij}\Delta e_j)^2\Big)^{1/2}, \qquad \mathrm{S/N}_i = F_i/\sigma_{F,i}, \tag{2.7}$$

$$f_i^{\mathrm{bkg}} = \frac{\sum_j x_{ij}^b \Delta e_j}{\sum_j x_{ij}^s \Delta e_j} \quad \text{(background fraction, 5–25 keV)}.$$

An observation is accepted only if all of the following hold (otherwise the next candidate of the same time group is tried): all files present and readable, 129 channels, EBOUNDS cover 5–25 keV; spectrum exposure ≥ 1000 s; S/N ≥ 10; DTF ≤ 0.10; header position `RA_OBJ/DEC_OBJ` within 0.1° of the source position; all values finite. $F_i > 0$ is guaranteed by the S/N rule, so the shape normalisation (Eq. (2.10)) is always defined.

2.4.6 Response check and response-normalised spectra

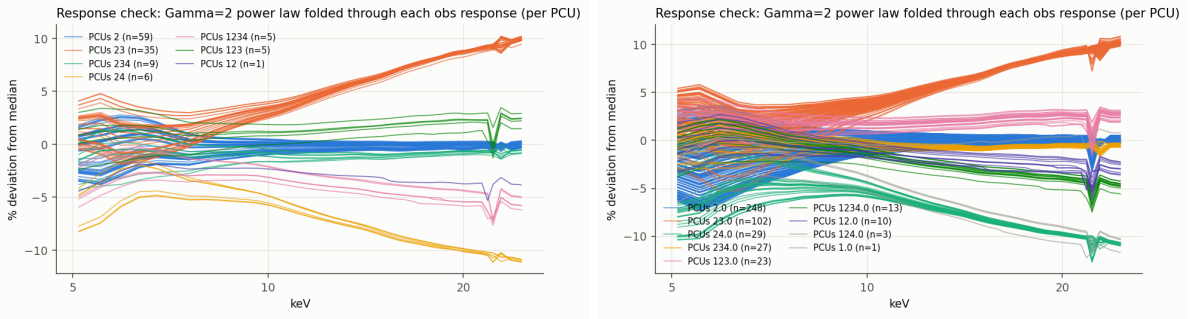
A fixed photon spectrum $N(E) = E^{-2}$ (photon index $\Gamma = 2$, no absorption, normalised to $1\,\mathrm{ph\,cm^{-2}\,s^{-1}\,keV^{-1}}$ at 1 keV) is folded through each observation's response matrix $R(E, c)$ (`spectra.powerlaw_folded`) and rebinned like the data:

$$P_{ic} = \sum_\ell R_i(E_\ell, c) \int_{E_\ell^{\mathrm{lo}}}^{E_\ell^{\mathrm{hi}}} E^{-2}\,dE, \qquad p_{ij} = \frac{\sum_c W_{jc} P_{ic}}{N_{\mathrm{PCU},i}\,\Delta e_j} \quad \text{(key } \texttt{pl2}\text{)}. \tag{2.8}$$

The across-observation scatter of $p_{ij}/\,\mathrm{median}_i\, p_{ij}$ measures how much the instrument response alone changes the count spectrum: median 3.7%, maximum 11–12.7%, driven by the PCU combination (Figure 2.4). BH and NS have similar PCU-combination mixtures (mean tilt 2.0% vs 2.4% in v1), so this is treated as noise rather than as a class shortcut. From v4b2 on, p_{ij} also defines the response-normalised representations R and H_R (Section 2.5), which can be compared across gain epochs.

2.4.7 Independent check with XSPEC

For 12 randomly chosen v2 observations (seed 42; 10 sources; net rates 5–950 $\mathrm{c\,s^{-1}}$), HEASoft 6.37.1/XSPEC (`data`, `backgrnd`, `response`, `ignore **-5.0 25.0-**`, `tclout rate`) and



(a) v1 (120 observations) **(b)** v2 (456 observations)

Figure 2.4: Response check: a fixed $\Gamma = 2$ power law folded through each observation's response, per PCU, shown as percentage deviation from the median over observations, against energy; colour = PCU combination. The spread (a few per cent, up to ∼10% at 25 keV) is set by which PCUs were on, not by the source class. Repo: `figures/step3_response_variation.png`, `figures/v2/step3_response_variation.png`; script `scripts/06_dataset_checks.py`.

Eq. (2.2) with $D = 1$ summed over the same noticed channels (10–52 or 10–53) agree to 3×10^{-10} (rate) and 4×10^{-10} (error) relative (`results/v2/xspec_crosscheck.csv`, `scripts/10_xspec_crosscheck.py`). This validates Eqs. (2.1) and (2.2); it does not validate the deadtime, the rebinning, or the background model itself.

2.5 Spectral representations

All representations are computed per observation from the rebinned spectrum x_i.; none uses information from other observations (the training-set standardisation of Section 4.1.1 is a separate step inside the model). Code: `07_train_eval.py`, `v3lib.v2_representations`, `12_v3b_extended_band.py`, `reps_R` in `22_v4b2_analysis.py`.

A (intensity kept). With scale $a = 0.1\,\mathrm{count\,s^{-1}\,keV^{-1}\,PCU^{-1}}$ (`ASINH_SCALE`),

$$A_{ij} = \operatorname{asinh}(x_{ij}/a)\,, \qquad j = 0, \ldots, 44. \tag{2.9}$$

$\operatorname{asinh}(u) \approx \operatorname{sgn}(u)\ln(2|u|)$ for $|u| \gg 1$ and $\approx u$ near 0, so A is log-like for bright bins and defined for the (rare) negative bins. A keeps the absolute count rate, which depends on distance, absorption and accretion rate as well as on the compact object.

B (shape).

$$B_{ij} = x_{ij}/F_i \quad [\mathrm{keV}^{-1}], \qquad \textstyle\sum_j B_{ij}\Delta e_j = 1. \tag{2.10}$$

H (two count-space colours). Four bands are formed from the grid bins whose geometric centre $\bar{e}_j = \sqrt{e_j e_{j+1}}$ lies in a nominal interval (`v3lib.band`); the exact bin ranges are in Table 2.5:

$$C_{i,m} = \sum_{j \in b_m} x_{ij}\Delta e_j \quad [\mathrm{count\,s^{-1}\,PCU^{-1}}], \qquad \boldsymbol{h}_i = (h_{i1}, h_{i2}) = \left(\frac{C_{i,2}}{C_{i,1}},\ \frac{C_{i,4}}{C_{i,3}}\right). \tag{2.11}$$

h_1 ("soft colour", ≈7–10 over 5–7 keV) and h_2 ("hard colour", ≈16–25 over 10–16 keV) are linear ratios. The pre-registration originally specified $\log_{10}$ colours; before any v2 model was trained, two BH spectra (95409-01-29-00 GX 339−4, 60113-01-34-00 XTE J1650−500) were

found with a slightly negative 16–25 keV net sum, so the linear ratio was adopted (logged in `reports/decision_log.md`). Denominators C_1, C_3 are positive for every observation (asserted in the code); numerators may be ≤ 0 and are kept.

HI, H+B, HI+A. $\mathrm{HI}_i = (h_{i1}, h_{i2}, \log_{10} F_i)$; $\mathrm{H{+}B}_i = (\boldsymbol{h}_i, B_{i0}, \ldots, B_{i44})$ (47 features); $\mathrm{HI{+}A}_i = (\mathrm{HI}_i, A_{i0}, \ldots, A_{i44})$ (48 features).

v3b (3–25 keV). On the 50-bin grid: $F_i^{(3)} = \sum_j x_{ij}\Delta e_j$ over all 50 bins; $\mathrm{B3}_{ij} = x_{ij}/F_i^{(3)}$; $\mathrm{H3}_i = (C_{i,0}/C_{i,1},\ C_{i,2}/C_{i,1},\ C_{i,4}/C_{i,3})$ where $C_{i,0}$ sums the five bins with centre in $[2.8, 5)$ keV (2.883–4.908 keV). The soft colour has 3–5 keV in the numerator and a denominator that is always positive.

R and H_R (response-normalised; v4b2 onwards). With p_{ij} of Eq. (2.8):

$$R_{ij} = \frac{x_{ij}}{p_{ij}}, \qquad P_{i,m} = \sum_{j \in b_m} p_{ij}\Delta e_j, \qquad \boldsymbol{h}_i^R = \left(\frac{C_{i,2}/P_{i,2}}{C_{i,1}/P_{i,1}},\ \frac{C_{i,4}/P_{i,4}}{C_{i,3}/P_{i,3}} \right). \tag{2.12}$$

H_R is the colour of the data relative to the colour a $\Gamma = 2$ power law would have through the same response; it removes first-order response differences (e.g. between gain epochs). Its predictions agree with H at Spearman 0.994 (Chapter 7).

Table 2.5: The four colour bands of H (and H_R): grid bins whose geometric centre lies in the nominal interval, and the actual energy range they cover (computed from `edges`).

Band	nominal interval (centre rule)	bins j	n bins	actual range [keV]
b_1	$[5, 7)$	0–4	5	4.908–6.948
b_2	$[7, 10)$	5–11	7	6.948–9.827
b_3	$[10, 16)$	12–26	15	9.827–16.082
b_4	$[16, 25.1)$	27–44	18	16.082–25.008
b_0 (v3b)	$[2.8, 5)$	5 bins of the 50-bin grid	5	2.883–4.908

Figure 2.5 shows the class-averaged spectra in the A, B and H representations.

2.6 HEXTE high-energy colours (v4b3)

Only HEXTE cluster B is used (`19_v4b3_hexte.py`), because it rocked between source and background until 2009-12-14 16:10 UT (MJD 55179.6736); observations on or after that date, or without cluster-B files, have no HEXTE features (60 of the 456 v2 observations). Header keyword `DEADAPP = T` (deadtime already applied) is required. The net cluster-B spectrum follows Eqs. (2.1) and (2.2) with $D = 1$; band rates $F_X(25\text{–}40)$ and $F_X(40\text{–}60)$ (count s^{-1} per cluster) use the overlap matrix of Eq. (2.5) on the RMF EBOUNDS. With $F_{\mathrm{PCA}}(16\text{–}25) = C_{i,4}$ of Eq. (2.11):

$$X_1 = \frac{F_X(25\text{–}40)}{F_{\mathrm{PCA}}(16\text{–}25)}, \qquad X_2 = \frac{F_X(40\text{–}60)}{F_X(25\text{–}40)}. \tag{2.13}$$

X_1 is a cross-instrument count ratio and is only used as a feature. A colour is set to missing when its denominator has S/N < 3 (X_1: 3 observations; X_2: 144, of which 23 BH). Missing colours are filled with the training-fold median, and one 0/1 indicator per colour is appended (inside the LOSO loop; `hx_features`); H+X therefore has 6 columns.

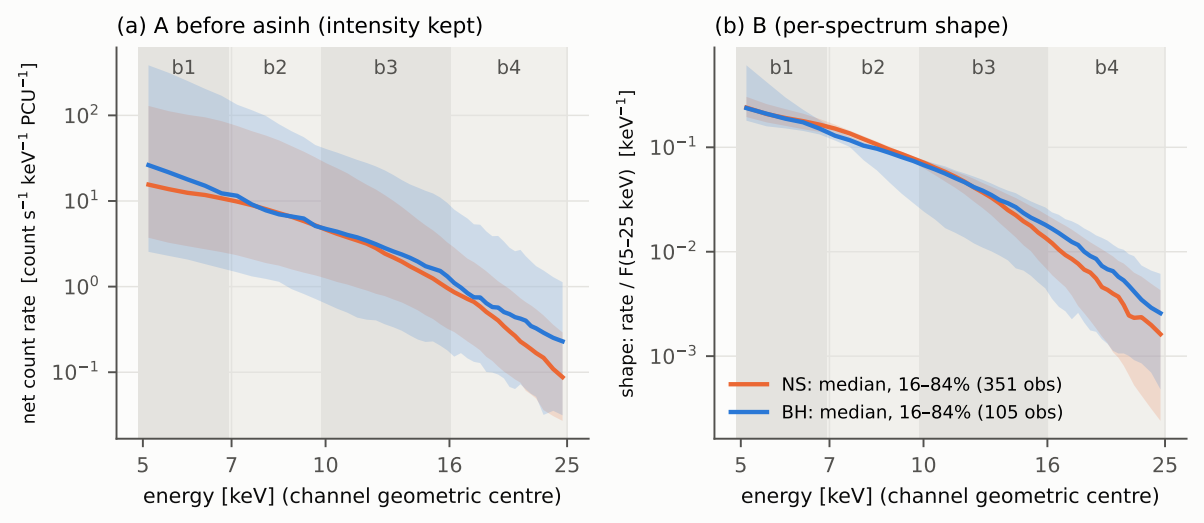


Figure 2.5: Median (line) and 16–84% range (shading) of the v2 count-space spectra, BH (blue, 105 observations) and NS (orange, 351 observations). (a) net rate per keV per PCU (the quantity transformed by asinh in representation A); (b) per-spectrum shape B. Grey and light bands mark the four colour bands b_1–b_4 of Table 2.5. Non-positive bins are omitted from the percentiles. The class medians differ mainly in the overall slope above 10 keV and overlap strongly, which is why two colours capture most of the shape information (Chapter 6). Generated for this report from `data/v2/processed/features.npz` (`report_en/figures/generated/make_figures.py`).

2.7 Type-I X-ray bursts

Type-I bursts are seconds-long thermonuclear flashes that only NSs produce; an observation containing one carries a label shortcut and a distorted spectrum. Three procedures are used.

Figure 2.6 summarises them.

Standard-1 burst detector (v4; `17_v4_burst_detect.py`). The 1 s light curve $r(t)$ is the sum of good-xenon counts of the spectrum's PCUs over complete 1 s bins inside the GTIs. With μ, s the mean and standard deviation of all bins: candidates are bins with $r > \mu + 4s$; candidates closer than 30 s are merged into one event. For an event with peak $r_{\rm pk}$ at $t_{\rm pk}$ and persistent level r_0 (median of the 60 s before the event, or of the whole curve if fewer than 60 bins), amplitude $A = r_{\rm pk} - r_0$ and levels $L_q = r_0 + qA$:

$$\text{burst} \iff r_{\rm pk} \geq 1.5\, r_0 \ \wedge\ t_{\rm rise} \leq 10\,\text{s} \ \wedge\ 3 \leq t_{1/2} \leq 300\,\text{s} \ \wedge\ n_{25} \geq 3, \tag{2.14}$$

where $t_{\rm rise} = t_{\rm pk} - \max\{t < t_{\rm pk} : r(t) < L_{0.25}\}$, $t_{1/2} = \min\{t > t_{\rm pk} : r(t) < L_{0.5}\} - t_{\rm pk}$, and n_{25} is the number of contiguous 1 s bins around the peak above $L_{0.25}$. The burst interval runs from the last time before the peak below $L_{0.25}$ to the first time after it below $L_{0.1}$. Thresholds are `V4_BURST` in `config.py`, following the search criterion of Galloway et al. (2008). Every candidate was also inspected by eye (`results/v4/burst_visual.csv`).

MINBAR matching (v7b1; `v7lib.load_minbar`, `gti_utc`, `minbar_flag`). The spectrum GTIs (`STDGTI`, mission elapsed time in TT) are converted to MJD(TT) = `MJDREF` + t/86400 and then to UTC with `astropy` (TT−UTC = 64.18 s; round-trip error < 1 ms). A MINBAR burst occupies $[T, T + \text{dur}]$ (dur missing ⇒ 300 s). Primary rule: an observation is contaminated if a burst interval overlaps a GTI *and* the burst has the same ObsID or originates within 1° (PCA field of view) of the source. The literal pre-registered rule (any burst from any origin) is kept as a sensitivity set.

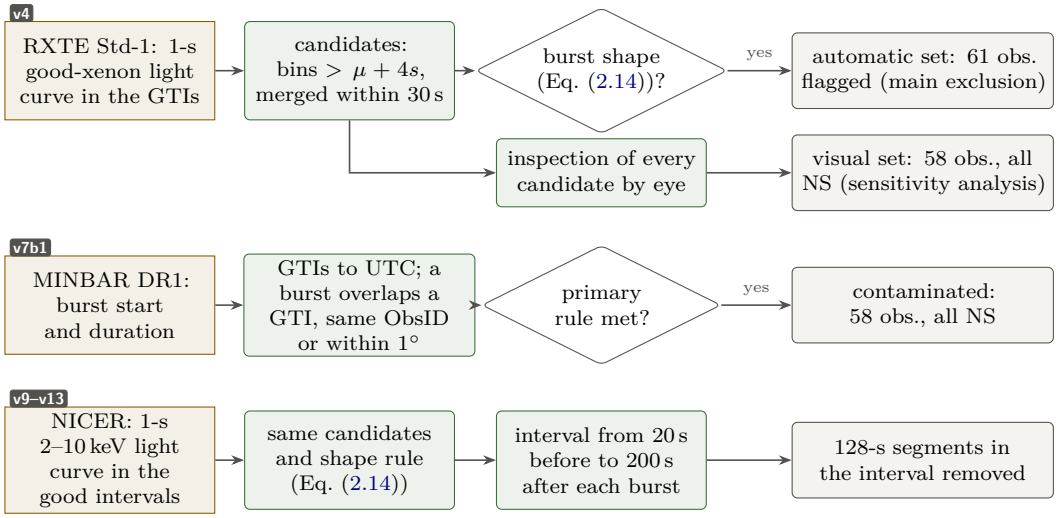


Figure 2.6: The three burst-screening procedures and the version that added each: the Standard-1 detector of v4 (automatic set used for the main exclusion, visual set as sensitivity analysis), the MINBAR catalogue match of v7b1 (primary rule shown; the literal pre-registered rule, any burst of any origin, is a sensitivity set), and the NICER rule of v9–v13, which removes segments rather than observations. Counts refer to the 456 v2 observations (Sections 7.3.1 and 9.6.1). Drawn for this report.

NICER burst rule (v9–v13; `v9lib.burst_events`). The same Eq. (2.14) is applied to the 2–10 keV 1 s light curve built from events inside the good intervals (Section 2.10); every 128 s segment that overlaps $[t_{\rm start} - 20\,{\rm s},\, t_{\rm end} + 200\,{\rm s}]$ of an automatically detected burst is removed.

2.8 Low-frequency timing from RXTE Standard-1 (v5–v10)

Figure 2.7 summarises the chain from light curve to features; this section gives the formulae.

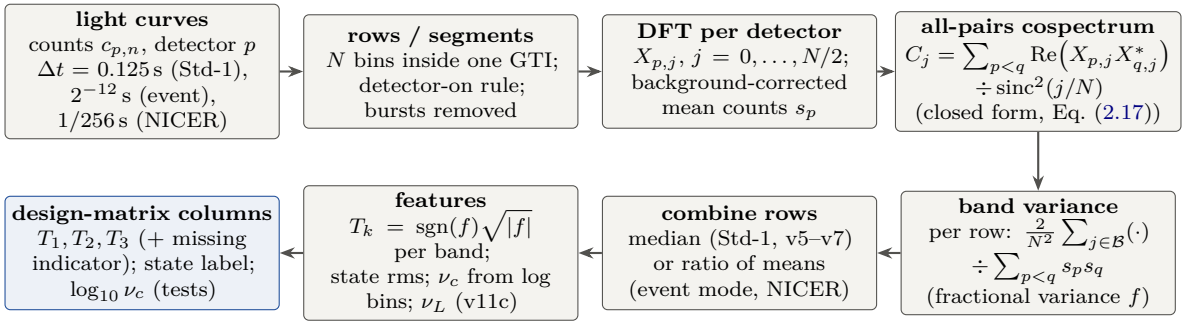


Figure 2.7: Timing-feature pipeline shared by RXTE Standard-1 (v5–v10), RXTE event mode (v8c) and NICER (v9–v13). Differences between the instruments are in the bin width, the segment length, the "detector" (PCU or NICER MPU), whether rows are combined by the median or by the ratio of means, and the frequency bands (Table 2.6). Drawn for this report.

2.8.1 Rows, detectors and background

A Standard-1 file stores, for each 128 s row, $N = 1024$ counts per PCU in $\Delta t = 0.125$ s bins. Only rows lying completely inside a spectrum GTI are used (start $\geq$ GTI start -1 ms and end $\leq$ GTI stop $+1$ ms; `v5lib.std1_rows`). PCU p is *on* in a row if each of the eight 128-bin blocks (16 s) has at least one count (`v5lib.pcus_on`). The background count rate per PCU is taken from the Standard-2 background spectrum over all channels, $b = \sum_c C^b_c / (t_b N_{{\rm PCU},b})$ count s^{-1}

(assumed equal for all PCUs). For a detector group or a single PCU with counts c_n and $n_{\rm det}$ detectors, the mean *source* counts per bin is $s = \bar{c} - n_{\rm det}\, b\, \Delta t$; a row is invalid if any $s \le 0$.

2.8.2 Fourier conventions and the binning correction

The discrete Fourier transform is $X_j = \sum_{n=0}^{N-1} c_n e^{-2\pi i j n/N}$, $j = 0, \ldots, N/2$ (`numpy.fft.rfft`), frequency $\nu_j = j/(N\Delta t)$ (for 128 s rows, $\nu_j = j/128$ Hz). Parseval's relation gives $\frac{1}{N}\sum_n (c_n - \bar{c})^2 = \frac{1}{N^2}\sum_{j\neq 0} |X_j|^2 \approx \frac{2}{N^2}\sum_{j=1}^{N/2} |X_j|^2$, so $\frac{2}{N^2}\sum_{j\in\mathcal{B}} |X_j|^2$ is the variance (in counts^2 per bin^2) contributed by band $\mathcal{B}$. Integrating counts over bins of width Δt multiplies the Fourier amplitude by $\mathrm{sinc}(j/N) = \sin(\pi j/N)/(\pi j/N)$ (van der Klis, 1989, Eq. 2.19); every power is divided by $\mathrm{sinc}^2(j/N)$.

2.8.3 Two-group cospectrum (v5)

The PCUs that are on are sorted; those in even positions form group A, those in odd positions group B (at least two PCUs are required). With x_n, y_n the summed counts of the groups, X_j, Y_j their transforms, and s_x, s_y their background-corrected mean source counts per bin (`v5lib.row_features`):

$$V_{\mathcal{B}} = \frac{2}{N^2}\sum_{j\in\mathcal{B}} \frac{\mathrm{Re}(X_j Y_j^*)}{\mathrm{sinc}^2(j/N)}, \qquad f_{\mathcal{B}} = \frac{V_{\mathcal{B}}}{s_x s_y}. \tag{2.15}$$

Because the Poisson (and deadtime) noise of different detectors is uncorrelated, the expected value of the cospectrum $\mathrm{Re}(X_j Y_j^*)$ in the absence of source variability is zero (Bachetti and Huppenkothen, 2018; Bachetti et al., 2015); no Poisson level has to be modelled. $f_{\mathcal{B}}$ is the fractional variance (rms^2 normalised to the source mean) in band $\mathcal{B}$. Per observation, the *median* over valid rows is taken (robust to rare rows with flares or bursts); fewer than 3 valid rows ⇒ missing. Features are signed square roots,

$$\begin{aligned} T_k &= \mathrm{sgn}(\mathrm{median}_r\, f_{k,r})\sqrt{|\mathrm{median}_r\, f_{k,r}|}, \qquad k \in \{1,2,3\}, \\ T_{\rm tot} &= \mathrm{sgn}(F)\sqrt{|F|}, \qquad F = \textstyle\sum_k \mathrm{median}_r\, f_{k,r}, \end{aligned} \tag{2.16}$$

(fractional rms, which may be slightly negative because of noise). The bands are listed in Table 2.6. As an implementation check the conventional auto-power estimator $(|Z_j|^2 - N\bar{z})/\,\mathrm{sinc}^2(j/N)$ of $z = x + y$ (Poisson level $N\bar{z}$, no deadtime) was also computed; for 276 S1 observations with 20–500 source c s^{-1} PCU^{-1} the two versions of T_2 agree at Spearman 0.970.

2.8.4 All-pairs estimator and single-PCU hybrid (v6)

Using every pair of the n PCUs that are on, rather than one A/B split, reduces the noise (`v6lib.allpairs_row`). With per-PCU transforms $X_{p,j}$ and source means s_p:

$$\sum_{p<q} \mathrm{Re}(X_{p,j} X_{q,j}^*) = \frac{\left|\sum_p X_{p,j}\right|^2 - \sum_p |X_{p,j}|^2}{2}, \qquad f_{\mathcal{B}} = \frac{\frac{2}{N^2}\sum_{j\in\mathcal{B}} \frac{\left|\sum_p X_{p,j}\right|^2 - \sum_p |X_{p,j}|^2}{2\,\mathrm{sinc}^2(j/N)}}{\frac{(\sum_p s_p)^2 - \sum_p s_p^2}{2}}. \tag{2.17}$$

Rows with a single PCU on use the auto-power estimator $f_{\mathcal{B}} = \frac{2}{N^2}\sum_{j\in\mathcal{B}} \frac{|X_j|^2 - N\bar{c}}{\mathrm{sinc}^2(j/N)}/s^2$, but only if the total count rate is ≤ 500 c s^{-1} (where the deadtime-free Poisson level is adequate); otherwise the row is invalid. T_k^* is the signed square root of the median over all valid rows (pair rows and single rows together). In synthetic tests the noise variance drops by 31%, 33% and 41% for 3, 4 and 5 PCUs (theory 33%, 33%, 40%) without bias; missing values over 8 350 observations fall from 942 to 341.

2.8.5 Characteristic frequency ν_c (v6)

Nine logarithmic Fourier-index bins are formed from unique(round(geomspace$(1, 512, 10)$)) $= \{1, 2, 4, 8, \ldots, 512\}$ as $[1, 1], [2, 3], [4, 7], \ldots, [256, 511]$ (Table 2.7). For each bin b, P_b is the median over rows of the fractional variance (Eq. (2.17)) in that bin; the bin centre is $\nu_b = \sqrt{j_{\rm lo} j_{\rm hi}}/128\,\mathrm{Hz}$ and its width in $\ln \nu$ is $\Delta_b = \ln((j_{\rm hi}+0.5)/(j_{\rm lo}-0.5))$. The νP_ν centroid is (`v6lib.nu_centroid`)

$$\ln \nu_c = \frac{\sum_b w_b \ln \nu_b}{\sum_b w_b}, \qquad w_b = \frac{\max(P_b, 0)}{\Delta_b}, \tag{2.18}$$

missing if $T^*_{\rm tot} < 0.05$ or $\sum_b w_b \le 0$. Since P_b/Δ_b is the power per unit $\ln \nu$ (i.e. νP_ν), ν_c is the frequency around which most of the variability power per logarithmic frequency interval lies: a model-free proxy for a break or characteristic frequency. Synthetic sinusoids at 0.05, 0.5 and 3 Hz are recovered within one log bin.

Table 2.6: Fourier-index bands and their frequencies. Standard-1 and NICER use 128 s rows or segments ($\nu_j = j/128\,\mathrm{Hz}$); RXTE event mode uses 16 s segments ($\nu_j = j/16\,\mathrm{Hz}$). From `config.py`.

Band	index j	frequency [Hz]	data	config name
T_1	2–12	0.0156–0.094	Std-1, NICER	`V5_BANDS_J`, `V9_T_BANDS_J`
T_2	13–128	0.102–1.000	Std-1, NICER	
T_3	129–511	1.008–3.992	Std-1, NICER	
state (v7a)	13–511	0.102–3.992	Std-1	`V7_RMS_J`
state (v9+)	13–1280	0.102–10.0	NICER	`V9_STATE_J`
white monitor	12288–16383	96–128	NICER	`V9_WHITE_J`
LC	16–63	1.0–3.94	event mode	`V8_HF_BANDS_J`
HF1	64–1023	4.0–63.9	event mode	
HF2	1024–8191	64–512	event mode	
HF3	8192–16383	512–1024	event mode	
NULL	24576–32767	1536–2048	event mode	

Table 2.7: Bins of the νP_ν centroid. Left: RXTE Standard-1 (v6; 9 bins, ν_b and Δ_b of Eq. (2.18)). Right: NICER (v9–v13; 13 octave bins $j \in [2^k, 2^{k+1}-1]$, centre $2^{k+0.5}/128\,\mathrm{Hz}$). Computed from the code (`v6lib.py`, `v9lib.py`).

Standard-1 (v6)				NICER (v9+)		
$[j_{\rm lo}, j_{\rm hi}]$	ν_b [Hz]	range [Hz]	Δ_b	k	$[j_{\rm lo}, j_{\rm hi}]$	ν_k [Hz]
$[1, 1]$	0.0078	0.0039–0.0117	1.099	0	$[1, 1]$	0.011
$[2, 3]$	0.0191	0.0117–0.0273	0.847	1	$[2, 3]$	0.022
$[4, 7]$	0.0413	0.0273–0.0586	0.762	2	$[4, 7]$	0.044
$[8, 15]$	0.0856	0.0586–0.121	0.726	3	$[8, 15]$	0.088
$[16, 31]$	0.174	0.121–0.246	0.709	4	$[16, 31]$	0.177
$[32, 63]$	0.351	0.246–0.496	0.701	5	$[32, 63]$	0.354
$[64, 127]$	0.704	0.496–0.996	0.697	6	$[64, 127]$	0.707
$[128, 255]$	1.411	0.996–1.996	0.695	7	$[128, 255]$	1.414
$[256, 511]$	2.826	1.996–3.996	0.694	8	$[256, 511]$	2.828
				9–12	$[512, 1023]$–$[4096, 8191]$	5.66, 11.3, 22.6, 45.3

2.8.6 State index (v7a)

The variability state of an observation is classified by its fractional rms in 0.1–4 Hz ($j = 13$–511), computed with Eq. (2.17) from rows with ≥ 2 PCUs only (no single-PCU rows, because their Poisson level would need a deadtime model), after removing every row that overlaps a burst interval $[t_{\rm start} - 20\,{\rm s},\ {\rm end}]$ (MINBAR field-of-view rule and the v4 detector; `v7lib.state_rms`). With $\rho = \mathrm{sgn}(\mathrm{median}_r\, f)\sqrt{|\,\mathrm{median}_r\, f|}$:

$$\text{state} = \begin{cases} \text{hard-like} & \rho > 0.10 \\ \text{soft-like} & \rho < 0.075 \\ \text{intermediate} & \text{otherwise} \end{cases} \qquad \text{(unknown if fewer than 3 rows)}, \tag{2.19}$$

the thresholds of Remillard and McClintock (2006, Table 2), which are defined for 0.1–10 Hz; the 4 Hz Nyquist limit of Standard-1 makes ρ slightly low and the hard-like class conservative. NICER (v9 onwards) uses the full 0.1–10 Hz band. "Soft-like/hard-like" are operational variability classes, not full spectral-state assignments.

2.9 High-frequency timing from RXTE event mode (v8c)

An event file qualifies if `DATAMODE` matches `^E_\d+us_\d+M_0_\d+s$`, `TDDES2` contains the six anode chains `X1L^X1R^X2L^X2R^X3L^X3R`, `TIMEDEL` $\leq 2^{-12}$ s and a `PCUID` column exists (`v8lib.mode_qualifies`); if several modes qualify, the one with the largest total size is used. Events are binned at $\Delta t = 2^{-12}$ s (Nyquist 2048 Hz) in 16 s segments ($N = 65536$) lying completely in (spectrum GTI) $\cap$ (event-file GTI) and not overlapping a burst interval. A PCU is on if each of the 16 one-second sub-intervals has an event; ≥ 2 PCUs are required. For segment r, with Eq. (2.17), $\mathrm{num}_{\mathcal{B},r} = \frac{2}{N^2}\sum_{j\in\mathcal{B}} \mathrm{cross}_j$ and $\mathrm{den}_r = \sum_{p<q} s_p s_q$. The pre-registered per-observation estimator was the mean of the per-segment ratios; because single 16 s segments of faint sources give unstable ratios (one Aql X-1 segment gave -195), it was replaced, after the implementation check IC4 and before any model, by the ratio of means with a delta-method standard error (`v8lib.hf_summary`):

$$\hat{R}_{\mathcal{B}} = \frac{\sum_r \mathrm{num}_{\mathcal{B},r}}{\sum_r \mathrm{den}_r}, \qquad \widehat{\mathrm{SE}}(\hat{R}_{\mathcal{B}}) = \frac{1}{\sum_r \mathrm{den}_r}\sqrt{\frac{n}{n-1}\sum_r \left(\mathrm{num}_{\mathcal{B},r} - \hat{R}_{\mathcal{B}}\,\mathrm{den}_r\right)^2}, \tag{2.20}$$

with n segments; features are $\mathrm{sgn}(\hat{R})\sqrt{|\hat{R}|}$ for LC, HF1, HF2, HF3. Quality: $n \geq 10$ and $|z_{\rm NULL}| = |\hat{R}_{\rm NULL}/\widehat{\mathrm{SE}}| \leq 5$ in the empty 1536–2048 Hz band. A post hoc correction subtracts the null-band power scaled by bandwidth from every band ($\hat{R}_{\mathcal{B}} - \hat{R}_{\rm NULL}\,|\mathcal{B}|/|\mathrm{NULL}|$), because a white cross-PCU correlated component was found (Chapter 9).

2.10 NICER/XTI features (v9–v13)

Figure 2.8 shows the NICER chain and the version that added each step.

2.10.1 Reading the events

Cleaned event files are read without HEASoft by a streaming FITS parser (`v9lib.PrefixParser`) that decompresses the gzip stream and extracts `TIME`, `PI` and `DET_ID` from the binary table rows. Three reading modes were used: v9 downloaded only the *prefix* of each file (HTTP range

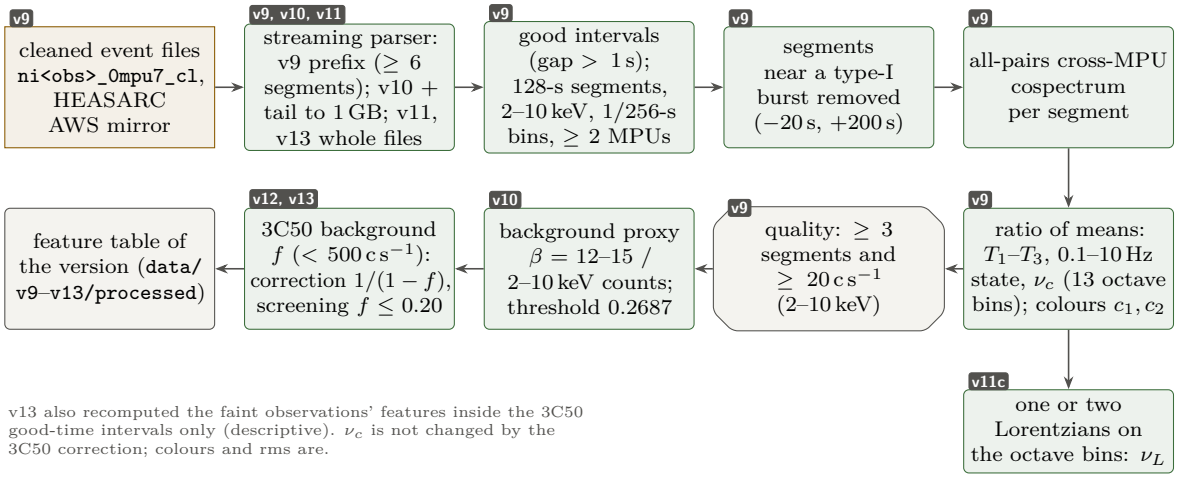


Figure 2.8: NICER/XTI feature pipeline (read the top row left to right, then the bottom row right to left). The v9 steps were frozen before the new observations of v11 were read, and the whole chain including the 3C50 step before those of v13; v10 added the full-length read and the background proxy, v11c the Lorentzian fits, v12 the 3C50 correction and screening. The colours are computed from the same segment windows as the timing features. Drawn for this report from Section 2.10 and `scripts/v9lib.py`–`v11lib.py`.

requests) until ≥ 6 gap-free 128 s segments, end of file, or 300 MB; v10 appended the *tail* up to 1 GB compressed and parsed prefix + tail as one stream; v11 and v13 *streamed* whole files (up to 1 GB) into memory without storing them. Parsing was checked to be identical to `astropy` and across the three modes (IC1 of v9–v11).

2.10.2 Good intervals, segments and detectors

Good intervals are maximal stretches without a gap > 1 s between consecutive events of any energy (`v9lib.good_intervals`). Each good interval $[a, b]$ is cut sequentially into 128 s segments starting at a. Only 2–10 keV events are used for timing (PI 200–999, PI in units of 10 eV). The detector unit is the MPU, $m = \lfloor \texttt{DET_ID}/10 \rfloor \in \{0, \ldots, 6\}$. Events are binned at $\Delta t = 1/256$ s ($N = 32768$, Nyquist 128 Hz). MPU m is on in a segment if each of the eight 16 s blocks has a 2–10 keV event; ≥ 2 MPUs are required. Segments overlapping a burst interval (Section 2.7) are removed.

2.10.3 Per-segment statistics and observation features

For each valid segment r (`v9lib.segment_stats`): s_m is the mean 2–10 keV counts per bin of MPU m (no background subtraction), the all-pairs cross-power is formed as in Eq. (2.17), and for each band $\mathrm{num}_{\mathcal{B},r} = \frac{2}{N^2} \sum_{j \in \mathcal{B}} \mathrm{cross}_j$, $\mathrm{den}_r = \sum_{m<m'} s_m s_{m'}$. Observation features use the ratio of means (Eq. (2.20)) for T_1, T_2, T_3, the state band 0.1–10 Hz and the 96–128 Hz white monitor, and the signed square root. In addition:

- **Colours** (`c1`, `c2`): counts of *all* events in the accepted segment windows in PI 200–399 (A, 2–4 keV), 400–599 (B, 4–6 keV) and 600–999 (C, 6–10 keV); $c_1 = B/A$, $c_2 = C/B$.
- **Count rate** `rate_2_10`: mean over segments of the 2–10 keV counts of the MPUs that are on, divided by 128 s.
- ν_c: 13 octave bins $k = 0, \ldots, 12$ with $j \in [2^k, 2^{k+1} - 1]$ (1/128–64 Hz, Table 2.7); $\ell_k = \sum_r \mathrm{num}_{k,r} / \sum_r \mathrm{den}_r$; $\nu_c = \exp(\sum_k w_k \ln \nu_k / \sum_k w_k)$ with $w_k = \max(\ell_k, 0)$ and

$\nu_k = 2^{k+0.5}/128\,\mathrm{Hz}$; missing if $\sqrt{\max(\sum_k \ell_k, 0)} < 0.05$.

- **State**: Eq. (2.19) applied to the 0.1–10 Hz rms.
- **Quality**: at least 3 valid segments and `rate_2_10` $\geq 20\,\mathrm{c\,s^{-1}}$ (background then $\lesssim 5\%$ by the pre-registered estimate).

2.10.4 Background proxy (v10)

$\beta_i = N(\mathrm{PI}\ 1200\text{–}1499)/N(\mathrm{PI}\ 200\text{–}999)$, the 12–15 keV to 2–10 keV event ratio of the whole file (`v10lib.bkg_ratio`); the source contributes little above 12 keV, so a high β indicates background. The threshold is the 99th percentile of β among observations with `rate_2_10` $\geq 500\,\mathrm{c\,s^{-1}}$ (where background is negligible); it was frozen at 0.2687 and reused in v11.

2.10.5 3C50 background correction and screening (v12, v13)

For observations with `rate_2_10` $< 500\,\mathrm{c\,s^{-1}}$, HEASoft `nibackgen3C50` (Remillard et al., 2022) was run inside WSL with default parameters and the NICER CALDB (`scripts/v12_bkg3c50.sh`); it returns a total spectrum $T(\mathrm{PI})$ and a background spectrum $B(\mathrm{PI})$. Band background fractions are $f_{\mathcal{X}} = \sum_{\mathcal{X}} B / \sum_{\mathcal{X}} T$ for $\mathcal{X} \in \{A, B, C, \mathrm{T} = 200\text{–}999\}$. Brighter observations get $f = 0$. The frozen correction (`corrected` in `59_v12_analysis.py`, copied verbatim into `63_v13_analysis.py`) is

$$T'_k = \frac{T_k}{1 - f_{\mathrm{T}}}\ (k = 1, 2, 3, \mathrm{state}), \qquad c'_1 = \frac{B(1 - f_B)}{A(1 - f_A)}, \qquad c'_2 = \frac{C(1 - f_C)}{B(1 - f_B)}, \tag{2.21}$$

followed by re-classification of the state; ν_c is not changed (it depends only on the shape of the power spectrum). The first relation holds because a fractional rms is normalised to the total mean; normalising to the source mean $S = (1 - f)(S + B)$ multiplies it by $1/(1 - f)$ (assuming a constant background). Screening keeps an observation if it is bright, or if 3C50 succeeded and $f_{\mathrm{T}} \leq 0.20$. 3C50 failures (mostly data processed with the oldest gain calibration, which the tool rejects) are excluded. In v13 a descriptive variant recomputed the faint observations' features from events inside the GTIs of the 3C50 total spectrum only (`gti_restricted` in `63_v13_analysis.py`).

2.10.6 Lorentzian fits (v11c)

The 13 octave-bin fractional variances ℓ_k (with delta-method errors, `v11lib.lb_se`) are fitted by one or two zero-centred Lorentzians whose integrated variance in a bin $[\nu_k^{\mathrm{lo}}, \nu_k^{\mathrm{hi}}]$ (bin edges $(j \mp 0.5)/128\,\mathrm{Hz}$) is

$$L_k(r^2, \Delta) = r^2\,\frac{2}{\pi}\left[\arctan\frac{\nu_k^{\mathrm{hi}}}{\Delta} - \arctan\frac{\nu_k^{\mathrm{lo}}}{\Delta}\right], \qquad \Delta \in [1/128,\ 128]\ \mathrm{Hz}, \tag{2.22}$$

by weighted least squares in log-parameters (`scipy.optimize.least_squares`, four starting points each, `v11lib.fit_lorentz`); BIC $= \chi^2 + k \ln n$ with $k = 2$ or 4 parameters; two components are preferred if $\mathrm{BIC}_1 - \mathrm{BIC}_2 > 6$. The width of the lowest-frequency component, ν_L, is the "break" frequency.

2.11 MAXI/GSC features (v7c, v8d, v9c)

Figure 2.9 shows the MAXI chain; the label rules are those of Table 2.2 (v7c).

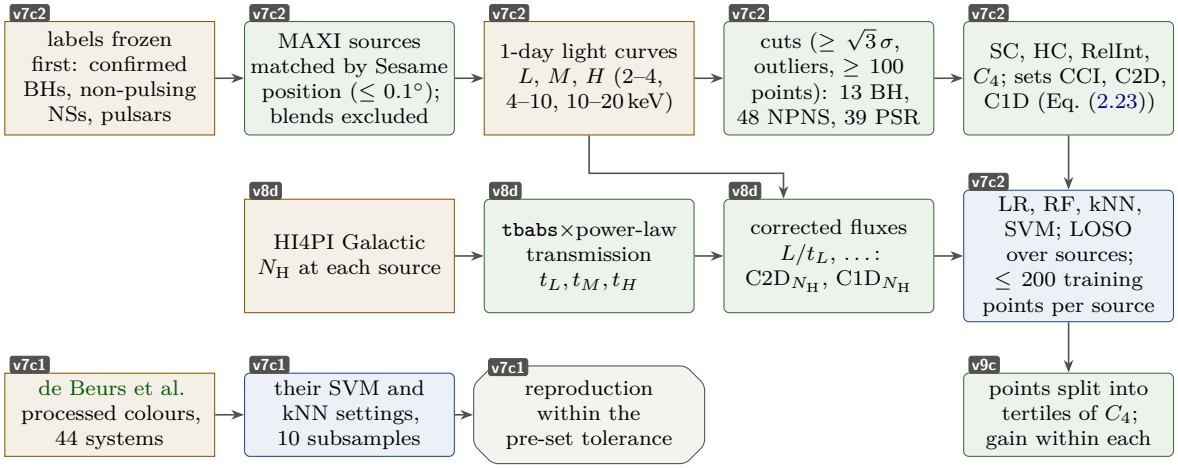


Figure 2.9: MAXI/GSC pipeline. v7c2 froze the labels before any light curve was downloaded and used the cuts of de Beurs et al. (2022); v8d added the Galactic absorption correction, which feeds the same models; v9c split the points into tertiles of the ≥ 4 keV colour C_4. The reproduction of de Beurs et al. (2022) (v7c1) uses their processed colours, not this pipeline. Drawn for this report from Sections 2.11 and 9.6.6.

The MAXI 1-day standard light curve of a source gives photon fluxes L (2–4 keV), M (4–10 keV), H (10–20 keV) and errors. Points are kept if every band has flux/error $\geq \sqrt{3}$ and positive errors, then points with $L+M+H \geq \overline{(L+M+H)} + 10\,\mathrm{sd}$ are dropped; sources need ≥ 100 points (cuts of de Beurs et al., 2022). Features per point (`39_v7c2_analysis.py`):

$$\mathrm{SC} = \frac{M-L}{M+L}, \quad \mathrm{HC} = \frac{H-L}{H+L}, \quad C_4 = \frac{H-M}{H+M}, \quad \mathrm{RelInt} = \frac{L+M+H}{Q_{99.99}(L+M+H)}, \tag{2.23}$$

where $Q_{99.99}$ is the 99.99th percentile over that source's points. Feature sets: CCI = (SC, HC, RelInt), C2D = (SC, HC) and C1D = (C_4), the only colour without 2–4 keV.

N_{H} correction (v8d; `45_v8d_nh.py`, `v8d_heasoft.sh`). The Galactic HI column N_{H} at each source position is the HI4PI (HI4PI Collaboration et al., 2016) weighted average within 0.1° (HEASoft `nh`, output `avwnh`). XSPEC computes photon fluxes of `tbabs*powerlaw` ($\Gamma = 2$; `abund wilm`, `xsect vern`; Wilms et al. 2000) on a grid $N_{\mathrm{H}} \in \{0\} \cup \{10^{19}, \ldots, 10^{23.5}\}\,\mathrm{cm}^{-2}$ (200 log steps); the transmission is $t_{\mathcal{X}}(N_{\mathrm{H}}) = \Phi_{\mathcal{X}}(N_{\mathrm{H}})/\Phi_{\mathcal{X}}(0)$ (e.g. 0.864, 0.979, 0.997 in L, M, H at $10^{22}\,\mathrm{cm}^{-2}$), interpolated linearly in $\log N_{\mathrm{H}}$. Corrected fluxes $L' = L/t_L$, $M' = M/t_M$, $H' = H/t_H$ give SC′, HC′, C_4', RelInt′ by Eq. (2.23) (sets C2D_NH, C1D_NH, CCI_NH).

2.12 Feature summary

Table 2.8 summarises all features, their units, defining equations and producing scripts.

Table 2.8: Summary of every feature that enters a model or test. Missing-value rule: "excluded" = the observation does not enter; "median+ind." = training-fold median and 0/1 indicator (Section 3.2.1); "term = 0" = CCTLR timing term set to zero.

Feature	Unit	Definition	Code	Non-positive / missing
x_{ij} (`rate`)	c/s/keV/PCU	Eq. (2.6)	`04_fetch_process.py`	kept as is
A_{ij}	–	Eq. (2.9)	`v2_representations`	asinh defined for < 0
B_{ij}	keV^{-1}	Eq. (2.10)	same	$F > 0$ guaranteed
h_1, h_2	–	Eq. (2.11)	same	denominators > 0; numerators kept
$\log_{10} F$	dex	Eq. (2.7)	same	$F > 0$
B3, H3	–	Section 2.5	`12_v3b_extended_band.py`	as above
R_{ij}, $h^R_{1,2}$	–	Eq. (2.12)	`reps_R`	as above
X_1, X_2	–	Eq. (2.13)	`19_v4b3_hexte.py`	S/N$<$ 3 $\rightarrow$ median+ind. (per colour)
T_1, T_2, T_3 (v5)	frac. rms	Eqs. (2.15) and (2.16)	`v5lib.obs_timing`	< 3 rows $\rightarrow$ median+ind.
T^*_k (v6)	frac. rms	Eq. (2.17)	`v6lib.obs_timing_v6`	median+ind.; CCTLR: term = 0
ν_c (RXTE)	Hz	Eq. (2.18)	`v6lib.nu_centroid`	missing $\rightarrow$ observation not in test
state rms (v7a)	frac. rms	Eq. (2.19)	`v7lib.state_rms`	< 3 rows $\rightarrow$ "unknown"
LC, HF1–3	frac. rms	Eq. (2.20)	`v8lib.hf_features`	QC fail $\rightarrow$ excluded
c_1, c_2 (NICER)	–	Section 2.10	`v9lib.obs_features`	excluded if QC fails
T_k, state (NICER)	frac. rms	Eq. (2.20)	same	excluded if QC fails
ν_c (NICER)	Hz	Section 2.10	same	missing $\rightarrow$ not in test
β	–	Section 2.10	`v10lib.bkg_ratio`	–
f_A, f_B, f_C, f_T	–	Eq. (2.21)	`58_v12_bkg.py`	3C50 failure $\rightarrow$ excluded
ν_L	Hz	Eq. (2.22)	`v11lib.fit_lorentz`	fit failure $\rightarrow$ missing
SC, HC, C_4, RelInt	–	Eq. (2.23)	`39_v7c2_analysis.py`	cuts in Section 2.11
SC′, HC′, C'_4	–	Section 2.11	`45_v8d_nh.py`	–

Chapter 3

Data model and machine-learning interface

This chapter fixes the objects that every analysis passes to a model, so that each analysis in the phase chapters can be re-implemented from its array table alone. The per-version tables (actual $n \times p$, column meanings, source files) are given in each phase chapter under "Data structure in this phase"; this chapter defines the common notation, the generic recipe, the catalogue of stored arrays, and the leakage controls.

3.1 Notation

An analysis uses n observations of S sources.

- $\boldsymbol{X} \in \mathbb{R}^{n \times p}$: **design matrix**; row i is one observation (one RXTE ObsID, one NICER ObsID, or one MAXI 1-day point), column ℓ one feature of Chapter 2.
- $\boldsymbol{y} \in \{0,1\}^n$: labels, $y_i = 1$ for BH, inherited from the source ($y_i = Y_{g_i}$).
- $\boldsymbol{g} \in \{1,\dots,S\}^n$: source (group) index of each row. $n_k = |\{i : g_i = k\}|$; $\mathcal{S}_{\rm BH}$ and $\mathcal{S}_{\rm NS}$ are the BH and NS source sets, $S_1 = |\mathcal{S}_{\rm BH}|$, $S_0 = |\mathcal{S}_{\rm NS}|$.
- $\boldsymbol{o}$: observation identifiers (`obs_id`; `obsid` for NICER); all joins between tables are by this key.
- $\boldsymbol{w} \in \mathbb{R}^n_{>0}$: sample weights used in a fit (class-balanced or source-equal; Section 4.1.2).
- $\phi : \{1,\dots,n\} \to \{1,\dots,S\}$: **fold assignment**; in leave-one-source-out (LOSO) $\phi(i) = g_i$, so fold k tests exactly the observations of source k.
- $\hat{\boldsymbol{s}} \in [0,1]^n$: **out-of-fold (OOF) score**; $\hat{s}_i$ is produced by a model that was fitted without any observation of source g_i. Scores are model outputs on $[0,1]$ ("BH score"), not calibrated probabilities.

Source-level score and decision. For source k,

$$\bar{s}_k = \frac{1}{n_k} \sum_{i:\, g_i = k} \hat{s}_i, \qquad \hat{Y}_k = \mathbf{1}[\bar{s}_k \geq t]\,, \qquad t = 0.5, \tag{3.1}$$

and the observation-level decision is $\hat{y}_i = \mathbf{1}[\hat{s}_i \geq t]$. The threshold t is fixed at 0.5 in every analysis except the nested procedures of v4a, where a fold-specific threshold t_k is selected inside the training sources and the stored score is shifted to $\hat{s}_i = \hat{s}_i^{\rm raw} - t_k + 0.5$ so that the fixed 0.5

rule reproduces it (Section 5.2). Observations of one source are never treated as independent evidence: there is no product of probabilities over observations.

3.2 From files to matrices

Algorithm 2 is the generic recipe used by every analysis script. Two properties are essential: (i) every transformation that uses information from more than one observation (standardisation, imputation, residualisation, threshold choice) is computed *inside* the training part of a fold; (ii) rows are matched by identifier, never by position, except where the code asserts that the orders are identical.

Algorithm 2 Building $(\boldsymbol{X}, \boldsymbol{y}, \boldsymbol{g})$ for one analysis (generic; concrete versions in the phase chapters)

Input: spectral archive `data/<v>/processed/features.npz`; optional feature tables keyed by `obs_id`

load `rate`, `edges`, `F`, `pl2`, `y`, `source_id`, `obs_id` (one row per accepted observation)
compute the per-observation representation(s) of Section 2.5: A, B, H, HI, R, H_R ▷ no cross-row information
optionally concatenate several samples (e.g. v2 + v4b2) by stacking rows; assert identical `edges`
for timing / HEXTE / state columns: `table.reindex(obs_id)`; non-finite → NaN
$\boldsymbol{X} \leftarrow$ column-stack of the chosen blocks; $\boldsymbol{y} \leftarrow$ `y`; $\boldsymbol{g} \leftarrow$ `source_id`
folds $\leftarrow$ `LeaveOneGroupOut().split`$(\boldsymbol{X}, \boldsymbol{y}, \boldsymbol{g})$; assert one test source per fold, disjoint sources, both classes in training
for each fold $(\mathcal{T}, \mathcal{V})$ **do**
    fill missing values with training medians + indicator (Eq. (3.2)), using $\boldsymbol{X}_{\mathcal{T}}$ only
    fit the model (scaler inside the model, weights from $\boldsymbol{y}_{\mathcal{T}}, \boldsymbol{g}_{\mathcal{T}}$) and score $\boldsymbol{X}_{\mathcal{V}}$
end for
write the OOF table (`representation, model, fold, source_id, obs_id, true_label, BH_score, predicted_label`)

3.2.1 Missing values

Spectral features are never missing (observations failing the quality rules are not in the sample). Timing features can be missing (fewer than 3 valid rows, e.g. only one PCU on). For a block $\boldsymbol{T} \in (\mathbb{R} \cup \{\mathrm{NaN}\})^{n \times q}$ and a fold with training rows $\mathcal{T}$ (`v5lib.fill_missing`):

$$m_\ell = \mathrm{median}_{i \in \mathcal{T},\ T_{i\ell} \text{ finite}}\, T_{i\ell} \quad (0 \text{ if none}), \qquad \tilde{T}_{i\ell} = \begin{cases} T_{i\ell} & \text{finite} \\ m_\ell & \text{otherwise,} \end{cases} \tag{3.2}$$

$$\mu_i = \mathbf{1}[\exists \ell :\ T_{i\ell} \text{ non-finite}]\,,$$

and $[\tilde{\boldsymbol{T}}, \boldsymbol{\mu}]$ (one indicator column) replaces $\boldsymbol{T}$ for both training and test rows. HEXTE colours use the same rule with one indicator per colour (Section 2.6). The CCTLR classifier needs no imputation: its timing term is set to zero for a row with any missing timing value (Section 4.11). NICER features are never imputed; observations failing the NICER quality rules are excluded. For the frozen external tests (v6a, v8b), the medians are taken from the S1 training set and applied unchanged to the external rows.

3.3 Catalogue of stored arrays

Table 3.1 lists every stored array from which a design matrix is built. Shapes and dtypes were read from the committed files with Python for this report.

Table 3.1: Catalogue of stored arrays (committed files). Shapes are the actual shapes in the repository. "c/s/keV/PCU" = count s^{-1} keV^{-1} PCU^{-1}. Every `features.npz` has the same keys: `rate`, `err`, `srate`, `brate`, `pl2` ($n \times 45$, float64, c/s/keV/PCU: net rate, its error, total rate, background rate, folded $\Gamma = 2$ power law), `edges` (46, float64, keV), `obs_id` (n, unicode), `source_id` (n, unicode = $\boldsymbol{g}$), `y` (n, int64 = $\boldsymbol{y}$), `F` (n, float64, c/s/PCU).

File	key / columns	shape	content
`data/processed/features.npz`	all keys	120×45	v1: 8 sources (4 BH), 60 BH rows
`data/v2/processed/features.npz`	all keys	456×45	v2 = S1: 31 sources (7 BH), 105 BH rows
`data/v3/processed/features_3_25.npz`	`rate`, `err`, `edges`, ids, `y`	456×50	v3b: 2.883–25.008 keV grid
`data/v3c/processed/features.npz`	all keys	210×45	14 BH candidates (`y`=1, never used as label in evaluation)
`data/v4b1/processed/features.npz`	all keys	7949×45	S3: every eligible epoch-5 pointing of the 31 sources (2369 BH rows)
`data/v4b2/processed/features.npz`	all keys	221×45	15 new sources (2 BH), all gain epochs
`data/v6/processed/features.npz`	all keys	180×45	12 external sources (2 BH; 4 are slow-pulsar stress sources)
`data/v7b1/processed/features_replaced.npz`	all keys	435×45	v2 with burst-contaminated observations replaced
`data/v7b2/processed/features.npz`	all keys	45×45	Cyg X-1, LMC X-1, LMC X-3
`data/v8b/processed/features.npz`	all keys	23×45	GS 1354−64, SS 433
`data/v4b3/processed/hexte_features.csv`	`X1`, `X2`, `status`, errors, S/N	456 rows	HEXTE colours of the v2 observations
`data/v5/processed/timing_features.csv`	`T1,T2,T3`, `T_total`, `status`, `in_v2/v4b2/v4b1`	8170 rows	v5 two-group estimator for S1 ∪ S2 ∪ S3
`data/v6/processed/timing_v6.csv`	`T1s,T2s,T3s,Ts_total`, `nu_c`, `lb0..lb8`, split halves	8350 rows	v6 all-pairs estimator and ν_c
`data/v6/processed/timing_v5_new.csv`	as v5	180 rows	v5 estimator for the v6 sources
`results/v7a/states.csv`	`rms`, `state`, `c1`, `c2`	456 rows	v7a state of every S1 observation

File	key / columns	shape	content
data/v8b/processed/external_states_timing.csv	state, T1..T3, role	439 rows	state and timing of external set X (incl. stress sources)
data/v8c/processed/hf_features.csv	LC, HF1..HF3, NULL, *_var, *_se, *_z, status	311 rows	event-mode features (148 S1 rows pass QC)
data/v10/processed/rxte_timing_table.csv	c1R,c2R, state, nu_c, T1s..T3s	895 rows	RXTE hard-state ν_c table for v10b
data/v7c/maxi_points.csv	mjd, L,M,H (+errors), source, cls	132 057 rows	MAXI daily points after cuts (BH, NPNS, PSR)
data/v8d/nh_hi4pi.csv; tbabs_transmission.csv	avwnh; t_L, t_M, t_H	100; 201 rows	HI4PI N_{H}; transmission table
data/v9/processed/nicer_features.csv	c1,c2,T1..T3,STATE, nu_c,lb0..lb12,rate_2_10,state	368 rows	v9 NICER (prefix)
data/v10/processed/nicer_full_features.csv	as v9 + bkg_ratio_12_15, split halves	368 rows	v10 NICER (full files; 366 pass QC)
data/v11/processed/nicer_new_features.csv	as v10 + lb*_se, group	318 rows	v11 never-used observations
data/v12/processed/bkg3c50.csv	T_*, B_*, f_A,f_B,f_C, f_T, status	503 rows	3C50 band rates and fractions
data/v12/processed/nicer_corrected_screened.csv	corrected c1, c2,T1..T3, STATE, f_used	534 rows	frozen v12 training set (used by v13)
data/v13/processed/nicer_new_features.csv; bkg3c50.csv	as v11; as v12	238; 148 rows	v13 third observation set

3.3.1 Worked example: the first rows of the real v2 matrices

Table 3.2 shows the first observations of data/v2/processed/features.npz (rows are stored source by source in the order of the source list, BH first) and the derived inputs of the representations. Table 3.3 shows the first rows of the NICER feature table. Figure 3.1 shows the whole 456 × 45 B matrix.

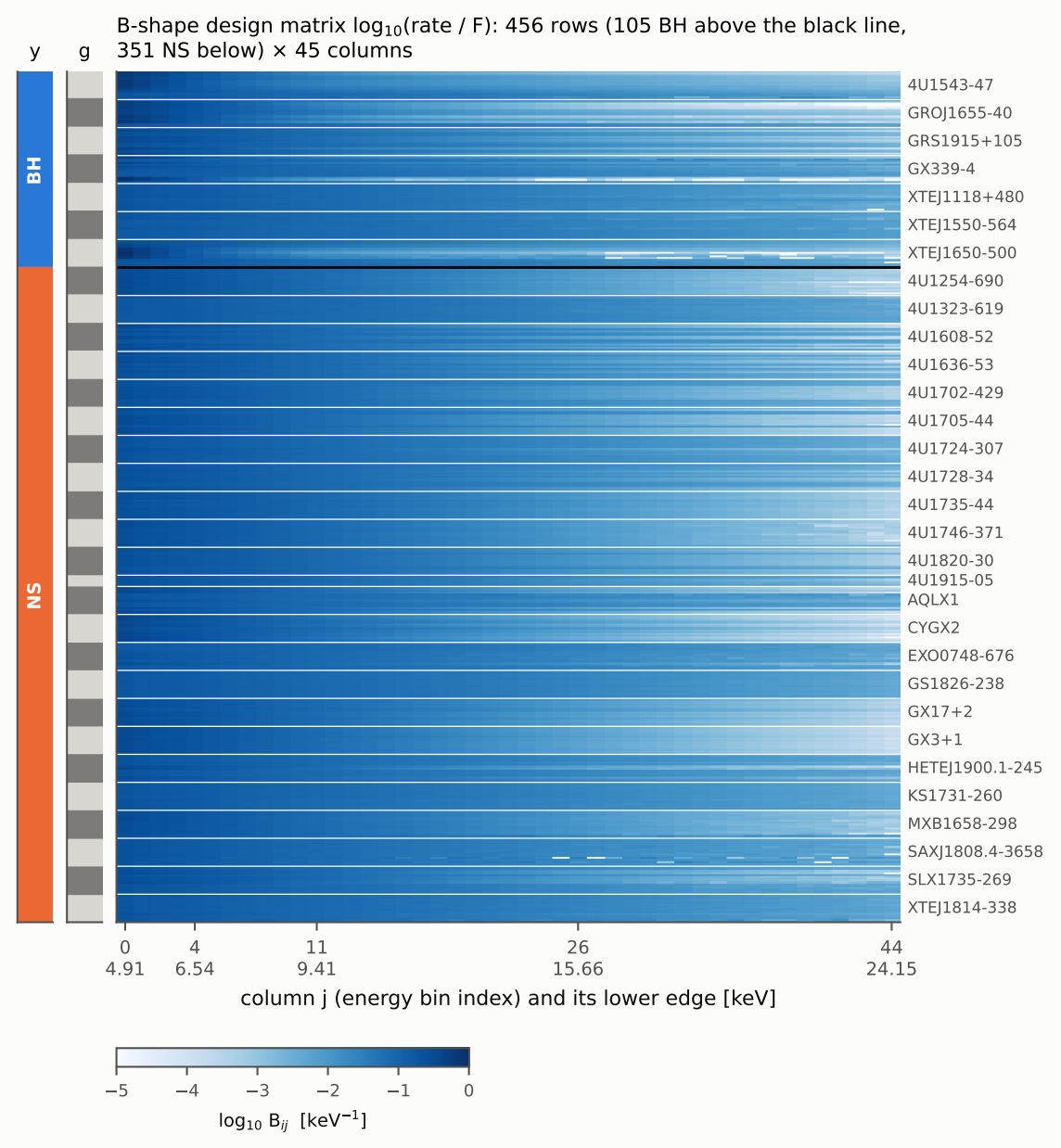


Figure 3.1: The real v2 design matrix for representation B: 456 rows (observations) × 45 columns (energy bins), colour = $\log_{10} B_{ij}$ (sequential scale, darker = larger). Rows are grouped by source (right labels; alternating grey in strip g), BH sources first (strip y: blue = 1) and NS below the black line (orange = 0). The steep fall of the shape with energy dominates every row; differences between sources are subtle and mostly in the high-energy columns, consistent with the finding that two colours capture the usable information. Generated for this report from `data/v2/processed/features.npz`.

Table 3.2: First rows of the v2 inputs (read from `data/v2/processed/features.npz` and `data/v5/processed/timing_features.csv`). x_0, x_{44}: net rate in the first (4.908–5.315 keV) and last (24.151–25.008 keV) grid bins [c/s/keV/PCU]; $A_0 = \mathrm{asinh}(x_0/0.1)$; $B_0 = x_0/F$; h_1, h_2: colours (Eq. (2.11)); F in c/s/PCU. Note the slightly negative x_{44} of 91702-01-22-00 (a soft-state spectrum), kept as is.

obs_id	source_id	y	F	h_1	h_2	$\log_{10} F$	x_0	A_0	B_0	x_{44}	T_1	T_2
90428-01-01-07	GROJ1655-40	1	36.31	0.9495	0.3864	1.560	6.115	4.807	0.1684	0.2296	0.1124	0.1923
91702-01-02-03	GROJ1655-40	1	607.22	0.4126	0.2450	2.783	284.152	8.645	0.4680	0.7872	0.0172	0.0162
91702-01-22-00	GROJ1655-40	1	728.45	0.3094	0.0235	2.862	402.680	8.994	0.5528	−0.0098	0.0309	0.0237
91702-01-22-03	GROJ1655-40	1	793.68	0.3312	0.0310	2.900	421.505	9.040	0.5311	0.0034	0.0305	0.0259
… (rows 4–104: the other BH observations) …												
50030-02-04-00	4U1636-53	0	183.57	0.6820	0.1084	2.264	55.527	7.013	0.3025	0.0765		
60032-01-20-000	4U1636-53	0	104.24	0.7208	0.1224	2.018	29.900	6.394	0.2868	0.0393		
91024-01-74-00	4U1636-53	0	80.50	0.8607	0.3250	1.906	16.855	5.820	0.2094	0.3327		

Table 3.3: First rows of `data/v10/processed/nicer_full_features.csv` (NICER, full files). $c_1 = C(4\text{–}6)/C(2\text{–}4)$, $c_2 = C(6\text{–}10)/C(4\text{–}6)$; T_k and "rms" are fractional rms (signed square roots, Eq. (2.20)); rms is the 0.1–10 Hz state index; rate in $\mathrm{c\,s^{-1}}$ (2–10 keV); n_{seg} = valid 128 s segments.

obsid	source	label	c_1	c_2	T_1	T_2	T_3	rms	ν_c [Hz]	rate	state	n_{seg}
0100320102	Cyg X-1	BH	0.1664	0.2826	0.0661	0.1351	0.1566	0.2276	1.124	2920.9	hard-like	12
2636010102	Cyg X-1	BH	0.0638	0.1857	0.0520	0.0645	0.0462	0.0843	0.378	3591.0	intermediate	10
4690020110	Cyg X-1	BH	0.2544	0.3400	0.1321	0.2211	0.1509	0.2837	0.560	757.9	hard-like	18
5100320107	Cyg X-1	BH	0.2517	0.3357	0.1113	0.2087	0.1589	0.2799	0.677	876.7	hard-like	20
5100320118	Cyg X-1	BH	0.2013	0.3062	0.0792	0.2133	0.1744	0.2990	1.057	1378.4	hard-like	16

3.4 The design matrix and the leave-one-source-out split

Figure 3.2 shows schematically what one LOSO fold sees. The test block is *all* observations of one source; the training block is all observations of every other source. The standardisation parameters $(\boldsymbol{\mu}, \boldsymbol{\sigma})$, imputation medians, sample weights and any selected hyper-parameter or threshold are computed from the training block only. After S folds every row has exactly one OOF score.

3.5 End-to-end data flow

Figure 3.3 maps raw archives to reported numbers, with the script numbers of `scripts/`. Appendix B gives the complete script → output table.

3.6 Leakage controls implemented in the code

1. **Folds by source.** `LeaveOneGroupOut` on $\boldsymbol{g}$; every script asserts that each test fold contains exactly one source, that training and test sources are disjoint, and that both classes are present in training (`results/fold_checks.csv`, `results/v2/fold_checks.csv`).
2. **Fixed folds across versions.** The v2 folds (`results/v2/folds_loso.csv`) are re-used and asserted identical in v3a, v4a and v5, so paired comparisons use the same test sets.
3. **Train-only preprocessing.** `StandardScaler` is part of the model object (`make_pipeline` or `V4Model.fit`), imputation medians come from training rows (Eq. (3.2)), and source-equal weights are computed from training sources only.
4. **Nested selection.** Hyper-parameters, representations and thresholds that are chosen at

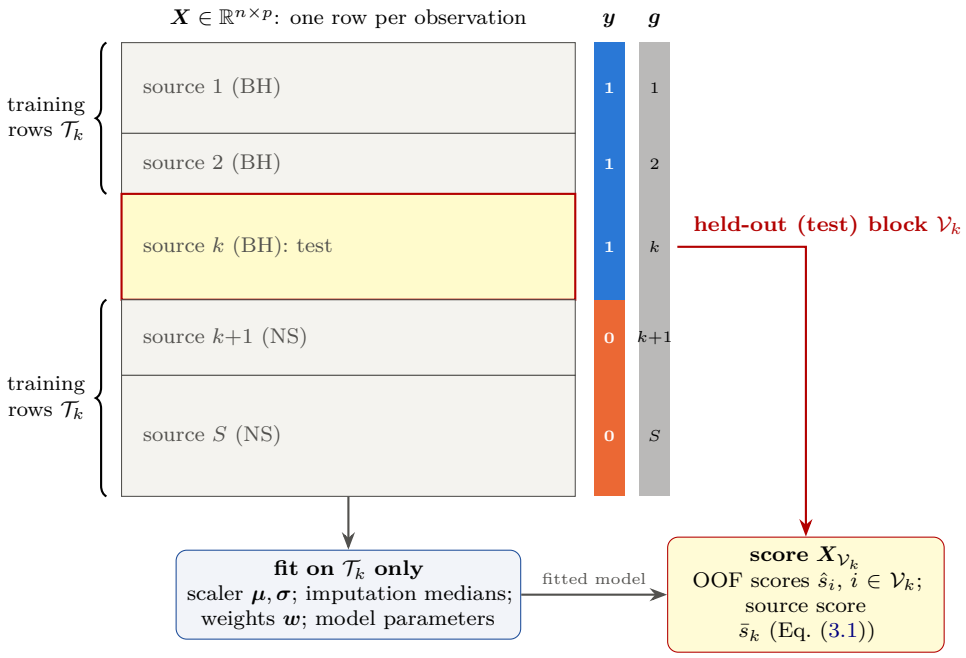


Figure 3.2: Schematic of the design matrix and one leave-one-source-out fold. Rows are observations grouped by source; the $\boldsymbol{y}$ column (blue = BH, orange = NS) and the source index $\boldsymbol{g}$ are attached to $\boldsymbol{X}$. The yellow block is the held-out source; every learned quantity – standardisation, imputation medians, sample weights, hyper-parameters, thresholds – is estimated from the grey training rows only. Repeating this for $k = 1, \ldots, S$ puts every row in the test block exactly once, so the OOF vector $\hat{\boldsymbol{s}} \in [0, 1]^n$ is complete and no observation of source k influences any quantity used to score source k. Drawn for this report.

all are chosen by an inner LOSO over the training sources (v4a, v7c2) – never on the test source (Section 5.4).

5. **Label-free residualisation.** The colour-conditional permutation tests regress timing on colour with a kNN that never sees labels, fitted leaving the source out (Section 5.6).
6. **Frozen models for external and prospective tests.** External (v6a, v8b) and new-observation (v11, v13) tests apply models whose parameters, imputation medians and decision rules were fixed before the test data were read; for the same-source prospective tests each source is scored by the model fitted *without* that source ("out-of-source", Section 5.10).
7. **No position-based joins without assertion.** Feature tables are joined by `obs_id`; where arrays are concatenated by position (e.g. `observations.csv` and `features.npz`) the code asserts equality of the ID order.

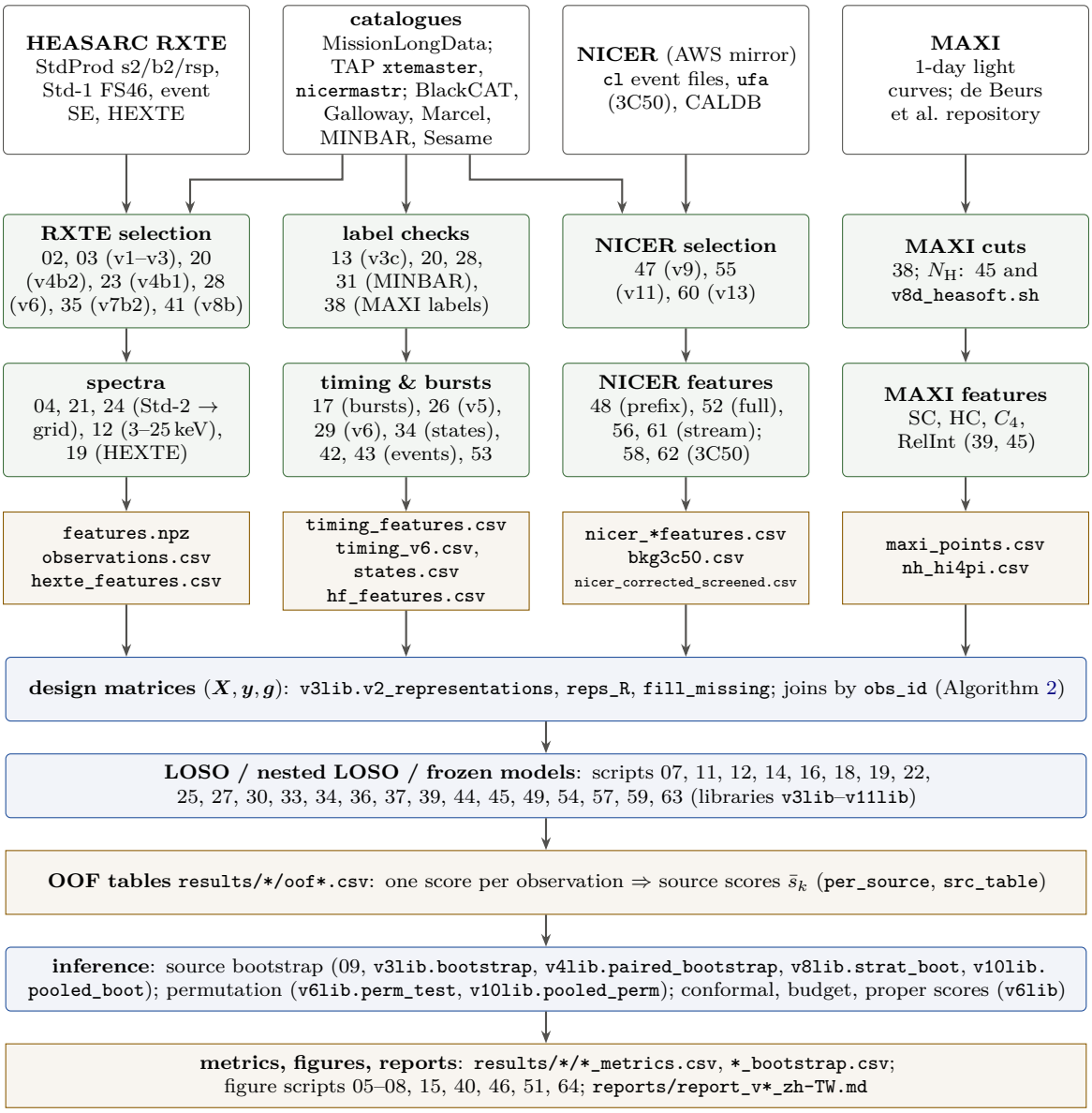


Figure 3.3: End-to-end data flow of the project. Top row (white): external archives and catalogues; green: selection and processing scripts (numbers = `scripts/NN_*.py`); amber: committed intermediate files; blue: model fitting and inference. Raw files downloaded from the top-row archives are not in git; everything from the amber row down is committed, so every number in this report can be traced to a file in `results/` or `data/`. The catalogues feed the RXTE and NICER selections and the label checks (which also define the MAXI labels in script 38). Drawn for this report.

Chapter 4

Machine-learning methods

Every model used in the project is defined here once, in the form in which it was actually run: scikit-learn 1.8.0 (`requirements-lock.txt`) with the settings of `scripts/config.py` and the wrappers `V4Model` (`v4lib.py`), `C2Model` (`v7lib.py`) and the pipelines of `07_train_eval.py` and `v3lib.py`. Equations describe the scikit-learn implementation, not a generic textbook variant. Table 4.1 is the inventory; Appendix A collects all hyper-parameters.

Table 4.1: Inventory of the models and where they are used. Role: P = model of a pre-registered primary comparison; B = baseline; D/E = descriptive/exploratory only.

Model	Versions	Code	Role
Dummy (prior)	v1, v2	`07_train_eval.py`	B
Logistic Regression	v1–v13	`07`, `v3lib.MODELS`, `V4Model`, `C2Model`, `CCTLR.lr`	P (v2–v11), B
Random Forest	v1–v13	as above	P (v8d, v12b, v13a), D
ExtraTrees	v4a	`V4Model`	E
HistGradientBoosting	v4a	`V4Model`	E
RBF SVM	v4a, v7c2; v7c1 (Platt, 3-class)	`V4Model`, `C2Model`, `37_v7c1_repro.py`	E / reproduction
kNN classifier	v4a, v7c2, v8d; v7c1 (3-class)	`V4Model`, `C2Model`, `37`	E / reproduction
QDA	v4a	`V4Model`	E
MLP (seed ensemble)	v4a	`V4Model`	E
Nested auto-selection	v4a 1d/1e	`v4lib.nested_fold` (Section 5.4)	P (v4)
CCTLR	v6c	`v6lib.CCTLR`	D (new method)
kNN regression	v6d, v9b–v13b	`v6lib.source_residuals`	inside the tests
Frozen ν^* rule	v11, v13	`57_v11_analysis.py`, `63_v13_analysis.py`	D

The hyper-parameters of each model, with the `config.py` names, are listed in Tables 4.2 to 4.9.

4.1 Shared components

4.1.1 Standardisation

For the scale-sensitive models (LR, SVM, kNN, QDA, MLP, the colour part of CCTLR and the kNN regression) each feature is standardised with the training rows $\mathcal{T}$ of the current fit only (`StandardScaler`):

$$\mu_\ell = \frac{1}{|\mathcal{T}|}\sum_{i\in\mathcal{T}} x_{i\ell}, \qquad \sigma_\ell = \Big(\frac{1}{|\mathcal{T}|}\sum_{i\in\mathcal{T}}(x_{i\ell}-\mu_\ell)^2\Big)^{1/2}, \qquad z_{i\ell} = \frac{x_{i\ell}-\mu_\ell}{\sigma_\ell} \quad (\sigma_\ell = 0 \Rightarrow \text{scale } 1), \tag{4.1}$$

and the same (μ_ℓ, σ_ℓ) are applied to test rows. Tree models (RF, ExtraTrees, HistGB) use raw features. Standardisation is learned; the per-spectrum normalisations of Section 2.5 (B, H, ...) are not.

4.1.2 Class-balanced weights

The BH:NS ratio is 7:24 sources in the main sample. With n training rows of which n_c are in class c, `class_weight='balanced'` gives every row of class c the weight

$$w_i^{\text{bal}} = \frac{n}{2\,n_{y_i}}, \qquad \sum_{i:y_i=c} w_i^{\text{bal}} = \frac{n}{2}. \tag{4.2}$$

In scikit-learn these class weights multiply any sample weights inside the loss (LR, SVM, HistGB) or inside the impurity (trees). For QDA (no weights) the same balance is imposed through equal priors, for kNN through a prior correction of the output (Eq. (4.10)), and for the MLP through w^{bal} passed as sample weights.

Source-equal weights (v4a 1c, v4b1, v5 S3, v6 S3). When sources contribute very different numbers of observations (v4b1: 6 to 1329 per source), balanced weights let the most-observed sources dominate. With S_c the number of training sources in class c and n_k the training rows of source k (`v4lib.source_equal_weights`):

$$w_i^{\text{src}} = \frac{n}{2\,S_{y_i}\,n_{g_i}}, \qquad \sum_{i:g_i=k} w_i^{\text{src}} = \frac{n}{2S_c} \;\; (k \in \text{class } c), \qquad \sum_{i:y_i=c} w_i^{\text{src}} = \frac{n}{2}. \tag{4.3}$$

Every source of a class has the same total weight, both classes sum to $n/2$ (the same overall scale as balanced weights), and $w^{\text{src}} = w^{\text{bal}}$ exactly when all sources have the same n_k (unit test in `test_v4lib.py`). When $\boldsymbol{w}^{\text{src}}$ is passed, `class_weight` is set to `None`.

Implementation fix: scale of the source-equal weights (2026-10-01, after the v4a results)

The first version normalised each class to total weight 1 (total 2 per fit) instead of $n/2$. Because the LR objective multiplies the summed weighted loss by C (Eq. (4.5)), this acted like an $\approx n/2$ times stronger L2 penalty (scores squeezed into 0.4–0.6); the HistGB internal `min_hessian_to_split` was also affected (RF is scale-invariant). The fix rescaled to $n/2$ per class; only v4a parts 1a–1c were re-run: 1a and 1b were bit-identical, 12 combinations of 1c changed, and no conclusion changed (pre-fix results in commit `9120afb`). v4b1 was run only with the fixed weights.

4.1.3 Score conventions

Every model returns a BH score $s(\boldsymbol{x}) \in [0,1]$: `predict_proba`$[:,1]$ for LR, RF, ExtraTrees, HistGB, QDA; the prior-corrected vote fraction for kNN; the seed-averaged probability for the MLP; the logistic function of the decision value for the SVM and CCTLR. Scores are rankings and thresholdable quantities, not calibrated probabilities, except where proper scoring rules are explicitly evaluated (v6g).

4.2 Dummy classifier

Role. Baseline in v1–v2 (`07_train_eval.py`). **Definition.** `DummyClassifier` with `strategy='prior'` returns the training class frequencies, $s(\boldsymbol{x}) = n_1/n$ for every $\boldsymbol{x}$, and predicts the majority class. **Why it matters.** Under LOSO the majority class of the training fold depends on which source is held out: with 4 BH and 4 NS sources of 15 observations each (v1), leaving out a BH leaves 45 BH : 60 NS, so the held-out BH is predicted NS, and vice versa; the Dummy balanced accuracy is therefore 0, not 0.5. With 7 BH and 24 NS sources (v2) NS is always the majority and the Dummy balanced accuracy is 0.5. "Chance level" under LOSO is thus not a fixed number, one reason why AUC is also reported.

Algorithm 3 Dummy (prior) classifier

Input: training labels $\boldsymbol{y}_{\mathcal{T}}$; test rows $\mathcal{V}$

1: $\pi \leftarrow \frac{1}{|\mathcal{T}|}\sum_{i\in\mathcal{T}} y_i$ ▷ no features; only hyper-parameter `strategy='prior'`

2: **return** $s_i = \pi$ and $\hat{y}_i = \mathbf{1}[\pi \geq 0.5]$ for every $i \in \mathcal{V}$

4.3 Logistic regression

Role. The workhorse: H-LR (two colours + LR) is the v2 reference against which v4 tests every alternative; H+T-LR is the primary model of v5, v6a, v8a/b, v9a, v10a, v11a. **Model.** On standardised inputs $\boldsymbol{z}$,

$$s(\boldsymbol{x}) = \sigma(\boldsymbol{w}^\top \boldsymbol{z} + b), \qquad \sigma(u) = \frac{1}{1+e^{-u}}. \tag{4.4}$$

Objective (`LogisticRegression(C=1, penalty='l2', solver='lbfgs', max_iter=5000)`):

$$(\hat{\boldsymbol{w}}, \hat{b}) = \arg\min_{\boldsymbol{w},b} \ \frac{1}{2}\|\boldsymbol{w}\|_2^2 + C\sum_{i\in\mathcal{T}} \omega_i \Big[-y_i \log s(\boldsymbol{x}_i) - (1-y_i)\log(1-s(\boldsymbol{x}_i))\Big], \tag{4.5}$$

with $\omega_i = w_i^{\text{bal}}$ (Eq. (4.2)) or w_i^{src} (Eq. (4.3)); the intercept is not penalised. scikit-learn minimises the equivalent $\frac{1}{\Omega}\sum_i \omega_i \ell_i + \frac{1}{2C\Omega}\|\boldsymbol{w}\|^2$ ($\Omega = \sum_i \omega_i$) with L-BFGS. $C = 1$ means the penalty weight $1/2$ competes with a weighted log-loss summed over $\approx n$ rows: for $n \approx 400$ and 2–6 features the penalty is weak. **Decision boundary.** In raw colours the 0.5 boundary of H-LR is the line $\sum_\ell (w_\ell/\sigma_\ell) h_\ell + b - \sum_\ell w_\ell \mu_\ell/\sigma_\ell = 0$ (used to draw Figure 6.11). **Assumptions.** Log-odds linear in the standardised features; a single monotone effect per feature. This is the main reason LR can use timing only as "more variability ⇒ more BH" everywhere in the colour plane (motivating CCTLR, Section 4.11), and it fails on MAXI where BHs lie on both sides of the NS colour track (Chapter 9).

Algorithm 4 LR fit and score (one fold)

Input: $\boldsymbol{X}_{\mathcal{T}}, \boldsymbol{y}_{\mathcal{T}}$, optional $\boldsymbol{w}$; test rows $\boldsymbol{X}_{\mathcal{V}}$
1: $(\boldsymbol{\mu}, \boldsymbol{\sigma}) \leftarrow$ Eq. (4.1) on $\boldsymbol{X}_{\mathcal{T}}$; $\boldsymbol{Z}_{\mathcal{T}}, \boldsymbol{Z}_{\mathcal{V}}$ standardised
2: $\omega_i \leftarrow w_i^{\text{bal}}$ if $\boldsymbol{w}$ is None else w_i
3: $(\boldsymbol{w}, b) \leftarrow$ L-BFGS minimisation of Eq. (4.5) (max. 5000 iterations)
4: **return** $s_i = \sigma(\boldsymbol{w}^\top \boldsymbol{z}_i + b)$ for $i \in \mathcal{V}$

Table 4.2: Logistic-regression hyper-parameters (`LR_PARAMS` in `config.py`).

Parameter	Value	Fixed / tuned	Grid (where tuned)
`C`	1.0	fixed (all versions)	v4a 1d/1e: $\{0.01, 0.1, 1, 10\}$ (`V4_GRID['LogReg']`)
`penalty`, `solver`	L2, lbfgs	fixed	–
`class_weight`	balanced (or `None` with $\boldsymbol{w}^{\text{src}}$)	fixed	–
`max_iter`	5000	fixed	–

4.4 Random forest

Role. Second reference model in every version; primary model of v8d (MAXI), v12b (chosen post hoc after v9–v11) and v13a (chosen before the v13 data). **Model.** An average of B classification trees,

$$s(\boldsymbol{x}) = \frac{1}{B}\sum_{b=1}^{B} \hat{p}_b(\boldsymbol{x}), \qquad \hat{p}_b(\boldsymbol{x}) = \frac{\sum_{i \in \text{leaf}_b(\boldsymbol{x})} \omega_i^{(b)} y_i}{\sum_{i \in \text{leaf}_b(\boldsymbol{x})} \omega_i^{(b)}}, \tag{4.6}$$

where $\omega_i^{(b)} = w_i^{\text{bal}} \cdot m_i^{(b)}$ (or $w_i^{\text{src}} m_i^{(b)}$), and $m_i^{(b)}$ is the number of times row i is drawn in the bootstrap sample of tree b (n draws with replacement). The balanced class weights are computed once on the whole training fold, not per bootstrap sample. **Tree growth.** CART: at each node draw $m = \max(1, \lfloor\sqrt{p}\rfloor)$ candidate features without replacement and choose the split that maximises the weighted decrease of the Gini impurity

$$G(\mathcal{N}) = 1 - \sum_{c \in \{0,1\}} \pi_c(\mathcal{N})^2, \quad \pi_c(\mathcal{N}) = \frac{\sum_{i \in \mathcal{N}, y_i = c} \omega_i}{\sum_{i \in \mathcal{N}} \omega_i}, \qquad \Delta = \Omega_{\mathcal{N}} G(\mathcal{N}) - \Omega_L G(L) - \Omega_R G(R); \tag{4.7}$$

grow without depth limit until nodes are pure or a child would have fewer than `min_samples_leaf` rows. With the two colours of H, $m = 1$: every split uses one randomly chosen colour; with H plus three timing features and a missing indicator ($p = 6$), $m = 2$. **Determinism.** `random_state=42`. **Assumptions/limitations.** Piecewise-constant, can follow non-monotone class regions (important for MAXI and for the timing×colour interaction), but with < 10 BH sources a leaf can memorise one source; scores of trees are not calibrated (v3: A-RF had good ranking but scores pushed below 0.5 – a threshold problem).

4.5 Extremely randomised trees

Role. Exploratory alternative in v4a. **Difference from RF.** No bootstrap ($m_i^{(b)} = 1$), and at each node, for each of the m candidate features one threshold is drawn uniformly between the feature's minimum and maximum in the node; the best of these random splits by Eq. (4.7) is used. Settings: 500 trees, `max_features='sqrt'`, `min_samples_leaf=2`, balanced, `random_state=42`; nested grid $\{1, 2, 5, 10\}$ for the leaf size (200/500 trees inner/outer). Lower variance than RF at the price of more bias; no scaling needed.

Algorithm 5 Random forest fit and score (one fold)

Input: $\boldsymbol{X}_{\mathcal{T}}, \boldsymbol{y}_{\mathcal{T}}$ (raw features), ω_i (balanced or source-equal), B, m, leaf size L

1: **for** $b = 1, \ldots, B$ **do** ▷ seeded
2: draw bootstrap counts $m^{(b)} \sim \text{Multinomial}(n; 1/n, \ldots, 1/n)$; weights $\omega_i^{(b)} = \omega_i m_i^{(b)}$
3: grow tree b: recursively split nodes on the best of m random features by Eq. (4.7), leaves $\geq L$ rows
4: **end for**
5: **return** s_i by Eq. (4.6) for $i \in \mathcal{V}$

Table 4.3: Random-forest hyper-parameters (`RF_PARAMS`, `V4_INNER_TREES`, `V4_OUTER_TREES`).

Parameter	Value	Fixed / tuned	Grid / note
`n_estimators`	500	fixed	v4a nested: 200 trees in the inner loop, 500 for the outer refit
`max_features`	`'sqrt'`	fixed	$m = \lfloor \sqrt{p} \rfloor$
`min_samples_leaf`	2	fixed	v4a 1d/1e: $\{1, 2, 5, 10\}$
`class_weight`	balanced (or `None` with $\boldsymbol{w}^{\text{src}}$)	fixed	–
`bootstrap`, `criterion`	True, Gini	default	–
`random_state`	42	fixed	`n_jobs`: 2 (v1–v3), 1 (`V4Model`)

Algorithm 6 ExtraTrees fit and score

Input: raw $\boldsymbol{X}_{\mathcal{T}}$, $\boldsymbol{y}_{\mathcal{T}}$, weights ω_i; $B = 500$ (inner 200), $m = \lfloor \sqrt{p} \rfloor$, leaf size L

1: **for** $b = 1, \ldots, B$ **do** ▷ all rows, no bootstrap
2: at each node: draw m features, one threshold $\sim U(\min, \max)$ each; keep the split with the largest Δ (Eq. (4.7)); stop at purity or leaf size L
3: **end for**
4: **return** $s_i = \frac{1}{B} \sum_b \hat{p}_b(\boldsymbol{x}_i)$ (Eq. (4.6) with $m_i^{(b)} = 1$)

Table 4.4: ExtraTrees hyper-parameters (`V4_FIXED['ExtraTrees']`, `V4_GRID['ExtraTrees']`).

Parameter	Value	Fixed / tuned	Grid
`n_estimators`	500	fixed (inner 200)	–
`max_features`	`'sqrt'`	fixed	–
`min_samples_leaf`	2	tuned in 1d/1e	$\{1, 2, 5, 10\}$
`class_weight`, `random_state`	balanced, 42	fixed	–

4.6 Histogram gradient boosting

Role. Exploratory in v4a. **Model.** An additive model of M regression trees on the log-odds scale, $F_M(\boldsymbol{x}) = F_0 + \eta \sum_{m=1}^{M} f_m(\boldsymbol{x})$, $s(\boldsymbol{x}) = \sigma(F_M(\boldsymbol{x}))$, with F_0 the weighted log-odds of the training classes. Each tree f_m is fitted to the weighted gradients and Hessians of the binary log loss at F_{m-1}, $g_i = \omega_i\,(\sigma(F_{m-1}(\boldsymbol{x}_i)) - y_i)$, $h_i = \omega_i\,\sigma(1-\sigma)$; leaf values are $-\sum_{i\in\text{leaf}} g_i / (\sum_{i\in\text{leaf}} h_i + \lambda)$ with $\lambda =$ `l2_regularization` $= 0$; trees are grown best-first on features pre-binned into at most 255 quantile bins, up to `max_leaf_nodes` leaves with at least `min_samples_leaf` rows. Settings (`V4_FIXED['HistGB']`): $\eta = 0.1$, $M = 200$, 15 leaves, 20 rows per leaf, no early stopping, balanced, `random_state=42`; nested grid $\eta \in \{0.05, 0.1\}\times$ leaves $\in \{7, 15\}$. **Limitation.** With 30 training sources and 20 rows per leaf, a single source (15 rows) cannot form a leaf, which partly protects against memorisation; with 2 colours the model is close to a smooth version of RF.

Algorithm 7 HistGradientBoosting fit and score

Input: raw $\boldsymbol{X}_{\mathcal{T}}$, $\boldsymbol{y}_{\mathcal{T}}$, weights ω_i; η, M, max leaves, min leaf rows

1: bin every feature into ≤ 255 quantile bins (training rows); $F_0 \leftarrow \text{logit}(\sum \omega_i y_i / \sum \omega_i)$
2: **for** $m = 1, \ldots, M$ **do**
3: $\quad g_i \leftarrow \omega_i(\sigma(F_{m-1}(\boldsymbol{x}_i)) - y_i)$, $h_i \leftarrow \omega_i\sigma(1-\sigma)$
4: $\quad$ grow a best-first tree on (g, h); leaf value $-\sum g / (\sum h + \lambda)$; $F_m \leftarrow F_{m-1} + \eta f_m$
5: **end for**
6: **return** $s_i = \sigma(F_M(\boldsymbol{x}_i))$

Table 4.5: HistGradientBoosting hyper-parameters (`V4_FIXED['HistGB']`, `V4_GRID['HistGB']`).

Parameter	Value	Fixed / tuned	Grid
`learning_rate` η	0.1	tuned in 1d/1e	$\{0.05, 0.1\}$
`max_leaf_nodes`	15	tuned in 1d/1e	$\{7, 15\}$
`max_iter` M	200	fixed	–
`min_samples_leaf`	20	fixed	–
`l2_regularization` λ	0	fixed	–
`early_stopping`	False	fixed	–
`class_weight`, `random_state`	balanced, 42	fixed	–

4.7 Support-vector machine with RBF kernel

Role. Exploratory in v4a (it was selected in 23 of 31 outer folds of the nested auto-selection) and descriptive in v7c2; a Platt-scaled three-class version reproduces de Beurs et al. (2022) in v7c1. **Training** (libsvm dual, labels $y'_i = 2y_i - 1$, standardised inputs $\boldsymbol{z}$):

$$\begin{aligned} &\max_{\boldsymbol{\alpha}} \sum_i \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j y'_i y'_j K(\boldsymbol{z}_i, \boldsymbol{z}_j) \quad \text{s.t. } 0 \le \alpha_i \le C\, c_{y_i},\ \sum_i \alpha_i y'_i = 0, \\ &K(\boldsymbol{z}, \boldsymbol{z}') = e^{-\gamma \|\boldsymbol{z} - \boldsymbol{z}'\|^2}, \end{aligned} \tag{4.8}$$

with class weights $c_y = n/(2n_y)$. **Kernel width.** `gamma='scale'` sets $\gamma = 1/(p\ \text{Var}(\boldsymbol{Z}))$, where $\text{Var}(\boldsymbol{Z})$ is the variance of *all* entries of the standardised training matrix (≈ 1, so $\gamma \approx 1/p$); in the v4a grid $\gamma = \kappa/(p\,\text{Var}(\boldsymbol{Z}))$ with factor $\kappa \in \{0.3, 1, 3\}$ (code `gamma_factor`). **Score.** The decision value $f(\boldsymbol{z}) = \sum_i \alpha_i y'_i K(\boldsymbol{z}_i, \boldsymbol{z}) + b$ is mapped by the logistic function,

$$s(\boldsymbol{x}) = \sigma(f(\boldsymbol{z})), \tag{4.9}$$

so $s = 0.5$ is exactly the SVM decision surface; Platt scaling is deliberately not used because it runs an internal random cross-validation. **v7c2 variant.** $C \in \{0.1, 1, 10\} \times \gamma \in \{0.1, 0.585, 3\}$ (absolute γ), standardised, balanced, same score; selected by inner grouped 5-fold source AUC (Section 5.4). **v7c1 variant.** Three classes (BH, non-pulsing NS, pulsar), one-vs-one, $C = 0.655298$, $\gamma = 0.585144$ on training-standardised features, unweighted, Platt probabilities (`probability=True, random_state=42`), exactly as the R/e1071 setting of de Beurs et al. (2022).

Algorithm 8 RBF-SVM fit and score (binary versions)

Input: $\boldsymbol{X}_{\mathcal{T}}$, $\boldsymbol{y}_{\mathcal{T}}$; C; γ or factor κ

1: standardise (Eq. (4.1)); if `'scale'`: $\gamma \leftarrow \kappa/(p\ \mathrm{Var}(\boldsymbol{Z}_{\mathcal{T}}))$ ($\kappa = 1$ for the fixed model)
2: solve the dual Eq. (4.8) with box constraints $C\, c_{y_i}$ (libsvm SMO)
3: **return** $s_i = \sigma(f(\boldsymbol{z}_i))$ (Eq. (4.9))

Table 4.6: SVM hyper-parameters (`V4_FIXED['SVM_RBF']`, `V4_GRID`, `V7_C2_SVM_GRID`, `V7_DEBEURS_SVM`).

Use	C	γ	selection
v4a fixed (1b)	1	`'scale'`	–
v4a nested (1d/1e)	$\{0.1, 1, 10\}$	$\{0.3, 1, 3\}\times$`'scale'`	inner LOSO, source AUC
v7c2 (MAXI)	$\{0.1, 1, 10\}$	$\{0.1, 0.585, 3\}$	inner grouped 5-fold, source AUC
v7c1 (reproduction)	0.655298	0.585144	fixed (from the paper), Platt, 3 classes

4.8 k-nearest-neighbour classifier

Role. Exploratory (v4a), descriptive (v7c2, v8d), reproduction (v7c1). **Model.** With $\mathcal{N}_k(\boldsymbol{z})$ the k training rows nearest to $\boldsymbol{z}$ in Euclidean distance on standardised features (uniform weights), $\hat{p}(\boldsymbol{x}) = \frac{1}{k}\sum_{i\in\mathcal{N}_k} y_i$. Because the training folds are imbalanced, the output is corrected to equal class priors with $\pi = n_1/n$ of the training fold (`V4Model.score`, `C2Model.score`):

$$s(\boldsymbol{x}) = \frac{\hat{p}/\pi}{\hat{p}/\pi + (1-\hat{p})/(1-\pi)}. \tag{4.10}$$

Hyper-parameters. $k = 15$ fixed (v4a 1b); grids $\{5, 15, 31\}$ (v4a nested) and $\{5, 15, 24, 35\}$ (v7c2, v8d, inner LOSO); k is capped at the number of training rows. **v7c1.** Three classes, $k = 24$, *raw* (unstandardised) features and plain vote fractions, as de Beurs et al. (2022). **Limitation.** A test source's nearest neighbours can all come from one training source of similar brightness or colour; this is exactly how the model can "recognise" a source type rather than a compact-object type.

Algorithm 9 kNN classifier (binary, prior-corrected)

Input: $\boldsymbol{X}_{\mathcal{T}}$, $\boldsymbol{y}_{\mathcal{T}}$, k

1: standardise (Eq. (4.1)); $\pi \leftarrow$ fraction of BH rows in $\mathcal{T}$; $k \leftarrow \min(k, |\mathcal{T}|)$
2: **for** $i \in \mathcal{V}$ **do**
3: $\quad\hat{p} \leftarrow$ BH fraction among the k nearest training rows; $s_i \leftarrow$ Eq. (4.10)
4: **end for**

Table 4.7: kNN hyper-parameters (`V4_FIXED['kNN']`, `V4_GRID['kNN']`, `V7_C2_KNN_GRID`, `V7_DEBEURS_KNN_K`).

Use	k	scaling	output	selection
v4a fixed	15	standardised	prior-corrected	–
v4a nested	$\{5, 15, 31\}$	standardised	prior-corrected	inner LOSO, source AUC
v7c2, v8d	$\{5, 15, 24, 35\}$	standardised	prior-corrected	inner LOSO, source AUC
v7c1	24	raw	3-class vote fractions	fixed (paper)

4.9 Quadratic discriminant analysis

Role. Exploratory (v4a). **Model.** Class-conditional Gaussians on standardised features with regularised covariances and equal priors $\pi_0 = \pi_1 = \frac{1}{2}$:

$$\hat{\Sigma}_c = (1 - r)\, S_c + r\, \boldsymbol{I}, \qquad s(\boldsymbol{x}) = \frac{\pi_1 \mathcal{N}(\boldsymbol{z}; \hat{\boldsymbol{\mu}}_1, \hat{\Sigma}_1)}{\sum_c \pi_c \mathcal{N}(\boldsymbol{z}; \hat{\boldsymbol{\mu}}_c, \hat{\Sigma}_c)}, \tag{4.11}$$

where S_c is the unbiased sample covariance of class c (scikit-learn applies the shrinkage to the squared singular values of the centred class data, which is equivalent). $r = 0.1$ fixed; nested grid $\{0.01, 0.1, 0.5\}$ (`V4_GRID['QDA']`). QDA takes no sample weights, so class balance enters only through the priors. **Note.** Three QDA combinations were the only v4a combinations whose source balanced-accuracy interval excluded 0 (Chapter 7); with 76 combinations this is within the expected number of chance exceedances.

Algorithm 10 QDA fit and score

Input: $\boldsymbol{X}_{\mathcal{T}}$, $\boldsymbol{y}_{\mathcal{T}}$, r

1: standardise; for $c \in \{0, 1\}$: class mean $\hat{\boldsymbol{\mu}}_c$, unbiased covariance S_c, $\hat{\Sigma}_c \leftarrow (1 - r)S_c + r\boldsymbol{I}$

2: **return** s_i by Eq. (4.11) with $\pi_0 = \pi_1 = \frac{1}{2}$

Table 4.8: QDA hyper-parameters (`V4_FIXED['QDA']`, `V4_GRID['QDA']`).

Parameter	Value	Fixed / tuned	Grid
`reg_param` r	0.1	tuned in 1d/1e	$\{0.01, 0.1, 0.5\}$
`priors`	$[0.5, 0.5]$	fixed	–

4.10 Multi-layer perceptron (seed ensemble)

Role. Exploratory (v4a). **Model.** One hidden layer of 32 ReLU units (grid: (16) or $(32, 16)$), logistic output, on standardised inputs. **Objective** (per minibatch $\mathcal{M}$; scikit-learn 1.8 with sample weights):

$$\mathcal{L} = \frac{\sum_{i \in \mathcal{M}} \omega_i\, \ell_i}{\sum_{i \in \mathcal{M}} \omega_i} + \frac{\alpha}{2} \frac{\sum_W \|W\|_F^2}{\sum_{i \in \mathcal{M}} \omega_i}, \qquad \omega_i = w_i^{\mathrm{bal}}, \tag{4.12}$$

with ℓ_i the binary cross-entropy, $\alpha = 10^{-3}$ (grid $\{10^{-4}, 10^{-2}\}$); Adam (learning rate 10^{-3}), batch size $\min(200, n)$, at most 500 epochs. **Early stopping.** A stratified 20% validation split (seeded with the model seed) is held out; training stops when the weighted validation accuracy has not improved by 10^{-4} for 10 epochs, and the weights with the best validation accuracy are restored. **Seed ensemble.** Because a small MLP is unstable, the score is the mean of

`predict_proba` over seeds $\{0,\dots,4\}$ (outer fits) or $\{0,1,2\}$ (inner fits of the nested procedure): $s(\boldsymbol{x}) = \frac{1}{|\mathcal{R}|}\sum_{r\in\mathcal{R}} s_r(\boldsymbol{x})$. A planned 1D-CNN on representation B was skipped as pre-declared because `torch` was not installed.

Algorithm 11 MLP seed ensemble

Input: $\boldsymbol{X}_{\mathcal{T}}$, $\boldsymbol{y}_{\mathcal{T}}$; architecture, α; seed set $\mathcal{R}$ (outer $\{0..4\}$, inner $\{0,1,2\}$)

1: standardise; $\omega_i \leftarrow w_i^{\text{bal}}$ (Eq. (4.2))
2: **for** $r \in \mathcal{R}$ **do**
3: stratified 80/20 split (seed r); Adam on Eq. (4.12); after each epoch the weighted validation accuracy; stop after 10 epochs without improvement $> 10^{-4}$ (max 500); restore the best weights
4: **end for**
5: **return** $s_i = \frac{1}{|\mathcal{R}|}\sum_r s_r(\boldsymbol{x}_i)$

Table 4.9: MLP hyper-parameters (`V4_FIXED['MLP']`, `V4_GRID['MLP']`, `V4_MLP_SEEDS_*`).

Parameter	Value	Fixed / tuned	Grid
`hidden_layer_sizes`	$(32,)$	tuned in 1d/1e	$\{(16,),(32,16)\}$
`alpha`	10^{-3}	tuned in 1d/1e	$\{10^{-4},10^{-2}\}$
`early_stopping`, `validation_fraction`	True, 0.2	fixed	–
`max_iter`	500	fixed	–
activation, solver, learning rate, batch	ReLU, Adam, 10^{-3}, $\min(200,n)$	defaults	–
seeds	outer $\{0,\dots,4\}$, inner $\{0,1,2\}$	fixed	–

4.11 Colour-conditional timing likelihood ratio (CCTLR, v6c)

Motivation. In LR, a timing feature has one slope everywhere in the colour plane, although the timing information is expected only where the colours overlap (hard states). CCTLR adds to the colour logit a generative log-likelihood ratio of the timing features *conditional on colour*, so that timing contributes only where the class-conditional timing distributions differ, and missing timing is simply "no evidence". It was designed for v6 and has no tunable hyper-parameter.
Model (`v6lib.CCTLR`): with standardised colours $\boldsymbol{h}$ and timing vector $\boldsymbol{T} = (T_1, T_2, T_3)$,

$$\ell(\boldsymbol{x}) = \underbrace{\boldsymbol{w}^\top \boldsymbol{h} + b}_{\text{H-LR logit}} + \Lambda(\boldsymbol{T} \mid \boldsymbol{h}), \qquad s(\boldsymbol{x}) = \sigma(\ell(\boldsymbol{x})), \tag{4.13}$$

$$\Lambda(\boldsymbol{T} \mid \boldsymbol{h}) = \sum_{k=1}^{3} \Big[\log t_4\big(T_k;\ \mu_{1k}(\boldsymbol{h}),\ \tau_{1k}\big) - \log t_4\big(T_k;\ \mu_{0k}(\boldsymbol{h}),\ \tau_{0k}\big)\Big],$$

$$\mu_{ck}(\boldsymbol{h}) = \beta_{ck0} + \beta_{ck1} h_1 + \beta_{ck2} h_2,$$

where $t_4(\cdot;\mu,\tau)$ is the Student-t density with 4 degrees of freedom, location μ and scale τ (robust to burst or dip outliers). For each class c and timing feature k, $\boldsymbol{\beta}_{ck}$ is a weighted least-squares fit on the training rows of class c with finite timing, with source-equal weights within the class $v_i = 1/(S_c\, n_{g_i})$ (`source_equal_within_class`) and a ridge of 10^{-9}; the scale is the weighted residual standard deviation, floored at 0.01:

$$\hat{\boldsymbol{\beta}}_{ck} = \big(\boldsymbol{Z}_c^\top \boldsymbol{V}_c \boldsymbol{Z}_c + 10^{-9}\boldsymbol{I}\big)^{-1} \boldsymbol{Z}_c^\top \boldsymbol{V}_c \boldsymbol{T}_{c,k}, \qquad \hat{\tau}_{ck} = \max\Big(0.01,\ \sqrt{\textstyle\sum_i v_i r_{ik}^2 / \sum_i v_i}\Big), \tag{4.14}$$

with $\boldsymbol{Z}_c = [\mathbf{1}, \boldsymbol{h}]$ of class c and residuals r_{ik}. The LR part uses `LR_PARAMS` (balanced, or the source-equal weights of Eq. (4.3) in S3). If any timing value of a row is missing, $\Lambda = 0$ and the score equals the H-LR probability (unit test in `test_v6lib.py`). **Limitation.** Adding a generative LR term to a discriminative logit double counts nothing only if timing is conditionally independent of the label given colour under the LR model – an idealisation; the colour model of the class means is linear. In practice CCTLR did not beat H+T-LR (Chapter 8).

Algorithm 12 CCTLR fit and score

Input: training colours $\boldsymbol{H}$, timing $\boldsymbol{T}$ (may contain NaN), labels $\boldsymbol{y}$, sources $\boldsymbol{g}$; test $(\boldsymbol{H}', \boldsymbol{T}')$

1: standardise $\boldsymbol{H}$ (Eq. (4.1)); fit H-LR (Eq. (4.5)) on $\boldsymbol{h}$
2: **for** $c \in \{0, 1\}$ and $k \in \{1, 2, 3\}$ **do**
3: rows: class c with all timing finite; weights $v_i = 1/(S_c n_{g_i})$; fit $\hat{\boldsymbol{\beta}}_{ck}, \hat{\tau}_{ck}$ by Eq. (4.14)
4: **end for**
5: **for** each test row **do**
6: $\Lambda \leftarrow 0$ if any T'_k missing, else the sum in Eq. (4.13)
7: $s \leftarrow \sigma(\boldsymbol{w}^\top \boldsymbol{h}' + b + \Lambda)$
8: **end for**

4.12 k-nearest-neighbour regression for colour residualisation

Role. Inside every colour-conditional permutation test (v6d, v9b, v10b, v11b, v12a, v13b; Section 5.6). It removes the part of a timing feature that is predictable from colour *without using labels*. For a feature F (e.g. $\log_{10} \nu_c$) and colours $\boldsymbol{h}$ standardised on the fit set (`v6lib.source_residuals`):

$$\hat{F}_i = \frac{1}{K} \sum_{j \in \mathcal{N}_K(\boldsymbol{h}_i;\ \{j:\ g_j \neq g_i\})} F_j, \qquad r_i = F_i - \hat{F}_i, \qquad \bar{r}_k = \frac{1}{|\{i : g_i = k\}|} \sum_{i: g_i = k} r_i, \tag{4.15}$$

with $K = 25$ (`V6_KNN`; capped at the number of available rows) and uniform weights; the neighbours of an observation are taken only from *other* sources (leave-one-source-out), so a source cannot explain itself. Rows with missing F are skipped. For the v6 external replication the regression is fitted on all S1 rows and applied to the external rows. **Caveat** (noted in v9): when BHs cluster in one colour region, their neighbours are partly other BHs, so part of the class difference is absorbed into $\hat{F}$; the test is then conservative.

Algorithm 13 Label-free colour residualisation (`source_residuals`)

Input: colours $\boldsymbol{H}$, feature F (may be NaN), sources $\boldsymbol{g}$, $K = 25$

1: standardise $\boldsymbol{H}$ on the fit set
2: **for** each source k **do**
3: fit a kNN regression (K neighbours) on rows with finite F and $g \neq k$; predict rows of k; $r_i \leftarrow F_i - \hat{F}_i$
4: **end for**
5: **return** source means $\bar{r}_k$ (Eq. (4.15))

4.13 Frozen characteristic-frequency rule (v11, v13)

A deliberately simple, pre-frozen decision rule: an observation is called BH if it is hard-like and $\nu_c < \nu^*$; a source is called BH if at least half of its new hard-like observations are called BH.

The threshold is the geometric mean of the class medians of ν_c in the previous data set,

$$\nu^{*} = \sqrt{\mathrm{median}_{\mathrm{BH}}\,\nu_c \cdot \mathrm{median}_{\mathrm{NS}}\,\nu_c}, \tag{4.16}$$

frozen before any new data were read: $\nu^{*} = \sqrt{0.5917 \times 2.8594} = 1.3007$ Hz from v10 (32 BH and 111 NS hard-like observations; `logs/v11/nu_star_frozen.txt`) and $\nu^{*} = \sqrt{0.6294 \times 2.8362} = 1.3361$ Hz from the v12 corrected and screened data (62 BH, 138 NS; `logs/v13/nu_star_frozen.txt`).

4.14 Models that were not used

Principal component analysis was never used: every occurrence of "PCA" in the code refers to the RXTE Proportional Counter Array. The pre-declared 1D convolutional network on representation B (v4) was skipped because `torch` was not available; no replacement was introduced. No probability calibration (Platt, isotonic) was used except in the de Beurs reproduction (v7c1).

Chapter 5

Evaluation and statistical inference

This chapter defines how every number in the result tables is produced: the cross-validation scheme, the source-level decision, the metrics, the source bootstrap and its paired differences, the permutation tests, conformal prediction, the observation-budget curves and the protocol of the frozen (prospective) tests. Figure 5.1 shows how they fit together and which version added each of them; Table 5.1 is a summary.

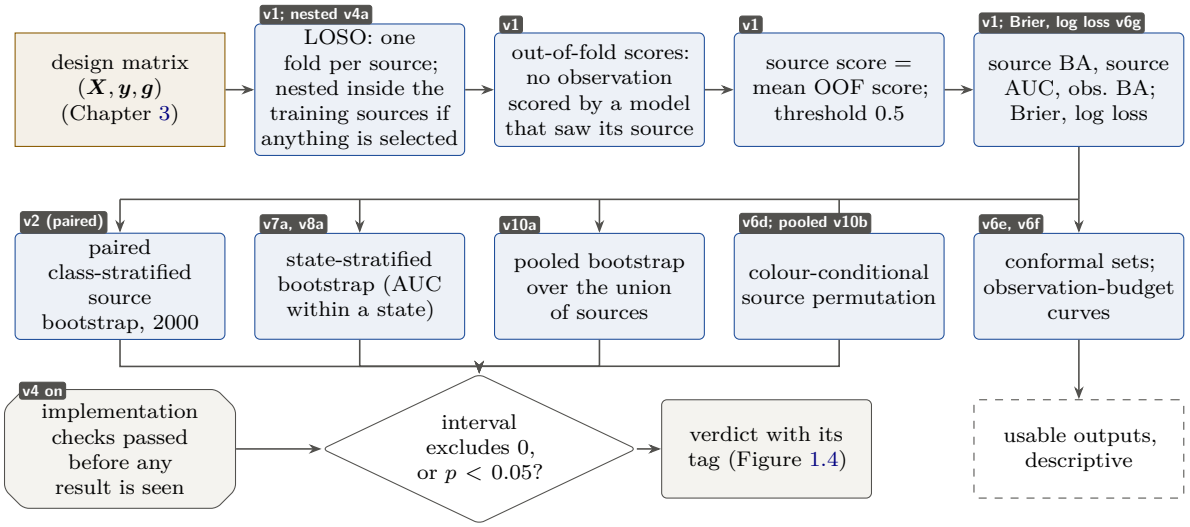


Figure 5.1: Evaluation chain, from the design matrix to a verdict, labelled with the version that introduced each element. Every inferential procedure in the middle row works on fixed out-of-fold scores or fixed features: nothing is refitted inside a resampling or a permutation. Bootstrap and permutation results are judged by the pre-registered rule when the comparison was declared primary or secondary (Figure 1.4); conformal sets and budget curves are always descriptive. Drawn for this report from Table 5.1 and the pre-registrations.

5.1 Leave-one-source-out cross-validation

Folds are produced by `LeaveOneGroupOut` on $\boldsymbol{g}$; fold k (in the sorted order of the source names) holds out all observations of source k. The OOF score of observation i is

$$\hat{s}_i = s\big(\boldsymbol{x}_i;\ \hat{\theta}^{(-g_i)}\big), \qquad \hat{\theta}^{(-k)} = \mathrm{Fit}\big(\{(\boldsymbol{x}_j, y_j, g_j) : g_j \neq k\}\big), \tag{5.1}$$

where Fit includes every learned step of Algorithm 2. Each script asserts per fold: exactly one test source, disjoint training and test sources, and both classes in training (Section 3.6).

Table 5.1: Summary of the inferential procedures. "Unit" is what is resampled or permuted; all procedures keep the out-of-fold scores or the features fixed (no refitting inside the resampling).

Procedure	Statistic	Unit, size	Decision rule / code
Class-stratified source bootstrap	BA, AUC (source, observation), Brier, log loss	sources within class; 2000, seed 42	95% percentile CI; paired difference "detected" iff its CI excludes 0 (`v4lib.paired_bootstrap`)
State-stratified bootstrap	observation AUC within a state	sources of the whole sample; 2000	as above (`v8lib.strat_boot`)
Pooled bootstrap (v10a)	equal-weight mean of component AUC gains	union of sources; 2000	as above (`v10lib.pooled_boot`)
Colour-conditional permutation	$D = \text{BH} - \text{NS}$ mean residual	source labels; 20 000, seed 42	two-sided $p < 0.05$; Holm across features (`v6lib.perm_test`, `holm`)
Pooled permutation (v10b)	mean of instrument D	labels within membership strata; 20 000	$p < 0.05$ (`v10lib.pooled_perm`)
Mondrian conformal (v6e)	per-class p-values	calibration sources	set $= \{y : p_y > \varepsilon\}$, $\varepsilon \in \{0.1, 0.2\}$ (`v6lib.conformal_*`)
Observation budget (v6f)	source BA/AUC with k obs/source	observations within source; 500 draws	median and 2.5–97.5% (`v6lib.budget_curves`)

Algorithm 14 Leave-one-source-out OOF scores (`v3lib.run_loso`, `v4lib.loso_scores`)

Input: $(\boldsymbol{X}, \boldsymbol{y}, \boldsymbol{g})$, model constructor $\mathcal{M}$, weighting option

1: **for** each source k (sorted) **do**
2: $\quad \mathcal{V} \leftarrow \{i : g_i = k\}$, $\mathcal{T} \leftarrow \{i : g_i \neq k\}$; assert $|\{g_i : i \in \mathcal{V}\}| = 1$, both classes in $\mathcal{T}$
3: $\quad$ fill missing values with $\mathcal{T}$-medians + indicator (Eq. (3.2)) if needed
4: $\quad \boldsymbol{w} \leftarrow$ balanced or source-equal weights from $(\boldsymbol{y}_\mathcal{T}, \boldsymbol{g}_\mathcal{T})$
5: $\quad m \leftarrow \mathcal{M}.\text{fit}(\boldsymbol{X}_\mathcal{T}, \boldsymbol{y}_\mathcal{T}, \boldsymbol{w})$; $\hat{s}_i \leftarrow m.\text{score}(\boldsymbol{x}_i)$ for $i \in \mathcal{V}$
6: **end for**
7: **return** $\hat{\boldsymbol{s}}$ and the source scores $\bar{s}_k$ (Eq. (3.1))

Leaky observation-wise split (diagnostic only). To show the size of the leak, v1 and v2 also ran a stratified 5-fold split over observations (`StratifiedKFold(5, shuffle=True, random_state=42)`), in which a source appears in both training and test folds. It is never used for any conclusion other than the size of the optimism (Figure 1.1).

5.2 Source-level aggregation and thresholds

The source score $\bar{s}_k$ is the mean of the OOF scores of its observations (Eq. (3.1)); the decision uses $t = 0.5$. In the nested procedures of v4a a threshold is chosen inside the training sources by maximising the inner source balanced accuracy (`v4lib.choose_threshold`). With $u_1 < \cdots < u_q$ the distinct inner source scores and candidates $\mathcal{C} = \{u_1 - 10^{-9}\} \cup \{(u_r + u_{r+1})/2\}_{r=1}^{q-1} \cup \{u_q + 10^{-9}\}$:

$$t^* = \arg\min_{t \in \mathcal{C}^*} |t - 0.5|, \qquad \mathcal{C}^* = \Big\{ t \in \mathcal{C} : \ \mathrm{BA}_{\mathrm{src}}(t) = \max_{t' \in \mathcal{C}} \mathrm{BA}_{\mathrm{src}}(t') \Big\}, \tag{5.2}$$

i.e. ties are broken towards 0.5 (unit tests: separable scores $(0.1, 0.2, 0.6, 0.7)$ give $t = 0.4$). The outer scores are stored as $\hat{s}_i = \hat{s}_i^{\mathrm{raw}} - t_k^* + 0.5$ so that the common rule $\hat{s} \geq 0.5$ applies; the outer *source AUC* of a nested procedure is computed on these shifted scores, an approximation because different folds may use different models (stated in the v4 pre-registration).

5.3 Metrics

Balanced accuracy. With BH recall (sensitivity) and NS recall (specificity),

$$\mathrm{BA} = \tfrac{1}{2}(\mathrm{TPR} + \mathrm{TNR}). \tag{5.3}$$

Source level: computed on $(\hat{Y}_k, Y_k)$ over sources; with 7 BH and 24 NS sources, "k of 31 correct" is misleading (all-NS gives 24/31) and BA is the headline. *Observation level*: over observations; inside the bootstrap it is computed from per-source counts c_k (correct observations) and n_k as $\mathrm{BA}_{\mathrm{obs}} = \frac{1}{2}(\sum_{k \in \mathrm{BH}} c_k / \sum_{k \in \mathrm{BH}} n_k + \sum_{k \in \mathrm{NS}} c_k / \sum_{k \in \mathrm{NS}} n_k)$, which is the pooled recall of the resampled multiset of sources.

ROC AUC (Mann–Whitney form, ties counted one half). For positive scores $\{a_r\}_{r=1}^{n_1}$ and negative scores $\{b_q\}_{q=1}^{n_0}$,

$$\mathrm{AUC} = \frac{1}{n_1 n_0} \sum_{r=1}^{n_1} \sum_{q=1}^{n_0} \Big[\mathbf{1}[a_r > b_q] + \tfrac{1}{2} \mathbf{1}[a_r = b_q] \Big]. \tag{5.4}$$

Source AUC uses $a = \{\bar{s}_k\}_{k \in \mathrm{BH}}$, $b = \{\bar{s}_k\}_{k \in \mathrm{NS}}$: the probability that a random BH source scores above a random NS source; it does not depend on the threshold. *Observation AUC* (used inside states) uses all observations. With observation weights w (`v7lib.auc_ba`), $\mathrm{AUC}_w = \sum_{r,q} w_r w_q [\mathbf{1}[a_r > b_q] + \frac{1}{2}\mathbf{1}[a_r = b_q]] / (\sum_r w_r \sum_q w_q)$; the reported state-stratified AUCs are unweighted.

Proper scores (v6g). Source-level, class-balanced, with the source score clipped to $[0.01, 0.99]$ (`v6lib.proper_score_bootstrap`):

$$\mathrm{LL} = \frac{1}{2}\Big[\frac{1}{S_1} \sum_{k \in \mathrm{BH}} -\log \tilde{s}_k + \frac{1}{S_0} \sum_{k \in \mathrm{NS}} -\log(1 - \tilde{s}_k) \Big], \qquad \mathrm{Brier} = \frac{1}{2}\Big[\frac{1}{S_1} \sum_{k \in \mathrm{BH}} (1 - \tilde{s}_k)^2 + \frac{1}{S_0} \sum_{k \in \mathrm{NS}} \tilde{s}_k^2 \Big], \tag{5.5}$$

lower is better. Unlike AUC they cannot saturate at a perfect ranking, which matters in v6a.

5.4 Nested leave-one-source-out selection

When a hyper-parameter, a representation, an algorithm or a threshold is chosen, the choice is made inside each outer training set by an inner LOSO over its sources (Figure 5.2). The outer test source never takes part.

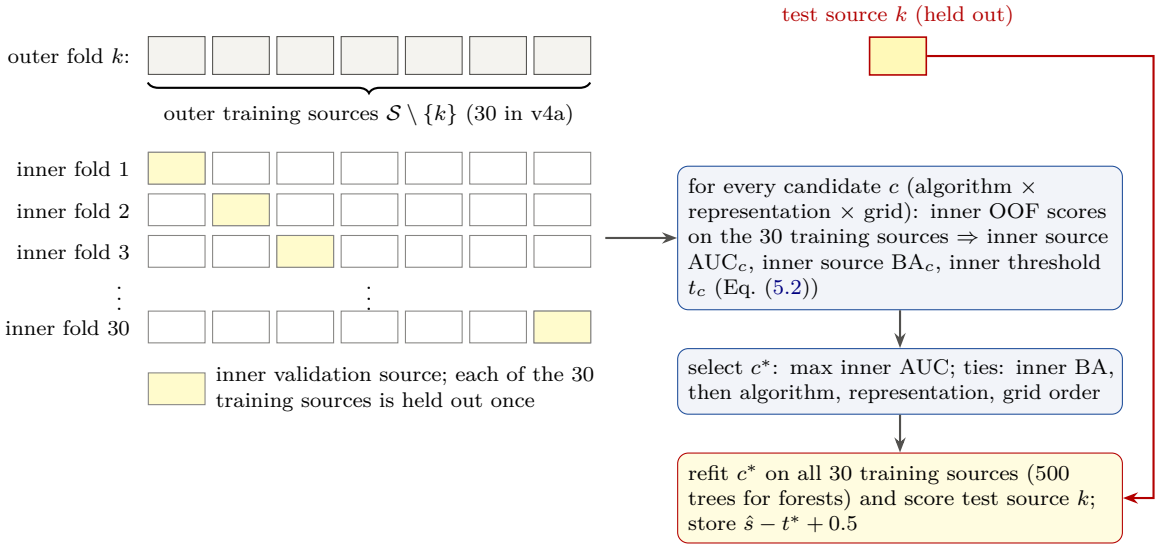


Figure 5.2: Nested LOSO as used in v4a (1d per algorithm and representation; 1e over all 140 configurations). For each outer fold, every candidate configuration is evaluated by an inner LOSO over the outer training sources only; the winner and its inner-optimal threshold are refitted on the outer training sources and applied once to the held-out source. The bias of "choosing the best of many" is therefore included in the outer estimate. Drawn for this report.

Algorithm 15 Nested automatic selection, v4a 1e (`v4lib.nested_fold`, `16_v4a_algorithms.py`)

Input: outer folds; representations $\mathcal{R}$ = (H, HI, B, A); algorithms $\mathcal{A}$ (8) with grids `V4_GRID` (35 settings)

1: **for** each outer fold k with training rows $\mathcal{T}$ **do**
2: **for** each $(r, a) \in \mathcal{R} \times \mathcal{A}$ and grid entry c **do**
3: inner LOSO over the sources of $\mathcal{T}$ (stage "inner": 200 trees, MLP seeds $\{0, 1, 2\}$) → inner source means
4: $\mathrm{AUC}_{rac} \leftarrow$ Eq. (5.4); $(t_{rac}, \mathrm{BA}_{rac}) \leftarrow$ Eq. (5.2)
5: **end for**
6: per (r, a): $c^*(r, a) \leftarrow \arg\max$ (AUC, BA, −grid index); refit (stage "outer": 500 trees, seeds $\{0..4\}$) and score fold k ▷ 1d
7: $(r, a)^* \leftarrow \arg\max$ (AUC, BA, −algorithm order, −representation order, −grid index) ▷ 1e
8: store $\hat{s}_i - t^* + 0.5$ for i in fold k
9: **end for**

Other nested selections. In v7c2 (MAXI) the kNN k was selected by inner LOSO source AUC; the SVM (C, γ) by inner `GroupKFold(5)` source AUC (a pre-run deviation for computing time; Chapter 9); the first configuration with strictly larger AUC wins (`v7lib.c2_outer`). Every other model in the project uses fixed hyper-parameters, so no other selection exists.

5.5 Source bootstrap

The unit of generalisation is the source, so uncertainty is estimated by resampling sources, stratified by class, while keeping the OOF scores fixed (Figure 5.3). The draws are identical in every version: BH and NS source names are sorted, `rng = numpy.random.default_rng(42)`, and draw b is $\mathcal{D}^{(b)} = (\text{rng.choice}(\mathcal{S}_{\text{BH}}, S_1),\ \text{rng.choice}(\mathcal{S}_{\text{NS}}, S_0))$, with replacement, $b = 1, \ldots, 2000$ (`09_bootstrap_sources.py`, `v3lib.bootstrap`, `v4lib.paired_bootstrap`, `v7lib.boot_draws`). Every metric M is recomputed on the multiset $\mathcal{D}^{(b)}$ (a source drawn twice counts twice); for AUC this means Eq. (5.4) over the drawn BH and NS source scores. The 95% interval is the percentile interval

$$\text{CI}_{95}(M) = [Q_{0.025}\{M^{(b)}\},\ Q_{0.975}\{M^{(b)}\}]. \tag{5.6}$$

Paired differences. Two models evaluated on the same sources use the same draws: $\Delta^{(b)} = M_A^{(b)} - M_B^{(b)}$, point estimate $M_A - M_B$ on the original sample, CI by Eq. (5.6) on $\{\Delta^{(b)}\}$, and $\texttt{frac_first_better} = \frac{1}{2000}\sum_b \mathbf{1}\left[\Delta^{(b)} > 0\right]$.

Decision rule used throughout (pre-registered from v4 on)

A difference is called **detected** only if its 95% bootstrap CI excludes 0. Otherwise it is **not detected under the pre-registered criterion** – never "no difference", and an interval touching 0 (e.g. $[0.000, +0.179]$) is *not* detected. Each metric is judged separately; source BA and source AUC are the headline metrics.

What the interval does and does not contain. Models are not refitted inside the bootstrap, so the interval reflects which sources happen to be in the sample, not the variability of model fitting or model selection. Because sources of one class are treated as exchangeable, very heterogeneous NS sub-classes (Z sources, atolls, AMXPs) or unusual BHs (GRS 1915+105) can make the interval optimistic. With 7 BH sources the BH side of every interval is coarse: a single BH source changes the source BA by $1/(2 \cdot 7) \approx 0.07$, which is why many intervals have endpoints exactly at 0.

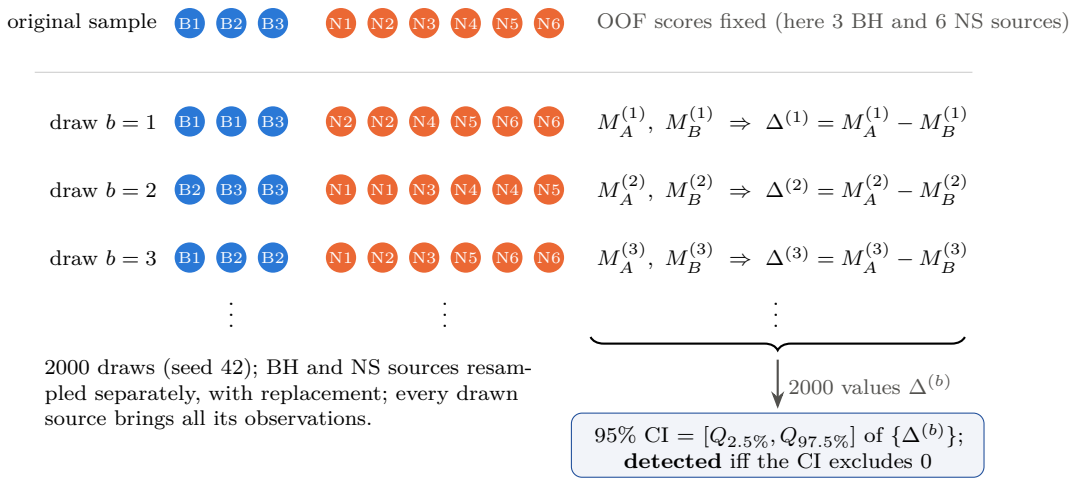


Figure 5.3: Class-stratified source bootstrap with paired differences. Each draw resamples the BH sources and the NS sources separately, with replacement and with all observations of a drawn source; both models are evaluated on the same drawn multiset, so the difference $\Delta^{(b)}$ is paired. No model is refitted. Drawn for this report.

State-stratified variant (v7a, v8, v9a, v10a, v11a, v12b, v13a). For observation-level metrics inside a state (soft-like, hard-like, ...), sources are drawn from the whole sample (so the

draws are the same as above), and the metric is computed on the drawn sources' observations that belong to the state (`v8lib.strat_boot`). If a draw contains no BH or no NS observation in the state, the metric is NaN and is ignored in the percentiles (the number of valid draws is reported). Differences between strata ("soft − hard") and between models are paired by draw.

Pooled variant (v10a). Sources of RXTE and NICER are mapped to one canonical name if they lie within 0.1° (`v10lib.canonical_map`); draws are taken over the union of sources; in each draw each component (RXTE v2, RXTE external, NICER) computes its own state-stratified AUC gain on its drawn sources, and the pooled statistic is the equal-weight mean; draws in which any component is undefined are dropped (`v10lib.pooled_boot`). A source present in two instruments is therefore always drawn (or not) in both.

5.6 Colour-conditional source permutation tests

Figure 5.4 shows the test as a flowchart; the formulae follow.

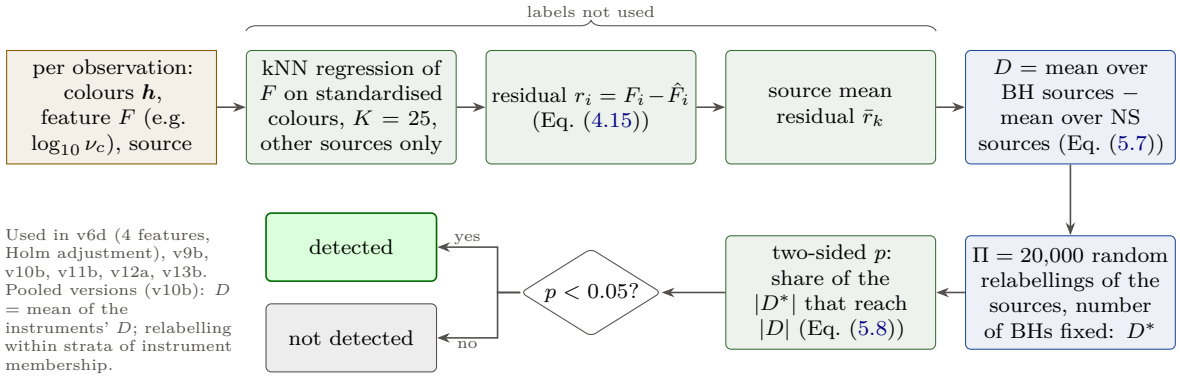


Figure 5.4: Colour-conditional source permutation test. The colour dependence of the feature is removed without labels (neighbours only from other sources), so the class enters only through D and its permutation null; the source, not the observation, is relabelled. Drawn for this report from Eqs. (4.15), (5.7) and (5.8) and `v6lib.perm_test`, `v10lib.pooled_perm`.

Question. At matched colour, does a timing feature F (e.g. $\log_{10}\nu_c$) differ between BH and NS sources? **Statistic.** With the source mean residuals $\bar{r}_k$ of the label-free kNN regression (Eq. (4.15)):

$$D = \frac{1}{S_1}\sum_{k\in\mathcal{S}_{\rm BH}} \bar{r}_k - \frac{1}{S_0}\sum_{k\in\mathcal{S}_{\rm NS}} \bar{r}_k. \tag{5.7}$$

Null distribution. Under H_0 (given colour, F does not depend on the class) the source labels are exchangeable: $\Pi = 20\,000$ random relabellings keeping the number of BH sources fixed (`v6lib.perm_test`, seed 42), giving $D^{*(1)}, \ldots, D^{*(\Pi)}$, and the two-sided p-value

$$p = \frac{1 + \#\{\pi :\ |D^{*(\pi)}| \geq |D| - 10^{-15}\}}{1 + \Pi}. \tag{5.8}$$

The source, not the observation, is permuted, so within-source correlation is respected. When several features are tested together (v6d: $T_1^*, T_2^*, T_3^*, \log_{10}\nu_c$), Holm's step-down adjustment is reported: for ordered $p_{(1)} \leq \cdots \leq p_{(m)}$, $\tilde{p}_{(r)} = \max_{q\leq r} \min\{1, (m-q+1)p_{(q)}\}$ (`v6lib.holm`). Decision: $p < 0.05$ (primary features); synthetic calibration: under H_0 the fraction of $p < 0.05$ was 0.033 over 60 data sets (`logs/v6/test_v6lib.log`).

Algorithm 16 Colour-conditional source permutation test

Input: observation colours $\boldsymbol{H}$, feature F, sources $\boldsymbol{g}$, source labels Y_k; $K = 25$, $\Pi = 20000$

1: $\bar{r}_k \leftarrow$ Algorithm 13; keep sources with a finite $\bar{r}_k$; $D \leftarrow$ Eq. (5.7)
2: rng $\leftarrow$ `default_rng(42)`
3: **for** $\pi = 1, \ldots, \Pi$ **do**
4: assign BH to a random subset of S_1 sources; $D^{*(\pi)} \leftarrow$ Eq. (5.7)
5: **end for**
6: **return** D and p by Eq. (5.8)

Pooled permutation across instruments (v10b, also v12a pooled). Each instrument ι has its own residuals and D_ι; the statistic is $D = \frac{1}{|I|}\sum_\iota D_\iota$. A source keeps one label in all instruments. The pre-registered null permuted labels over the union of sources; a synthetic test before any v10b statistic showed that sources present in only one instrument then absorb BH labels and widen the null, so labels are permuted *within strata of instrument membership* (RXTE only, NICER only, both), keeping the number of BH per stratum fixed (`v10lib.pooled_perm`). In synthetic data this gave a fraction 0.065 of $p < 0.05$ under the null (200 data sets) and power 0.98 at a -1.5 SD BH shift (`logs/v10/test_v10lib.csv`).

5.7 Source-level conformal prediction (v6e)

A conformal predictor returns, for a new source, a set of labels – {BH}, {NS}, {BH, NS} ("cannot tell") or {} – with a finite-sample coverage guarantee under exchangeability. The class-conditional (Mondrian) version guarantees coverage within each class separately, which matters with 7 BHs. **Nonconformity** of a calibration source with score $\bar{s}$: $\alpha = 1 - \bar{s}$ if BH, $\alpha = \bar{s}$ if NS. For a test source with score s and calibration sources of class y (n_y of them),

$$p_{\mathrm{BH}}(s) = \frac{1 + \#\{k \in \mathrm{cal}_{\mathrm{BH}} : 1 - \bar{s}_k \geq 1 - s\}}{1 + n_{\mathrm{BH}}}, \qquad p_{\mathrm{NS}}(s) = \frac{1 + \#\{k \in \mathrm{cal}_{\mathrm{NS}} : \bar{s}_k \geq s\}}{1 + n_{\mathrm{NS}}}, \quad (5.9)$$

$$\Gamma_\varepsilon(s) = \{y : p_y(s) > \varepsilon\},$$

(`v6lib.conformal_sets`). If the test source is exchangeable with the calibration sources of its class, $\Pr(Y \in \Gamma_\varepsilon) \geq 1 - \varepsilon$ for that class. **LOSO version** (S1): each source is the test point and the other sources' OOF scores are the calibration set (`conformal_loso`). **Split version** (external set E): the calibration set is the S1 LOSO OOF source scores of the same (H_R) model and the test points are the frozen-model scores of the external sources. Reported: per-class coverage, mean set size, number of correct singletons and of ambiguous sets, at $\varepsilon \in \{0.1, 0.2\}$.

Figure 5.5 illustrates the source-level Mondrian procedure.

5.8 Observation-budget curves (v6f)

How many pointings does a new source need? For $k \in \{1, 2, 3, 5, 8, 13, 21\}$ and 500 draws (seed 42), each source's score is the mean OOF score of $\min(k, n_s)$ of its observations drawn without replacement; source BA (threshold 0.5) and source AUC (rank form of Eq. (5.4)) are computed for each draw and summarised by the median and the 2.5–97.5% range (`v6lib.budget_curves`; S3 OOF scores, so a source has up to 1329 observations to draw from).

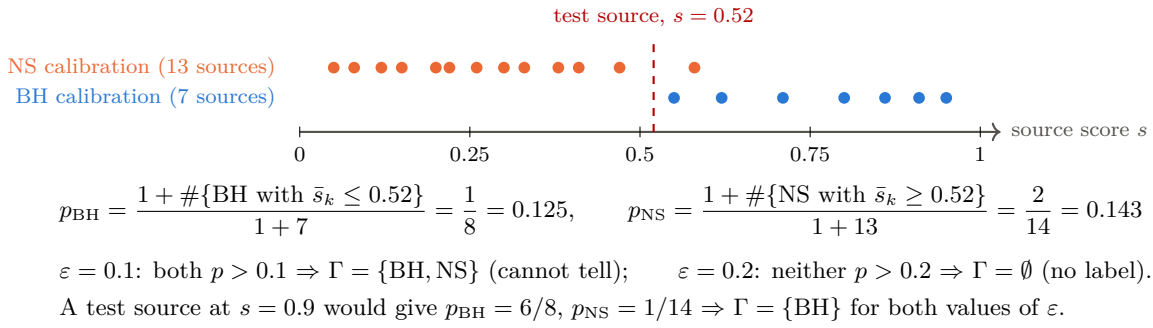


Figure 5.5: Illustration (made-up calibration scores) of the class-conditional conformal p-values of Eq. (5.9). A test source is compared with the calibration sources of each class separately; a label enters the prediction set if the test source is not too atypical for that class. Drawn for this report.

5.9 Agreement, bridging and reliability statistics

- **Agreement** between two models (v3a, v4): Spearman correlation of their OOF scores over common observations, and the fraction of identical observation-level predictions (`v3lib.agreement`, `v4lib.agreement_vs`).
- **Bridge check** (v4b2): H_R may replace H only if, on the v2 sample, H_R-LR and H-LR have paired source-BA and source-AUC intervals containing 0 and OOF scores with Spearman ≥ 0.90.
- **Split-half reliability** (v6, v8c, v10c): the feature is computed separately from odd and even rows/segments; the Spearman correlation $\rho_{1/2}$ between halves is stepped up to the full length with Spearman–Brown, $\rho_{\mathrm{full}} = 2\rho_{1/2}/(1+\rho_{1/2})$.

5.10 Frozen and prospective protocol

Several tests were designed so that nothing about the model could adapt to the test data. Table 5.2 lists what was frozen and when. Figure 5.6 shows the common protocol.

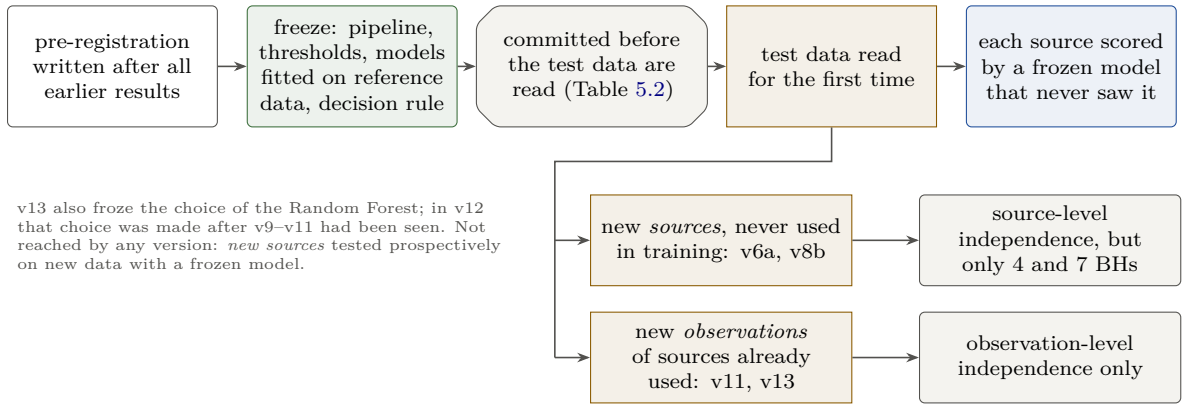


Figure 5.6: Frozen and prospective protocol. Everything that could adapt to the test data was fixed and committed before the data were read. The two kinds of test data give different levels of independence: new sources (v6a, v8b) and new observations of sources already used (v11, v13). Drawn for this report from Table 5.2 and the pre-registrations.

Two levels of independence must be distinguished: *source-level* independence (the test sources were never used in training: v6a, v8b, v11d) and *observation-level* independence (new observations

of sources already used: v11, v13). Only the second was achievable for NICER BHs, because every dynamically confirmed BH with usable RXTE or NICER timing data had already been used.

Table 5.2: What was frozen in the frozen and prospective tests, and when. "Out-of-source" = each source's new observations are scored by the model fitted without that source.

Test	Frozen before the test data were read	Model application	Record
v6a	H_R-LR, H_R+T-LR, CCTLR(H_R) fitted once on S1; imputation medians from S1	all external sources scored by the same S1 models	pre-reg. v6; models refitted deterministically in v8b (IC2: max diff 1.1×10^{-16})
v8b	the v6 models; v7a state rule	external set X (incl. GS 1354−64, SS 433)	pre-reg. v8 (`a481fa0`)
v11	NICER feature pipeline (`v9lib.obs_features`), state thresholds, models fitted on the v10 NICER full data, $\nu^* = 1.3007$ Hz	out-of-source for the 60 v9 sources; all-data model for never-used sources	pre-reg. `6fea016`; ν^* `58ded89`
v13	v12 pipeline incl. 3C50 correction and screening (Eq. (2.21)), RF selected as primary, models fitted on the 534 v12 observations, $\nu^* = 1.3361$ Hz	out-of-source	pre-reg. `42fe1d2` (with ν^*)

5.11 Implementation checks and multiple comparisons

From v4 on, every version defined *implementation checks* (IC) that had to pass before any result was looked at: reproduction of earlier OOF predictions to $\leq 10^{-8}$–10^{-15}, synthetic signals recovered by the timing estimators within 10%, pure-Poisson data giving $|z| < 3$, sanity checks with known sources (XTE J1118+480 hard-like, Z sources soft-like), and agreement with independent tools (XSPEC). A failed IC stops the sub-analysis or downgrades it to exploratory (v8c). Unit tests are in `scripts/test_v4lib.py`–`test_v11lib.py`.

Multiplicity. Each sub-analysis declares one primary comparison; all others are descriptive and are not used to claim a difference. No family-wise correction is applied across descriptive comparisons; where many are made (v4a: 76 combinations × 3 metrics = 228 intervals; v5: 45), the report states how many intervals would exclude 0 by chance ($\approx 2.5\%$ one-sided per interval, but strongly correlated) and draws no conclusion from them. Within a permutation family (v6d) Holm adjustment is reported.

Chapter 6

Phase 1 (v1–v3): spectral baselines

Phase at a glance: v1–v3

Question. Can one RXTE/PCA 5–25 keV spectrum classify the compact object of a source never seen in training, and where in the spectrum is the information?
Pre-declared primaries. v1: none formal (LOSO evaluation of A and B; leakage diagnostic); v2: paired B−A difference (source bootstrap); v3a: H+B − H ("is there information beyond two colours?"), judged on observation BA, source BA and source AUC.
Data. RXTE/PCA Standard-2 Standard Products, gain epoch 5 (2000–2011); v1: 8 sources (4 BH, 4 NS) × 15 observations; v2/v3: 31 sources (7 BH, 24 NS), 456 observations, 1842 ks; v3c: 14 BH candidates (210 observations, prediction only).
Verdict. Better than chance on unseen sources (source BA 0.80–0.87, source AUC 0.88–0.95 for the balanced models), but *the usable information is in two count-space colours*: the full shape (H+B − H), 3–5 keV and intensity add nothing **detectable**. Errors concentrate on hard-state BHs and hard NSs, which overlap in colour.

6.1 Motivation

Published classifiers of RXTE spectra reported 87–94% accuracy (Garg et al., 2026; Pattnaik et al., 2021), but with either sigma-clipped source accuracies or observation-wise random splits, and with BH candidates labelled as BHs. Neither published its processed spectra. Phase 1 rebuilt the data from the HEASARC Standard Products, used only confirmed labels, and asked the question in its strict form: how well can an *unseen source* be classified, and is the information more than a simple hardness measure? Figure 6.1 maps the phase: for each version, what it added, what it reused, the pre-declared comparison and the verdict.

6.2 Pre-registration

Table 6.1 summarises the three pre-registrations of this phase.

All three were written before the models of their version were trained, but after the results of the earlier versions; the v3 document states explicitly that v3a tests a question raised by v2.

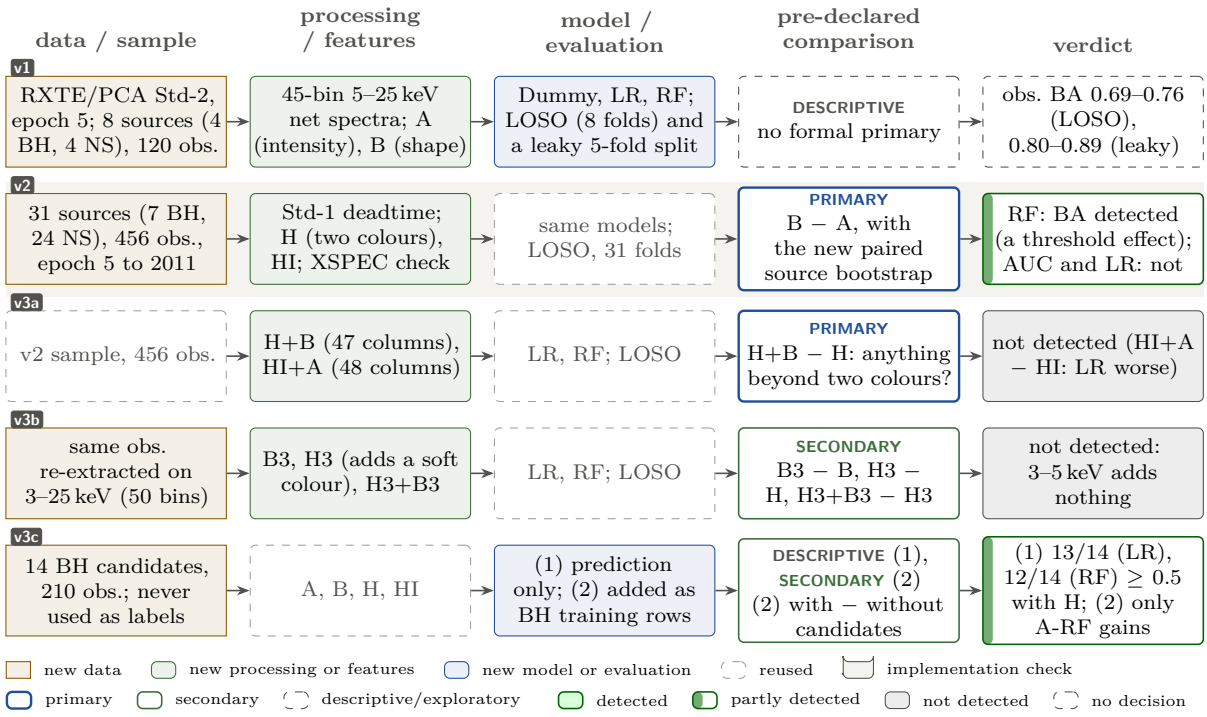


Figure 6.1: Phase 1 analysis map. Each row is one version or sub-analysis, read from left to right: the data, the processing or features, the models and evaluation, the pre-declared comparison and its verdict (95% class-stratified source-bootstrap interval). Solid boxes are new in that version; dashed boxes reuse an element of an earlier version. Values from Tables 6.8, 6.10 and 6.11 and Section 6.6. Drawn for this report.

Table 6.1: Phase 1 pre-registrations (`reports/preregistration.md`, `_v2.md`, `_v3.md`).

Ver.	Written	Primary / decision rule	Secondary / descriptive
v1	2026-09-29, after selection ran and during download, before any model	LOSO evaluation of A and B with Dummy, LR, RF; source rule $\bar{s} \geq 0.5$; no formal primary comparison	leaky observation-wise 5-fold (to show the leak)
v2	2026-09-29, after the v1 results, before any v2 download	source bootstrap (2000, seed 42) of v1 and v2; paired B−A difference	hardness baselines H and HI ("does the model only learn hardness?"); XSPEC cross-check
v3	2026-09-30, after v1/v2 results and the B–H agreement statistics	v3a: H+B − H (LR and RF); "information beyond hardness" only if the CI is entirely > 0	v3a: HI+A − HI, source AUC for all; v3b: B3−B, H3−H, H3+B3−H3; v3c: candidate predictions and label sensitivity

6.3 Data

6.3.1 Instrument, mode, band and epoch

RXTE/PCA Standard-2 Standard Products (Section 2.4), 5–25 keV on the 45-bin grid (v3b: 2.883–25.008 keV), net count rate per keV per PCU after background subtraction and deadtime correction (v1: good-xenon lower bound; v2: full Standard-1 recipe). Gain epoch 5 only: MJD 51677–54094 for v1 (2000-05-13 to 2006-12-25, before PCU1 lost its propane layer), MJD 51677–55931 for v2/v3 (to the end of the mission).

6.3.2 Sources

v1 used 4 BlackCAT dynamical BHs and 4 Galloway+2008 bursters with ≥ 30 eligible pointings (Table 6.2). v2 applied the label rule mechanically: every BlackCAT Table-4 BH and every Galloway+2008 appendix-A burster with a MissionLongData catalogue and ≥ 10 eligible pointings, giving 7 BH and 24 NS (Table 6.3); XTE J1859+226 (0 eligible in epoch 5) and V4641 Sgr (9) were dropped by rule. The NS set mixes atolls, two Z sources (GX 17+2, Cyg X-2), three accreting millisecond pulsars (SAX J1808.4−3658, XTE J1814−338, HETE J1900.1−2455) and globular-cluster sources. v3c used 14 BlackCAT BH candidates with a catalogue (Table 6.10); XTE J1748−288 and GRS 1739−278 had none.

Table 6.2: v1 sources and catalogue-level attrition (`data/sources.csv`, `results/attrition_catalog.csv`, `results/attrition_download.csv`). Columns: catalogue rows, after the epoch window (MJD 51677–54094), exposure ≥ 1000 s, `STD1RATE` ≥ 5, pointing $\leq 0.1°$ (= eligible); 15 accepted per source, all first-ranked candidates. Exposure and median net 5–25 keV rate of the accepted observations from `data/observations.csv`.

Source	label	cat.	window	exp.	rate	eligible	accepted	exp. [ks]	F [c/s/PCU]
GRO J1655−40	BH	593	512	477	440	440	15	43.7	596.4
GRS 1915+105	BH	1809	887	699	699	699	15	52.1	1151.6
XTE J1550−564	BH	414	138	99	79	63	15	50.3	74.4
4U 1543−47	BH	108	108	97	58	58	15	36.7	70.6
4U 1636−53	NS	1580	558	490	490	490	15	48.5	115.8
4U 1608−52	NS	1128	423	236	213	213	15	49.2	149.4
4U 1728−34	NS	440	188	180	180	174	15	59.1	221.8
Aql X-1	NS	602	270	248	226	226	15	61.9	103.7
Total	4/4						120	401.6	

Table 6.3: v2 sources (31 included, 2 BHs dropped by the ≥ 10 rule) and attrition (`data/v2/sources.csv`, `results/v2/attrition_catalog.csv`, `.../attrition_download.csv`). "Galloway" = appendix-A section of Galloway et al. (2008). Columns as in Table 6.2 (window MJD 51677–55931); "tried" = candidates processed; "acc." = accepted.

Source	label	evidence	cat.	window	exp.	elig.	tried	acc.	exp. [ks]
GRO J1655−40	BH	BlackCAT T4	593	512	477	440	15	15	43.7
GRS 1915+105	BH	BlackCAT T4	1809	1296	1033	1033	15	15	30.3
XTE J1550−564	BH	BlackCAT T4	414	138	99	63	15	15	50.3
4U 1543−47	BH	BlackCAT T4	108	108	97	58	15	15	36.7
GX 339−4	BH	BlackCAT T4	1405	1315	692	652	15	15	27.1
XTE J1650−500	BH	BlackCAT T4	184	184	128	95	15	15	40.8
XTE J1118+480	BH	BlackCAT T4	99	71	51	41	15	15	38.7
XTE J1859+226	BH (dropped)	BlackCAT T4	136	2	2	0	–	–	–

Source	label	evidence	cat.	window	exp.	elig.	tried	acc.	exp. [ks]
V4641 Sgr	BH (dropped)	BlackCAT T4	75	63	44	9	–	–	–
4U 1636−53	NS	A.8	1580	1482	1332	1332	15	15	54.1
4U 1608−52	NS	A.7	1128	1002	567	540	15	15	24.7
4U 1728−34	NS	A.16	440	296	282	276	15	15	37.1
Aql X-1	NS	A.44	602	462	432	400	15	15	40.2
EXO 0748−676	NS	A.2	751	616	526	510	16	15	39.1
4U 1254−69	NS	A.5	37	27	24	24	15	15	154.8
4U 1323−62	NS	A.6	32	20	19	19	15	15	93.0
MXB 1659−298	NS	A.9	81	36	33	30	15	15	45.4
4U 1702−429	NS	A.10	228	211	201	201	15	15	80.7
4U 1705−44	NS	A.11	508	429	402	402	15	15	37.4
4U 1724−307	NS	A.15	127	79	77	77	15	15	47.8
KS 1731−260	NS	A.18	86	45	42	33	15	15	89.9
SLX 1735−269	NS	A.19	82	66	65	64	15	15	57.1
4U 1735−44	NS	A.20	224	172	161	161	15	15	54.6
GX 3+1	NS	A.28	108	77	75	75	15	15	61.6
4U 1746−37	NS	A.32	62	45	43	43	15	15	86.4
SAX J1808.4−3658	NS (AMXP)	A.36	497	438	294	234	15	15	89.8
XTE J1814−338	NS (AMXP)	A.37	93	93	80	73	15	15	66.1
GX 17+2	NS (Z)	A.38	212	101	90	90	15	15	90.4
4U 1820−30	NS	A.39	266	140	133	133	15	15	84.7
GS 1826−238	NS	A.40	160	142	140	118	15	15	97.2
HETE J1900.1−2455	NS (AMXP)	A.43	358	358	349	344	15	15	42.2
4U 1916−053	NS	A.45	54	12	11	11	11	6	50.9
Cyg X-2	NS (Z)	A.48	591	449	403	403	15	15	49.2
Total (31 included)	7/24							456	1841.8

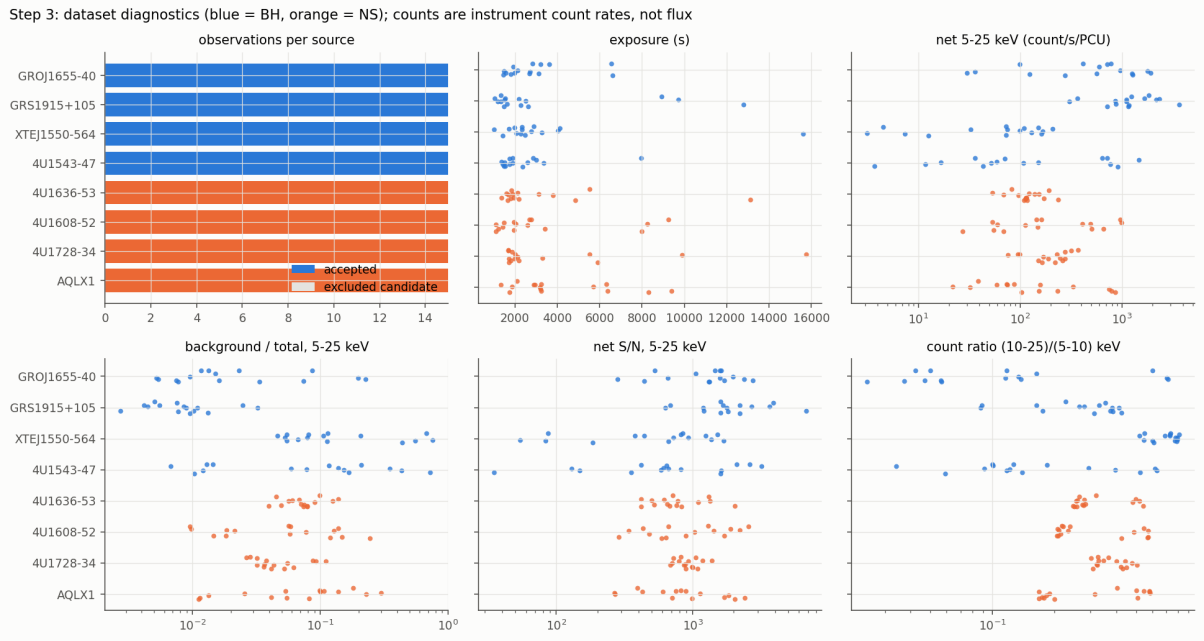
Of the 462 candidates processed in v2, 6 were excluded: one EXO 0748−676 observation with a short spectrum exposure (replaced by the next candidate of the same time group) and five 4U 1916−053 ObsIDs of proposal 30066 for which the Standard-2 products were not found (HTTP 404), leaving 6 observations of that source in 11 time groups.

6.3.3 Data quality

All integrity checks of `06_dataset_checks.py` pass (unique ObsIDs, labels consistent, all values finite, units `count`, BACKSCAL ratio 1, header position $\leq 0.045°$ from the source). Non-positive net bins: 7 of 5400 in v1 (3 observations, all above 21 keV), 55 of 20 520 in v2 (18 observations, high-energy end of soft spectra); kept as is. Minimum S/N: 34.9 (v1), 19.1 (v2). Deadtime (v2 recipe): median 1.38%, maximum 8.8%. Exposure: 401.6 ks (v1); 1841.8 ks in v2, of which 267.6 ks BH and 1574.2 ks NS. Figures 6.2 and 6.3 show the per-source distributions of the quantities that define the sample.

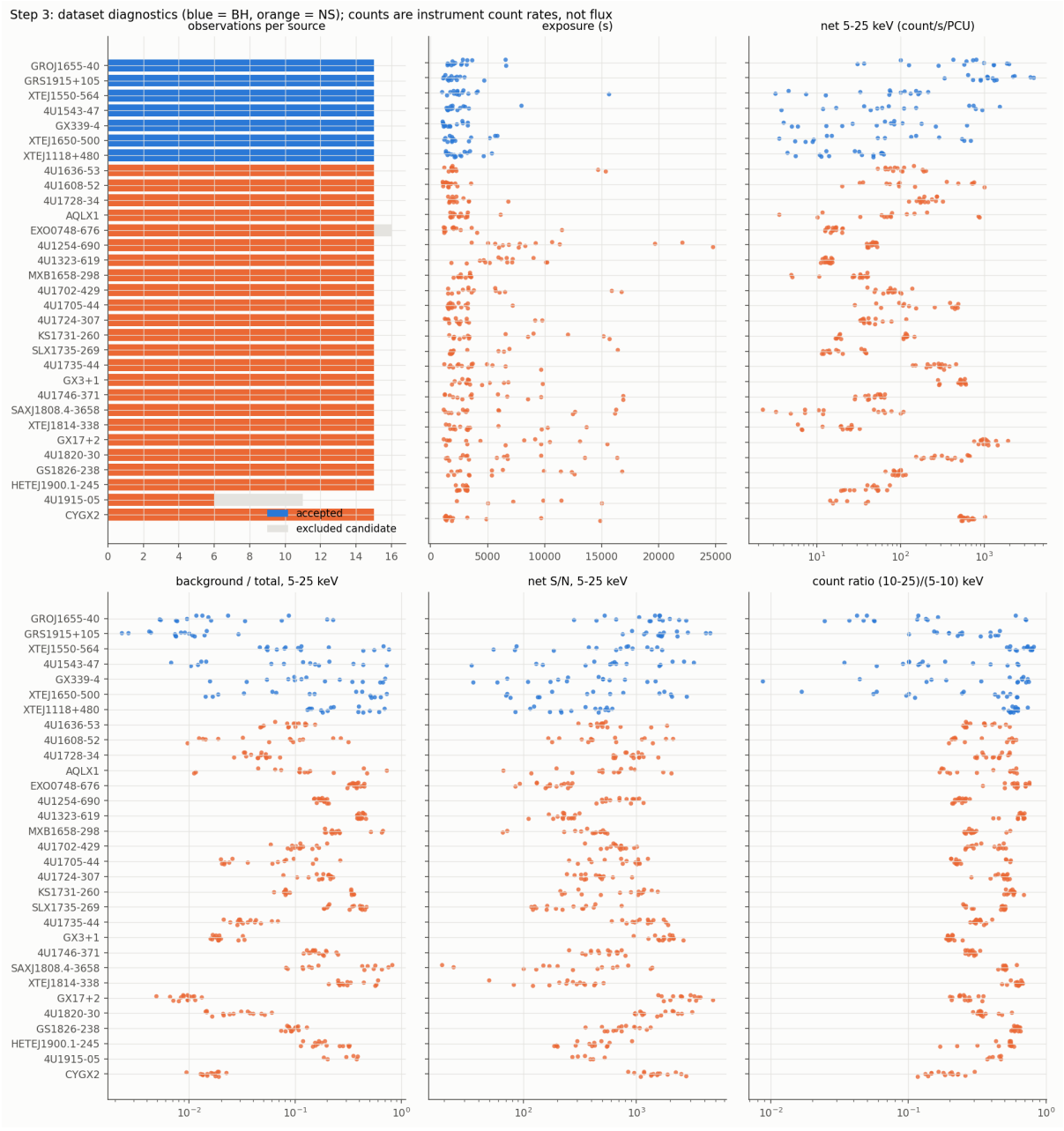
6.3.4 BH candidates (v3c)

The 16 candidates were drafted from memory, then verified against the BlackCAT web pages and Corral-Santana et al. (2016) and frozen on 2026-09-30 03:47 +08:00 before any v3c download (all 16 kept; `reports/v3c_candidate_verification.md`): none is dynamically confirmed and none has NS evidence; MAXI J1659−152 and Swift J1753.5−0127 are in BlackCAT's "strictly speaking not dynamical but strong mass evidence" class. The only post-freeze change was the catalogue file name of Swift J1753.5−0127 (`SWIFTJ1753.5-01`). 14 candidates had catalogues,



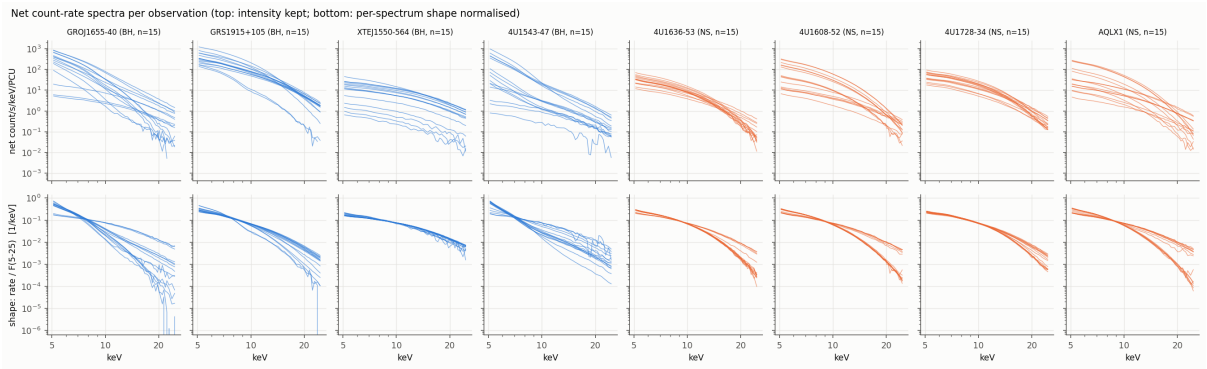
(a) v1, 8 sources

Figure 6.2: Per-source diagnostics of the accepted observations (blue = BH, orange = NS): observations per source (accepted vs excluded candidates), spectrum exposure, net 5–25 keV count rate per PCU, background fraction, net S/N and the count ratio (10–25)/(5–10) keV (log axes where indicated). BH and NS overlap in every quantity; the BHs are on average brighter in v1 (GRS 1915+105, GRO J1655−40), a possible intensity shortcut that representation A could exploit. Repo: `figures/step3_dataset_diagnostics.png`, `figures/v2/step3_dataset_diagnostics.png`; script `scripts/06_dataset_checks.py`. Panel (b), v2, is in the continuation of this figure.



(b) v2, 31 sources

Figure 6.2: (continued) Per-source diagnostics of the v2 sample (31 sources); panels as in (a).



(a) v1

Figure 6.3: Net count-rate spectra of every accepted observation, one panel pair per source (upper: count rate per keV per PCU, log–log, non-positive bins omitted; lower: per-spectrum shape normalised to the 5–25 keV rate). The spread within a source (state changes) is comparable to the spread between sources, which is why a single observation is a hard input. Repo: `figures/step3_spectra_by_source.png`, `figures/v2/step3_spectra_by_source.png`; script `scripts/06_dataset_checks.py`. Panel (b), v2, is in the continuation of this figure.

each with ≥ 16 eligible pointings; 15 per source were accepted (210 in total, 561.7 ks), header pointings within 0.035°. Five lie in the crowded Galactic-centre field (three with data: SLX 1746−331, GRS 1758−258, 1E 1740.7−2942), where other bright sources may contaminate the 1° PCA field of view. Figure 6.4 shows their diagnostics.

6.4 Data structure in this phase

Table 6.4 lists every array passed to a model in Phase 1 (Chapter 3). Rows of all v2 matrices are in the order of `data/v2/processed/features.npz`; folds are `LeaveOneGroupOut` on `source_id` (`results/v2/folds_loso.csv`, 31 folds; v1: 8 folds, `results/folds_loso.csv`). In a v2 fold, the training set has 441 rows (450 when 4U 1916−053 is held out); when a BH source is held out it contains 90 BH and 351 NS rows, so the balanced weights are $441/(2 \cdot 90) = 2.45$ for BH and $441/(2 \cdot 351) = 0.63$ for NS rows.

Compared with v1, v2 adds rows (31 sources instead of 8; 456 instead of 120 observations) and the H/HI representations; v3a adds the nested concatenations, v3b five low-energy columns, v3c 210 unlabelled rows used only as extra BH training rows or as prediction targets.

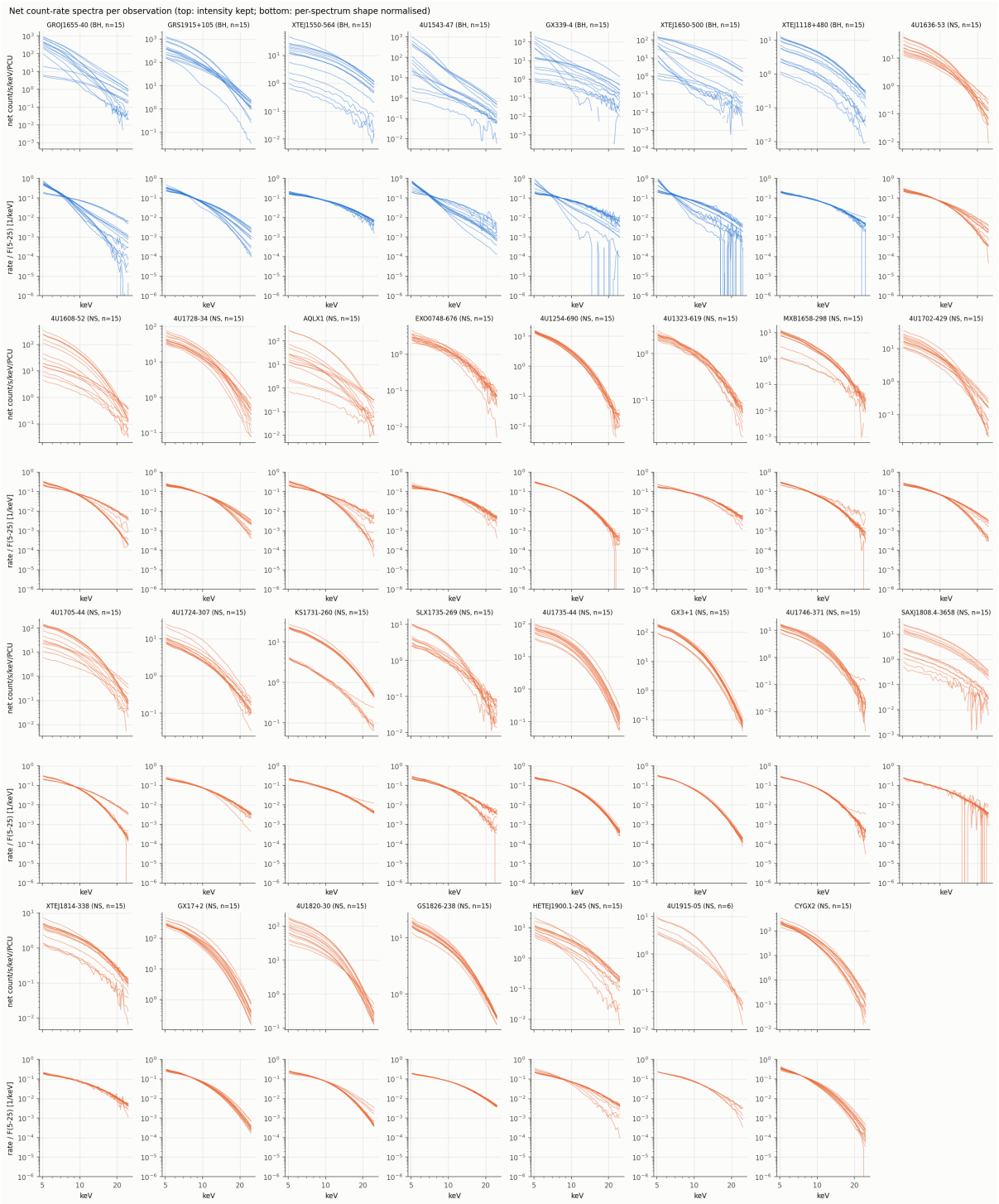
6.5 Models in this phase

Table 6.5 lists the models.

6.6 Results

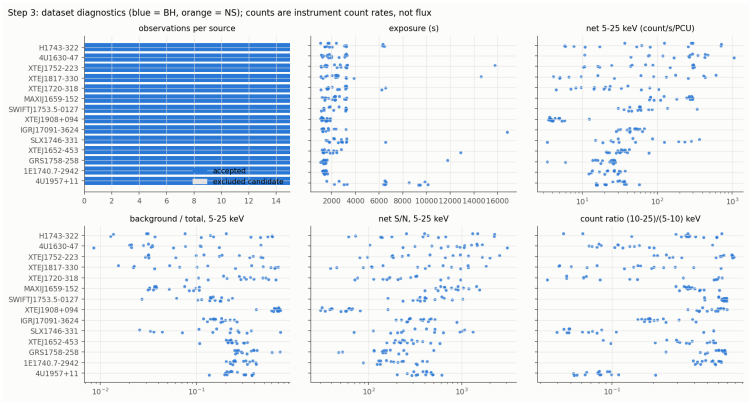
6.6.1 The pilot (v1, 8 sources)

DESCRIPTIVE With 4 BH and 4 NS sources, the observation-level BA under LOSO was 0.70 (A-LR), 0.69 (A-RF), 0.76 (B-LR) and 0.74 (B-RF); 6–7 of 8 sources were classified correctly, but every model had one or two sources that were mostly wrong, and *which* sources failed changed

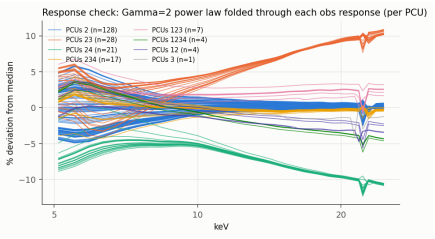


(b) v2

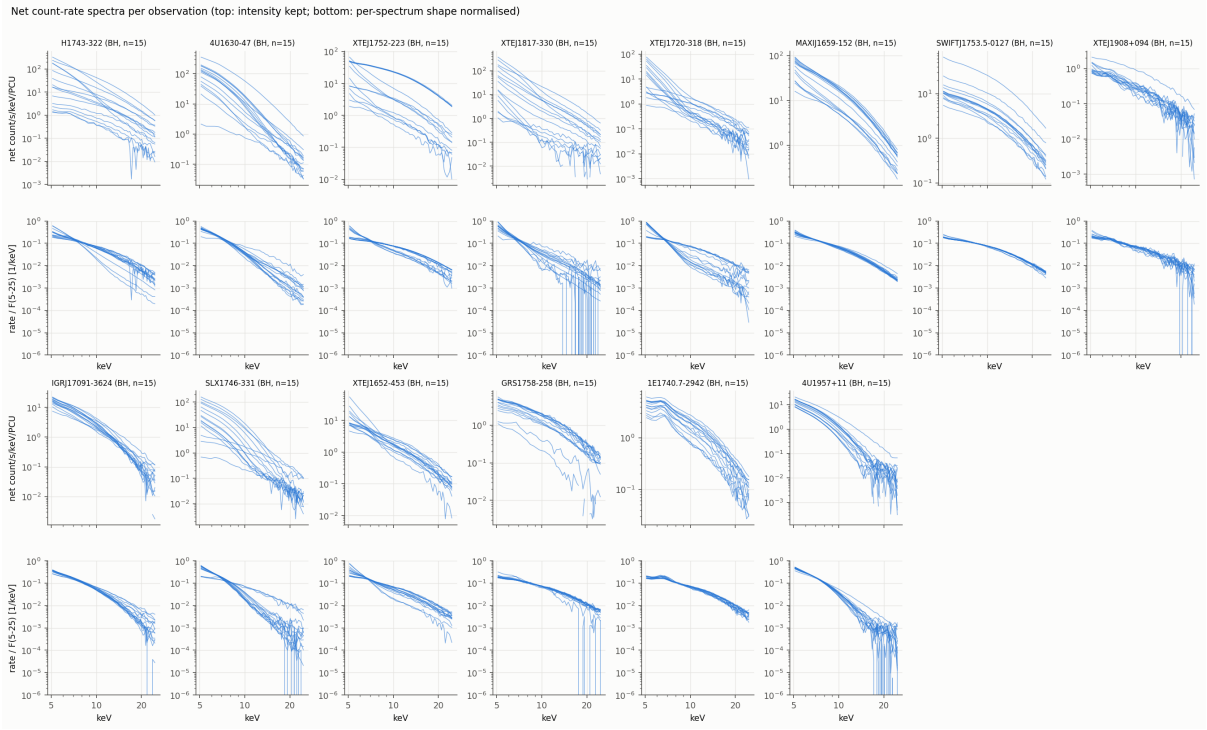
Figure 6.3: (continued) Net count-rate spectra of the v2 sample, one panel pair per source as in (a).



(a) diagnostics



(b) response check



(c) spectra

Figure 6.4: The 14 v3c BH candidates (all plotted in the BH colour although they are unlabelled): per-source diagnostics as in Figure 6.2, the response check of Figure 2.4, and spectra as in Figure 6.3. XTE J1908+094 is the faintest (median 4.1 $\mathrm{c\,s^{-1}\,PCU^{-1}}$); 4U 1630−47 the brightest (269). Repo: `figures/v3c/step3_*.png`; script `scripts/06_dataset_checks.py` with `XRB_VERSION=v3c`.

Table 6.4: Arrays passed to the models in Phase 1. All float arrays are float64; $\boldsymbol{y}$ is int64; $\boldsymbol{g}$ is a unicode array of source identifiers. "npz" = `data/v2/processed/features.npz` unless stated.

Analysis	Array	Shape	Columns (unit)	Source
v1	$\boldsymbol{X}_A$, $\boldsymbol{X}_B$	120×45	A_{ij} (dimensionless), B_{ij} (keV^{-1}), $j = 0..44$	`data/processed/features.npz`: `rate`, `F`
	$\boldsymbol{y}$, $\boldsymbol{g}$	120	60 BH; 8 sources	same: `y`, `source_id`
v2	$\boldsymbol{X}_A$, $\boldsymbol{X}_B$	456×45	as v1	npz: `rate`, `F`
	$\boldsymbol{X}_H$	456×2	h_1, h_2 (Eq. (2.11))	npz: `rate`, `edges`
	$\boldsymbol{X}_{HI}$	456×3	$h_1, h_2, \log_{10} F$	npz
	$\boldsymbol{y}$, $\boldsymbol{g}$	456	105 BH; 31 sources	npz
v3a	$\boldsymbol{X}_{H+B}$	456×47	$h_1, h_2, B_0..B_{44}$	npz
	$\boldsymbol{X}_{HI+A}$	456×48	$h_1, h_2, \log_{10} F, A_0..A_{44}$	npz
v3b	$\boldsymbol{X}_{B3}$	456×50	$x_{ij}/F_i^{(3)}$ on the 50-bin grid	`data/v3/processed/features_3_25.npz`
	$\boldsymbol{X}_{H3}$	456×3	$C_0/C_1, C_2/C_1, C_4/C_3$	same
	$\boldsymbol{X}_{H3+B3}$	456×53	H3 and B3	same
v3c (1)	train $\boldsymbol{X}$	$456 \times p$	A, B, H or HI of the 31 confirmed sources	npz
	predict $\boldsymbol{X}_c$	$210 \times p$	same representation of the candidates	`data/v3c/processed/features.npz`
v3c (2)	train (fold k)	$(456 - n_k + 210) \times p$	confirmed training rows + all candidate rows, candidates with $y = 1$	both npz
	test (fold k)	$n_k \times p$	held-out confirmed source	npz
leaky split	$\boldsymbol{X}_A$, $\boldsymbol{X}_B$, $\boldsymbol{X}_H$, $\boldsymbol{X}_{HI}$	as above	`StratifiedKFold(5, shuffle, 42)` over rows	npz

Table 6.5: Models of Phase 1 (all with fixed hyper-parameters, no tuning; Chapter 4).

Model	Representations	Weights	Hyper-parameters
Dummy (prior)	A, B, H, HI	–	Section 4.2
LR (Section 4.3)	A, B, H, HI, H+B, HI+A, B3, H3, H3+B3	balanced	$C = 1$, L2, lbfgs
RF (Section 4.4)	same	balanced	500 trees, `sqrt`, leaf 2, seed 42

with the representation (Table 6.6). The Dummy scored BA = 0 for the reason explained in Section 4.2. The source bootstrap (added in v2) gives source BA 0.75 [0.50, 1.00] (A-LR), 0.875 [0.625, 1.00] (A-RF), 0.75 [0.375, 1.00] (B-LR, B-RF), and the B−A observation-BA differences +0.058 [−0.158, +0.217] (LR) and +0.050 [−0.250, +0.283] (RF): eight sources cannot distinguish A from B (`results/bootstrap_source_level.csv`). The leaky observation-wise split gave 0.80–0.89 (Figure 1.1).

Table 6.6: v1 per-source LOSO results: source-mean BH score (fraction of the source's 15 observations classified correctly). Bold = source misclassified by the 0.5 rule. From `results/per_source_results.csv`.

Source	label	A-LR	A-RF	B-LR	B-RF
GRO J1655-40	BH	0.73 (0.67)	0.69 (0.80)	0.92 (1.00)	0.94 (1.00)
GRS 1915+105	BH	**0.48** (0.67)	0.81 (0.87)	**0.17** (0.13)	**0.19** (0.13)
XTE J1550-564	BH	**0.36** (0.27)	0.53 (0.33)	0.56 (0.73)	0.72 (0.73)
4U 1543-47	BH	0.81 (0.87)	**0.45** (0.27)	0.97 (1.00)	0.90 (1.00)
4U 1636-53	NS	0.17 (1.00)	0.19 (1.00)	0.09 (1.00)	0.21 (1.00)
4U 1608-52	NS	0.25 (0.80)	0.30 (0.87)	0.28 (0.73)	0.23 (0.93)
4U 1728-34	NS	0.27 (0.87)	0.16 (0.87)	0.10 (1.00)	0.38 (0.73)
Aql X-1	NS	0.42 (0.47)	0.44 (0.53)	**0.50** (0.47)	**0.57** (0.40)

Figure 6.5 shows the v1 LOSO scores per observation and source.

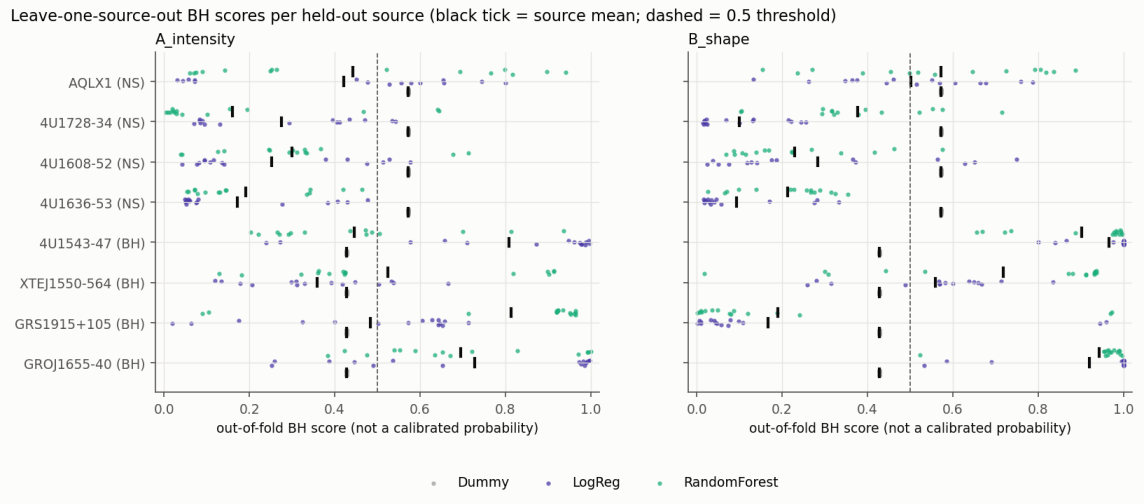


Figure 6.5: v1: out-of-fold BH scores of every observation of each held-out source (left A, right B; grey Dummy, purple LR, green RF), black tick = source mean, dashed line = 0.5. GRS 1915+105 is classified by its intensity (A) but not by its shape (B); XTE J1550−564 the other way round. Repo: `figures/step4_loso_scores.png`; script `scripts/07_train_eval.py`.

6.6.2 How much is in two colours? (v2, v3a, v3b)

Representations on 31 sources (v2). Table 6.7 gives every representation and model, with source AUC from v3a. The balanced models reach source BA 0.77–0.87 and source AUC 0.88–0.95; the two-colour H-LR has source BA 0.795 [0.589, 0.958] and source AUC 0.923 [0.804, 0.994] – this is the reference against which v4 tests every alternative. RF on the intensity representation A fails at the 0.5 threshold (source BA 0.62, 2/7 BH correct) although its ranking is good (AUC 0.88).

Table 6.7: v2/v3a LOSO results on 31 sources (456 observations). 95% intervals: 2000 class-stratified source bootstrap replicates. From `results/v3/v3a_metrics.csv` and `results/v3/v3a_bootstrap.csv` (v2 rows are reproduced exactly; `results/v2/metrics_*.csv`).

Repr.	Model	obs. BA	source BA	source AUC	BH ok	NS ok
A	LR	0.724 [0.606, 0.817]	0.866 [0.702, 0.979]	0.946 [0.833, 1.000]	6/7	21/24
A	RF	0.606 [0.546, 0.661]	0.622 [0.458, 0.786]	0.881 [0.738, 0.988]	2/7	23/24
B	LR	0.761 [0.592, 0.903]	0.774 [0.568, 0.938]	0.875 [0.690, 1.000]	5/7	20/24
B	RF	0.761 [0.621, 0.877]	0.857 [0.643, 1.000]	0.893 [0.732, 1.000]	5/7	24/24
H	LR	0.776 [0.633, 0.893]	0.795 [0.589, 0.958]	0.923 [0.804, 0.994]	5/7	21/24
H	RF	0.755 [0.606, 0.882]	0.857 [0.643, 1.000]	0.905 [0.750, 1.000]	5/7	24/24
HI	LR	0.782 [0.673, 0.881]	0.795 [0.589, 0.958]	0.929 [0.810, 1.000]	5/7	21/24
HI	RF	0.754 [0.609, 0.874]	0.836 [0.643, 1.000]	0.946 [0.839, 1.000]	5/7	23/24
H+B	LR	0.758 [0.589, 0.899]	0.774 [0.568, 0.938]	0.875 [0.690, 1.000]	5/7	20/24
H+B	RF	0.768 [0.625, 0.895]	0.857 [0.643, 1.000]	0.893 [0.732, 1.000]	5/7	24/24
HI+A	LR	0.727 [0.599, 0.832]	0.795 [0.589, 0.958]	0.923 [0.792, 1.000]	5/7	21/24
HI+A	RF	0.721 [0.615, 0.811]	0.693 [0.500, 0.857]	0.935 [0.821, 1.000]	3/7	23/24
Dummy	–	0.500	–	–	0/7	24/24

Paired comparisons. Table 6.8 collects the paired differences. PRIMARY (v3a) H+B − H: every interval contains 0 or lies below it (LR source AUC -0.048 $[-0.179, +0.048]$; RF -0.012 $[-0.077, +0.048]$) ⇒ information beyond the two colours is **not detected** under the pre-registered criterion. SECONDARY HI+A − HI: adding 45 intensity bins makes LR worse at the observation level (-0.055 $[-0.100, -0.017]$, CI entirely below 0). PRIMARY (v2) B−A: for RF the shape beats intensity ($+0.154$ $[+0.030, +0.270]$ observation BA; $+0.235$ $[+0.042, +0.429]$ source BA) but not in AUC ($+0.012$ $[-0.149, +0.161]$): the A-RF failure is a threshold, not a ranking, problem. POST HOC B−H and A−HI were added after training (logged): B−H is within ±0.02 of 0 for both models. The descriptive agreement is high: B and H give OOF scores with Spearman 0.91 (LR) / 0.83 (RF) and identical labels for 88% / 93% of observations; H+B vs H: 0.92 / 0.85 and 88% / 93% (`results/v3/v3a_agreement.csv`).

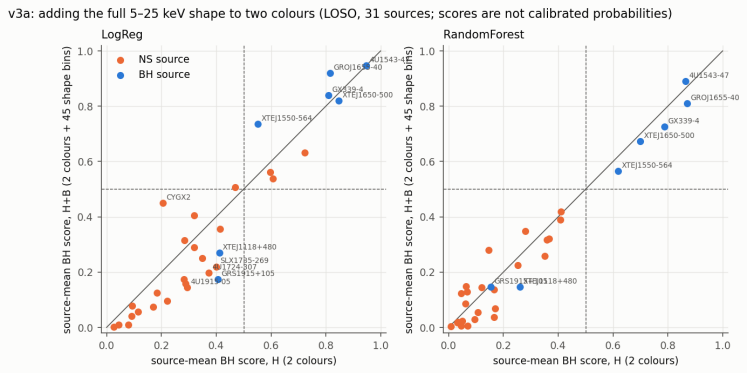
3–5 keV (v3b). The extended grid has 50 bins from 2.883 keV; all 456 observations were re-extracted from the same raw files with the same deadtime factors, the 5–25 keV part reproduces v2 to a relative difference of 6.9×10^{-16}, and no 3–5 keV bin is non-positive (minimum $F(3\text{–}5) = 1.37\,\mathrm{c\,s^{-1}\,PCU^{-1}}$; `results/v3/v3b_extraction_checks.csv`). None of the three pre-declared comparisons has an interval entirely above 0; adding a soft colour even lowers the LR observation BA (-0.027 $[-0.048, -0.008]$). The soft colour's highest per-source medians belong to three BHs (4U 1543−47, XTE J1650−500, GRO J1655−40), but the other four BHs lie among the NSs (Figure 6.6b); differences at 3–5 keV are expected to be dominated by the interstellar column N_{H}, which is tied to the line of sight, not to the compact object (stated in the pre-registration; not corrected in v3).

Figure 6.6 shows how closely the H+B and H models agree, source by source and observation by observation.

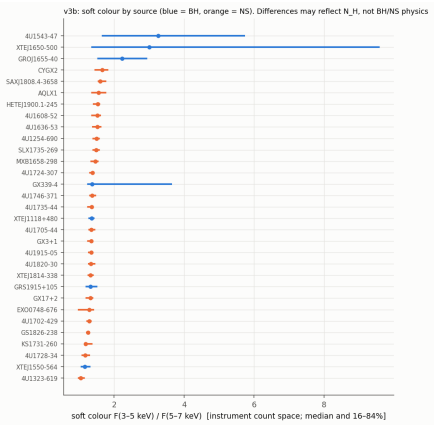
Which sources fail. Table 6.9 lists every source's mean score under all twelve combinations. XTE J1118+480 (BH, almost always in a hard state in RXTE data) is classified NS by all of them (0.14–0.41); GRS 1915+105 (BH) is NS for every shape/colour model (0.15–0.43) and BH only with A-LR (intensity, 0.62); LR tends to call the hard NSs SAX J1808.4−3658, HETE J1900.1−2455 (both AMXPs) and KS 1731−260 BHs, while RF classifies all 24 NSs correctly with B and H. Figures 6.7 and 6.8 show the observation-level picture: misclassified NS observations cluster at hard count ratios ((10–25)/(5–10) keV $\gtrsim 0.4$) and low rates, where the

Table 6.8: Phase 1 paired differences (first minus second), 95% source-bootstrap CI. "*" = CI excludes 0. From `results/v3/v3a_bootstrap.csv` and `results/v3/v3b_bootstrap.csv`.

Comparison	Model	obs. BA	source BA	source AUC
PRIMARY v3a: information beyond two colours				
H+B − H	LR	−0.018 [−0.078, +0.031]	−0.021 [−0.062, 0.000]	−0.048 [−0.179, +0.048]
H+B − H	RF	+0.013 [−0.009, +0.038]	0.000 [0.000, 0.000]	−0.012 [−0.077, +0.048]
SECONDARY v3a: intensity bins beyond colours + intensity				
HI+A − HI	LR	−0.055 [−0.100, −0.017]*	0.000 [0.000, 0.000]	−0.006 [−0.077, +0.054]
HI+A − HI	RF	−0.033 [−0.114, +0.061]	−0.143 [−0.340, +0.021]	−0.012 [−0.083, +0.048]
PRIMARY v2: shape vs intensity				
B − A	LR	+0.037 [−0.092, +0.137]	−0.092 [−0.268, +0.021]	−0.071 [−0.238, +0.024]
B − A	RF	+0.154 [+0.030, +0.270]*	+0.235 [+0.042, +0.429]*	+0.012 [−0.149, +0.161]
POST HOC v2: full shape vs colours; full intensity vs colours + intensity				
B − H	LR	−0.015 [−0.076, +0.035]	−0.021 [−0.062, 0.000]	−0.048 [−0.179, +0.048]
B − H	RF	+0.006 [−0.015, +0.026]	0.000 [0.000, 0.000]	−0.012 [−0.077, +0.048]
A − HI	LR	−0.059 [−0.137, +0.018]	+0.071 [−0.042, +0.235]	+0.018 [−0.030, +0.077]
A − HI	RF	−0.148 [−0.254, −0.024]*	−0.214 [−0.408, −0.021]*	−0.065 [−0.173, +0.018]
SECONDARY v3b: adding 3–5 keV				
B3 − B	LR	+0.013 [−0.011, +0.037]	+0.021 [0.000, +0.062]	−0.030 [−0.119, +0.018]
B3 − B	RF	−0.001 [−0.027, +0.021]	−0.021 [−0.062, 0.000]	−0.060 [−0.202, +0.024]
H3 − H	LR	−0.027 [−0.048, −0.008]*	0.000 [0.000, 0.000]	−0.018 [−0.071, 0.000]
H3 − H	RF	−0.020 [−0.066, +0.020]	−0.092 [−0.235, 0.000]	−0.054 [−0.161, +0.006]
H3+B3 − H3	LR	+0.015 [−0.042, +0.071]	+0.021 [0.000, +0.062]	−0.065 [−0.214, +0.048]
H3+B3 − H3	RF	+0.027 [−0.012, +0.076]	+0.071 [0.000, +0.214]	−0.030 [−0.089, 0.000]



(a) v3a: H vs H+B source scores



(b) v3b: soft colour by source

Figure 6.6: (a) Source-mean BH score with two colours (x) versus two colours plus 45 shape bins (y), LR left, RF right; blue BH, orange NS; dashed lines at 0.5. Points lie on the diagonal: the extra 45 columns change almost nothing. (b) Median and 16–84% range of the soft colour $F(3\text{–}5)/F(5\text{–}7)$ per source (blue BH, orange NS); three BHs are the softest but four BHs sit among the NSs, and the ordering may reflect $N_{\rm H}$. Repo: `figures/v3/v3a_H_vs_HplusB_source_scores.png` (`scripts/11_v3a_increment.py`), `figures/v3/v3b_soft_colour_by_source.png` (`scripts/12_v3b_extended_band.py`).

hard BHs also lie; misclassified BH observations are either hard or the very bright, soft GRS 1915+105 observations.

Table 6.9: v2/v3a per-source mean OOF BH score for every representation and model (LR, RF). Bold = misclassified by the 0.5 rule. n = observations. From `results/v3/v3a_per_source.csv`.

			A		B		H		HI		H+B		HI+A	
Source	lab.	n	LR	RF	LR	RF	LR	RF	LR	RF	LR	RF	LR	RF
GRO J1655-40	BH	15	0.68	0.52	0.92	0.82	0.82	0.87	0.85	0.83	0.92	0.81	0.74	0.67
GRS 1915+105	BH	15	0.62	**0.45**	**0.18**	**0.15**	**0.41**	**0.15**	**0.43**	**0.35**	**0.17**	**0.15**	**0.34**	**0.49**
XTE J1550-564	BH	15	0.61	**0.44**	0.75	0.58	0.55	0.62	0.62	0.63	0.74	0.56	0.67	**0.49**
4U 1543-47	BH	15	0.87	**0.33**	0.94	0.88	0.95	0.86	0.93	0.77	0.95	0.89	0.90	**0.50**
GX 339-4	BH	15	0.70	**0.50**	0.84	0.72	0.81	0.79	0.80	0.70	0.84	0.73	0.78	0.58
XTE J1650-500	BH	15	0.87	0.52	0.82	0.68	0.85	0.70	0.81	0.70	0.82	0.67	0.85	0.64
XTE J1118+480	BH	15	**0.34**	**0.24**	**0.28**	**0.14**	**0.41**	**0.26**	**0.36**	**0.23**	**0.27**	**0.15**	**0.33**	**0.23**
4U 1636-53	NS	15	0.25	0.14	0.09	0.03	0.22	0.09	0.22	0.04	0.10	0.03	0.21	0.07
4U 1608-52	NS	15	0.27	0.13	0.25	0.14	0.35	0.17	0.37	0.10	0.25	0.14	0.29	0.09
4U 1728-34	NS	15	0.39	0.29	0.07	0.03	0.17	0.17	0.23	0.17	0.07	0.04	0.29	0.27
Aql X-1	NS	15	0.37	0.25	0.36	0.33	0.41	0.37	0.38	0.26	0.36	0.32	0.37	0.23
EXO 0748-676	NS	15	0.22	0.20	0.32	0.23	0.28	0.25	0.22	0.26	0.31	0.22	0.21	0.20
4U 1254-69	NS	15	0.16	0.05	0.06	0.02	0.11	0.03	0.06	0.02	0.06	0.02	0.08	0.05
4U 1323-62	NS	15	0.21	0.34	0.41	0.30	0.32	0.15	0.24	0.26	0.40	0.28	0.20	0.34
MXB 1659-298	NS	15	0.33	0.15	0.16	0.14	0.29	0.12	0.18	0.13	0.16	0.14	0.21	0.14
4U 1702-429	NS	15	0.28	0.09	0.16	0.10	0.28	0.06	0.27	0.02	0.17	0.09	0.26	0.05
4U 1705-44	NS	15	0.16	0.05	0.12	0.03	0.18	0.05	0.19	0.02	0.12	0.02	0.15	0.03
4U 1724-307	NS	15	0.32	0.24	0.19	0.07	0.37	0.17	0.34	0.09	0.20	0.07	0.30	0.17
KS 1731-260	NS	15	**0.61**	0.45	**0.55**	0.31	**0.60**	0.36	**0.64**	0.19	**0.56**	0.32	**0.65**	0.35
SLX 1735-269	NS	15	0.37	0.17	0.21	0.15	0.40	0.06	0.32	0.07	0.22	0.15	0.33	0.13
4U 1735-44	NS	15	0.03	0.02	0.00	0.00	0.03	0.01	0.03	0.01	0.00	0.00	0.01	0.01
GX 3+1	NS	15	0.03	0.04	0.08	0.14	0.09	0.07	0.11	0.04	0.08	0.13	0.03	0.02
4U 1746-37	NS	15	0.12	0.02	0.01	0.00	0.08	0.05	0.05	0.01	0.01	0.01	0.07	0.01
SAX J1808.4-3658	NS	15	**0.63**	0.40	**0.59**	0.36	**0.72**	0.28	**0.64**	0.28	**0.63**	0.35	**0.64**	0.37
XTE J1814-338	NS	15	0.34	0.35	**0.51**	0.42	0.47	0.41	0.40	**0.51**	**0.51**	0.42	0.37	0.47
GX 17+2	NS	15	0.17	**0.60**	0.04	0.01	0.09	0.07	0.16	0.40	0.04	0.00	0.10	**0.58**
4U 1820-30	NS	15	0.11	0.04	0.01	0.02	0.04	0.03	0.06	0.05	0.01	0.02	0.06	0.03
GS 1826-238	NS	15	**0.51**	0.19	0.30	0.14	0.32	0.05	0.39	0.15	0.29	0.12	0.47	0.19
HETE J1900.1-2455	NS	15	0.48	0.33	**0.52**	0.27	**0.61**	0.35	**0.57**	0.22	**0.54**	0.26	**0.53**	0.28
4U 1916-053	NS	6	0.24	0.11	0.14	0.05	0.29	0.11	0.22	0.08	0.14	0.06	0.21	0.09
Cyg X-2	NS	15	0.08	0.10	0.44	0.39	0.21	0.41	0.25	0.35	0.45	0.39	0.09	0.20

Figure 6.9 compares the per-source scores of the intensity (A) and shape (B) representations.

6.6.3 Label sensitivity: BH candidates (v3c)

DESCRIPTIVE **Predictions only** (models trained on all 31 confirmed sources; candidates have no verified label, so no accuracy is computed): with H, 13/14 candidates have a source-mean score ≥ 0.5 for LR and 12/14 for RF (Table 6.10); hard-state BH XTE J1118+480. 1E 1740.7−2942, GRS 1758−258 and IGR J17091−3624 are near the threshold and model-dependent. SECONDARY **Candidates added to training as BHs** (LOSO only over confirmed sources, so the test labels remain confirmed): Table 6.11. Only A-RF improves clearly (source BA +0.244 [+0.051, +0.429], observation BA +0.111 [+0.035, +0.182]): adding 210 BH rows (105 → 315) fixes its threshold problem. The balanced H and B models hardly change.

Figure 6.10 shows the candidate scores.

6.6.4 What the classifier looks like

DESCRIPTIVE Figure 6.11 draws the two colours of all 456 confirmed observations with the 0.5

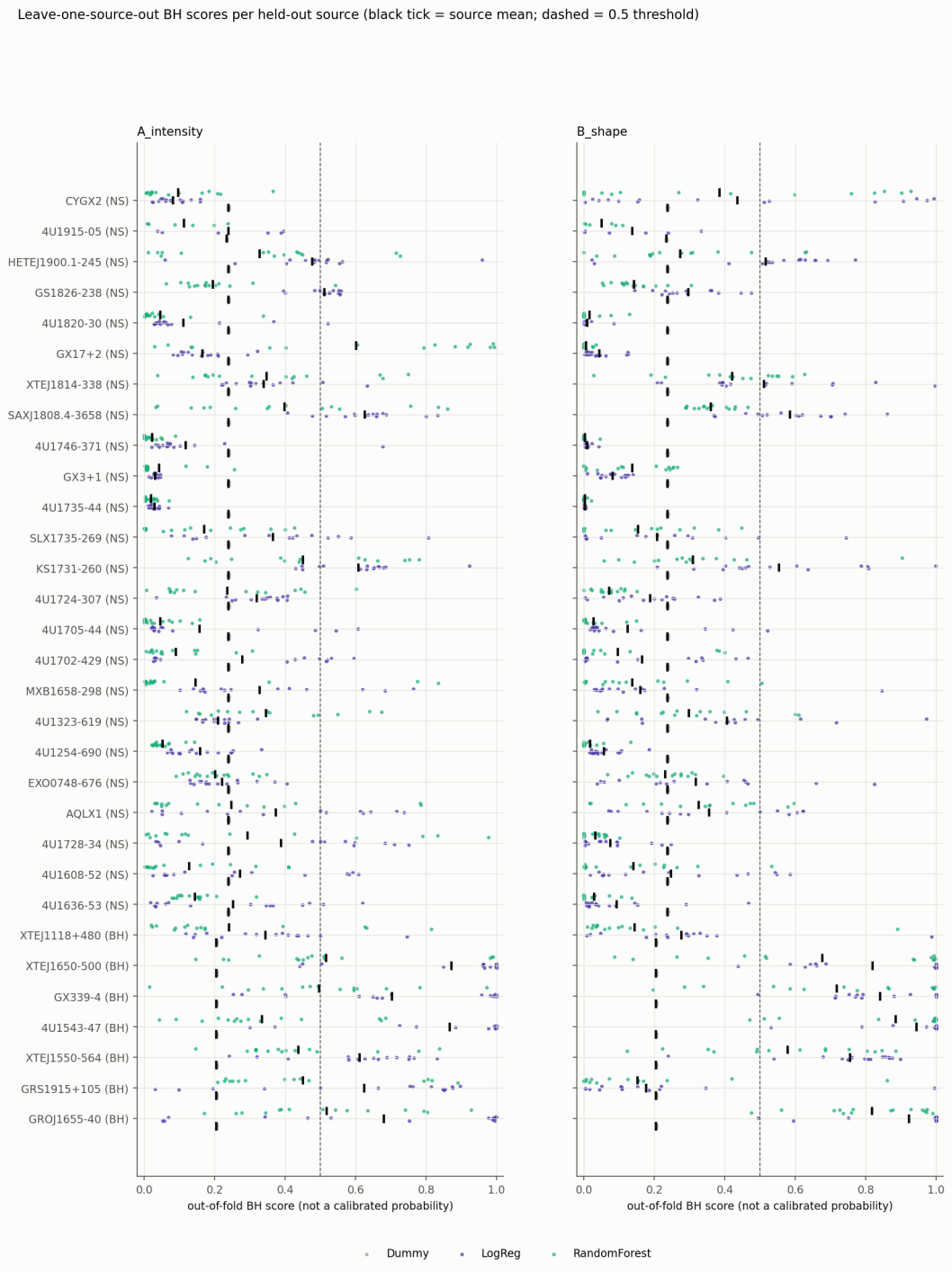
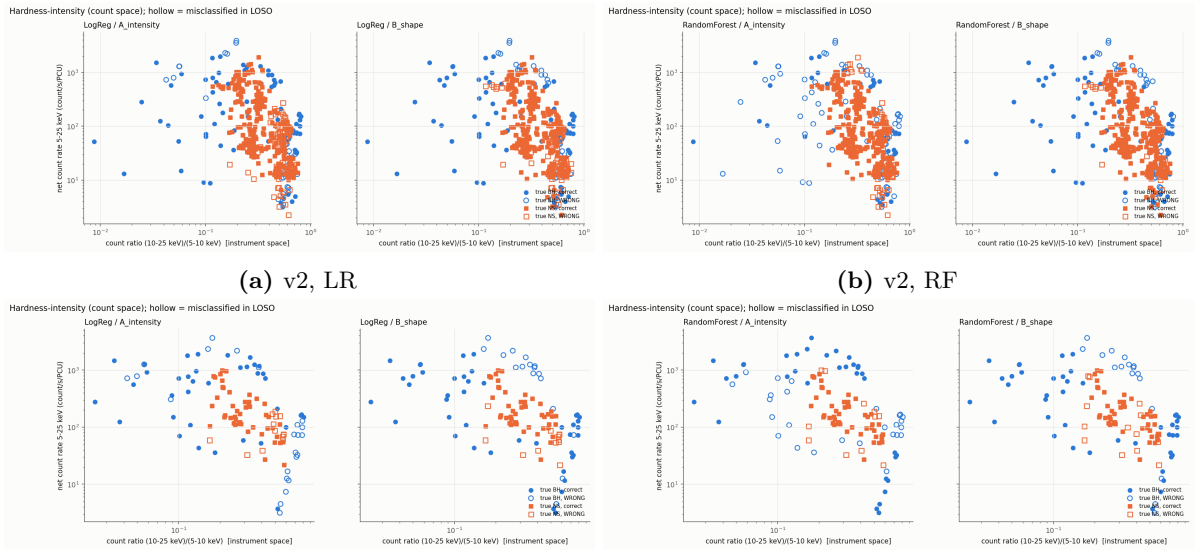


Figure 6.7: v2: out-of-fold BH score of every observation of each held-out source (left: A, right: B; grey Dummy, purple LR, green RF), black tick = source mean, dashed = 0.5. The BH sources (bottom seven) are spread over the whole score range; XTE J1118+480 and GRS 1915+105 are below 0.5 for B. Repo: `figures/v2/step4_loso_scores.png`; script `scripts/07_train_eval.py` (`XRB_VERSION=v2`).



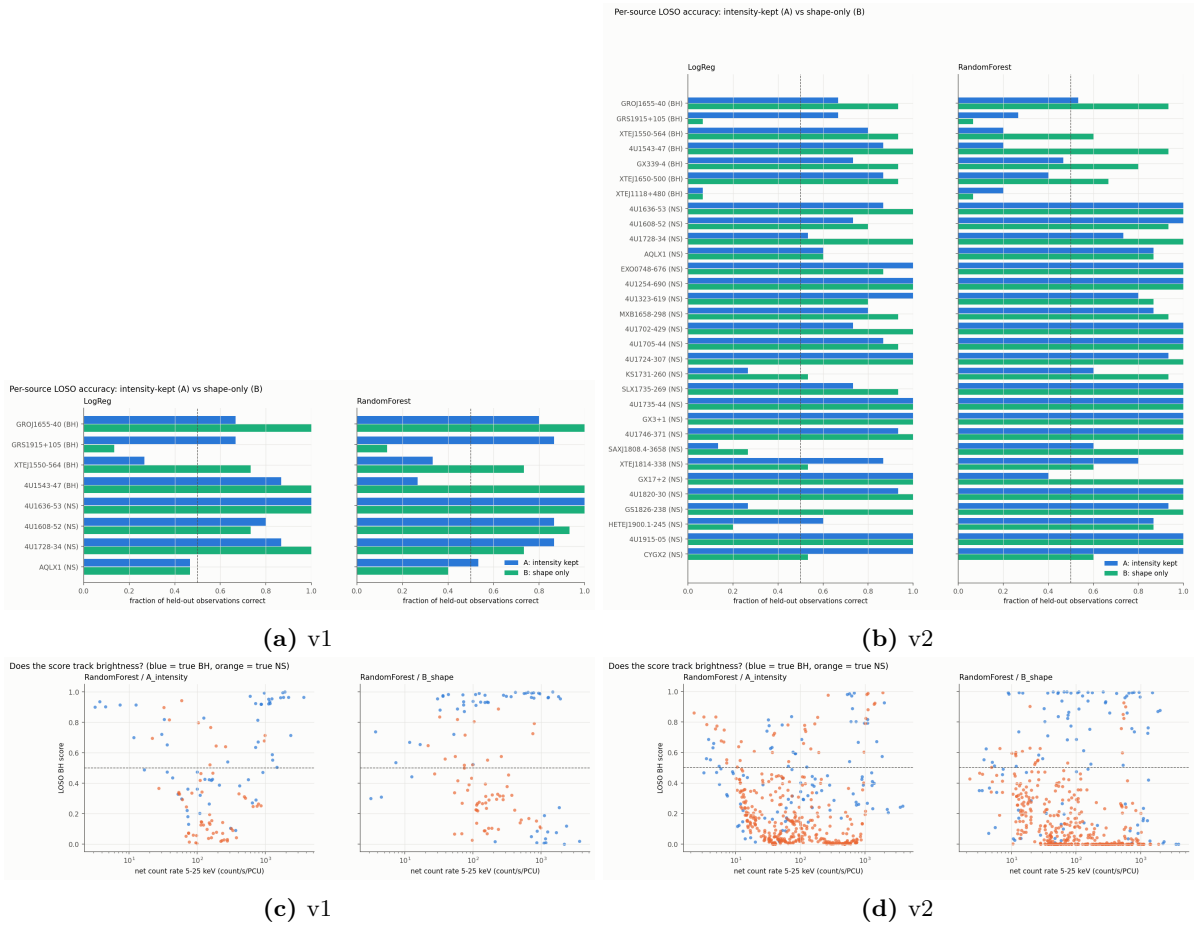
(a) v2, LR **(b)** v2, RF

(c) v1, LR **(d)** v1, RF

Figure 6.8: Hardness–intensity diagrams in count space: count ratio (10–25)/(5–10) keV (x, log) versus net 5–25 keV rate (y, log), each panel pair for A (left) and B (right); filled = correct, hollow = misclassified under LOSO; circles BH, squares NS. Misclassified NSs cluster at hard ratios ($\gtrsim 0.4$) and low rates where hard BHs also lie; misclassified BHs are hard observations or the very bright soft GRS 1915+105 observations (top left). Repo: `figures/step5_hid_errors_*.png`, `figures/v2/step5_hid_errors_*.png`; script `scripts/08_error_analysis.py`.

Table 6.10: v3c: mean predicted BH score of each candidate (models trained on the 31 confirmed sources) for H, B, A, HI with LR and RF, and the fraction of its 15 observations ≥ 0.5 for H. "elig." = eligible pointings; F = median net 5–25 keV rate [c s^{-1} PCU^{-1}]. Predictions, not accuracies. From `results/v3/v3c_candidate_predictions.csv`, `data/v3c/sources.csv`, `data/v3c/observations.csv`.

			H		B		A		HI		H: frac. ≥ 0.5	
Candidate	elig.	F	LR	RF	LR	RF	LR	RF	LR	RF	LR	RF
XTE J1817-330	131	60.2	0.96	0.93	0.99	0.93	0.95	0.60	0.95	0.88	1.00	1.00
XTE J1652-453	41	33.2	0.93	0.76	0.94	0.74	0.75	0.24	0.92	0.60	1.00	0.80
XTE J1720-318	88	20.3	0.91	0.80	0.89	0.81	0.82	0.52	0.88	0.73	1.00	0.80
4U 1630-47	607	269.1	0.84	0.85	0.96	0.86	0.91	0.49	0.84	0.79	1.00	0.93
XTE J1908+094	16	4.1	0.82	0.51	0.81	0.57	0.71	0.67	0.71	0.56	1.00	0.53
SLX 1746-331	57	47.1	0.81	0.84	0.91	0.79	0.83	0.53	0.78	0.76	0.87	0.93
MAXI J1659-152	55	246.1	0.77	0.67	0.60	0.74	0.78	0.41	0.82	0.48	0.87	0.87
4U 1957+11	89	29.8	0.76	0.87	0.99	0.83	0.98	0.51	0.61	0.77	0.93	1.00
H1743-322	434	61.6	0.73	0.67	0.66	0.67	0.74	0.43	0.69	0.58	0.80	0.73
XTE J1752-223	160	69.6	0.71	0.84	0.87	0.76	0.77	0.59	0.78	0.84	0.67	0.87
GRS 1758-258	590	23.3	0.63	0.52	0.62	0.51	0.47	0.37	0.57	0.46	0.93	0.53
IGR J17091-3624	185	49.9	0.60	0.53	0.69	0.62	0.60	0.09	0.51	0.30	0.73	0.47
1E 1740.7-2942	572	26.4	0.58	0.37	0.55	0.44	0.41	0.26	0.51	0.26	0.73	0.33
Swift J1753.5-0127	332	58.3	0.47	0.19	0.43	0.28	0.49	0.58	0.52	0.47	0.20	0.07



(a) v1 **(b)** v2

(c) v1 **(d)** v2

Figure 6.9: Top: fraction of each source's held-out observations classified correctly with A (blue) and B (green), for LR and RF; dashed = 0.5. Bottom: RF OOF score versus net count rate for A and B (blue BH, orange NS). With A the RF score depends on brightness in a non-monotone way: the faintest ($\lesssim 10$) and the brightest ($\gtrsim 500\,\mathrm{c\,s^{-1}\,PCU^{-1}}$) observations of *both* classes tend to get high BH scores and many BH observations at intermediate rates get low ones – the forest partly learned brightness ranges of the training sources (possible explanation, not verified); with B the BH and NS scores separate at every rate. Repo: `figures/step5_per_source_A_vs_B.png`, `figures/step5_score_vs_rate.png` and their `figures/v2/` versions; script `scripts/08_error_analysis.py`.

Table 6.11: v3c: effect of adding the 210 candidate observations (labelled BH) to every training fold, evaluated on the 31 confirmed sources and paired with v2 (same folds and test observations). From `results/v3/v3c_metrics.csv` and `results/v3/v3c_bootstrap.csv`.

Repr.	Model	source BA (v2 → +cand)	source AUC (v2 → +cand)	Δ source BA [95% CI]
A	LR	0.866 → 0.887	0.946 → 0.964	+0.021 [0.000, +0.062]
A	RF	0.622 → 0.866	0.881 → 0.964	+0.244 [+0.051, +0.429]*
B	LR	0.774 → 0.815	0.875 → 0.929	+0.042 [0.000, +0.104]
B	RF	0.857 → 0.836	0.893 → 0.940	−0.021 [−0.062, 0.000]
H	LR	0.795 → 0.795	0.923 → 0.935	0.000 [0.000, 0.000]
H	RF	0.857 → 0.845	0.905 → 0.940	−0.012 [−0.146, +0.152]
HI	LR	0.795 → 0.795	0.929 → 0.935	0.000 [0.000, 0.000]
HI	RF	0.836 → 0.958	0.946 → 0.982	+0.122 [−0.021, +0.315]

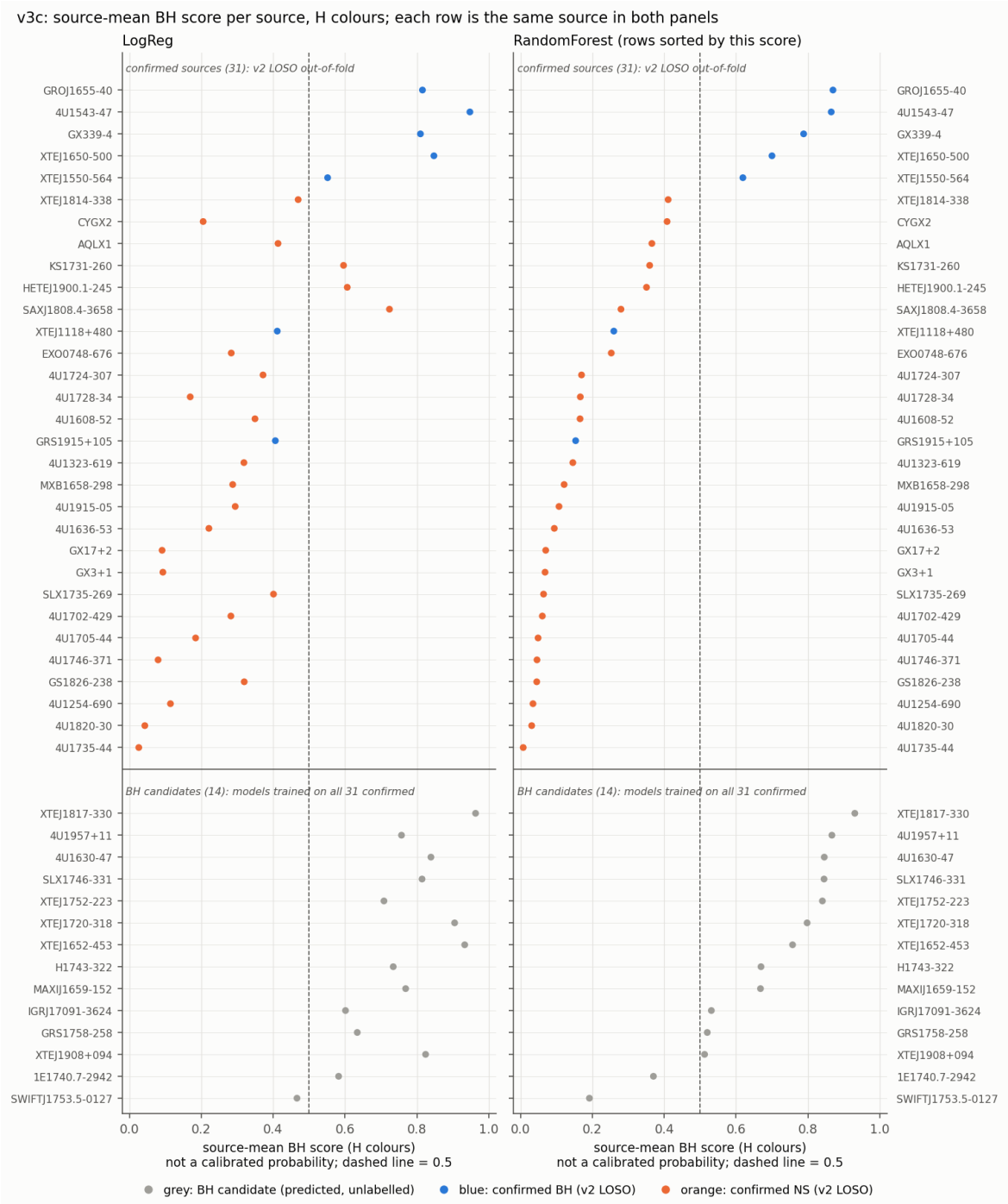


Figure 6.10: v3c with H: LOSO source means of the 31 confirmed sources (blue BH, orange NS, top block) and source-mean BH scores of the 14 candidates (grey, predicted, bottom block), LR left, RF right; each row is the same source in both panels, sorted within each block by the RF score; dashed = 0.5. Most candidates sit with the confirmed BHs; Swift J1753.5−0127 sits with the NSs (0.47 LR, 0.19 RF), and 1E 1740.7−2942 is also below 0.5 for RF (0.37; `results/v3/v3c_candidate_source_summary.csv`). Repo: `figures/v3/v3c_candidate_scores.png`, drawn from the saved predictions by `scripts/v3c_plot.py` (`scripts/14_v3c_candidates.py --plot-only`); this version replaces an earlier one whose LR panel had mismatched source labels.

boundaries of LR and RF fitted on all 31 sources (for visualisation only; every reported number is LOSO). NSs form one track from lower left to upper right; soft-state BHs (4U 1543−47, XTE J1650−500, GRO J1655−40) lie to the upper left, far from it; the medians of GX 339−4, XTE J1118+480 and XTE J1550−564 sit at the dense upper-right end of the NS track, and GRS 1915+105 near the boundary. The fitted LR boundary in raw colours is printed by `scripts/15_colour_colour.py` (`logs/final/15_colour_colour.log`).

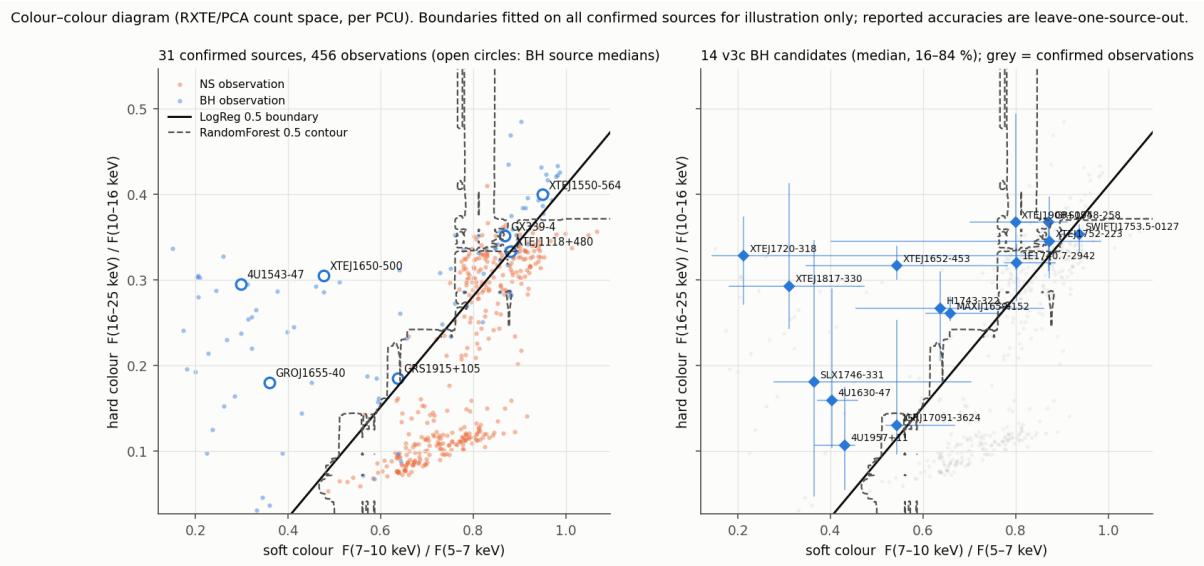


Figure 6.11: The two count-space colours of H: soft $F(7\text{–}10)/F(5\text{–}7)$ (x) and hard $F(16\text{–}25)/F(10\text{–}16)$ (y). Left: 456 observations of the 31 confirmed sources (orange NS, blue BH; open circles = BH source medians), the LR 0.5 boundary (solid) and the RF 0.5 contour (dashed), both fitted on all sources for illustration only. Right: the 14 v3c candidates (median and 16–84% range) on the confirmed observations (grey). Look at the upper-right end of the NS track: hard BHs and hard NSs overlap there, which is where every spectral model errs and where Phases 3–6 look for timing information. Repo: `figures/final/colour_colour_decision_boundary.png`; script `scripts/15_colour_colour.py`.

6.6.5 XSPEC cross-check

DESCRIPTIVE For the 12 observations of Section 2.4.7, the Python net rates and errors equal the XSPEC values to $\leq 3 \times 10^{-10}$ relative (e.g. 4U 1820−30 92030-02-12-00: 953.94154 vs 953.94154 c s^{-1}; `results/v2/xspec_crosscheck.csv`).

6.7 Interpretation

Statistically. On 31 sources the spectral classifiers are far better than chance, but every comparison that asks whether *more* spectral information helps is not detected: the full shape adds nothing to two colours, 3–5 keV adds nothing, and 45 intensity bins add nothing to one intensity number. The intervals are wide because only 7 BH sources exist, so "not detected" is not "absent"; but the point estimates are themselves near zero or negative, and the B and H models agree on 88–93% of individual observations. The leaky split would have reported 0.77–0.88 BA.

Physically. In soft states the BH disc is cool and accompanied by a power-law tail, so the soft colour is low and the hard colour relatively high; NS soft states have a boundary-layer/surface component that cuts off at 10–20 keV. These two populations separate in the colour plane, as in

Done and Gierliński (2003) (whose colours were response-corrected and defined differently; no formal comparison was made). In hard states both are Comptonisation-dominated and their 5–25 keV shapes are similar, so the colours overlap – the errors are the hard-state BHs (XTE J1118+480, the hard observations of GX 339−4, XTE J1550−564) and hard NSs (AMXPs, KS 1731−260). GRS 1915+105, long near Eddington with a curved soft spectrum, resembles Z/atoll soft states in shape. The Pattnaik et al. (2021) LOSO result (87%) is of the same order as this project's; the new finding is that two hardness ratios reproduce it, which suggests that their Random Forest also learned hardness (not tested directly).

6.8 Deviations and caveats

- **H colours linear instead of** $\log_{10}$ (before any v2 training; two BH spectra with slightly negative 16–25 keV sums).
- POST HOC **B−H and A−HI paired differences** added after the v2 results (they answer the pre-declared question "does the model only learn hardness?").
- **Output-path bug**: `08_error_analysis.py` overwrote two v1 figures during the v2 run; restored from git and fixed (no effect on numbers).
- **XSPEC invocation**: commands are passed as a script file because inline passing through `wsl.exe` expanded `$xspec_tclout` to an empty string (fixed before the comparison).
- **v3c**: candidate list drafted from memory, verified and frozen before download; one catalogue-name fix; two candidates without catalogues skipped by rule.
- **v2 source rule executed incompletely** (found in v4b2): 13 further Galloway+2008 NSs had catalogues and ≥ 10 eligible pointings but were missed; the v1–v3 numbers were kept and describe a smaller NS sample (Chapter 7).
- **Reproducibility**: v3a re-ran the v2 models exactly (`results/v3/v3a_reproduction_check.csv`: maximum score difference 1.1×10^{-16} for LR and 2.2×10^{-16} for RF, 0 label changes); a second machine (macOS) reproduced `features.npz` to $\leq 10^{-16}$ and RF source means to ≤ 0.001.
- **Caveats**: count space (no response unfolding; PCU-combination tilt of ±10% at 25 keV); background-model systematics not included; Standard Products use generic screening; no per-observation state labels in this phase; bootstrap intervals do not include refitting variability.

6.9 Takeaway

A single 5–25 keV RXTE spectrum classifies unseen sources better than chance, but what it offers is captured by two count-space colours; the failures are hard-state BHs and hard NSs that overlap in those colours. Phase 2 asks whether this ceiling is a property of the models or of the data: more flexible algorithms, more observations, more sources and a wider energy band are tested against the two-colour LR.

Chapter 7

Phase 2 (v4): is the spectral information saturated?

Phase at a glance: v4 (v4a, burst check, v4b3, v4b2, v4b1)

Question. Phase 1 found that two count-space colours carry all the detectable information in a 5–25 keV spectrum. Is that a limit of the *models* (more flexible algorithms), of the *data quality* (type-I burst contamination), of the *band* (HEXTE 25–60 keV) or of the *sample* (more sources from earlier gain epochs; all observations of each source)?
Single baseline. The v2 two-colour logistic regression H-LR (source BA 0.795 [0.589, 0.958], source AUC 0.923 [0.804, 0.994], observation BA 0.776). "Improvement" ⇔ the paired 95% interval lies entirely above 0.
Pre-declared primary. v4a part 1e: a fully nested procedure that chooses algorithm, representation, hyper-parameters and threshold inside the training sources, compared with H-LR. Each v4b sub-phase and the burst check also had one pre-declared main comparison.
Data. v4a, burst check: the v2 sample (31 sources, 456 observations); v4b3: the 396 observations with HEXTE cluster-B products; v4b2: 46 sources (9 BH, 37 NS), 677 observations, gain epochs 2–5; v4b1: all 7,949 eligible epoch-5 observations of the 31 v2 sources.
Verdict. Every primary comparison: **not detected**. Nested automatic selection was *worse* in point estimate (source BA −0.062, source AUC −0.173); bursts, HEXTE colours and 17× more observations changed nothing. The only detected difference is a threshold effect on the extended 46-source sample (RF source BA +0.068 [+0.014, +0.122] over the same-sample H-LR, source AUC not different). **The v4 pre-registration was written after all v1–v3 results were known.**

7.1 Motivation

Phase 1 left one negative result that could have several causes. "The full spectral shape adds nothing to two colours" (H+B − H not detected, Section 6.6) can mean that the 5–25 keV spectrum really contains no further source-transferable information, or that the classifiers were too rigid (LR, RF with fixed settings), that observations were contaminated by thermonuclear bursts (whose spectra are not those of the persistent emission), that the informative part of the spectrum lies above 25 keV (high-energy cut-off of NS Comptonisation versus the unbroken BH power law), or that 31 sources (7 BH) are simply too few. Phase 2 tests each explanation separately against the same fixed baseline, H-LR, so that its answer is "the ceiling is (or is not) a property of the data". The analyses are grouped by theme below: *flexible algorithms* (v4a),

data quality (burst check), *wider band* (v4b3) and *more data* (v4b2: more sources and earlier epochs; v4b1: all observations of the same sources). Figure 7.1 maps the five sub-analyses: what each added to the v2 analysis and what it tested against H-LR.

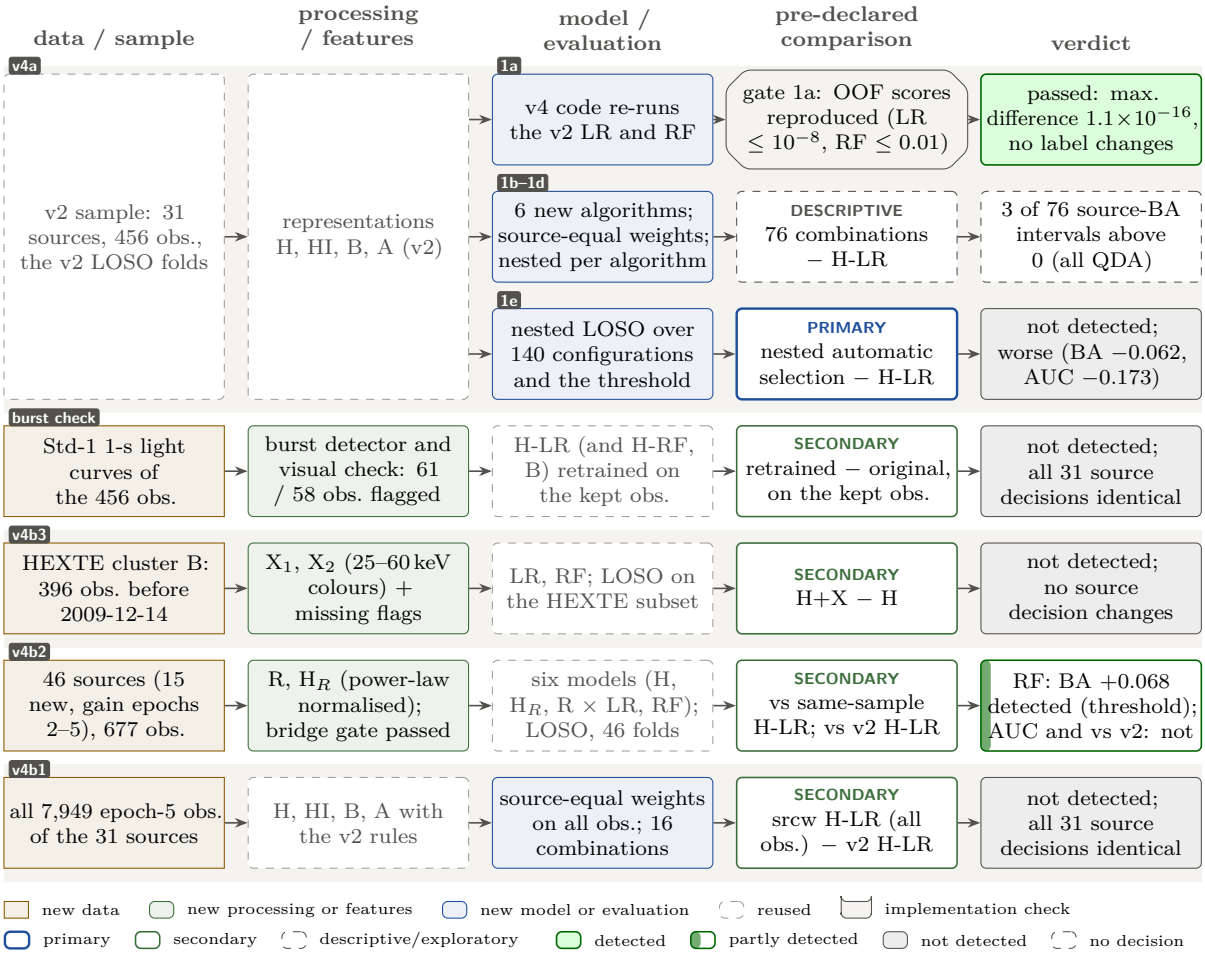


Figure 7.1: Phase 2 analysis map (reading as in Figure 6.1: solid boxes are new in that sub-analysis, dashed boxes reuse earlier elements). The three v4a procedures share the v2 sample and representations; the 1a reproduction gate had to pass before any v4 result was looked at. Every comparison is paired with H-LR on the same bootstrap resamples. Values from Tables 7.6, 7.9, 7.10, 7.12 and 7.14 and Figure 7.4. Drawn for this report.

7.2 Pre-registration

`reports/preregistration_v4.md` was written on 2026-09-30, before any v4 model was trained and before any v4 download, *after all v1–v3 results were known*; every v4 question is an extension aimed at a known result, and the document says so. Common rules (Table 7.1): LOSO on the v2 folds (or on the extended source set), source score = mean OOF BH score, threshold 0.5 except where a threshold is explicitly chosen inside the training sources, headline metrics source BA and source AUC plus observation BA, class-stratified source bootstrap (2000, seed 42, fixed OOF predictions) with all comparisons to H-LR *paired* on the same resamples, "improvement" only if the interval is entirely > 0, each metric judged separately.

Table 7.1: Phase 2 pre-registration (`reports/preregistration_v4.md`, 2026-09-30, written after all v1–v3 results). Baseline everywhere: H-LR. Status tags as used in the Results section.

Part	Main comparison (status)	Descriptive / secondary
v4a 1a	reproduction of v2 LR/RF OOF scores (gate: LR $\leq 10^{-8}$, RF ≤ 0.01, 0 label changes)	–
v4a 1e	PRIMARY nested automatic selection over 8 algorithms $\times$ 4 representations $\times$ grids (140 configurations) − H-LR; source BA and source AUC	observation BA difference; selected configuration per fold; agreement with H-LR (1f)
v4a 1b–1d	–	DESCRIPTIVE 1b: six new algorithms with fixed settings; 1c: source-equal weights for LR, RF, HistGB; 1d: nested LOSO per algorithm $\times$ representation (32 procedures). 76 combinations; the best of them is *not* to be taken as the conclusion
burst check	SECONDARY H-LR (and H-RF, B-LR, B-RF) after excluding observations with an automatically detected type-I burst, paired with the original OOF on the kept observations; v2 (B−H) and v3a (H+B−H) comparisons recomputed	visual set as sensitivity analysis
v4b3	SECONDARY H+X − H on the HEXTE subset (LR and RF)	colour-overlap box (defined after seeing the v2 colour plot: POST HOC definition)
v4b2	gate: bridge check H_R-LR vs H-LR on the 456 v2 observations (both CIs contain 0, Spearman $\geq$ 0.90); then SECONDARY extended-sample models vs the same-sample H-LR and vs v2 H-LR	gain epoch as diagnostic only, never a feature
v4b1	SECONDARY source-equal-weight H-LR on all observations − v2 H-LR	16 combinations (H, B, A, HI $\times$ LR, RF $\times$ two weightings); restriction to the 456 v2 observations

7.3 Data

Table 7.2 summarises the five samples. Only v4b1, v4b2 and v4b3 downloaded new files; v4a and the burst check reuse the committed v2 feature file and the Standard-1 files already on disk for the 456 v2 observations.

Table 7.2: Samples of Phase 2. Sources and observation counts from `results/v4*/` OOF files and `data/v4b*/`; "epochs" = PCA gain epochs (Section 2.4).

Analysis	BH src	NS src	obs. (BH)	epochs	Notes
v4a	7	24	456 (105)	5	identical to v2 (`data/v2/processed/features.npz`), same 31 folds
burst, auto set	7	24	395	5	61 observations removed (12 of them BH)
burst, visual set	7	24	398	5	58 observations removed (all NS)
v4b3 HEXTE	7	24	396 (99)	5	observations before MJD 55179.6736 with cluster-B source and background spectra
v4b2 extended	9	37	677 (131)	2–5	v2 sample unchanged + 221 observations of 15 new sources
v4b1 all obs.	7	24	7,949 (2,369)	5	every eligible pointing of the 31 v2 sources

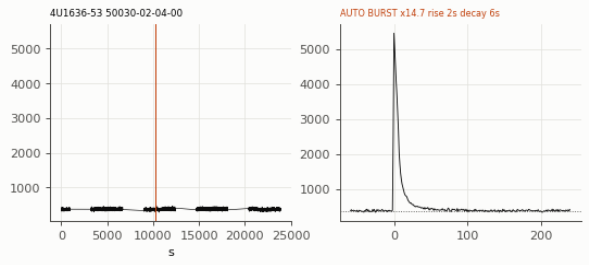
7.3.1 Type-I burst screening

The detector is defined in Section 2.7: Standard-1 good-xenon counts summed over PCUs, 1-s bins inside the spectral GTIs, candidates $> 4\sigma$ above the observation mean (after Galloway et al., 2008), merged within 30 s, and an automatic shape rule (peak $\geq 1.5\times$ persistent, rise ≤ 10 s, half-decay 3–300 s, ≥ 3 adjacent bins above 25% amplitude). Every candidate was plotted and inspected (Figure 7.2).

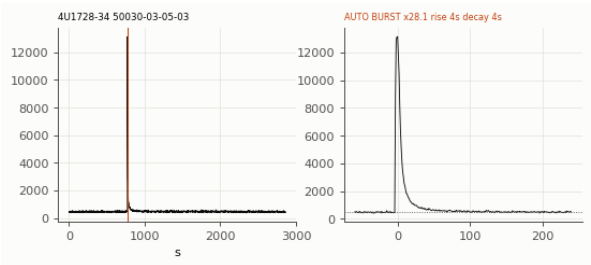
- 187 of the 456 observations had at least one candidate (410 candidates in total); the automatic rule flagged 61 observations (`results/v4/burst_flags.csv`).
- Visual inspection (`results/v4/burst_visual.csv`): of the 61 automatic flags, **12 are BH observations** (the periodic flares of GRS 1915+105 and second-scale flares of hard-state BHs: GRS 1915+105 2, GX 339−4 2, XTE J1118+480 4, XTE J1550−564 4) and one is NS noise, so 48 are real bursts; the rule missed 10 further observations (8 clear bursts with GTI truncation, rise > 10 s or half-decay < 3 s, and 2 "possible"). The visual set has 58 observations, all NS (3 marked "possible").
- As pre-registered, the *automatic* set defines the main exclusion (395 observations kept) and the visual set is the sensitivity analysis (398 kept). No source was removed; 4U 1916−053 kept 4 of its 6 observations (below the pre-declared 5, kept and recorded) and the dipper 4U 1323−619 lost 9 of 15 (`results/v4/burst_exclusion_counts.csv`).

7.3.2 HEXTE cluster B (v4b3)

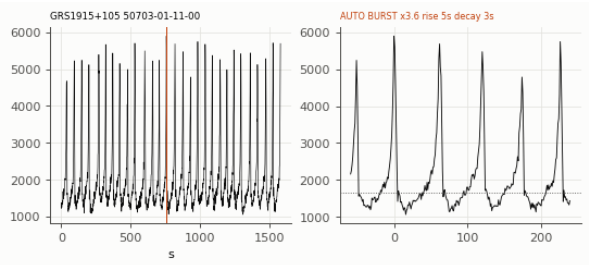
The HEXTE Standard Products contain source and background spectra of both clusters and their responses (Section 2.6). Cluster A has no background products after July 2006, and cluster B stopped rocking on 2009-12-14 (MJD 55179.6736), so the pre-registered rule used cluster



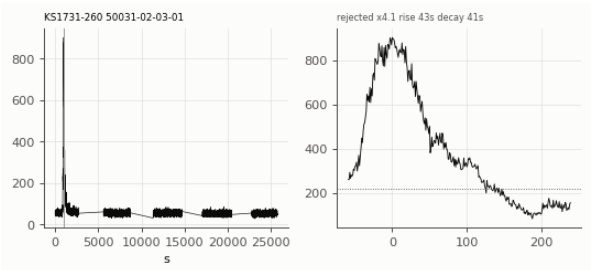
(a) 4U 1636−53, 50030-02-04-00: automatic and visual burst (fast rise, exponential decay).



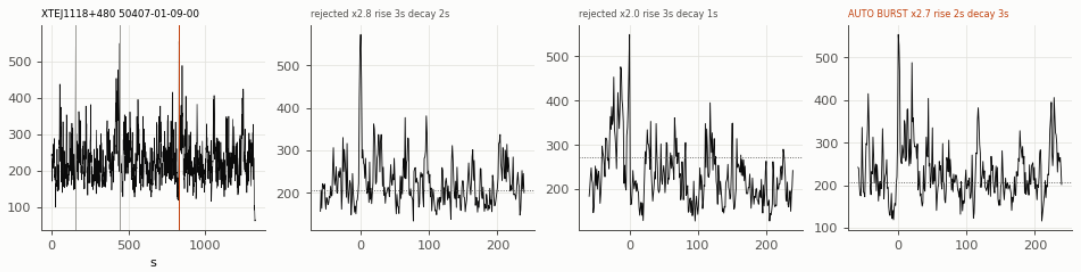
(b) 4U 1728−34, 50030-03-05-03: automatic and visual burst.



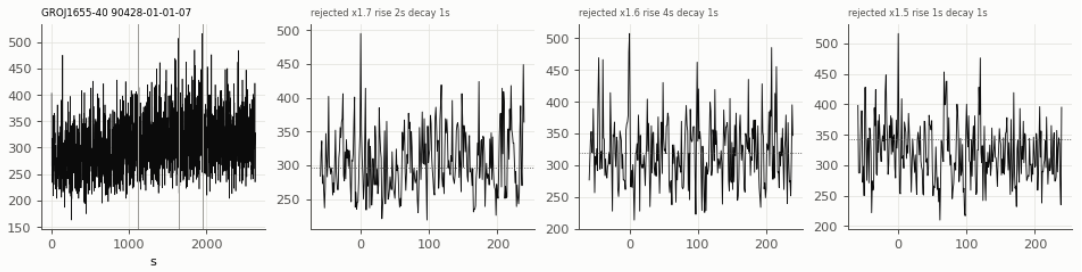
(c) GRS 1915+105, 50703-01-11-00 (BH): flagged by the automatic rule (peak ×3.6, rise 5 s, decay 3 s), rejected visually – periodic flaring, not a type-I burst.



(d) KS 1731−260, 50031-02-03-01: long burst (rise 43 s) missed by the automatic rule (rise $\leq$ 10 s required); burst by visual inspection.



(e) XTE J1118+480, 50407-01-09-00 (hard-state BH): three candidates, one passing the automatic rule; rejected visually (BH variability).



(f) GRO J1655−40, 90428-01-01-07 (BH): three 4σ candidates, none passing the shape rule; visual: noise or variability.

Figure 7.2: Six of the burst-candidate plots, chosen to show each outcome (true positive, BH false positive, missed long burst, rejected candidates). Each plot: left, the whole 1-s Standard-1 light curve of the observation (counts s^{-1} summed over the PCUs on; GTI gaps joined by lines; red marks = candidates); then one zoom per candidate with the persistent level (dotted) and the automatic verdict (peak ratio, rise and half-decay times) in its title. Repo: `figures/v4/burst_candidates/` (one PNG per observation with candidates), produced by `scripts/17_v4_burst_detect.py`; verdicts in `results/v4/burst_visual.csv`.

B only (net = `s1` − `b1`, scaled by exposure and `BACKSCAL`) and gave no HEXTE features to observations after that date: 60 of the 456 v2 observations (6 BH, 54 NS, 18 sources). The remaining 396 observations all had products; every header has `DEADAPP = T` (deadtime already applied) and every spectrum uses `xh97mar20c_pwb013.rmf`. The two colours are

$$X_1 = \frac{F_{\mathrm{HEXTE}}(25\text{–}40)}{F_{\mathrm{PCA}}(16\text{–}25)}, \qquad X_2 = \frac{F_{\mathrm{HEXTE}}(40\text{–}60)}{F_{\mathrm{HEXTE}}(25\text{–}40)} \tag{7.1}$$

(count space; X_1 is a cross-instrument ratio used only as a feature). A colour whose denominator has $S/N < 3$ is set missing: X_1 in 3 observations (2 BH, 1 NS), X_2 in 144 (23 BH, 121 NS); one observation (92023-03-37-00) had zero background exposure. Missing values are replaced by the training-fold median and flagged by a 0/1 indicator (Section 3.2.1). The class distributions (Figure 7.3) overlap almost completely.

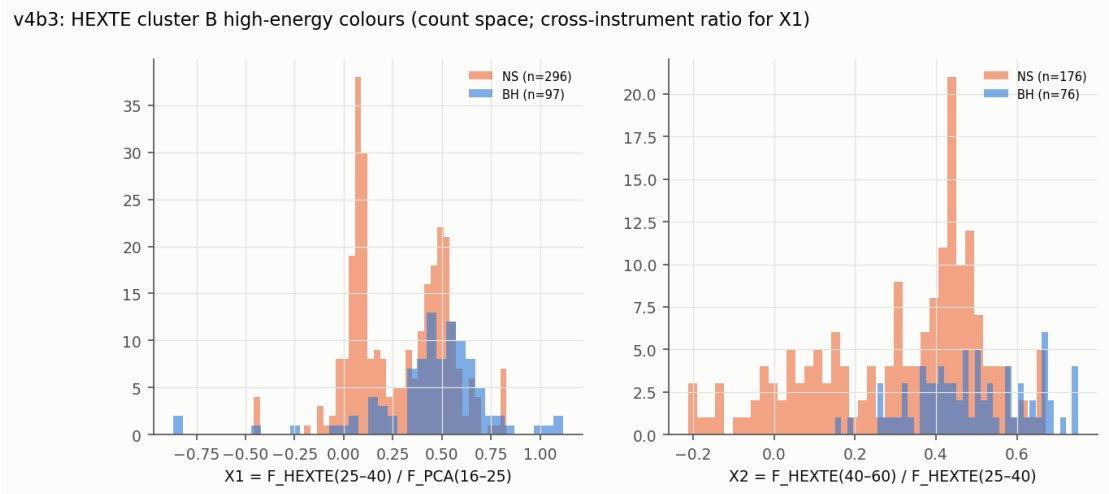


Figure 7.3: HEXTE cluster-B colours (Eq. (7.1)) of the v2 observations with valid values, BH blue, NS orange. Left: X_1 (393 observations: 97 BH, 296 NS); right: X_2 (252 observations: 76 BH, 176 NS). The NS X_1 distribution is bimodal (soft states near 0, hard states near 0.5); the BHs lie mostly in the hard NS mode, and the X_2 distributions overlap. Repo: `figures/v4b3/v4b3_hexte_colours.png`, `scripts/19_v4b3_hexte.py`; values in `data/v4b3/processed/hexte_features.csv`.

7.3.3 Source-list verification and earlier gain epochs (v4b2)

The v4b2 selection (`scripts/20_v4b2_select.py`) matched every Galloway et al. (2008) burster (48) and every BlackCAT dynamical BH (18) to the 291 RXTE MissionLongData catalogues by *position*: CDS Sesame/SIMBAD coordinates against the median pointing of each catalogue, accepted within 0.1° (`data/v4b2/source_verification.csv`). Results:

- 46 of the 66 objects have a matching catalogue; 20 do not (9 BHs, e.g. GRS 1009−45, A 0620−00, V404 Cyg, and 11 NSs, including the Rapid Burster and IGR J17473−2721, whose catalogues exist but point $> 0.1°$ away).
- **The v2 source rule had been executed incompletely**: 13 NSs have catalogues and ≥ 10 eligible pointings but were not in the v2 list (4U 0513−40, 4U 0919−54, XTE J1709−267, XTE J1710−281, SAX J1747.0−2853, 1A 1744−361, SAX J1748.9−2021 in NGC 6440, EXO 1745−248 in Terzan 5, SAX J1750.8−2900, GRS 1747−312, XTE J1759−220, Ser X-1, 4U 2129+12 in M15). The v2 results were not changed; this is a limitation of Phases 1–2 (Section 7.8).

- With all gain epochs allowed (MJD < 55931; epoch boundaries MJD 50163.7729, 50188.9618, 51259.7340, 51677.0000 from the HEASARC energy–channel table), two BHs become eligible: XTE J1859+226 (112 eligible pointings, all in epoch 4) and V4641 Sgr (11, epochs 4–5).

The 456 v2 observations were kept unchanged; each of the 15 new sources contributed 15 observations (V4641 Sgr: 11) drawn with the v2 time-stratified rule over the whole mission. All 221 new observations passed the quality rules (`results/v4b2/attrition_download.csv`), giving 46 sources (9 BH, 37 NS) and 677 observations (131 BH). Table 7.13 lists the new sources with their epochs. The new representation R divides each bin by the expected counts of a $\Gamma = 2$, $N_{\rm H} = 0$ power law folded through the observation's response (`pl2`, Section 2.4.6); H_R are the colours of R (Eq. (2.12)).

7.3.4 All observations of the same sources (v4b1)

For the 31 v2 sources, every eligible epoch-5 pointing was processed with the v2 rules: 7,975 candidates, 7,949 accepted (2,369 BH, 5,580 NS); 26 excluded (14 without Standard-2 products at HEASARC, 7 high deadtime, 5 short exposure; `data/v4b1/observations.csv`). Per source the counts range from 6 (4U 1916−053) to 1,329 (4U 1636−53); GRS 1915+105 has 1,031 (Table 7.16). The raw files ($\approx 2\,$GB, $\approx 30{,}000$ files) were stored outside the repository (`XRB_EXTERNAL_RAW`); download threads were raised from 4 to 6. **The effective sample size is still 31 sources (7 BH)**: the number of observations grows 17×, but the source-level power hardly changes.

7.4 Data structure in this phase

Table 7.3 lists the arrays. The v4a and burst analyses use the v2 rows and folds (`results/v4/folds_loso.csv` equals `results/v2/folds_loso.csv`); v4b3 uses the 396-row subset and its own 31 LOSO folds; v4b2 uses 46 folds (`results/v4b2/folds_loso.csv`); v4b1 uses 31 folds over 7,949 rows (`results/v4b1/folds_loso.csv`). In the nested procedures (1d, 1e) each outer training set (30 sources) is itself split by LOSO into 30 inner folds (Figure 5.2); for 1e the inner OOF scores of all 140 configurations are computed once per outer fold and shared with 1d.

7.5 Models in this phase

Phase 2 introduces six algorithms (Table 7.4); their decision functions, objectives and pseudo-code are in Chapter 4. Every algorithm has a pre-declared way of handling the class imbalance. The 1d/1e grids are in Table 7.5. A 1D convolutional network on B was pre-declared as optional and skipped because the environment has no `torch` (stated in the pre-registration).

7.6 Results

All intervals are 95% class-stratified source-bootstrap intervals (2000 resamples, seed 42, fixed OOF predictions, no refitting; Section 5.5); every difference is paired with the baseline on the same resamples.

Table 7.3: Arrays passed to the models in Phase 2 (float64 features, int64 labels, string groups). n_k = rows of the held-out source.

Analysis	Array	Shape	Columns	Source
v4a	$\boldsymbol{X}_H$, $\boldsymbol{X}_{HI}$, $\boldsymbol{X}_B$, $\boldsymbol{X}_A$	$456 \times \{2, 3, 45, 45\}$	as in Table 6.4 (`v3lib.v2_representations`)	`data/v2/processed/features.npz`
	$\boldsymbol{y}, \boldsymbol{g}$	456	105 BH; 31 sources	same
	inner (1d/1e)	$(456 - n_k) \times p$, 30 groups	per outer fold k; inner LOSO over the 30 training sources	same
	$\boldsymbol{w}$ (1c)	$456 - n_k$	source-equal weights, Eq. (4.3)	computed per fold
burst	$\boldsymbol{X}$	$395 \times p$ / $398 \times p$	H, B, H+B after removing flagged observations	same + `results/v4/burst_flags.csv`
v4b3	$\boldsymbol{X}_H$	396×2	h_1, h_2	`data/v2/processed/features.npz` (subset)
	$\boldsymbol{X}_{H+X}$	396×6	$h_1, h_2, \tilde{X}_1, \tilde{X}_2, m_1, m_2$; $\tilde{X}$ = median-imputed in the training fold, m = missing indicators	+ `data/v4b3/processed/hexte_features.csv`
v4b2	$\boldsymbol{X}_H, \boldsymbol{X}_{H_R}$	677×2	count-space colours; power-law-normalised colours	`data/v2/.../features.npz` + `data/v4b2/processed/features.npz` (221 rows)
	$\boldsymbol{X}_R$	677×45	$x_{ij}/\mathrm{pl2}_{ij}$	`rate`, `pl2`
	$\boldsymbol{y}, \boldsymbol{g}$	677	131 BH; 46 sources	both npz
v4b1	$\boldsymbol{X}_H$, $\boldsymbol{X}_{HI}$, $\boldsymbol{X}_B$, $\boldsymbol{X}_A$	$7949 \times \{2, 3, 45, 45\}$	as v2	`data/v4b1/processed/features.npz`
	$\boldsymbol{y}, \boldsymbol{g}$	7,949	2,369 BH; 31 sources	same

Table 7.4: Models of Phase 2 with their fixed (1b) settings, imbalance handling and score (`scripts/v4lib.py`, `scripts/config.py`, v4 block).

Model	Fixed settings (1b)	Imbalance	Score
LR (§4.3)	scaler + $C = 1$, L2, lbfgs, `max_iter`=5000	balanced; srcw (1c, v4b1)	`predict_proba`
RF (§4.4)	500 trees, `sqrt`, leaf 2, seed 42	balanced; srcw	`predict_proba`
ExtraTrees (§4.5)	500 trees, `sqrt`, leaf 2, seed 42	balanced	`predict_proba`
HistGB (§4.6)	lr 0.1, 200 iterations, 15 leaves, leaf 20, $\ell_2 = 0$, no early stopping, seed 42	balanced; srcw (1c)	`predict_proba`
SVM-RBF (§4.7)	scaler + $C = 1$, γ=`'scale'`	balanced	$\sigma(f(\boldsymbol{x}))$, no Platt (Eq. (4.9))
kNN (§4.8)	scaler + $k = 15$, uniform	prior correction (Eq. (4.10))	corrected probability
QDA (§4.9)	scaler + `reg_param`=0.1	priors $(0.5, 0.5)$	`predict_proba`
MLP (§4.10)	scaler + hidden (32,), $\alpha = 10^{-3}$, early stopping (20%), 500 iterations; seeds 0–4	equal class weights (`sample_weight`)	mean of 5 seeds

Table 7.5: Grids of the nested procedures 1d (per algorithm and representation) and 1e (all 35 grid points × 4 representations = 140 configurations). Other settings as in Table 7.4. Selection uses inner source AUC, then inner source BA at the chosen threshold, then the order algorithm → representation → grid (Algorithm 15); threshold by Eq. (5.2).

Algorithm	Grid	points
LR	$C \in \{0.01, 0.1, 1, 10\}$	4
RF	`min_samples_leaf` $\in \{1, 2, 5, 10\}$; 200 trees inside, 500 when refitted	4
ExtraTrees	`min_samples_leaf` $\in \{1, 2, 5, 10\}$; 200/500 trees	4
HistGB	learning rate $\in \{0.05, 0.1\}\times$ `max_leaf_nodes` $\in \{7, 15\}$	4
SVM-RBF	$C \in \{0.1, 1, 10\}\times$ γ-factor $\in \{0.3, 1, 3\}$ (multiplying the `'scale'` value)	9
kNN	$k \in \{5, 15, 31\}$	3
QDA	`reg_param` $\in \{0.01, 0.1, 0.5\}$	3
MLP	hidden $\in \{(16,), (32, 16)\} \times \alpha \in \{10^{-4}, 10^{-2}\}$; seeds 0–2 inside, 0–4 outside	4
Total		35

7.6.1 Flexible algorithms (v4a)

Reproduction (1a). The v4 code reproduced the v2 OOF scores of LR and RF on all four representations: maximum absolute difference 1.1×10^{-16} (RF on H: 2.2×10^{-16}), 0 label changes (`results/v4/v4a_reproduction_check.csv`); the v4 paired bootstrap also reproduces the v3a intervals bit for bit (`scripts/test_v4lib.py`).

Primary (1e). PRIMARY Table 7.6 gives the result: nested automatic selection reached source BA 0.732 and source AUC 0.750, *below* H-LR in both point estimates; the differences are **not detected** under the pre-registered criterion (both intervals contain 0, and both lie mostly below 0). Its OOF scores correlate with those of H-LR at Spearman 0.673 and agree in label on 75.7% of observations (`results/v4/v4a_agreement.csv`).

Table 7.6: PRIMARY v4a 1e: nested automatic selection vs the v2 H-LR baseline (456 observations, 31 sources; `results/v4/v4a_bootstrap.csv`, rows `part = 1e_primary`).

	source BA	source AUC	observation BA
H-LR (v2 baseline)	0.795 [0.589, 0.958]	0.923 [0.804, 0.994]	0.776 [0.633, 0.893]
nested automatic selection	0.732 [0.527, 0.896]	0.750 [0.464, 0.976]	0.691 [0.529, 0.822]
difference	−0.062 [−0.146, +0.021]	−0.173 [−0.399, +0.012]	−0.084 [−0.175, +0.004]
verdict	not detected	not detected	not detected

What the nested procedure chose. Table 7.7 lists the configuration selected in every outer fold. In 23 of 31 folds the inner LOSO chose **SVM-RBF on the intensity representation A** ($C = 10$; γ-factor 1 in 20 folds, 0.3 in 3); the others chose HistGB (on HI 3 times, on H once), LR on A ($C = 0.1$, twice), an MLP on A and ExtraTrees on HI. The intensity representations (A, HI) were chosen in 30 of 31 folds. The inner source AUC of the chosen configuration was 0.958–1.000 in every fold, while the outer source AUC is 0.750: the gap is the optimism of picking the best of 140 models on 30 sources, which the nested design is meant to expose. Since folds use different models, the outer source AUC is computed on the shifted scores $\bar{s} - t_k + 0.5$ (Section 5.4) and is an approximation (stated in the pre-registration). Eleven sources were called BH: 5 of the 7 BHs (GRS 1915+105 and XTE J1118+480 were missed, as by H-LR) and 6 of the 24 NSs (4U 1728−34, GS 1826−238, GX 17+2, HETE J1900.1−2455, SAX J1808.4−3658, XTE J1814−338), i.e. source BA $(5/7 + 18/24)/2 = 0.732$.

Table 7.7: v4a 1e: configuration chosen by the inner LOSO in each outer fold (one fold per held-out source). t_k = threshold chosen on the inner source scores; inner AUC/BA = inner source-level AUC and balanced accuracy of the chosen configuration on the 30 training sources; $\bar{s}$ = mean raw score of the held-out source; prediction BH iff $\bar{s} \geq t_k$; **bold** = wrong. Sorted by label and name. From `results/v4/v4a_nested_selection.csv`.

Held-out source	label	algorithm	rep.	configuration	t_k	inner AUC	inner BA	$\bar{s}$	pred.
4U 1543−47	BH	SVM-RBF	A	C=10.0, γ-factor=0.3	0.588	0.972	0.917	0.702	BH
GRO J1655−40	BH	SVM-RBF	A	C=10.0, γ-factor=1.0	0.401	0.958	0.917	0.868	BH
GRS 1915+105	BH	HistGB	H	lr=0.05, leaves=15	0.355	0.993	0.979	0.081	**NS**
GX 339−4	BH	SVM-RBF	A	C=10.0, γ-factor=1.0	0.399	0.958	0.917	0.719	BH
XTE J1118+480	BH	ET	HI	leaf=1	0.419	1.000	1.000	0.176	**NS**
XTE J1550−564	BH	HistGB	HI	lr=0.05, leaves=15	0.255	0.958	0.938	0.632	BH
XTE J1650−500	BH	MLP	A	hidden=(32, 16), α=0.0001	0.446	0.979	0.979	0.675	BH
4U 1254−69	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.406	0.969	0.913	0.104	NS
4U 1323−62	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.430	0.975	0.935	0.347	NS
4U 1608−52	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.440	0.975	0.935	0.224	NS
4U 1636−53	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.417	0.969	0.913	0.150	NS
4U 1702−429	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.440	0.975	0.935	0.221	NS
4U 1705−44	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.410	0.969	0.913	0.130	NS
4U 1724−307	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.408	0.969	0.913	0.308	NS
4U 1728−34	NS	LR	A	C=0.1	0.385	0.969	0.913	0.424	**BH**
4U 1735−44	NS	HistGB	HI	lr=0.1, leaves=15	0.177	0.969	0.935	0.000	NS
4U 1746−37	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.406	0.969	0.913	0.089	NS
4U 1820−30	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.406	0.969	0.913	0.038	NS
4U 1916−053	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.407	0.969	0.913	0.164	NS
Aql X-1	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.437	0.975	0.935	0.366	NS
Cyg X-2	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.408	0.969	0.913	0.280	NS
EXO 0748−676	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.430	0.969	0.913	0.283	NS
GS 1826−238	NS	SVM-RBF	A	C=10.0, γ-factor=0.3	0.407	0.981	0.935	0.502	**BH**
GX 17+2	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.409	0.981	0.935	0.495	**BH**
GX 3+1	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.406	0.969	0.913	0.063	NS
HETE J1900.1−2455	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.434	0.975	0.935	0.443	**BH**
KS 1731−260	NS	LR	A	C=0.1	0.590	0.975	0.929	0.590	NS
MXB 1659−298	NS	SVM-RBF	A	C=10.0, γ-factor=0.3	0.398	0.969	0.913	0.215	NS
SAX J1808.4−3658	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.455	0.994	0.978	0.570	**BH**
SLX 1735−269	NS	SVM-RBF	A	C=10.0, γ-factor=1.0	0.410	0.969	0.913	0.217	NS
XTE J1814−338	NS	HistGB	HI	lr=0.1, leaves=15	0.275	0.994	0.978	0.704	**BH**

Descriptive combinations (1b–1d). DESCRIPTIVE Table 7.8 lists all 76 descriptive combinations (31 fixed-setting rows of 1b – the H-LR row of 1b is the baseline itself and is omitted –, 12 source-equal-weight rows of 1c, 32 per-algorithm nested rows of 1d, plus the 1e row for reference) and Figure 7.4 plots them.

- Three source-BA intervals lie entirely above 0, all involving QDA: A-QDA (fixed), A-QDA (nested) and HI-QDA (nested), each 0.929, +0.134 [+0.021, +0.298], misclassifying only XTE J1118+480. Their source-AUC differences (+0.024, +0.060, +0.036) all have intervals containing 0.
- No source-AUC interval lies entirely above 0. Two nested combinations are *detectably worse* than H-LR on both source metrics: H-QDA (source BA −0.155 [−0.318, −0.021], AUC −0.101 [−0.238, −0.012]) and HI-MLP (source BA −0.301 [−0.557, −0.024], AUC −0.315 [−0.583, −0.089]).
- Many tree models on H and B (RF, ExtraTrees, HistGB, fixed or source-weighted) give source BA +0.062 [0.000, +0.125]: they classify all 24 NSs correctly but still miss 2 BHs; the lower limit is exactly 0, so by the rule this is not an improvement.
- With 76 combinations × 3 metrics = 228 comparisons, about 5–6 one-sided 2.5% exceedances are expected even if nothing differs. The three QDA results are strongly

dependent (same algorithm, related representations), and the 1e procedure never chose QDA in its inner loop. They are therefore not interpreted.

Table 7.8: DESCRIPTIVE All v4a combinations against H-LR (456 observations, 31 sources). Part 1b = fixed settings, 1c = source-equal weights (`[srcw]`), 1d = nested per algorithm and representation, 1e = primary. Differences are paired source-bootstrap differences with 95% intervals; * = interval excludes 0 (in either direction). From `results/v4/v4a_metrics.csv` and `results/v4/v4a_bootstrap.csv`.

Part	Rep.	Model	src BA	src AUC	Δ src BA [95% CI]	Δ src AUC [95% CI]
1b	H	RandomForest	0.857	0.905	+0.062 [+0.000, +0.125]	−0.018 [−0.113, +0.060]
1b	H	ExtraTrees	0.815	0.899	+0.021 [−0.042, +0.083]	−0.024 [−0.143, +0.060]
1b	H	HistGB	0.857	0.899	+0.062 [+0.000, +0.125]	−0.024 [−0.131, +0.060]
1b	H	SVM_RBF	0.815	0.857	+0.021 [+0.000, +0.062]	−0.065 [−0.214, +0.048]
1b	H	kNN	0.815	0.875	+0.021 [−0.042, +0.083]	−0.048 [−0.214, +0.060]
1b	H	QDA	0.786	0.905	−0.009 [−0.173, +0.104]	−0.018 [−0.089, +0.036]
1b	H	MLP	0.812	0.917	+0.018 [−0.158, +0.223]	−0.006 [−0.107, +0.089]
1b	HI	LogReg	0.795	0.929	+0.000 [+0.000, +0.000]	+0.006 [−0.048, +0.060]
1b	HI	RandomForest	0.836	0.946	+0.042 [−0.042, +0.125]	+0.024 [−0.060, +0.119]
1b	HI	ExtraTrees	0.836	0.958	+0.042 [−0.042, +0.125]	+0.036 [−0.036, +0.137]
1b	HI	HistGB	0.765	0.929	−0.030 [−0.193, +0.104]	+0.006 [−0.077, +0.101]
1b	HI	SVM_RBF	0.866	0.946	+0.071 [−0.083, +0.256]	+0.024 [−0.065, +0.125]
1b	HI	kNN	0.836	0.952	+0.042 [−0.042, +0.125]	+0.030 [−0.048, +0.131]
1b	HI	QDA	0.857	0.952	+0.062 [−0.152, +0.277]	+0.030 [−0.060, +0.137]
1b	HI	MLP	0.661	0.869	−0.134 [−0.310, +0.000]	−0.054 [−0.161, +0.012]
1b	B	LogReg	0.774	0.875	−0.021 [−0.062, +0.000]	−0.048 [−0.179, +0.048]
1b	B	RandomForest	0.857	0.893	+0.062 [+0.000, +0.125]	−0.030 [−0.131, +0.054]
1b	B	ExtraTrees	0.857	0.881	+0.062 [+0.000, +0.125]	−0.042 [−0.161, +0.048]
1b	B	HistGB	0.857	0.893	+0.062 [+0.000, +0.125]	−0.030 [−0.143, +0.054]
1b	B	SVM_RBF	0.836	0.881	+0.042 [−0.042, +0.125]	−0.042 [−0.161, +0.048]
1b	B	kNN	0.836	0.798	+0.042 [−0.042, +0.125]	−0.125 [−0.339, +0.036]
1b	B	QDA	0.786	0.923	−0.009 [−0.173, +0.104]	+0.000 [−0.083, +0.077]
1b	B	MLP	0.783	0.881	−0.012 [−0.146, +0.152]	−0.042 [−0.137, +0.024]
1b	A	LogReg	0.866	0.946	+0.071 [−0.042, +0.235]	+0.024 [−0.048, +0.113]
1b	A	RandomForest	0.622	0.881	−0.173 [−0.387, +0.033]	−0.042 [−0.179, +0.077]
1b	A	ExtraTrees	0.765	0.899	−0.030 [−0.286, +0.226]	−0.024 [−0.155, +0.101]
1b	A	HistGB	0.908	0.940	+0.113 [−0.021, +0.289]	+0.018 [−0.107, +0.143]
1b	A	SVM_RBF	0.795	0.917	+0.000 [−0.223, +0.226]	−0.006 [−0.089, +0.083]
1b	A	kNN	0.815	0.893	+0.021 [−0.193, +0.235]	−0.030 [−0.167, +0.095]
1b	A	QDA	0.929	0.946	+0.134 [+0.021, +0.298]*	+0.024 [−0.071, +0.131]
1b	A	MLP	0.792	0.839	−0.003 [−0.187, +0.202]	−0.083 [−0.232, +0.042]
1c	H	LogReg[srcw]	0.795	0.923	+0.000 [+0.000, +0.000]	+0.000 [+0.000, +0.000]
1c	H	RandomForest[srcw]	0.857	0.905	+0.062 [+0.000, +0.125]	−0.018 [−0.113, +0.060]
1c	H	HistGB[srcw]	0.857	0.899	+0.062 [+0.000, +0.125]	−0.024 [−0.137, +0.060]
1c	HI	LogReg[srcw]	0.795	0.929	+0.000 [+0.000, +0.000]	+0.006 [−0.048, +0.060]
1c	HI	RandomForest[srcw]	0.836	0.940	+0.042 [−0.042, +0.125]	+0.018 [−0.071, +0.119]
1c	HI	HistGB[srcw]	0.744	0.917	−0.051 [−0.223, +0.083]	−0.006 [−0.095, +0.095]
1c	B	LogReg[srcw]	0.774	0.875	−0.021 [−0.062, +0.000]	−0.048 [−0.179, +0.048]
1c	B	RandomForest[srcw]	0.857	0.893	+0.062 [+0.000, +0.125]	−0.030 [−0.131, +0.054]
1c	B	HistGB[srcw]	0.857	0.911	+0.062 [+0.000, +0.125]	−0.012 [−0.095, +0.060]
1c	A	LogReg[srcw]	0.866	0.952	+0.071 [−0.042, +0.235]	+0.030 [−0.048, +0.131]
1c	A	RandomForest[srcw]	0.622	0.881	−0.173 [−0.387, +0.033]	−0.042 [−0.185, +0.083]
1c	A	HistGB[srcw]	0.765	0.940	−0.030 [−0.286, +0.226]	+0.018 [−0.107, +0.143]
1d	H	LogReg	0.824	0.923	+0.030 [−0.083, +0.193]	+0.000 [−0.036, +0.030]
1d	H	RandomForest	0.836	0.952	+0.042 [−0.042, +0.125]	+0.030 [−0.030, +0.125]
1d	H	ExtraTrees	0.836	0.940	+0.042 [−0.042, +0.125]	+0.018 [−0.060, +0.119]
1d	H	HistGB	0.723	0.905	−0.071 [−0.235, +0.062]	−0.018 [−0.095, +0.048]
1d	H	SVM_RBF	0.836	0.774	+0.042 [−0.042, +0.125]	−0.149 [−0.393, +0.024]
1d	H	kNN	0.815	0.911	+0.021 [−0.042, +0.083]	−0.012 [−0.083, +0.060]
1d	H	QDA	0.640	0.821	−0.155 [−0.318, −0.021]*	−0.101 [−0.238, −0.012]*
1d	H	MLP	0.765	0.887	−0.030 [−0.193, +0.083]	−0.036 [−0.137, +0.054]
1d	HI	LogReg	0.783	0.905	−0.012 [−0.146, +0.152]	−0.018 [−0.083, +0.042]
1d	HI	RandomForest	0.774	0.875	−0.021 [−0.125, +0.083]	−0.048 [−0.190, +0.054]
1d	HI	ExtraTrees	0.774	0.810	−0.021 [−0.104, +0.062]	−0.113 [−0.327, +0.036]

Part	Rep.	Model	src BA	src AUC	Δ src BA [95% CI]	Δ src AUC [95% CI]
1d	HI	HistGB	0.824	0.893	+0.030 [−0.104, +0.206]	−0.030 [−0.179, +0.089]
1d	HI	SVM_RBF	0.774	0.887	−0.021 [−0.125, +0.083]	−0.036 [−0.143, +0.065]
1d	HI	kNN	0.795	0.893	+0.000 [−0.104, +0.104]	−0.030 [−0.119, +0.060]
1d	HI	QDA	0.929	0.958	+0.134 [+0.021, +0.298]*	+0.036 [−0.054, +0.143]
1d	HI	MLP	0.494	0.607	−0.301 [−0.557, −0.024]*	−0.315 [−0.583, −0.089]*
1d	B	LogReg	0.857	0.905	+0.062 [+0.000, +0.125]	−0.018 [−0.113, +0.060]
1d	B	RandomForest	0.857	0.911	+0.062 [+0.000, +0.125]	−0.012 [−0.089, +0.060]
1d	B	ExtraTrees	0.857	0.911	+0.062 [+0.000, +0.125]	−0.012 [−0.107, +0.078]
1d	B	HistGB	0.753	0.845	−0.042 [−0.104, +0.000]	−0.077 [−0.232, +0.030]
1d	B	SVM_RBF	0.857	0.881	+0.062 [+0.000, +0.125]	−0.042 [−0.161, +0.048]
1d	B	kNN	0.857	0.869	+0.062 [+0.000, +0.125]	−0.054 [−0.196, +0.048]
1d	B	QDA	0.804	0.875	+0.009 [−0.146, +0.193]	−0.048 [−0.214, +0.071]
1d	B	MLP	0.702	0.833	−0.092 [−0.277, +0.042]	−0.089 [−0.226, +0.000]
1d	A	LogReg	0.866	0.923	+0.071 [−0.062, +0.247]	+0.000 [−0.083, +0.095]
1d	A	RandomForest	0.732	0.839	−0.062 [−0.280, +0.152]	−0.083 [−0.244, +0.054]
1d	A	ExtraTrees	0.744	0.935	−0.051 [−0.307, +0.205]	+0.012 [−0.089, +0.125]
1d	A	HistGB	0.866	0.911	+0.071 [−0.062, +0.244]	−0.012 [−0.143, +0.113]
1d	A	SVM_RBF	0.732	0.911	−0.062 [−0.280, +0.164]	−0.012 [−0.113, +0.083]
1d	A	kNN	0.732	0.899	−0.062 [−0.277, +0.164]	−0.024 [−0.143, +0.089]
1d	A	QDA	0.929	0.982	+0.134 [+0.021, +0.298]*	+0.060 [−0.006, +0.173]
1d	A	MLP	0.670	0.810	−0.125 [−0.351, +0.089]	−0.113 [−0.315, +0.042]
1e	auto	auto	0.732	0.750	−0.062 [−0.146, +0.021]	−0.173 [−0.399, +0.012]

What the algorithms learn in the colour plane. Figure 7.5 draws the 0.5 decision boundaries of five algorithms fitted on all 456 observations (visualisation only; no evaluation uses it). LR draws one oblique line; RF and HistGB carve axis-parallel boxes around the training points, kNN draws islands and SVM-RBF a curved boundary. All of them agree on the two well-separated populations (soft BHs at low soft colour, soft NSs at low hard colour) and differ only in the region where hard BHs and hard NSs overlap – exactly where no algorithm has information to transfer to unseen sources.

7.6.2 Data quality: type-I bursts

SECONDARY Removing the 61 automatically flagged observations changes nothing at the source level (Table 7.9): the 31 H-LR source decisions are identical (source BA 0.795, source AUC 0.917 on the kept observations, differences 0.000 [0, 0]) and the observation BA moves by −0.012 [−0.062, +0.017]. The Phase 1 conclusions also survive: after exclusion, B − H and H+B − H are still not detected for LR and RF. In the visual-set sensitivity analysis two *observation-level* RF differences have intervals above 0 (B − H +0.027 [+0.004, +0.049]; H+B − H +0.029 [+0.005, +0.051]), with source-level differences of 0; the visual-set H-RF itself lost a little observation BA relative to the original (−0.009 [−0.017, −0.002]). These are small, observation-level and confined to the sensitivity analysis.

7.6.3 Wider band: HEXTE 25–60 keV (v4b3)

SECONDARY On the 396-observation HEXTE subset (31 sources, 99 BH observations), adding the two HEXTE colours does not change a single source decision (Table 7.10): source BA differences 0.000 [0, 0] for LR and RF, source AUC −0.048 [−0.179, +0.024] for LR and 0.000 [−0.036, +0.036] for RF. In the colour-overlap box (soft colour ≥ 0.70 and hard colour ≥ 0.20, sources with ≥ 3 box observations: 24 sources, 203 observations) – the region where HEXTE should help if it helps anywhere – nothing is detected either. The box was defined after seeing the v2 colour plot (POST HOC definition). All **not detected** under the pre-registered criterion.

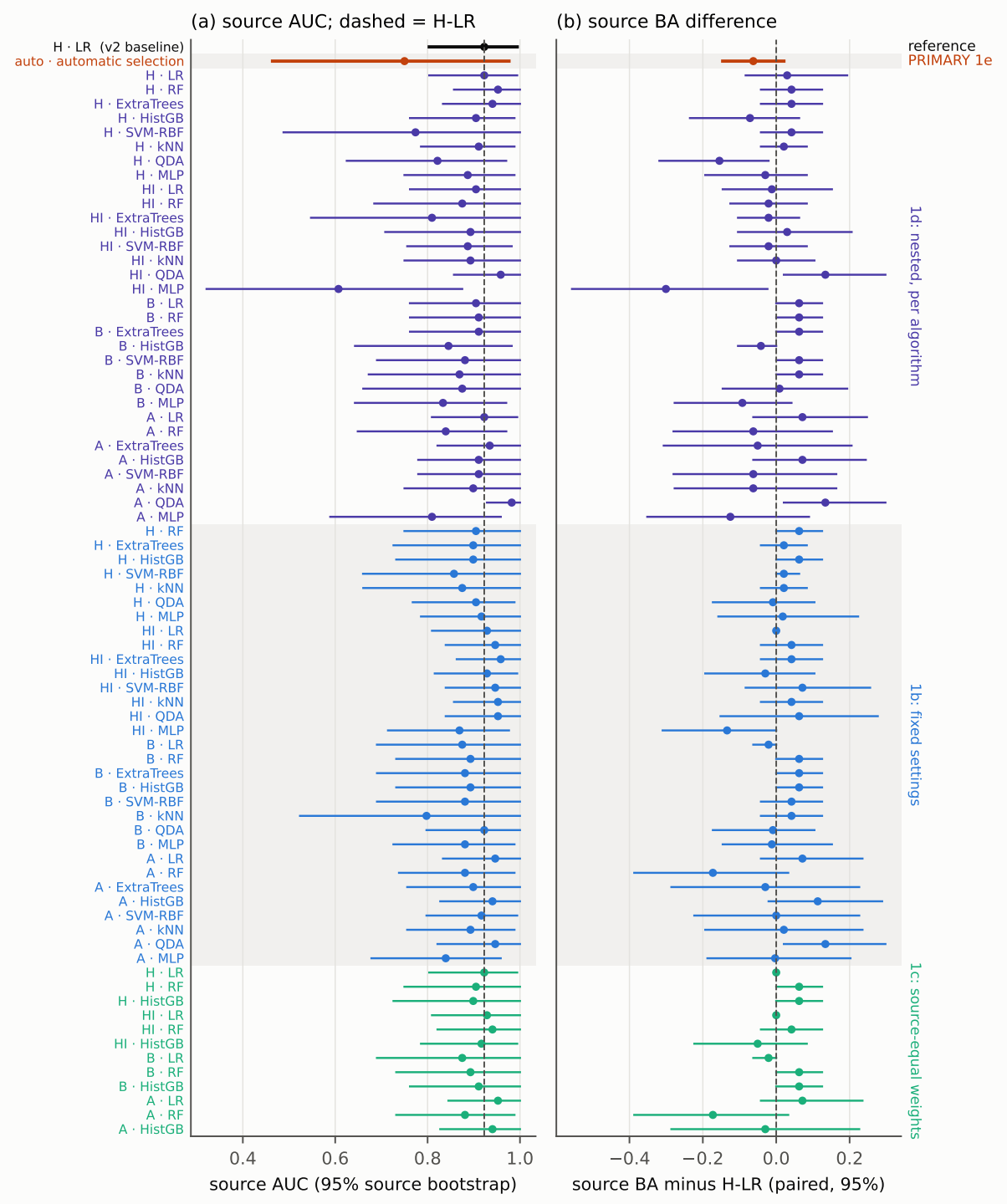


Figure 7.4: All v4a combinations. Left: source-level AUC with 95% source-bootstrap interval (dashed line = H-LR, 0.923); right: paired difference in source balanced accuracy against H-LR. Orange = the primary nested automatic selection (1e), purple = nested per algorithm (1d), blue = fixed settings (1b), green = source-equal weights (1c), black = the H-LR baseline. The primary is the lowest-AUC entry except HI-MLP (nested); the three source-BA intervals above 0 are A-QDA (fixed), HI-QDA and A-QDA (nested). Redrawn for this report from `results/v4/v4a_bootstrap.csv` with the same rows, order, colours and intervals as the repository figure `figures/v4/v4a_forest_source_auc.png` of `scripts/16_v4a_algorithms.py`, whose 77 row labels are unreadable at page size; rows are labelled representation · algorithm and grouped by sub-analysis (right margin).

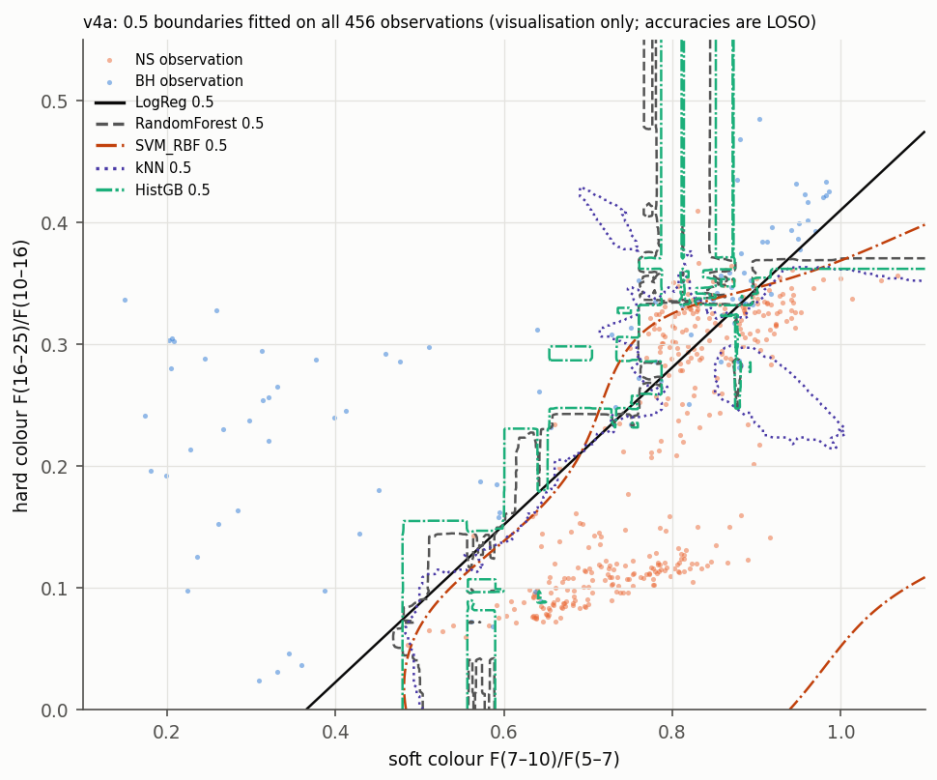


Figure 7.5: 0.5 decision boundaries in the H colour plane (soft colour $F(7–10)/F(5–7)$, hard colour $F(16–25)/F(10–16)$, count space) of LR, RF, SVM-RBF, kNN and HistGB fitted on all 456 v2 observations (blue BH, orange NS). Visualisation only; all accuracies in this chapter are LOSO. Repo: `figures/v4/v4a_colour_boundaries.png`, `scripts/16_v4a_algorithms.py`.

Table 7.9: SECONDARY Effect of excluding observations with type-I bursts (auto set: 395 kept; visual set: 398 kept). "excl. − orig." compares the re-trained model with the original v2/v3a OOF scores on the *same* kept observations; "B − H" etc. are recomputed after exclusion (original 456-observation values in Table 6.8). From `results/v4/burst_exclusion_bootstrap.csv`.

Set	Comparison	Δ source BA	Δ source AUC	Δ obs. BA
auto	H-LR excl. − orig.	0.000 [0, 0]	0.000 [0, 0]	−0.012 [−0.062, +0.017]
auto	H-RF excl. − orig.	0.000 [0, 0]	+0.006 [0, +0.036]	−0.010 [−0.024, +0.002]
auto	B − H (LR)	−0.042 [−0.104, 0]	−0.048 [−0.179, +0.048]	+0.028 [−0.032, +0.088]
auto	B − H (RF)	−0.071 [−0.214, 0]	−0.036 [−0.131, +0.024]	+0.014 [−0.011, +0.041]
auto	H+B − H (LR)	−0.042 [−0.104, 0]	−0.048 [−0.179, +0.048]	+0.022 [−0.045, +0.093]
auto	H+B − H (RF)	−0.071 [−0.214, 0]	−0.030 [−0.113, +0.024]	+0.005 [−0.018, +0.028]
visual	H-LR excl. − orig.	0.000 [0, 0]	0.000 [0, 0]	−0.002 [−0.010, +0.005]
visual	H-RF excl. − orig.	0.000 [0, 0]	−0.012 [−0.048, 0]	−0.009 [−0.017, −0.002]*
visual	B − H (LR)	−0.042 [−0.104, 0]	−0.060 [−0.208, +0.048]	0.000 [−0.062, +0.053]
visual	B − H (RF)	0.000 [0, 0]	−0.018 [−0.095, +0.048]	+0.027 [+0.004, +0.049]*
visual	H+B − H (LR)	−0.042 [−0.104, 0]	−0.060 [−0.208, +0.048]	+0.002 [−0.060, +0.054]
visual	H+B − H (RF)	0.000 [0, 0]	−0.006 [−0.071, +0.060]	+0.029 [+0.005, +0.051]*

* interval excludes 0.

Table 7.10: SECONDARY v4b3: H vs H+X (H + HEXTE colours + missing indicators), LOSO on the HEXTE subset; the overlap box uses the same OOF scores restricted to box observations of sources with ≥ 3 of them. From `results/v4b3/v4b3_bootstrap.csv`.

Subset	Model	source BA	source AUC	obs. BA
HEXTE (396 obs.)	H-LR	0.795 [0.589, 0.958]	0.923 [0.804, 0.994]	0.776
	H+X-LR	0.795 [0.589, 0.958]	0.875 [0.690, 0.994]	0.779
	difference	0.000 [0, 0]	−0.048 [−0.179, +0.024]	+0.003 [−0.078, +0.097]
	H-RF	0.857 [0.643, 1.000]	0.911 [0.762, 1.000]	0.744
	H+X-RF	0.857 [0.643, 1.000]	0.911 [0.762, 1.000]	0.751
	difference	0.000 [0, 0]	0.000 [−0.036, +0.036]	+0.007 [−0.027, +0.041]
overlap box	H-LR	0.651	0.748	0.629
(24 src, 203 obs.)	H+X-LR	0.681	0.739	0.659
	difference	+0.029 [0, +0.088]	−0.008 [−0.210, +0.160]	+0.030 [−0.099, +0.147]
	H-RF	0.756	0.815	0.625
	H+X-RF	0.685	0.798	0.635
	difference	−0.071 [−0.214, 0]	−0.017 [−0.092, +0.059]	+0.010 [−0.057, +0.077]

7.6.4 More sources and earlier epochs (v4b2)

Bridge check (gate). On the 456 v2 observations, H_R-LR and H-LR give OOF scores with Spearman $\rho = 0.994$, source BA difference 0.000 [0, 0] and source AUC difference −0.018 [−0.065, 0] (`results/v4b2/v4b2_bridge_check.csv`): both intervals contain 0 and $\rho \geq 0.90$, so the gate was passed and the power-law normalisation can be used across gain epochs.

Extended sample. SECONDARY Table 7.11 gives the six models on the 46-source sample and Table 7.12 the paired differences. The analysis script compared every model with the same-sample H_R-LR (the reference named in the v4b2 rules); against it, H-RF and H_R-RF gain +0.123 [+0.027, +0.248] in source BA. Against the H-LR recomputed on the same sample (the baseline required by §0 of the pre-registration), H-RF and H_R-RF gain source BA +0.068 [+0.014, +0.122] – **detected** (95% CI excludes 0) by the pre-declared rule – but their source AUC differences (−0.009 and −0.012) are not. The gain comes entirely from classifying all 37 NSs correctly (H-LR: 32/37) while missing the same 2 BHs, i.e. from the position of the threshold, not from a better ranking of sources. Evaluated on the original 456 v2 observations (models trained on the extended sample), no combination beats the v2 H-LR (H-RF and H_R-RF: +0.062 [0, +0.125]). The same-sample H-LR comparison was required by the pre-registration (§0) but had been omitted from the analysis script; it was added after the v4b2 results were seen.

Table 7.11: SECONDARY v4b2: 46 sources (9 BH, 37 NS), 677 observations, LOSO, with 95% source-bootstrap intervals. "Correct" = sources on the right side of 0.5. From `results/v4b2/v4b2_bootstrap.csv`.

Model	source BA	source AUC	obs. BA	BH / NS correct
H-LR	0.821 [0.668, 0.946]	0.946 [0.868, 0.997]	0.792 [0.674, 0.893]	7/9, 32/37
H-RF	0.889 [0.722, 1.000]	0.937 [0.832, 1.000]	0.779 [0.665, 0.878]	7/9, 37/37
H_R-LR	0.766 [0.599, 0.917]	0.928 [0.832, 0.991]	0.765 [0.646, 0.874]	6/9, 32/37
H_R-RF	0.889 [0.722, 1.000]	0.934 [0.826, 1.000]	0.770 [0.653, 0.869]	7/9, 37/37
R-LR	0.862 [0.709, 0.986]	0.883 [0.667, 1.000]	0.749 [0.643, 0.839]	7/9, 35/37
R-RF	0.709 [0.542, 0.875]	0.922 [0.808, 0.994]	0.662 [0.590, 0.728]	4/9, 36/37

The new sources. Table 7.13 shows how the 15 sources that were never part of any earlier training set were scored. Both new BHs are called BH by every H and H_R model (XTE J1859+226 0.84–0.94, V4641 Sgr 0.63–0.93). Among the 13 new NSs, the two LR models call 4U 2129+12

Table 7.12: v4b2 paired differences. Upper block: against the H-LR recomputed on the 46-source sample (pre-registration §0; added after the v4b2 results were seen). Middle: against the same-sample H_R-LR (the reference computed by the analysis script). Lower: models trained on the extended sample, evaluated on the 456 v2 observations, against the v2 H-LR. * = interval excludes 0; † = entirely above 0 ("improvement"). From `results/v4b2/v4b2_bootstrap.csv`.

Comparison	Δ source BA	Δ source AUC	Δ obs. BA
H-RF − H-LR	+0.068 [+0.014, +0.122]†	−0.009 [−0.069, +0.039]	−0.013 [−0.062, +0.038]
H_R-LR − H-LR	−0.056 [−0.167, 0]	−0.018 [−0.057, 0]	−0.028 [−0.072, +0.005]
H_R-RF − H-LR	+0.068 [+0.014, +0.122]†	−0.012 [−0.072, +0.036]	−0.022 [−0.073, +0.030]
R-LR − H-LR	+0.041 [−0.014, +0.095]	−0.063 [−0.243, +0.045]	−0.044 [−0.116, +0.032]
R-RF − H-LR	−0.113 [−0.293, +0.053]	−0.024 [−0.123, +0.063]	−0.130 [−0.217, −0.035]*
H-LR − H_R-LR	+0.056 [0, +0.167]	+0.018 [0, +0.057]	+0.028 [−0.005, +0.072]
H-RF − H_R-LR	+0.123 [+0.027, +0.248]†	+0.009 [−0.045, +0.066]	+0.015 [−0.051, +0.085]
H_R-RF − H_R-LR	+0.123 [+0.027, +0.248]†	+0.006 [−0.042, +0.060]	+0.005 [−0.062, +0.078]
R-LR − H_R-LR	+0.096 [0, +0.233]	−0.045 [−0.231, +0.078]	−0.016 [−0.103, +0.080]
R-RF − H_R-LR	−0.057 [−0.210, +0.081]	−0.006 [−0.096, +0.078]	−0.103 [−0.181, −0.018]*
on v2 obs.: H-LR − v2 H-LR	0.000 [0, 0]	0.000 [−0.030, +0.030]	−0.004 [−0.025, +0.016]
on v2 obs.: H-RF − v2 H-LR	+0.062 [0, +0.125]	−0.024 [−0.131, +0.060]	−0.024 [−0.095, +0.049]
on v2 obs.: H_R-LR − v2 H-LR	−0.071 [−0.214, 0]	−0.018 [−0.065, 0]	−0.040 [−0.106, +0.006]
on v2 obs.: H_R-RF − v2 H-LR	+0.062 [0, +0.125]	−0.030 [−0.131, +0.054]	−0.027 [−0.103, +0.052]
on v2 obs.: R-LR − v2 H-LR	+0.021 [−0.042, +0.083]	−0.077 [−0.304, +0.060]	−0.066 [−0.159, +0.027]
on v2 obs.: R-RF − v2 H-LR	−0.173 [−0.387, +0.033]	−0.042 [−0.179, +0.083]	−0.152 [−0.268, −0.030]*

(in M15) and XTE J1710−281 BH (scores 0.52–0.54); RF classifies all of them correctly.

Table 7.13: v4b2 new sources: eligible pointings over all gain epochs, sampled observations, observations per epoch (epoch:count), and LOSO source mean scores; **bold** = wrong side of 0.5. From `data/v4b2/sources.csv`, `results/v4b2/v4b2_sample_by_epoch.csv`, `results/v4b2/v4b2_oof_predictions.csv`.

Source	label	elig.	obs.	epochs	H-LR	H-RF	H_R-LR	H_R-RF	R-LR	R-RF
V4641 Sgr	BH	11	11	4:2, 5:9	0.93	0.63	0.93	0.68	0.65	0.60
XTE J1859+226	BH	112	15	4:15	0.94	0.85	0.93	0.84	0.75	0.56
1A 1744−361	NS	42	15	5:15	0.15	0.01	0.17	0.05	0.22	0.04
4U 0513−40	NS	592	15	3:1, 4:1, 5:13	0.17	0.07	0.18	0.10	0.28	0.23
4U 0919−54	NS	167	15	3:1, 5:14	0.33	0.07	0.35	0.07	0.30	0.11
4U 2129+12	NS	58	15	2:2, 3:2, 5:11	**0.54**	0.33	**0.52**	0.34	0.33	0.07
EXO 1745−248	NS	91	15	5:15	0.12	0.05	0.13	0.05	0.18	0.20
GRS 1747−312	NS	102	15	4:3, 5:12	0.27	0.11	0.27	0.11	0.26	0.15
SAX J1747.0−2853	NS	40	15	4:2, 5:13	0.28	0.05	0.30	0.07	0.34	0.14
SAX J1748.9−2021	NS	82	15	5:15	0.24	0.10	0.26	0.10	0.22	0.11
SAX J1750.8−2900	NS	90	15	5:15	0.36	0.10	0.38	0.11	0.29	0.09
Ser X-1	NS	83	15	3:2, 4:1, 5:12	0.08	0.03	0.08	0.05	0.11	0.04
XTE J1709−267	NS	37	15	3:2, 5:13	0.25	0.08	0.24	0.07	0.25	0.11
XTE J1710−281	NS	103	15	3:1, 5:14	**0.52**	0.39	**0.54**	0.43	0.32	0.40
XTE J1759−220	NS	39	15	5:15	0.24	0.09	0.25	0.12	0.30	0.08

Gain epoch (diagnostic only). BH scores correlate negatively with the epoch (Spearman ≈ -0.27 to -0.30 for the H and H_R models, -0.10 to -0.12 for R), NS scores not at all ($|\rho| \leq 0.04$; `results/v4b2/v4b2_epoch_score_correlation.csv`). Almost all epoch-4 BH observations belong to XTE J1859+226, so this reflects source identity, not a response effect (Figure 7.6).

7.6.5 All observations of each source (v4b1)

SECONDARY The pre-declared main comparison is the source-equal-weight H-LR trained on all 7,949 observations against the v2 H-LR (15 observations per source). It is **not detected** under the pre-registered criterion: source BA difference 0.000 [0, 0], source AUC 0.000 [−0.048, +0.036], observation BA −0.009 [−0.114, +0.123] (Table 7.14). The 31 source decisions are *identical* to v2 (wrong: the BHs GRS 1915+105 and XTE J1118+480, the NSs HETE J1900.1−2455, KS 1731−260 and SAX J1808.4−3658), and on the 456 shared observations the two models' scores have Spearman 0.999 and agree in label on 99.1% (Table 7.15). The source means lie on the diagonal (Figure 7.7): the LR boundary in the colour plane is the same whether it is fitted on 15 or on up to 1,329 observations per source.

Without source-equal weights (`class_weight=balanced`, i.e. each observation weighted equally within its class) H-LR is worse (source BA 0.744; XTE J1550−564 drops to 0.49), presumably because the sources with the most observations (4U 1636−53, GRS 1915+105) then dominate the boundary.

DESCRIPTIVE Of the 16 combinations, four have source-BA intervals entirely above 0 – A-RF with both weightings, HI-LR (balanced) and HI-RF (source weights) – all at 0.929, +0.134 [+0.021, +0.298], misclassifying only XTE J1118+480; none has a source-AUC interval above 0 (closest: A-RF balanced +0.060 [−0.006, +0.167]). All four include intensity (A or HI). The same models with 15 observations per source (v4a 1b) did not show this (A-RF 0.622, HI-RF 0.836, HI-LR 0.795), and in v4a the nested selection's preference for intensity models failed on unseen sources; the transferability of intensity is therefore unclear. With 48 comparisons (plus 48 on the shared observations) and four dependent results with identical point estimates and the

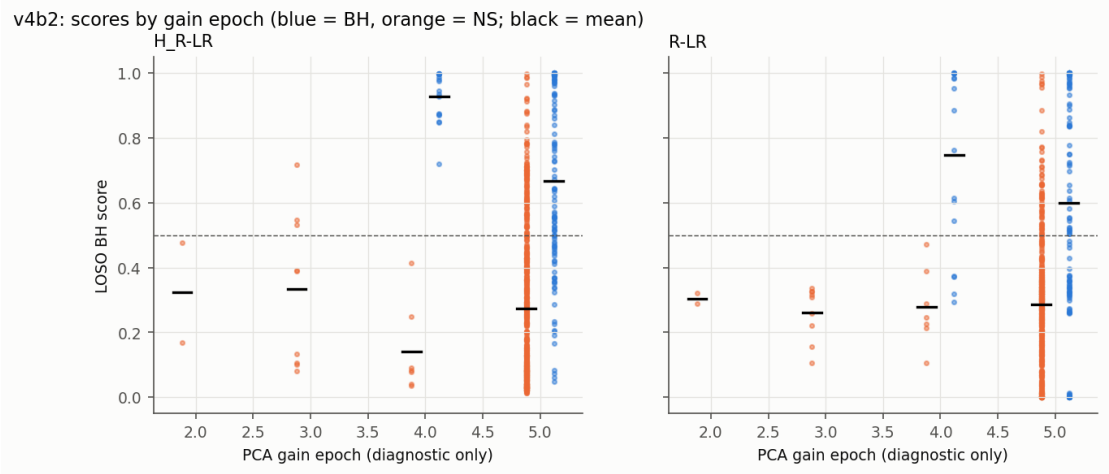


Figure 7.6: LOSO BH scores of the 677 v4b2 observations by PCA gain epoch (diagnostic only; BH blue right of each epoch, NS orange left, black bar = mean). Left: H_R-LR; right: R-LR. The epoch-4 BH points are almost all XTE J1859+226; epochs 2–3 contain only a few NS observations of the new sources. Repo: `figures/v4b2/v4b2_scores_by_epoch.png`, `scripts/22_v4b2_analysis.py`.

same single error, these are not interpreted.

Table 7.14: v4b1: all 7,949 eligible observations of the 31 v2 sources. srcw = source-equal weights (Eq. (4.3)), bal = `class_weight=balanced`. Differences are paired with the v2 H-LR (456 observations) by source. Only the main comparison and the combinations mentioned in the text are shown; all 16 are in `results/v4b1/v4b1_bootstrap.csv`.

Model (7,949 obs.)	source BA	source AUC	obs. BA	Δ source BA	Δ source AUC
v2 H-LR (456 obs.)	0.795	0.923	0.776	–	–
H-LR [srcw] (main)	0.795	0.923	0.766	0.000 [0, 0]	0.000 [−0.048, +0.036]
H-LR [bal]	0.744	0.917	0.692	−0.051 [−0.214, +0.062]	−0.006 [−0.083, +0.071]
H-RF [srcw]	0.857	0.929	0.727	+0.062 [0, +0.125]	+0.006 [−0.083, +0.095]
HI-LR [bal]	0.929	0.964	0.791	+0.134 [+0.021, +0.298]*	+0.042 [−0.036, +0.143]
HI-RF [srcw]	0.929	0.952	0.874	+0.134 [+0.021, +0.298]*	+0.030 [−0.071, +0.143]
A-RF [srcw]	0.929	0.976	0.855	+0.134 [+0.021, +0.298]*	+0.054 [−0.024, +0.161]
A-RF [bal]	0.929	0.982	0.864	+0.134 [+0.021, +0.298]*	+0.060 [−0.006, +0.167]

* interval entirely above 0 (descriptive combination; not interpreted).

Table 7.16: v4b1 per-source LOSO mean scores (n = all eligible observations). "v2" = v2 H-LR with 15 observations (6 for 4U 1916−053). s = source-equal weights, b = balanced. **Bold** = wrong side of 0.5. From `results/v4b1/v4b1_per_source.csv`.

Source	label	n	v2	H-LR s	H-LR b	H-RF s	HI-LR b	HI-RF s	A-RF s	A-RF b
GRS 1915+105	BH	1,031	**0.41**	**0.47**	**0.48**	**0.23**	0.58	0.56	0.69	0.69
GX 339−4	BH	648	0.81	0.78	0.63	0.87	0.68	0.83	0.64	0.66
GRO J1655−40	BH	433	0.82	0.87	0.92	0.90	0.89	0.88	0.77	0.84
XTE J1650−500	BH	95	0.85	0.89	0.88	0.71	0.81	0.80	0.71	0.71
XTE J1550−564	BH	63	0.55	0.58	**0.49**	0.79	0.59	0.83	0.77	0.76
4U 1543−47	BH	58	0.95	0.94	0.93	0.88	0.87	0.87	0.75	0.76
XTE J1118+480	BH	41	**0.41**	**0.38**	**0.36**	**0.11**	**0.26**	**0.08**	**0.19**	**0.18**
4U 1636−53	NS	1,329	0.22	0.14	0.14	0.06	0.08	0.01	0.03	0.02
4U 1608−52	NS	540	0.35	0.41	0.41	0.15	0.36	0.05	0.08	0.08
EXO 0748−676	NS	509	0.28	0.22	0.16	0.19	0.04	0.27	0.33	0.35
4U 1705−44	NS	402	0.18	0.14	0.15	0.03	0.09	0.01	0.01	0.01
Cyg X-2	NS	402	0.21	0.19	0.37	0.45	0.31	0.33	0.19	0.17
Aql X-1	NS	400	0.41	0.44	0.48	0.35	0.36	0.15	0.15	0.14

Source	label	n	v2	H-LR s	H-LR b	H-RF s	HI-LR b	HI-RF s	A-RF s	A-RF b
HETE J1900.1−2455	NS	344	**0.61**	**0.67**	**0.68**	0.17	0.48	0.05	0.11	0.10
4U 1728−34	NS	276	0.17	0.15	0.13	0.05	0.16	0.05	0.11	0.11
SAX J1808.4−3658	NS	234	**0.72**	**0.71**	**0.72**	0.18	0.46	0.14	0.17	0.15
4U 1702−429	NS	201	0.28	0.17	0.17	0.04	0.11	0.00	0.01	0.01
4U 1735−44	NS	161	0.03	0.01	0.01	0.01	0.01	0.00	0.00	0.00
4U 1820−30	NS	130	0.04	0.03	0.02	0.02	0.03	0.02	0.01	0.01
GS 1826−238	NS	118	0.32	0.26	0.16	0.05	0.23	0.08	0.10	0.10
GX 17+2	NS	90	0.09	0.07	0.10	0.03	0.17	0.33	0.38	0.41
4U 1724−307	NS	77	0.37	0.34	0.30	0.06	0.16	0.02	0.02	0.01
GX 3+1	NS	75	0.09	0.06	0.10	0.02	0.06	0.00	0.00	0.00
XTE J1814−338	NS	73	0.47	0.47	0.36	0.38	0.15	0.44	0.25	0.21
SLX 1735−269	NS	64	0.40	0.46	0.46	0.09	0.15	0.02	0.09	0.07
4U 1746−37	NS	43	0.08	0.04	0.04	0.02	0.01	0.00	0.01	0.01
KS 1731−260	NS	33	**0.60**	**0.55**	0.48	0.21	0.47	0.06	0.10	0.08
MXB 1659−298	NS	30	0.29	0.27	0.33	0.15	0.05	0.09	0.11	0.12
4U 1254−69	NS	24	0.11	0.08	0.12	0.03	0.01	0.00	0.00	0.01
4U 1323−62	NS	19	0.32	0.28	0.18	0.07	0.04	0.11	0.16	0.15
4U 1916−053	NS	6	0.29	0.28	0.27	0.08	0.06	0.02	0.03	0.02

7.6.6 Phase 1–2 comparisons at a glance

Figure 7.8 collects the main paired differences of Phases 1 and 2 on one scale. Every source-AUC interval contains 0 and most point estimates are at or below 0; the only source-BA interval entirely above 0 among them is the v4b2 threshold effect.

7.7 Interpretation

What the data support directly.

1. On the 31 confirmed sources, no pre-declared comparison beats H-LR: not algorithm choice (v4a 1e), not HEXTE colours (v4b3), not the extended sample evaluated on the original observations (v4b2), and not 17 times more observations per source (v4b1, whose 31 source decisions are identical to those of v2).
2. Letting the training data choose the algorithm, representation and threshold made the outer result *worse* in point estimate (−0.06 in source BA, −0.17 in source AUC). The inner AUC of 0.96–1.00 against an outer 0.75 shows how large the selection optimism is with 30 training sources.
3. Type-I burst contamination does not drive the Phase 1 conclusions. An automatic Standard-1 burst detector confuses BH variability with bursts (12 of 61 flags) and misses slow bursts, so visual confirmation is necessary.
4. HEXTE cluster B adds nothing detectable to the two PCA colours on this sample.
5. On 46 sources, RF beats the same-sample H-LR in source BA but not in source AUC: it places the threshold better for the larger NS population (37/37 against 32/37) without ranking sources better.
6. Four descriptive v4b1 combinations with intensity reach source BA 0.929 (only XTE J1118+480 wrong) without a source-AUC difference; the same models with 15 observations per source did not.

Possible explanations (not verified).

Table 7.15: DESCRIPTIVE v4b1 models restricted to the 456 v2 observations: agreement with the v2 H-LR and paired differences (`results/v4b1/v4b1_agreement_on_v2_obs.csv`, `results/v4b1/v4b1_bootstrap_on_v2_obs.csv`).

Model	Spearman	label agreement	Δ source BA	Δ obs. BA
H-LR [srcw]	0.999	0.991	0.000 [0, 0]	−0.013 [−0.026, −0.0005]*
H-LR [bal]	0.968	0.928	−0.071 [−0.214, 0]	−0.072 [−0.150, −0.005]*
H-RF [srcw]	0.729	0.781	+0.062 [0, +0.125]	−0.029 [−0.095, +0.046]
HI-LR [bal]	0.844	0.846	+0.134 [+0.021, +0.298]*	−0.007 [−0.074, +0.077]
HI-RF [srcw]	0.641	0.785	+0.134 [+0.021, +0.298]*	+0.027 [−0.054, +0.114]
HI-RF [bal]	0.586	0.783	+0.134 [+0.021, +0.298]*	+0.007 [−0.065, +0.082]
A-RF [srcw]	0.655	0.763	+0.134 [+0.021, +0.298]*	+0.027 [−0.065, +0.142]
A-RF [bal]	0.616	0.763	+0.134 [+0.021, +0.298]*	+0.017 [−0.085, +0.137]

* interval excludes 0 (for the observation-level rows: entirely below 0).

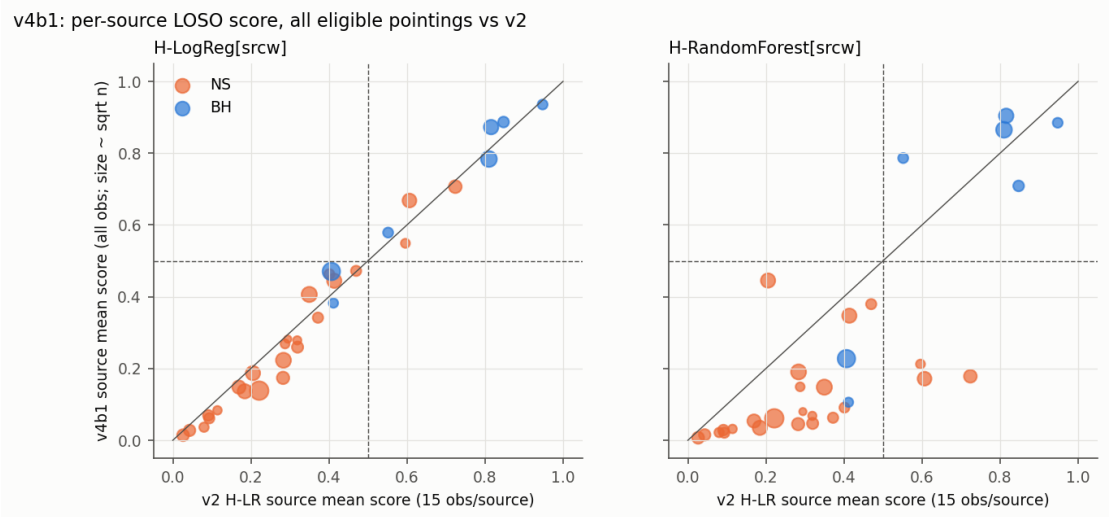


Figure 7.7: Per-source LOSO mean scores with all eligible observations (v4b1, vertical) against the v2 H-LR with 15 observations per source (horizontal); marker size $\propto \sqrt{n}$, BH blue, NS orange, dashed lines at 0.5. Left: H-LR with source-equal weights – every source stays on the diagonal and in the same quadrant. Right: H-RF with source-equal weights – the forest pushes most NSs to lower scores and moves the NSs that H-LR misclassifies below 0.5, but GRS 1915+105 (large blue circle near 0.23) and XTE J1118+480 stay below 0.5. Repo: `figures/v4b1/v4b1_source_scores.png`, `scripts/25_v4b1_analysis.py`.

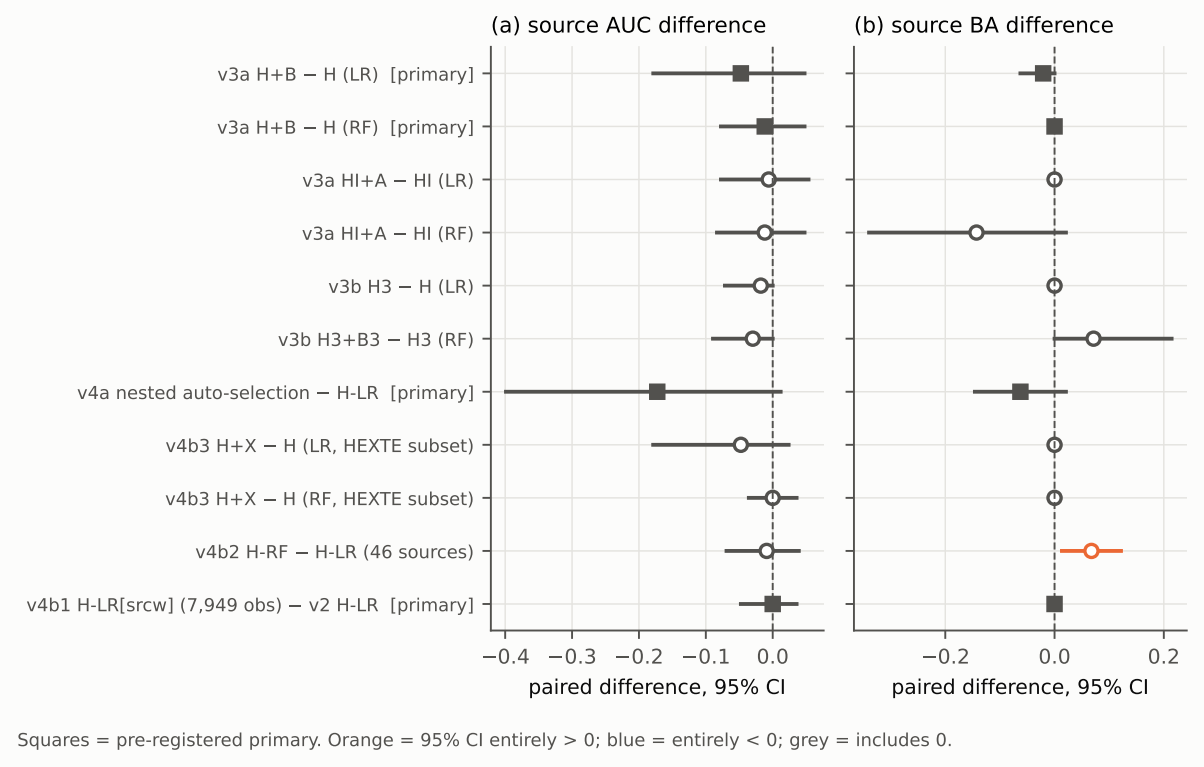


Figure 7.8: Main paired differences of Phases 1–2 against their two-colour baselines with 95% source-bootstrap intervals: (a) source AUC, (b) source balanced accuracy. Rows marked [primary] were the pre-declared primary comparisons of v3a, v4a and v4b1 (the v4b1 main comparison was pre-declared as its main comparison; the pre-registration calls only v4a 1e "primary"). Generated for this report from `results/v3/v3a_bootstrap.csv`, `results/v3/v3b_bootstrap.csv`, `results/v4/v4a_bootstrap.csv`, `results/v4b3/v4b3_bootstrap.csv`, `results/v4b2/v4b2_bootstrap.csv` and `results/v4b1/v4b1_bootstrap.csv` by `report_en/figures/generated/make_figures.py`; plotted values in `G5_phase12_forest.csv`.

- The two colours already compress what is source-transferable in 5–25 keV; flexible models add variance, not information, on 31 sources.
- SVM-RBF on intensity A won the inner selection probably because intensity is tied to source identity (distance, luminosity, RXTE target selection), so inside 30 training sources it can "recognise" groups of sources, which does not transfer to a new source.
- HEXTE cluster B has a small effective area; 40–60 keV is below $S/N = 3$ in 144 of 396 observations. A high-energy cut-off difference between hard-state BHs and NSs may need much higher S/N (stacking, or NuSTAR).
- The extended-sample RF advantage agrees with the "threshold problem" seen in v3: when many more NSs enter, LR gives NSs near the boundary scores slightly above 0.5.
- The v4b1 intensity result may arise because more observations per source cover each source's count-rate range more completely; this is a selection property of the sample, not necessarily spectral information. v4a points the other way (intensity models chosen inside, failing outside), so the transferability of intensity is open.
- Observation-weighted (balanced) H-LR is worse probably because sources with very many observations (4U 1636−53, GRS 1915+105) dominate the boundary; source-equal weights avoid this.

Hypotheses that need more data. More dynamically confirmed BHs (v4b2 added only two); a pre-defined spectral-state split (do BH and NS separate within the hard state? – taken up in Phase 4); stacked HEXTE or NuSTAR spectra above 25 keV; intensity on an independent source sample with a different distance distribution and instrument (taken up with MAXI in Phase 4).

Physically. The decision boundaries (Figure 7.5) agree wherever the populations are separated and disagree only in the hard-colour region where hard-state BHs and hard NSs (atolls in the island state, AMXPs) overlap. Both are dominated there by thermal Comptonisation with similar electron temperatures in 5–25 keV; the difference expected from the literature – on average lower Comptonisation temperatures (lower cut-off energies) in NSs, attributed to the additional seed photons from the surface (Burke et al., 2017) – lies mostly above 25 keV, which is why HEXTE was tried; at HEXTE's sensitivity it is not visible in single observations. The quantity that differs fundamentally (a surface or not) is not imprinted on a single PCA spectrum strongly enough to be learned from 7–9 BH sources, which motivates the turn to timing in Phase 3.

7.8 Deviations and caveats

- **Written after v1–v3**: the v4 pre-registration was written after all earlier results; every question targets a known result.
- POST HOC **Source-equal-weight scale bug** (found after the v4a results): the first `source_equal_weights` made each class sum to 1 instead of $n/2$ as for balanced weights, so LR's effective regularisation was $\approx n/2$ times stronger (HistGB was also affected). Fixed to the balanced scale (relative weights unchanged, Section 4.1.2); only 1a–1c were re-run: 1b identical, 12 rows of 1c changed, conclusion unchanged (pre-fix results in commit `9120afb`). v4b1 ran after the fix.
- **1e tie-break**: the pre-registration only said "first in the grid"; the implementation fixed the full order (algorithm, representation, grid) before the run.
- **1D-CNN skipped** as pre-declared (no `torch`).

- **v4b2**: a duplicate source (XB 1254−690 = 4U 1254−69) was fixed in the matching code before download; the 13 additional NSs result from executing the same rule correctly, not from a rule change; the download script `21_v4b2_fetch_process.py` is a copy of `04_fetch_process.py` that also scans AO1–AO4. The same-sample H-LR comparison was added after the v4b2 results were seen (required by §0 but missing in the script).
- **v4b1**: raw files outside the repository, 6 download threads instead of 4; processing unchanged. Two descriptive analyses (restriction to the v2 observations; per-source score plot) were decided before the v4b1 results.
- **Approval**: the user approved v4 with "start and finish full if no big problem", so no stop was made before the v4b1–v4b3 downloads, although the pre-registration foresaw one per sub-phase.
- **Bootstrap** intervals keep the OOF predictions fixed and do not include the variability of model fitting or of the nested selection; the nested source AUC is an approximation (shifted scores).
- **Multiplicity**: 76 descriptive v4a combinations and 16 (+16) v4b1 combinations, uncorrected; the intervals above 0 (QDA; intensity models) are not interpreted.
- **Bursts**: the automatic rule has false positives and misses; the visual verdicts were made by one person.
- **HEXTE**: cluster B only; no features after 2009-12-14; X_1 is a cross-instrument count ratio; the overlap box is a post hoc definition.
- **v4b2**: many new NSs lie in the Galactic-centre region or in globular clusters (Terzan 5, Terzan 6, NGC 6440, M15, NGC 1851), where the $\approx 1^\circ$ PCA field may contain other sources; gain epoch and source identity are confounded (XTE J1859+226 is entirely in epoch 4), so an epoch effect cannot be estimated.
- **v2 source rule**: executed incompletely (13 NSs missed); the v1–v3 numbers were not changed and describe a smaller NS sample.
- **v4b1**: observation-level metrics are dominated by the sources with most observations (4U 1636−53: 1,329; GRS 1915+105: 1,031); observations of the same source are strongly correlated.

7.9 Takeaway

The two-colour ceiling is a property of the data, not of the models: more flexible algorithms (with honest nested selection), burst cleaning, a 25–60 keV extension and 17 times more observations all leave the classification of unseen sources where H-LR put it. Adding sources helps only through the threshold. Single-spectrum information appears exhausted at about source BA 0.8 and source AUC 0.92 on these sources, with hard-state BHs and hard NSs inseparable. Phase 3 therefore turns to a different physical observable: fast X-ray variability.

Chapter 8

Phase 3 (v5–v6): fast variability as a second axis

Phase at a glance: v5–v6

Question. Phase 2 showed that a single 5–25 keV spectrum is exhausted at two colours, with hard-state BHs and hard NSs inseparable. Does the aperiodic variability between 1/128 and 4 Hz, measured from the same RXTE observations, add information *at fixed colour*, and does that hold for sources the model has never seen?
Pre-declared primaries. v5: H+T-LR − H-LR on the 31 v2 sources (S1, LOSO); v6a: a model *frozen* after training on S1, H_R+T-LR − H_R-LR, evaluated on 23 external sources (E: 4 BH, 19 NS) never used in training. Both judged on source BA and source AUC.
Data. RXTE/PCA Standard-1 (0.125 s, no energy resolution), no new download for v5; v6 adds 12 sources (2 HMXB BHs, 6 AMXPs, 4 slow pulsars as a stress set; 180 observations).
Verdict. Both primaries **not detected** under the strict rule, for different reasons: in v5 the intervals touch 0 (source BA +0.113 [−0.021, +0.277], AUC +0.065 [0.000, +0.179]); in v6a both models already reach source AUC 1.000 on E and the BA gain +0.105 has lower limit 0.000. Every non-saturating metric points the same way (v6a: Brier −0.059 [−0.093, −0.027], log loss −0.127 [−0.217, −0.044]). The colour-conditional permutation test is significant on the 31 v2 sources (T1* Holm $p = 0.0024$) but **does not replicate** on E. The new classifier CCTLR does not beat the linear H+T-LR. Both pre-registrations were written after all earlier results and self-approved under the user's standing instruction.

8.1 Motivation

The Phase 1–2 errors are hard-state sources whose colours overlap. In the hard state both BHs and NSs show strong band-limited noise, but at a given colour the amplitude and the characteristic frequencies need not be the same: the characteristic frequencies of the broad-band noise scale with the size of the emitting region and therefore, at fixed state, are expected to be lower for more massive compact objects (a possible explanation, not tested here). The well-known high-frequency difference (NS power persisting above ~ 500 Hz while BH power falls off above 10–50 Hz; Sunyaev and Revnivtsev, 2000) is not accessible with Standard-1 data. The PCA Standard-1 mode, however, is available for every observation already processed: 128-s rows of 1024 counts in 0.125-s bins per PCU, giving frequencies from 1/128 to 4 Hz. The rms of BH transients changes with state (Muñoz-Darias et al., 2011); whether < 4 Hz variability separates BHs from NSs *beyond* colour had no clear literature answer, so v5 was declared

exploratory. v6 then asked the confirmatory question with frozen models on new sources and added better estimators, a structured classifier, correct inference for clustered data and usable outputs (conformal sets, observation budgets).

Figure 8.1 maps the sub-analyses of v5 and v6: what each added and what it tested.

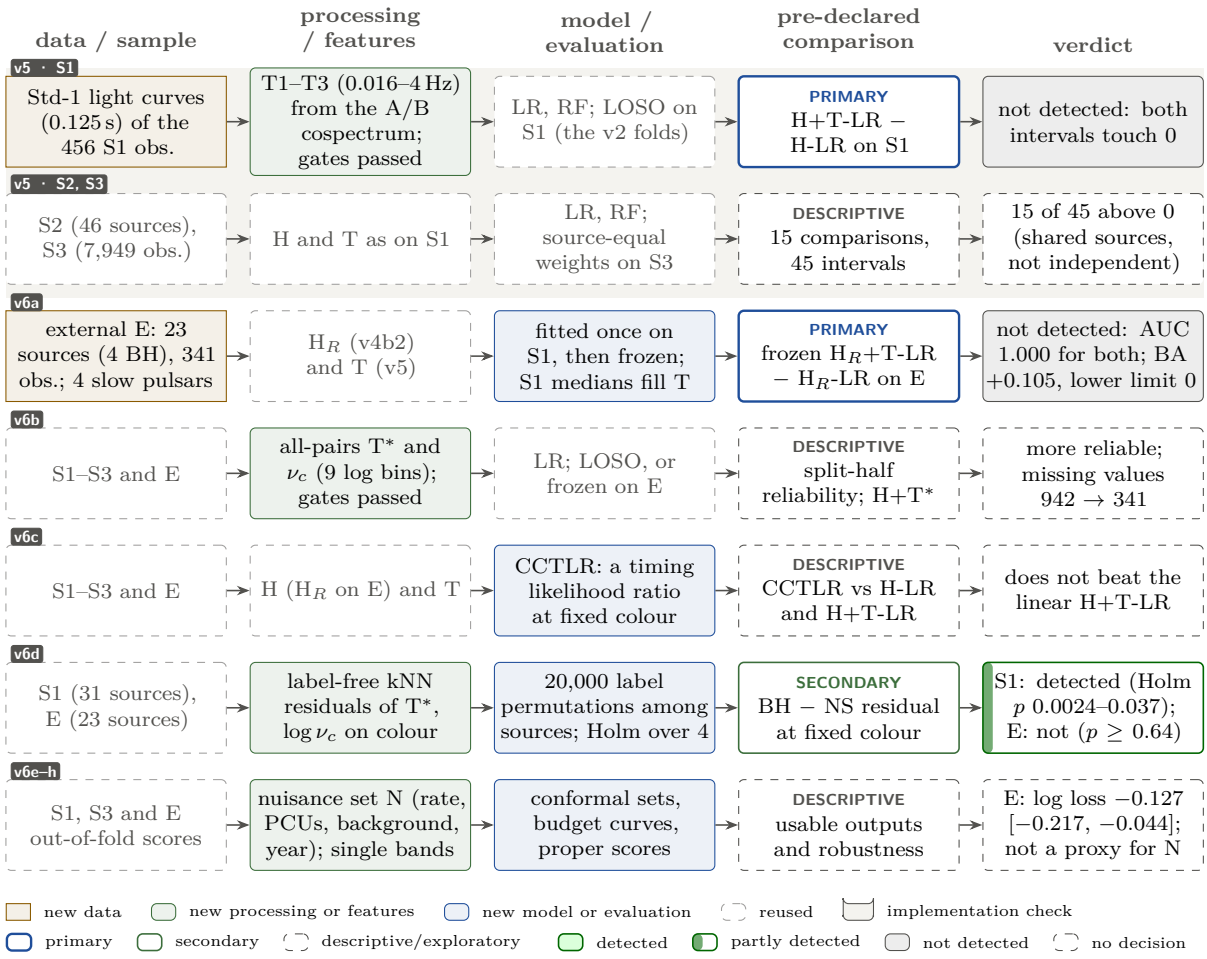


Figure 8.1: Phase 3 analysis map (reading as in Figure 6.1). v5 adds the timing block T to the Phase 1–2 samples; v6 adds the external set E with models frozen after training on S1, better timing estimators, a structured classifier, the colour-conditional permutation test and usable outputs. Both pre-registrations were written after all earlier results. Values from Tables 8.4, 8.7, 8.9, 8.10 and 8.13 and Section 8.6.6. Drawn for this report.

8.2 Pre-registration

Table 8.1 summarises the two pre-registrations.

Common rules are those of v4 (LOSO, source mean score, threshold 0.5, class-stratified source bootstrap with 2000 resamples and seed 42, paired differences, "improvement" only if the interval is entirely above 0). All models use the fixed v2 hyper-parameters; there is no selection step and hence no nested loop. The v6 document states that the analyst had seen results of other models on the 15 v4b2 sources in E, which is why E_{new} is reported separately.

Table 8.1: Phase 3 pre-registrations (`reports/preregistration_v5.md`, `reports/preregistration_v6.md`). Both were written and approved by the analyst under the user's standing instruction ("if v4 is finish, come up with v5 and run it yourself"; for v6: "deep think and come up with some innovations"), without external review.

Ver.	Written	Primary / decision rule	Secondary / descriptive
v5	2026-10-01, after all v1–v4 results, before any v5 feature	PRIMARY S1: H+T-LR − v2 H-LR, source BA and source AUC, interval entirely > 0; implementation checks as gate (synthetic recovery $< 10\%$, cospectrum vs auto-power $\rho \geq 0.90$, within-BH rms–hardness median ≥ 0)	T-LR, T-RF; H+T-RF − H-RF; overlap box; burst-excluded; S2 (46 sources) and S3 (7,949 observations); $\approx 12 \times 3$ comparisons
v6	2026-10-01 ($\approx$03:00), after all v1–v5 results, before any v6 download or feature	PRIMARY v6a: frozen H_R+T-LR − frozen H_R-LR on external E; no judgement if E has < 3 BHs; transparency: E_{new} (never-seen sources) sensitivity	v6b all-pairs estimator and ν_c; v6c CCTLR; v6d colour-conditional source permutation test (4 features, Holm); v6e conformal sets; v6f budget curves; v6g proper scores; v6h nuisance control and band ablation; stress set of slow pulsars

8.3 Data

8.3.1 Timing inputs and coverage

The timing features are defined in Section 2.8: v5 uses the A/B cospectrum (Eq. (2.15)) in three bands T1 (0.016–0.094 Hz, $j = 2$–12), T2 (0.102–1.00 Hz, $j = 13$–128), T3 (1.01–3.99 Hz, $j = 129$–511), each the signed square root of the median fractional variance over valid 128-s rows (Eq. (2.16)); v6 uses the all-pairs estimator T^* with a single-PCU auto-power fallback (Section 2.8.4) and the νP_ν centroid frequency ν_c on 9 logarithmic bins (Section 2.8.5). No new download was needed for v5: the Standard-1 files had been fetched with the spectra of v2, v4b2 and v4b1 (SHA256 recorded).

Table 8.2 gives the coverage of the timing features in each sample.

Table 8.2: Coverage of the timing features (observation missing if < 3 valid 128-s rows; most missing observations have only one PCU on). From `data/v5/processed/timing_features.csv` and `data/v6/processed/timing_v6.csv`.

Sample	obs.	v5 T missing	v6 T^* missing	ν_c available
S1 = v2 (31 sources)	456	34 (7 BH, 27 NS)	8	250
v4b2 new sources (15)	221	34	2	81
S2 = S1 + v4b2 (46 sources)	677	68	10	331
S3 = v4b1 (31 sources)	7,949	908 (323 BH, 585 NS)	338	4,126
v6 new sources (12)	180	25	1	150
all distinct observations	8,350	942 of 8,170 (v5)	341	4,357

8.3.2 External sources (v6a)

The v6 selection (`scripts/28_v6_select.py`, `data/v6/source_verification.csv`) took dynamical BHs in high-mass X-ray binaries from Casares and Jonker (2014, §3, Table 2) and

accreting millisecond pulsars (pulsation-confirmed NSs) and slow pulsars from Patruno and Watts (2021, Table 1), resolved positions with CDS Sesame and matched them to the RXTE catalogues within 0.1°; eligibility and the 15-observation time-stratified sampling are those of v4b2 (all gain epochs). Table 8.3 lists every candidate and the decision. All 12 included sources had 15 accepted observations (`results/v6/attrition_download.csv`).

Table 8.3: v6 candidate sources and decisions (`data/v6/source_verification.csv`). "elig." = eligible pointings over all gain epochs. The pointing offsets of LMC X-1 and Swift J1756.9−2508 (catalogue median pointing 0.26–0.31° from the source) are quoted from `reports/report_v6_zh-TW.md`.

Source	label	role	catalogue	elig.	decision
Cyg X-1	BH	E_{new}	CYGX1	1,540	included (HMXB)
LMC X-3	BH	E_{new}	LMCX3	943	included (HMXB)
LMC X-1	BH	E_{new}	–	–	excluded: no catalogue within 0.1°
M33 X-7	BH	E_{new}	–	–	excluded: no catalogue
XTE J1751−305	NS	E_{new}	XTEJ1751-305	51	included (AMXP)
XTE J0929−314	NS	E_{new}	XTEJ0929-314	23	included (AMXP)
XTE J1807−294	NS	E_{new}	XTEJ1807-294	104	included (AMXP)
IGR J00291+5934	NS	E_{new}	IGRJ00291+5934	89	included (AMXP)
IGR J17511−3057	NS	E_{new}	IGRJ17511-3057	74	included (AMXP)
IGR J17498−2921	NS	E_{new}	IGRJ17498-2921	34	included (AMXP)
Swift J1756.9−2508	NS	E_{new}	–	–	excluded: no catalogue within 0.1°
Swift J1749.4−2807	NS	E_{new}	–	–	excluded: no catalogue
NGC 6440 X-2	NS	E_{new}	–	–	excluded: name not resolved by Sesame
IGR J18245−2452	NS	E_{new}	PSRB1821-24	0	excluded: 0 eligible pointings
2A 1822−371	NS	stress	4U1822-371	94	included (slow pulsar)
4U 1626−67	NS	stress	4U1626-67	55	included (slow pulsar, 0.13 Hz spin)
GRO J1744−28	NS	stress	GROJ1744-28	306	included (slow pulsar)
Her X-1	NS	stress	HERX1	388	included (slow pulsar)

The external set E is the union of the 15 v4b2 sources (221 observations; never used to train the frozen models) and the 8 new E_{new} sources: 23 sources (4 BH: Cyg X-1, LMC X-3, XTE J1859+226, V4641 Sgr; 19 NS), 341 observations. The four slow pulsars (60 observations) form a stress set outside E: their spin frequencies (0.13–2.1 Hz) lie inside the timing bands, so they test whether coherent pulsations are mistaken for BH-like broad-band noise.

8.3.3 Implementation checks

All pre-declared gates passed before any model was trained (`logs/v5/test_v5lib.log`, `logs/v6/test_v6lib.log`, `results/v5/v5_implementation_checks.csv`):

- **Synthetic recovery (v5).** A common sinusoid of fractional rms 0.2 at 0.05, 0.5 and 3 Hz was recovered in T1, T2, T3 as 0.198, 0.198, 0.199 (cospectrum) and 0.199, 0.198, 0.198 (auto-power); pure Poisson gives median $|T| < 0.02$ in every band (−0.006, −0.008, −0.016); with 50% background the T2 rms is 0.199.
- **Cross-check on real data.** For the 276 S1 observations with 20–500 $c\,s^{-1}\,PCU^{-1}$, the cospectrum T2 and the classical auto-power-minus-Poisson T2 agree with Spearman $\rho = 0.970$ (gate 0.90).
- **Plausibility.** Within each of the 7 BH sources, the total rms correlates positively with the hard colour (Spearman 0.18 for GRO J1655−40 to 0.88 for XTE J1550−564, median 0.61; `results/v5/v5_check_bh_rms_hardness.csv`), as expected from higher rms in the hard state.

- **All-pairs estimator (v6b).** Under pure Poisson noise the all-pairs estimator is unbiased and its variance is 31%, 33% and 41% lower than that of the A/B split for 3, 4 and 5 PCUs. For equal count rates the expected reduction is $1 - n_A n_B / \binom{n}{2}$, i.e. 33%, 33% and 40% (the A/B variance scales as $1/(n_A n_B)$, the all-pairs variance as $1/\binom{n}{2}$ because distinct pairs are uncorrelated under independent Poisson noise). A sinusoid of rms 0.2 is recovered as 0.188–0.199 with 1 or 4 PCUs; ν_c of a 0.05, 0.5 and 3 Hz sinusoid is 0.045, 0.705 and 2.81 Hz (within one logarithmic bin). On real data T* agrees with the v5 T (Spearman 0.977, 0.977, 0.968 for T1–T3, 7,228 observations) and is more reliable in odd/even-row split halves (Table 8.4). Missing values drop from 942 to 341.
- **Permutation test calibration.** Under a synthetic null the fraction of $p < 0.05$ was 0.033 (gate ≤ 0.12); under a synthetic alternative $p < 0.01$. Conformal LOSO coverage at $\varepsilon = 0.2$ was 0.816 (gate ≥ 0.78).

Table 8.4: Split-half reliability of the timing features on S1 (odd vs even 128-s rows; 422 observations with both halves): Spearman correlation between halves and median absolute half-difference. From `results/v6/v6_estimator_reliability.csv`.

Band	ρ all-pairs (T*)	ρ v5 A/B (T)	MAD all-pairs	MAD v5 A/B
T1	0.890	0.854	0.0036	0.0035
T2	0.873	0.867	0.0042	0.0046
T3	0.736	0.709	0.0065	0.0077

8.4 Data structure in this phase

Table 8.5 lists the arrays. The timing block is always $[T_1, T_2, T_3, m]$, where missing T_k are replaced by the training-fold median of that column and m is a single 0/1 indicator "any timing feature missing" (Eq. (3.2)); for the frozen v6a models the medians are those of S1. Rows, labels and groups are those of the corresponding Phase 1–2 sample, so S1, S2 and S3 reuse the v2, v4b2 and v4b1 folds.

8.5 Models in this phase

Table 8.6 lists the models and procedures.

The frozen v6a H_R+T-LR has the standardised-feature coefficients c_1 −3.05, c_2 +1.62, T_1 +1.76, T_2 +0.45, T_3 +0.16, missing indicator +0.34 (`logs/v6/30_v6_analysis.log`): after the colours, the lowest band carries the largest weight, and its sign (more low-frequency rms ⇒ more BH-like) is the one v5 suggested.

8.6 Results

8.6.1 Timing as a new axis on the v2 sources (v5)

PRIMARY On S1, adding the three timing features to the two colours raises every point estimate (Table 8.7): source BA 0.795 → 0.908, source AUC 0.923 → 0.988, observation BA 0.776 → 0.878. By the pre-declared rule the result is **not detected** under the pre-registered criterion: the source-BA difference +0.113 [−0.021, +0.277] (positive in 92.2% of resamples) and the

Table 8.5: Arrays passed to the models in Phase 3. h_1, h_2 = count-space colours (Eq. (2.11)); h_1^R, h_2^R = power-law-normalised colours (Eq. (2.12)); T_k = v5 timing features, T_k^* = v6 all-pairs features, m = missing indicator. Timing columns from `data/v5/processed/timing_features.csv` (v6 new sources: `data/v6/processed/timing_v5_new.csv`) or `data/v6/processed/timing_v6.csv`, joined on `obs_id`.

Analysis	Array	Shape	Columns
v5 S1	$\boldsymbol{X}_{H+T}$	456×6	$h_1, h_2, T_1, T_2, T_3, m$
	$\boldsymbol{X}_{T}$	456×4	T_1, T_2, T_3, m
v5 S2	$\boldsymbol{X}_{H+T}$	677×6	as S1; 46 groups
v5 S3	$\boldsymbol{X}_{H+T}$, $\boldsymbol{w}$	7949×6	as S1; source-equal weights
v6a train (S1)	$\boldsymbol{X}_{H_R}$, $\boldsymbol{X}_{H_R+T}$	456×2, 456×6	h_1^R, h_2^R (+ T_1, T_2, T_3, m); fitted once, frozen
v6a test (E)	same	341×2, 341×6	23 external sources; S1 medians for missing T
stress set	same	60×6	4 slow pulsars
v6c CCTLR	$(\boldsymbol{h}, \boldsymbol{T})$	$n \times 2$, $n \times 3$	logit of H-LR plus $\Lambda(\boldsymbol{T} \mid \boldsymbol{h})$, missing $T \Rightarrow \Lambda = 0$ (Eq. (4.13))
v6d	residual table	31 (E: 23) sources	per-source mean residual of $T_1^*, T_2^*, T_3^*, \log_{10}\nu_c$ after label-free kNN regression on (h_1^R, h_2^R)
v6e	source scores	31 / 23	LOSO OOF source scores (calibration) and test scores
v6f	draws	500×7	$k \in \{1, 2, 3, 5, 8, 13, 21\}$ random observations per source from the S3 OOF scores
v6h	$\boldsymbol{X}_{H+N}$, $\boldsymbol{X}_{H+N+T}$	456×6, 456×10	$N = \log_{10} F$, PCUs on, background fraction, year
	$\boldsymbol{X}_{H+T_k}$	456×4	band ablation h_1, h_2, T_k, m; also $\boldsymbol{X}_{H+T^*}$ (456×6)

Table 8.6: Models and procedures of Phase 3. No hyper-parameter or threshold is selected anywhere.

Model / procedure	Inputs	Settings
LR (§4.3)	H, H+T, T, H_R, H_R+T, H+N(+T), H+T_k, H+T^*	$C = 1$, balanced; srcw on S3
RF (§4.4)	H+T, T (S1, S2, S3)	500 trees, `sqrt`, leaf 2, seed 42
CCTLR (§4.11)	H (H_R on E) and T	Student-t_4, WLS with source-equal weights, $\sigma \geq 0.01$
kNN regression (§4.12)	standardised H_R	$k = 25$, uniform, leave-one-source-out, labels not used
permutation test (§5.6)	source mean residuals	20,000 source-level permutations, seed 42, Holm over 4
Mondrian conformal (§5.7)	source scores	$\varepsilon \in \{0.1, 0.2\}$
budget curves (§5.8)	S3 OOF scores	500 draws, seed 42

source-AUC difference +0.065 [0.000, +0.179] (96.4%) both include 0. The reported observation-BA difference, +0.102 [+0.001, +0.249], lies above 0. The AUC lower limit of exactly 0 is a ceiling effect: in resamples where both models rank all sources perfectly the difference cannot be positive. Timing alone is much weaker than colour (T-LR source AUC 0.744): it complements colour, it does not replace it.

Table 8.7: PRIMARY v5 on S1 (456 observations, 31 sources, LOSO, v2 folds); 95% source-bootstrap intervals. "> 0" = fraction of resamples with a positive difference. From `results/v5/v5_bootstrap.csv`.

Model	source BA	source AUC	obs. BA
H-LR (v2 baseline)	0.795 [0.589, 0.958]	0.923 [0.804, 0.994]	0.776 [0.633, 0.893]
H+T-LR (primary)	0.908 [0.744, 1.000]	0.988 [0.940, 1.000]	0.878 [0.780, 0.947]
H-RF (v2)	0.857 [0.643, 1.000]	0.905 [0.750, 1.000]	0.755 [0.606, 0.882]
H+T-RF	0.929 [0.786, 1.000]	0.976 [0.905, 1.000]	0.852 [0.741, 0.933]
T-LR	0.693 [0.529, 0.908]	0.744 [0.518, 0.929]	0.661 [0.543, 0.799]
T-RF	0.622 [0.467, 0.786]	0.845 [0.667, 0.970]	0.677 [0.576, 0.789]
H+T-LR − H-LR	+0.113 [−0.021, +0.277]	+0.065 [0.000, +0.179]	+0.102 [+0.001, +0.249]
> 0	0.922	0.964	0.977

Which sources change. Table 8.8 gives every S1 source. From H-LR to H+T-LR the hard BH XTE J1118+480 moves from 0.41 to 0.94, and the three hard NSs that H-LR called BH move below 0.5 (HETE J1900.1−2455 0.61 → 0.42, KS 1731−260 0.60 → 0.20, SAX J1808.4−3658 0.72 → 0.38); XTE J1550−564 rises from 0.55 to 0.87. One NS becomes wrong: 4U 1724−307 (in Terzan 2), 0.37 → 0.53. GRS 1915+105 stays wrong (0.41 → 0.40). Figure 8.2 shows why: at hard colour 0.3–0.45, where the classes overlap in colour, BHs have much larger T1 than NSs, while in T3 the classes are similar.

Table 8.8: v5 per-source LOSO mean scores on S1. "miss" = observations with missing timing features (imputed). **Bold** = wrong side of 0.5. From `results/v5/v5_per_source_S1.csv`.

Source	label	n	miss	H-LR	H+T-LR	H-RF	H+T-RF	T-LR	T-RF
4U 1543−47	BH	15	0	0.95	0.97	0.86	0.85	**0.37**	**0.32**
GRO J1655−40	BH	15	0	0.82	0.86	0.87	0.74	**0.40**	**0.21**
GRS 1915+105	BH	15	4	**0.41**	**0.40**	**0.15**	**0.27**	**0.43**	**0.36**
GX 339−4	BH	15	3	0.81	0.83	0.79	0.74	0.58	**0.47**
XTE J1118+480	BH	15	0	**0.41**	0.94	**0.26**	0.77	0.89	0.86
XTE J1550−564	BH	15	0	0.55	0.87	0.62	0.84	0.86	0.89
XTE J1650−500	BH	15	0	0.85	0.86	0.70	0.72	**0.45**	**0.40**
4U 1254−69	NS	15	1	0.11	0.09	0.03	0.03	0.41	0.20
4U 1323−62	NS	15	0	0.32	0.10	0.15	0.13	0.45	0.20
4U 1608−52	NS	15	4	0.35	0.19	0.17	0.03	0.41	0.23
4U 1636−53	NS	15	3	0.22	0.07	0.09	0.05	0.36	0.24
4U 1702−429	NS	15	0	0.28	0.07	0.06	0.02	0.37	0.04
4U 1705−44	NS	15	1	0.18	0.08	0.05	0.01	0.37	0.11
4U 1724−307	NS	15	0	0.37	**0.53**	0.17	0.44	**0.60**	**0.62**
4U 1728−34	NS	15	1	0.17	0.03	0.17	0.02	0.34	0.18
4U 1735−44	NS	15	3	0.03	0.01	0.01	0.02	0.40	0.15
4U 1746−37	NS	15	0	0.08	0.02	0.05	0.01	0.39	0.11
4U 1820−30	NS	15	0	0.04	0.01	0.03	0.01	0.37	0.18
4U 1916−053	NS	6	0	0.29	0.09	0.11	0.04	0.41	0.14
Aql X-1	NS	15	4	0.41	0.31	0.37	0.15	0.47	0.33
Cyg X-2	NS	15	2	0.21	0.25	0.41	0.40	0.38	0.17

Source	label	n	miss	H-LR	H+T-LR	H-RF	H+T-RF	T-LR	T-RF
EXO 0748−676	NS	15	1	0.28	0.04	0.25	0.13	0.41	0.20
GS 1826−238	NS	15	0	0.32	0.13	0.05	0.12	0.49	0.22
GX 17+2	NS	15	0	0.09	0.03	0.07	0.02	0.34	0.47
GX 3+1	NS	15	0	0.09	0.07	0.07	0.03	0.41	0.35
HETE J1900.1−2455	NS	15	2	**0.61**	0.42	0.35	0.24	0.45	0.27
KS 1731−260	NS	15	0	**0.60**	0.20	0.36	0.06	0.37	0.17
MXB 1659−298	NS	15	0	0.29	0.28	0.12	0.32	0.48	0.37
SAX J1808.4−3658	NS	15	2	**0.72**	0.38	0.28	0.31	0.34	0.31
SLX 1735−269	NS	15	3	0.40	0.32	0.06	0.22	0.49	0.41
XTE J1814−338	NS	15	0	0.47	0.10	0.41	0.06	0.41	0.12

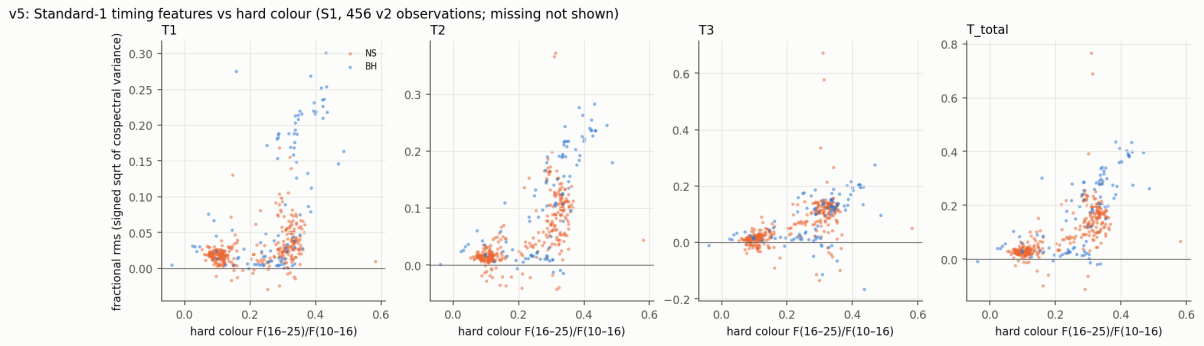


Figure 8.2: v5 timing features against the hard colour $F(16–25)/F(10–16)$ for the 422 S1 observations with timing (BH blue, NS orange; fractional rms = signed square root of the cospectral variance). Panels: T1 (0.016–0.094 Hz), T2 (0.10–1.0 Hz), T3 (1.0–4.0 Hz), T_{total}. At hard colour $\gtrsim 0.3$ the BH T1 values (0.1–0.3) lie well above most NS values ($\lesssim 0.1$); in T3 the classes overlap. Repo: `figures/v5/v5_timing_vs_colour.png`, `scripts/27_v5_analysis.py`.

Consistency across subsets and samples. DESCRIPTIVE Table 8.9 lists the 15 descriptive comparisons (45 differences). Every H+T comparison has a positive point estimate and every T-only comparison a negative one; 15 of the 45 intervals lie entirely above 0, against about one expected by chance if timing carried no information. These comparisons share sources and features, however, and are not 15 independent pieces of evidence; S2 and S3 contain the S1 sources and are *not* independent confirmations.

In the colour-overlap box the H+T-LR source AUC is 0.916 against 0.731 for H-LR (RF: 0.992 against 0.798); excluding the 61 burst-flagged observations leaves the result unchanged, so it is not produced by bursts. On S3 (all observations, source-equal weights) H+T-LR classifies all 31 sources correctly (source BA 1.000, AUC 1.000, observation BA 0.835).

8.6.2 Confirmation on unseen sources with frozen models (v6a)

PRIMARY Table 8.10 gives the result on the 23 external sources. Both frozen models rank every external BH above every external NS (source AUC 1.000), so the AUC difference is 0.000 [0, 0]; the source-BA difference is +0.105 [0.000, +0.237]. The primary is therefore **not detected** under the pre-registered criterion – a saturation of the headline metrics on E that was not anticipated when the rule was written. What the timing features change is the *threshold calibration*: with colour only, five NSs score 0.56–0.67 (XTE J1807−294 0.67, XTE J0929−314 and XTE J1751−305 0.57, 4U 2129+12 and XTE J1710−281 0.56); with timing they fall to 0.25–0.34, while one NS, the AMXP IGR J00291+5934, rises from 0.48 to 0.59 (Table 8.11 and Figure 8.3). The non-saturating metrics all improve with intervals excluding 0 (pre-declared

Table 8.9: DESCRIPTIVE v5 descriptive comparisons (paired source bootstrap). Overlap box = soft colour ≥ 0.70 and hard colour ≥ 0.20 (234 observations, 24 sources incl. 7 BH; sources with ≥ 3 box observations). † = interval entirely above 0. From `results/v5/v5_bootstrap.csv`.

Sample	Comparison	Δ source BA	Δ source AUC	Δ obs. BA
S1	H+T-RF − H-LR	+0.134 [+0.021, +0.298]†	+0.054 [−0.024, +0.167]	+0.076 [−0.038, +0.227]
S1	T-LR − H-LR	−0.101 [−0.348, +0.176]	−0.179 [−0.441, +0.060]	−0.114 [−0.305, +0.133]
S1	T-RF − H-LR	−0.173 [−0.417, +0.134]	−0.077 [−0.286, +0.119]	−0.099 [−0.276, +0.121]
S1	H+T-RF − H-RF	+0.071 [0, +0.214]	+0.071 [0, +0.202]	+0.097 [−0.002, +0.229]
box	H+T-LR − H-LR	+0.118 [−0.113, +0.349]	+0.185 [+0.008, +0.403]†	+0.191 [−0.038, +0.406]
box	H+T-RF − H-RF	+0.172 [0, +0.357]	+0.193 [0, +0.445]	+0.175 [−0.014, +0.342]
burst-excl.	H+T-LR − H-LR	+0.113 [−0.021, +0.277]	+0.065 [0, +0.173]	+0.118 [+0.015, +0.261]†
burst-excl.	H+T-RF − H-RF	+0.051 [−0.042, +0.214]	+0.077 [0, +0.214]	+0.092 [+0.005, +0.208]†
S2	H+T-LR − H-LR (S2)	+0.123 [+0.027, +0.261]†	+0.042 [0, +0.108]	+0.089 [+0.001, +0.214]†
S2	H+T-RF − H-LR (S2)	+0.123 [+0.027, +0.261]†	+0.045 [0, +0.120]	+0.041 [−0.058, +0.162]
S3	H+T-LR[srcw] − v2 H-LR	+0.205 [+0.042, +0.411]†	+0.077 [+0.006, +0.196]†	+0.060 [−0.041, +0.223]
S3	H+T-RF[srcw] − v2 H-LR	+0.134 [+0.021, +0.298]†	+0.077 [+0.006, +0.196]†	+0.023 [−0.106, +0.230]
S3	H+T-LR[srcw] − H-LR[srcw]	+0.205 [+0.042, +0.411]†	+0.077 [0, +0.196]	+0.069 [+0.015, +0.148]†
S3	H+T-RF[srcw] − H-LR[srcw]	+0.134 [+0.021, +0.298]†	+0.077 [0, +0.196]	+0.032 [−0.054, +0.163]

as descriptive): observation BA +0.080 [+0.003, +0.162], source Brier −0.059 [−0.093, −0.027], source log loss −0.127 [−0.217, −0.044].

Table 8.10: PRIMARY v6a: models trained once on S1 and frozen, evaluated on the external set E (23 sources: 4 BH, 19 NS; 341 observations) and on E_{new} (8 never-seen sources: 2 BH, 6 NS). Proper scores are class-balanced source-level log loss and Brier (lower is better; Eq. (5.5)). Upper block: point values; lower block: paired differences with 95% source-bootstrap intervals. From `results/v6/v6_bootstrap.csv` and `results/v6/v6_proper_scores.csv`.

Set	Model	source BA	source AUC	obs. BA	log loss	Brier
E	H_R-LR	0.868	1.000	0.798	0.358	0.107
	H_R+T-LR	0.974	1.000	0.878	0.231	0.048
	CCTLR (H_R)	0.974	0.987	0.825	0.230	0.052
E_{new}	H_R-LR	0.750	1.000	0.661	0.559	0.187
	H_R+T-LR	0.917	1.000	0.856	0.297	0.074
	CCTLR (H_R)	0.917	0.917	0.756	0.329	0.091

Difference	E: H_R+T − H_R (**primary**)	E: CCTLR − H_R	E_{new}: H_R+T − H_R
source BA	+0.105 [0.000, +0.237]	+0.105 [0.000, +0.237]	+0.167 [−0.167, +0.417]
source AUC	0.000 [0.000, 0.000]	−0.013 [−0.079, 0.000]	0.000 [0.000, 0.000]
obs. BA	+0.080 [+0.003, +0.162]	+0.027 [−0.051, +0.096]	+0.194 [+0.056, +0.322]
log loss	−0.127 [−0.217, −0.044]	−0.129 [−0.221, −0.037]	−0.262 [−0.389, −0.118]
Brier	−0.059 [−0.093, −0.027]	−0.055 [−0.091, −0.018]	−0.113 [−0.173, −0.046]

The four external BHs are all called BH (Cyg X-1 0.76, LMC X-3 0.84, XTE J1859+226 0.80, V4641 Sgr 0.82), but the two transients score lower with timing than with colour alone (0.94 and 0.93). On E_{new} – sources about which the analyst had seen no result of any kind – source BA rises from 0.750 to 0.917 and observation BA by +0.194 [+0.056, +0.322]. In the stress set all four slow pulsars stay NS, but 4U 1626−67, whose 0.13 Hz spin falls in the T2 band, rises from 0.15 to 0.40.

Table 8.11: v6a per-source mean scores of the frozen models on the external and stress sources. "miss" = observations without v5 timing features (S1 median imputed). **Bold** = wrong side of 0.5. From `results/v6/v6a_external_per_source.csv`.

Source	label	set	n	miss	H_R-LR	H_R+T-LR	CCTLR
Cyg X-1	BH	E_{new}	15	1	0.73	0.76	0.86
LMC X-3	BH	E_{new}	15	6	0.70	0.84	0.65
IGR J00291+5934	NS	E_{new}	15	4	0.48	**0.59**	**0.67**
IGR J17498−2921	NS	E_{new}	15	3	0.42	0.16	0.08
IGR J17511−3057	NS	E_{new}	15	8	0.50	0.13	0.26
XTE J0929−314	NS	E_{new}	15	0	**0.57**	0.25	0.16
XTE J1751−305	NS	E_{new}	15	0	**0.57**	0.34	0.30
XTE J1807−294	NS	E_{new}	15	0	**0.67**	0.25	0.18
V4641 Sgr	BH	v4b2	11	1	0.93	0.82	0.76
XTE J1859+226	BH	v4b2	15	0	0.94	0.80	0.85
1A 1744−361	NS	v4b2	15	3	0.20	0.14	0.09
4U 0513−40	NS	v4b2	15	5	0.20	0.08	0.12
4U 0919−54	NS	v4b2	15	5	0.38	0.20	0.22
4U 2129+12	NS	v4b2	15	1	**0.56**	0.31	0.16
EXO 1745−248	NS	v4b2	15	2	0.14	0.15	0.14
GRS 1747−312	NS	v4b2	15	1	0.29	0.16	0.20
SAX J1747.0−2853	NS	v4b2	15	1	0.35	0.16	0.04
SAX J1748.9−2021	NS	v4b2	15	1	0.29	0.15	0.12
SAX J1750.8−2900	NS	v4b2	15	10	0.41	0.26	0.27
Ser X-1	NS	v4b2	15	1	0.10	0.08	0.01
XTE J1709−267	NS	v4b2	15	0	0.27	0.13	0.03
XTE J1710−281	NS	v4b2	15	1	**0.56**	0.28	0.14
XTE J1759−220	NS	v4b2	15	2	0.28	0.08	0.03
2A 1822−371	NS	stress	15	0	0.05	0.01	0.00
4U 1626−67	NS	stress	15	2	0.15	0.40	0.44
GRO J1744−28	NS	stress	15	0	0.19	0.11	0.15
Her X-1	NS	stress	15	1	0.27	0.07	0.07

8.6.3 A structured classifier: CCTLR (v6c)

DESCRIPTIVE The colour-conditional timing likelihood-ratio classifier (Section 4.11) adds to the H-LR logit a generative log-likelihood ratio of the timing features given colour and class, so timing can only act where the two class-conditional distributions differ at that colour, and missing timing gives zero evidence. Table 8.12 compares it with H-LR and H+T-LR. It improves on H-LR about as much as H+T-LR does (S1 log loss −0.192 [−0.344, −0.054]) but does *not* beat H+T-LR: on S1 the differences are source BA 0.000 [0, 0], source AUC −0.012 [−0.048, 0], observation BA −0.032 [−0.081, +0.010], log loss +0.003 [−0.044, +0.053]; on S3 it misclassifies two sources (source BA 0.908 = (6/7 + 23/24)/2) where H+T-LR misclassifies none.

Figure 8.4 shows where in the colour plane the CCTLR timing term contributes evidence.

8.6.4 Is the timing difference a class difference? (v6d)

SECONDARY The colour-conditional source permutation test (Section 5.6) removes the colour dependence of each timing feature by label-free kNN regression ($k = 25$, leave-one-source-out), averages residuals per source and compares the BH and NS source means; the null distribution permutes labels among sources (20,000 permutations). Table 8.13 shows that on S1 all four

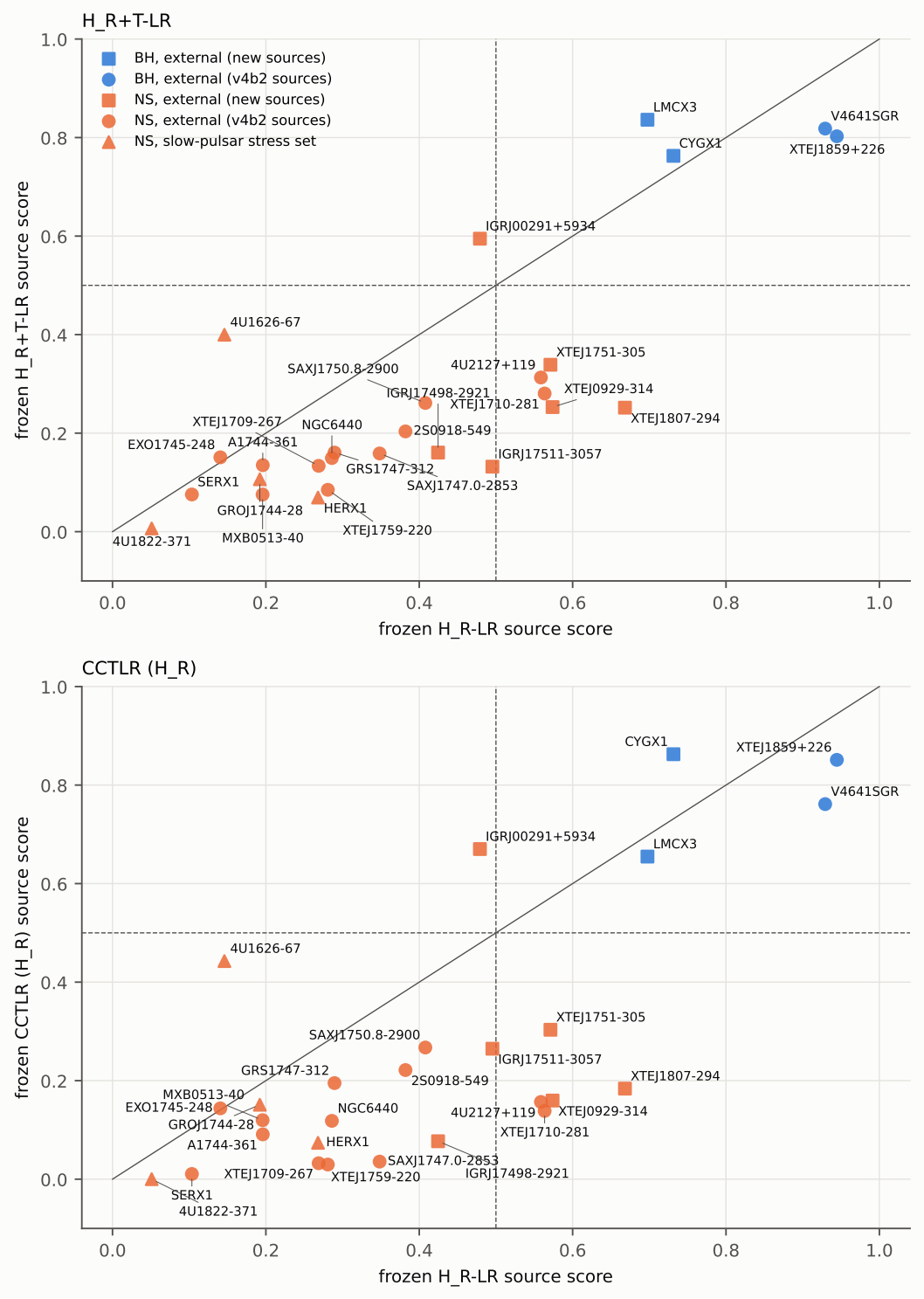


Figure 8.3: Source scores of the frozen models on sources never used in training: horizontal, colour-only H_R-LR; vertical, H_R+T-LR (top) and CCTLR (bottom). Squares = E_{new}, circles = v4b2 sources, triangles = slow-pulsar stress set; BH blue, NS orange; dashed lines at 0.5. Points in the lower-right quadrant are NSs misclassified by colour and corrected by timing; IGR J00291+5934 moves the other way, and 4U 1626−67 is pushed up by its coherent 0.13 Hz pulsation. Redrawn for this report from `results/v6/v6a_external_per_source.csv` (same points and encoding as the repository figure `figures/v6/v6a_external_sources.png` of `scripts/30_v6_analysis.py`, whose source labels overlap); labels are offset, with leader lines, so that none overlap.

Table 8.12: DESCRIPTIVE Source BA / source AUC of H-LR, H+T-LR and CCTLR (S1–S3: LOSO; E: frozen models with H_R colours, H_R-LR as the colour-only model). From `results/v6/v6_bootstrap.csv`.

Sample	H-LR	H+T-LR	CCTLR
S1 (31 sources)	0.795 / 0.923	0.908 / 0.988	0.908 / 0.976
S2 (46 sources)	0.821 / 0.946	0.944 / 0.988	0.931 / 0.991
S3 (7,949 obs., srcw)	0.795 / 0.923	1.000 / 1.000	0.908 / 0.994
E (frozen, H_R)	0.868 / 1.000	0.974 / 1.000	0.974 / 0.987

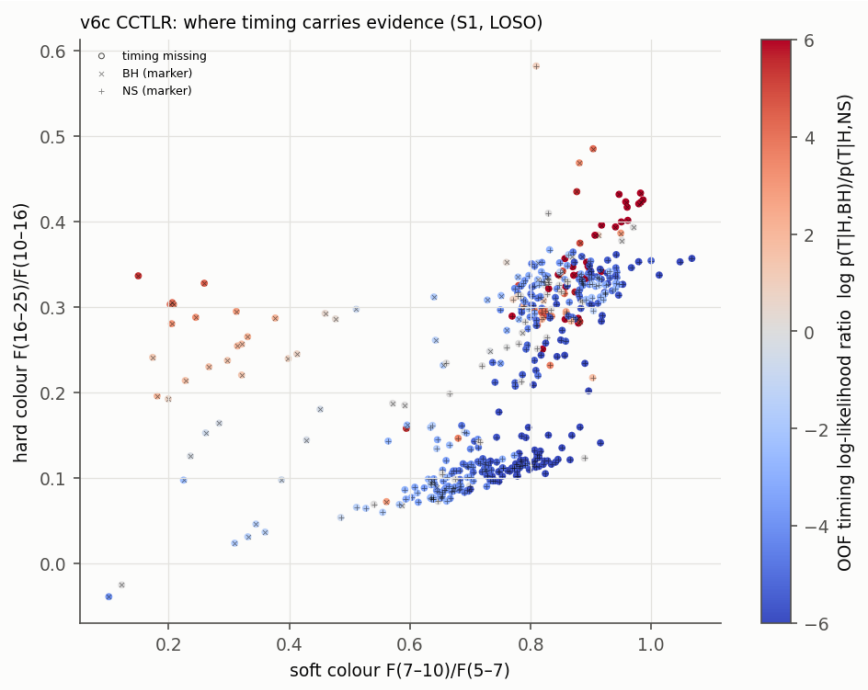


Figure 8.4: Where the timing features carry evidence: each S1 observation in the colour plane, coloured by its out-of-fold CCTLR timing log-likelihood ratio $\Lambda = \log p(\boldsymbol{T} \mid \boldsymbol{h}, \mathrm{BH})/p(\boldsymbol{T} \mid \boldsymbol{h}, \mathrm{NS})$ (red BH-like, blue NS-like, clipped at ± 6); $\times$ = BH, $+$ = NS markers, grey open circles = timing missing ($\Lambda = 0$). The hard-state corner (hard colour $\gtrsim 0.35$) and the soft BH region at low soft colour are red; the NS banana branch at hard colour ≈ 0.1 is blue. Repo: `figures/v6/v6c_timing_evidence_map.png`, `scripts/30_v6_analysis.py`.

features differ between the classes at fixed colour (Holm-adjusted $p = 0.0024$ to 0.037): BHs have more low-frequency rms and a νP_ν centroid lower by a factor $10^{0.31} \approx 2$. On the external set the same test, with the kNN fitted on all S1 observations and labels permuted among the 23 E sources, finds **no difference** (Holm $p \geq 0.64$). The mechanism claim does not replicate.

Table 8.13: SECONDARY v6d colour-conditional source permutation test. D = mean source residual of BH sources minus that of NS sources; two-sided $p = (1 + \#\{|D_\pi| \geq |D|\})/(1 + 20000)$; Holm over the four features within each sample. ν_c is defined for 27 S1 and 21 E sources. From `results/v6/v6d_permutation_test.csv`.

	S1 (31 sources, 7 BH)			E (23 sources, 4 BH)		
Feature	D	p	$p_{\rm Holm}$	D	p	$p_{\rm Holm}$
T_1^* (0.016–0.094 Hz)	+0.057	0.0006	0.0024	+0.004	0.746	1.000
T_2^* (0.10–1.0 Hz)	+0.044	0.0021	0.0063	+0.001	0.939	1.000
T_3^* (1.0–4.0 Hz)	+0.020	0.0371	0.0371	+0.020	0.159	0.636
$\log_{10} \nu_c$	−0.313	0.0037	0.0075	−0.037	0.783	1.000

8.6.5 Usable outputs: conformal sets and observation budgets (v6e, v6f)

DESCRIPTIVE **Conformal sets.** Source-level Mondrian conformal prediction (Section 5.7) returns {BH}, {NS}, {BH, NS} ("cannot tell") or the empty set. Table 8.14 summarises it. On S1 at $\varepsilon = 0.2$, H+T-LR gives 26 correct singletons and no ambiguous set (3 empty sets), against 24 correct singletons and 2 ambiguous sets (Aql X-1, XTE J1118+480) for H-LR. On E, calibrated on the S1 OOF scores, the colour-only model's NS coverage collapses to 0.632 (7 of the 19 NSs get a wrong {BH} set): the calibration does not transfer. With timing it is 0.895, with BH coverage 1.000 and 21 of 23 correct singletons.

Table 8.14: DESCRIPTIVE Source-level Mondrian conformal sets. Coverage = fraction of sources of a class whose set contains the true label; "correct" = singleton sets equal to the true label. S1: LOSO calibration; E: split conformal with the S1 OOF scores as calibration and the frozen-model scores as test. From `results/v6/v6e_conformal_summary.csv`.

Sample	Model	ε	cov. BH	cov. NS	mean size	correct	ambiguous	empty
S1	H-LR	0.1	1.000	0.917	1.81	4	25	0
	H-LR	0.2	0.857	0.833	1.06	24	2	0
	H+T-LR	0.1	1.000	0.917	1.74	6	23	0
	H+T-LR	0.2	0.857	0.833	0.90	26	0	3
	CCTLR	0.2	0.857	0.833	0.94	26	0	2
E	H_R-LR	0.1	1.000	0.947	1.78	4	18	0
	H_R-LR	0.2	1.000	0.632	1.09	14	2	0
	H_R+T-LR	0.1	1.000	0.947	1.78	4	18	0
	H_R+T-LR	0.2	1.000	0.895	0.96	21	0	1
	CCTLR (H_R)	0.2	1.000	0.684	0.78	17	0	5

Observation budgets. On S3, drawing k random observations per source from the OOF scores (500 draws, Figure 8.5), one observation per source already gives H+T-LR a median source AUC of 0.970 [0.896, 1.000] against 0.866 [0.756, 0.946] for H-LR; H+T-LR reaches a median AUC of 1.000 at $k = 5$, while H-LR saturates at 0.917–0.923 (the value obtained with all observations). The median source BA with one observation is 0.908 for H+T-LR and 0.774 for H-LR (`results/v6/v6f_budget_curves.csv`).

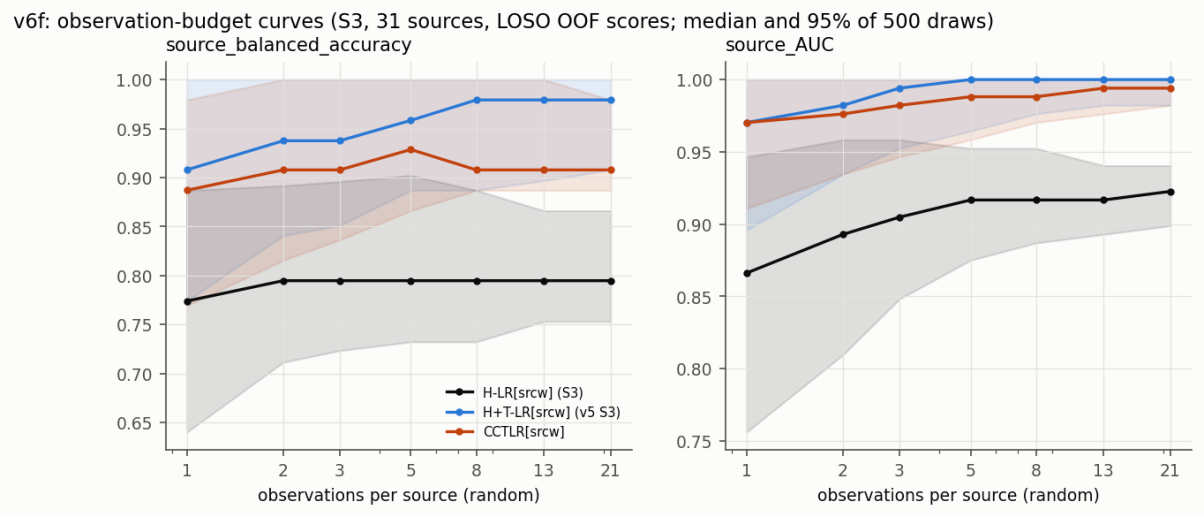


Figure 8.5: Observation-budget curves on S3: source balanced accuracy (left) and source AUC (right) when each source is represented by k randomly drawn observations (LOSO OOF scores of the source-equal-weight models; median and central 95% of 500 draws). Black H-LR, blue H+T-LR, red CCTLR. Repo: `figures/v6/v6f_budget_curves.png`, `scripts/30_v6_analysis.py`.

8.6.6 Robustness: nuisance variables and frequency bands (v6h)

DESCRIPTIVE Timing amplitudes depend on brightness (Poisson noise), the number of PCUs, the background fraction and the epoch. Adding these nuisance variables N to the colours does not help detectably (H+N − H log loss −0.063 [−0.151, +0.018]), whereas adding timing on top of them does (Table 8.15): the timing information is not a proxy for N. The band ablation places the information in T1 and T2; T3 (1–4 Hz) adds nothing (log loss −0.004 [−0.017, +0.007]). Replacing the v5 features by the all-pairs T* gives source BA 0.929 (+0.134 [+0.021, +0.298] over H-LR) and source AUC 0.976.

Table 8.15: DESCRIPTIVE v6h on S1 (LOSO). Differences against H-LR unless stated; † = interval entirely above 0 (BA, AUC) or entirely below 0 (log loss). From `results/v6/v6_bootstrap.csv` and `results/v6/v6_proper_scores.csv`.

Model	src BA / AUC	Δ source BA	Δ obs. BA	Δ log loss
H+N-LR	0.887 / 0.952	+0.092 [0, +0.256]	+0.020 [−0.046, +0.094]	−0.063 [−0.151, +0.018]
H+N+T-LR	1.000 / 1.000	+0.205 [+0.042, +0.411]†	+0.155 [+0.036, +0.299]†	−0.268 [−0.394, −0.154]†
H+N+T − H+N		+0.113 [0, +0.277]	+0.135 [+0.034, +0.268]†	−0.204 [−0.347, −0.103]†
H+T1-LR	0.929 / 0.988	+0.134 [+0.021, +0.298]†	+0.103 [−0.000, +0.251]	−0.195 [−0.323, −0.086]†
H+T2-LR	0.887 / 0.988	+0.092 [−0.021, +0.256]	+0.092 [+0.007, +0.207]†	−0.165 [−0.252, −0.086]†
H+T3-LR	0.795 / 0.929	0.000 [0, 0]	−0.016 [−0.045, +0.011]	−0.004 [−0.017, +0.007]
H+T*-LR	0.929 / 0.976	+0.134 [+0.021, +0.298]†	+0.105 [−0.004, +0.252]	−0.193 [−0.323, −0.075]†

8.7 Interpretation

What the data support directly.

1. Both pre-declared primaries are **not detected**: in v5 because the intervals touch 0, in v6a because both models already reach source AUC 1.000 on the external sources and the BA interval has lower limit 0.

2. On external sources the timing features correct the *threshold miscalibration* of the colour-only model: five NSs at 0.56–0.67 move to 0.25–0.34 (one NS moves the other way), and the non-saturating metrics – observation BA, Brier score and log loss – all improve with intervals excluding 0.
3. The timing information sits at 0.016–1 Hz (T1, T2); 1–4 Hz adds nothing. It survives control for brightness, number of PCUs, background fraction and year.
4. "At fixed colour, BHs have stronger low-frequency variability and a lower characteristic frequency" holds on the 31 v2 sources (Holm $p = 0.002$–0.04) but **does not replicate** on the 23 external sources.
5. The all-pairs estimator is more precise and reduces missing values from 942 to 341; the new CCTLR classifier does not beat the simple linear H+T-LR.
6. With timing, source-level conformal sets keep their nominal coverage on external sources; with colour only they do not.

Possible explanations (not verified).

- The BH part of E differs from the training BHs: two are high-mass X-ray binaries (Cyg X-1; LMC X-3, mostly in the soft state), XTE J1859+226 is entirely in gain epoch 4. With 4 BHs the permutation test has little power, so "not replicated" may be lack of power; the S1 effect may also be driven by a few sources (XTE J1118+480, GX 339−4).
- On E the timing features mainly pull NSs down while slightly lowering the BH scores, which is consistent with weaker low-frequency variability of NSs at a given colour but suggests that the BH side of the relation is unstable.
- CCTLR's class-conditional model (linear in colour, Student-t residuals) may be mis-specified for 31 sources; a linear LR is more stable at this sample size.
- 4U 1626−67 rises because its coherent 0.13 Hz pulsation is counted as broad-band variability; for strongly pulsating sources the pulse frequency would have to be removed first.

Physically. The direction found on S1 – more variability at 0.016–1 Hz and a centroid frequency lower by a factor ≈ 2 for BHs at the same colour – is what one expects if the characteristic frequencies of the broad-band noise scale inversely with the mass of the compact object, but the frequencies probed here (below 4 Hz) are far below those of the canonical BH/NS timing differences, and the effect did not replicate. The robust part of the result is practical rather than physical: timing calibrates the colour classifier on new sources.

Hypotheses that need more data. More external confirmed BHs (especially hard-state LMXB transients); an explicit spectral-state split and higher frequencies from event-mode data (taken up in Phase 4); another instrument with new observations (NICER, Phase 5); full power-spectrum fits with Lorentzian characteristic frequencies (Phase 6); a confirmation in which a proper scoring rule is the pre-declared primary.

8.8 Deviations and caveats

- **Written after earlier results, self-approved**: v5 after v1–v4, v6 after v1–v5, both designed and approved by the analyst under the user's standing instruction, without external review; timing was proposed to address the known colour overlap.
- **v5 unit test**: the "PCU switched off within a row" test case was wrong (switch-off at bin 900 leaves counts in the last 128-bin block, so by the rule the PCU is still on); the test was corrected before any real feature was computed. The rule itself did not change.

- **v5 reporting**: `frac_first_better` (fraction of resamples with a positive difference) is reported; it was already an output of `v4lib.paired_bootstrap`, not a new analysis.
- **v6**: no change of source list, bands, estimators, models or decision rules; all exclusions followed the rules (`data/v6/source_verification.csv`). A figure-annotation bug was fixed before the first run; `run_v6.ps1` clears `XRB_VERSION` with `$env:XRB_VERSION = $null` (same effect).
- **Saturation**: the headline metrics saturated on E (AUC = 1), which the design did not anticipate; the proper-score improvements are descriptive.
- **Small BH samples**: 7 BHs in S1, 9 in S2, 4 in E (2 in E_{new}); one source such as XTE J1118+480 changes the source BA by about 0.07.
- **Not independent**: S2 and S3 share the S1 sources; the 15 v4b2 sources in E had been seen by the analyst in other models' results (hence E_{new}).
- **Instrument**: Standard-1 has no energy resolution (rms over ≈2–60 keV, colours in 5–25 keV) and a 4 Hz upper limit; the background is assumed equal for all PCUs and taken from the Standard-2 background model; the cospectrum amplitude is not deadtime-corrected (a few per cent at high rates); 34 S1 observations (7 BH) with one PCU are imputed.
- **Population differences**: training BHs are all LMXB transients, two external BHs are HMXBs; the external NSs are mostly AMXPs; XTE J1751−305, IGR J17511−3057 and IGR J17498−2921 lie 0.3–0.7° from each other or from v4b2 sources and may contaminate each other in the ≈ 1° PCA field.
- **Bootstrap** intervals keep the predictions fixed and do not include fitting variability.

8.9 Takeaway

Low-frequency variability is the first quantity in this project that adds to the two colours: every point estimate moves in its favour, on the 31 training sources and, with frozen models, on 23 unseen ones. By the strict pre-declared rule neither primary is detected – in v6a because the headline metrics saturate – and the colour-conditional mechanism (lower characteristic frequency for BHs) does not replicate externally. What does hold externally is calibration: timing moves the colour-only classifier's misplaced NSs back below the threshold and keeps conformal coverage. Phase 4 makes the state dependence explicit (does the BH/NS separation fail precisely in the hard state?), adds event-mode frequencies above 4 Hz, rechecks burst contamination against the MINBAR catalogue, and tests long-term colours on an independent all-sky instrument (MAXI).

Chapter 9

Phase 4 (v7–v8): where the colours fail, and why

Phase at a glance: v7–v8 (+ v9c MAXI strata)

Question. Phases 1–3 suggested that two colours separate BHs from NSs in soft states but not in hard states, and that low-frequency variability fills the gap. Phase 4 makes this explicit and tests the alternatives: is the failure a hard-state failure (variability-defined states)? Is any of it burst contamination (official MINBAR catalogue)? Do more BHs (persistent HMXBs), higher frequencies (event mode, 4–1024 Hz) or a different instrument (MAXI/GSC, 2–20 keV) change the picture?
Pre-declared primaries (one per sub-analysis, ★ in the repository reports): v7b1 burst exclusion; v7a soft-like − hard-like AUC; v7b2 adding three persistent BHs; v7c1 reproduction of de Beurs et al. (2022); v7c2 two LR comparisons on MAXI; v8a timing inside the hard-like state; v8b the same on external sources; v8c event-mode frequencies (exploratory after a failed gate); v8d MAXI 2–4 keV gain after N_{H} correction.
Data. RXTE S1 (456 observations) with variability states; MINBAR DR1; Cyg X-1, LMC X-1, LMC X-3 (45 observations), GS 1354−64 and SS 433 (23); 201 event-mode files (2.44 GB); MAXI/GSC one-day light curves of 13 BHs, 48 non-pulsing NSs and 39 pulsars.
Verdict. Detected: colours work in soft-like observations (AUC 0.954) and fail in hard-like ones (0.646; difference +0.309 [+0.099, +0.510]); inside the hard-like state, low-frequency timing raises the AUC to 0.910 (+0.264 [+0.048, +0.455]); on MAXI the 2–4 keV colour survives an N_{H} correction (+0.377 [+0.160, +0.595]). **Not detected**: any effect of burst contamination; any effect of adding persistent BHs; the hard-state timing gain on external sources (+0.064 [−0.185, +0.367]); the two LR comparisons on MAXI. Event-mode high frequencies are exploratory and add nothing beyond Standard-1 timing. Both pre-registrations were written after all earlier results; v8a was predictable from v5.

9.1 Motivation

By the end of Phase 3 three statements were supported only indirectly: the colour classifier fails in the hard state, timing helps precisely there, and type-I bursts do not matter. Each could be tested directly. Spectral states can be assigned from variability alone (fractional rms; Remillard and McClintock, 2006), independently of the colours used for classification; the official MINBAR burst catalogue (Galloway et al., 2020) provides an independent burst record; the persistent HMXB BHs add hard-state BHs (Cyg X-1) and soft-state BHs (LMC X-1, LMC X-3) that are

not transients; event-mode data reach the 500–1000 Hz range where Sunyaev and Revnivtsev (2000) found NS power and no BH power; and MAXI offers long-term colours of many more sources from a different instrument, with a published machine-learning result (de Beurs et al., 2022) to reproduce. The analyses are grouped below by theme: data quality, states, timing within the hard state, more BHs, higher frequencies, and an independent instrument (including the v9c MAXI strata, which were run with Phase 5 but belong to this theme).

Figure 9.1 maps the eleven sub-analyses by theme.

9.2 Pre-registration

`reports/preregistration_v7.md` was written on 2026-10-05 after all v1–v6 results and approved by the user at 07:23 (in Chinese, "do it your way and continue to the end", adopting the three pending recommendations). `reports/preregistration_v8.md` was written on 2026-10-05 (≈21:30) after all v1–v7 results, committed and pushed (`a481fa0`) before any v8 download and self-approved under the user's instruction to "go directly to the next step". The v9c strata were specified in `reports/preregistration_v9.md` §6 as descriptive. Decision rule throughout: "difference" iff the 95% class-stratified source-bootstrap interval excludes 0 (2000 resamples, seed 42, fixed predictions).

Table 9.1 lists the pre-declared comparisons.

Table 9.1: Phase 4 pre-declared comparisons (one primary per sub-analysis) and the main descriptive ones. Observation-level AUCs within a state are computed on the pooled observations of the resampled sources (Section 9.3.1).

Part	Primary	Descriptive / sensitivity
v7b1	H-LR retrained without MINBAR-contaminated observations − v2 H-LR (source BA, source AUC)	literal rule; MINBAR ∪ v4 visual; same-segment replacement; H-RF, B-LR, B-RF; B−H within sets
v7a	H-LR observation AUC, soft-like − hard-like (testable if each stratum has ≥ 3 BH sources)	other models; colour-overlap box; hard-like-only retraining (exploratory)
v7b2	H-LR trained on 34 sources, evaluated on the 31 v2 sources − v2 H-LR	34-source performance; +LMC X-3 only; colour positions
v7c1	reproduction of de Beurs et al. (2022): total correct within 2 sources and BH correct within 1 of the paper	per-source comparison with their published SVM medians
v7c2	LR on MAXI: (i) CCI − C2D (intensity increment); (ii) C2D − C1D (2–4 keV increment), source AUC and BA	RF, kNN, SVM; secondary task BH vs pulsars
v7c3	–	RXTE vs MAXI scores of common sources
v8a	S1 hard-like observation AUC: H+T-LR − H-LR (existing OOF)	H+T*, RF, H+N(+T), bands, state gap and difference-in-differences, burst-excluded, hard-only retraining, source level
v8b	external set X, hard-like observations: frozen H_R+T-LR − H_R-LR	soft-like, all states, source level
v8c	S1-HF: H+HF-LR − H-LR (only if IC4 passes; otherwise exploratory)	H+T+HF, RF, HF alone, without AMXPs, external X-HF, SR2000-type rule
v8d	MAXI: RF[$C2D_{NH}$] − RF[$C1D_{NH}$] source AUC	LR, kNN; difference-in-differences; $\log N_H$ as a feature and alone
v9c	– (descriptive)	MAXI 2–4 keV gain within C4 tertiles

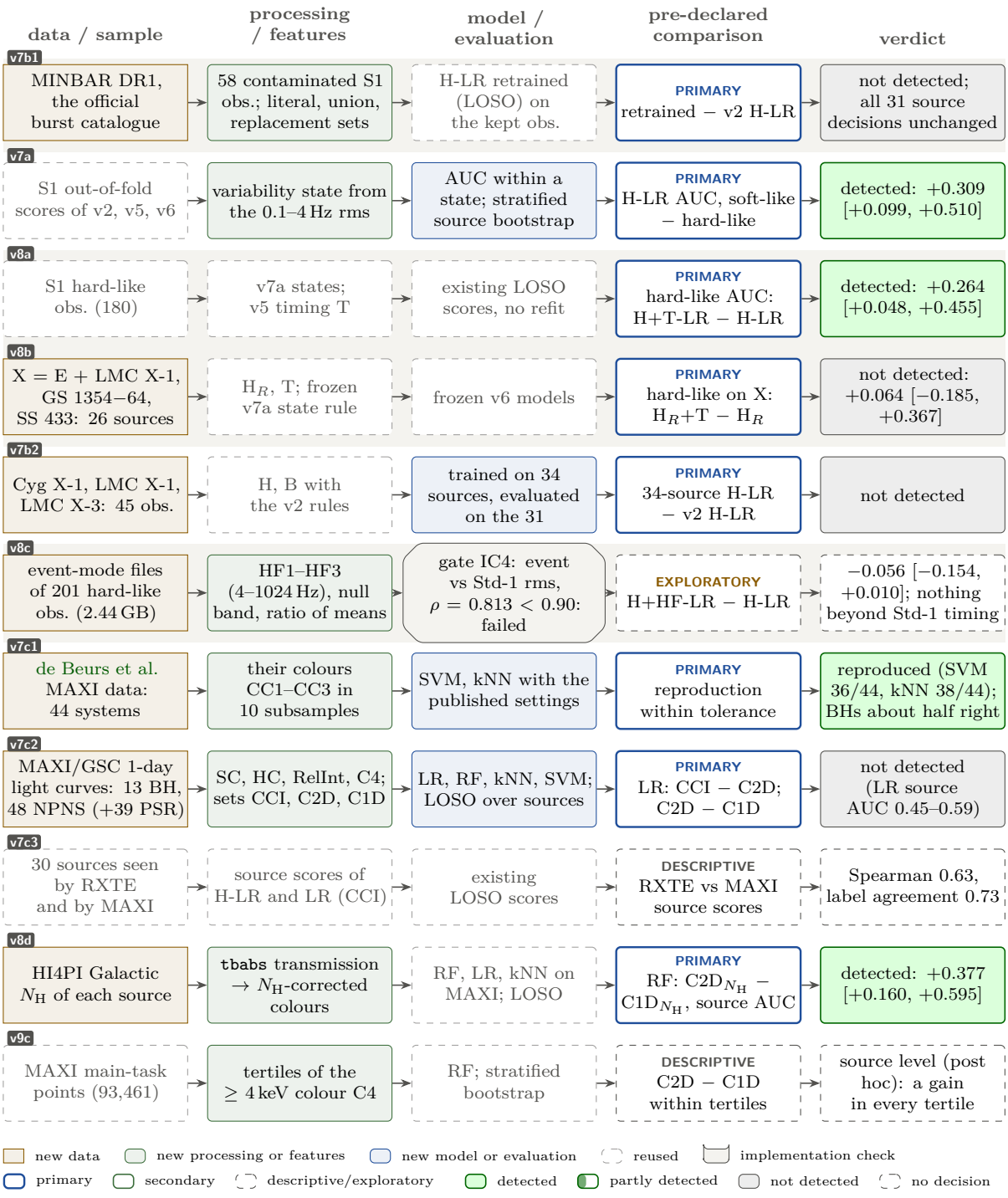


Figure 9.1: Phase 4 analysis map (reading as in Figure 6.1). Shaded and white bands separate the themes: data quality (v7b1), states (v7a), timing within the hard state (v8a, v8b), more BHs (v7b2), higher frequencies (v8c) and an independent instrument (v7c1–v7c3, v8d, v9c). Most RXTE analyses re-use the existing out-of-fold scores and add only a state label or a mask; v8c was downgraded to exploratory when its implementation check failed. Values from Tables 9.7 to 9.9, 9.11, 9.13, 9.14, 9.16, 9.17 and 9.19 to 9.21. Drawn for this report.

9.3 Data

9.3.1 Variability states (v7a)

The state index (Section 2.8.6) is the 0.1–4 Hz fractional rms from the all-pairs Standard-1 cospectrum in 128-s rows inside the spectral GTIs with ≥ 2 PCUs, after removing burst rows (MINBAR intervals in the field of view and v4 candidates from 20 s before the burst; 143 rows), as the median over rows (≥ 3 rows, otherwise "unknown"). Following Table 2 of Remillard and McClintock (2006), rms > 0.10 is *hard-like*, rms < 0.075 *soft-like*, otherwise *intermediate*. The band stops at 4 Hz (RM06 use 0.1–10 Hz), so the rms is lower and the hard-like assignment conservative; the BH thresholds are applied to NSs as well. The pre-declared sanity check passed: XTE J1118+480 15/15 hard-like and the Z sources Cyg X-2 and GX 17+2 27/28 (96%) soft-like (`results/v7a/sanity_check.csv`, Figure 9.2). The thresholds were not tuned.

Table 9.2 gives the resulting state counts and the state composition of each BH source.

Table 9.2: Variability states of the 456 S1 observations (`results/v7a/state_counts.csv`) and the fraction of each BH source's observations per state (`results/v7a/bh_state_fractions.csv`).

State	BH obs. (sources)	NS obs. (sources)	total
soft-like (rms < 0.075)	38 (5)	183 (22)	221
intermediate	1 (1)	20 (11)	21
hard-like (rms > 0.10)	59 (7)	121 (17)	180
unknown (< 3 rows)	7 (2)	27 (12)	34

BH source	hard-like	intermediate	soft-like	unknown
4U 1543−47	0.33	0	0.67	0
GRO J1655−40	0.20	0	0.80	0
GRS 1915+105	0.40	0.07	0.27	0.27
GX 339−4	0.53	0	0.27	0.20
XTE J1118+480	1.00	0	0	0
XTE J1550−564	1.00	0	0	0
XTE J1650−500	0.47	0	0.53	0

9.3.2 Burst contamination from MINBAR (v7b1)

MINBAR DR1 (Galloway et al., 2020) was downloaded from Monash Bridges (CC BY 4.0; 7111 bursts, 2288 from RXTE/PCA; md5 checked; SHA256 in `logs/downloads.jsonl`). The spectral GTIs (`STDGTI`, MET in TT) were converted to UTC MJD (TT−UTC = 64.18 s; round trip < 1 ms); the v4 burst peaks lie 1–9 s after the MINBAR start times, confirming the time systems agree. A burst interval is [start, start + duration] (duration missing $\Rightarrow$ 300 s). **Primary rule (revised after the matching, before any classification result)**: the interval overlaps a GTI *and* the burst was recorded in the same ObsID or originated within 1° (the PCA field of view) of the target. The pre-registered literal rule (any origin) also counts bursts seen by WFC/JEM-X 8–58° away (9 observations, 3 of them BH) and is kept as a sensitivity set.

58 of the 456 observations are contaminated (all NS, 19 sources; 4U 1323−62 10, GS 1826−24 7, MXB 1658−298 6); the literal rule gives 67. Table 9.3 compares MINBAR with the v4 Standard-1 detector: agreement 99.1% with the v4 visual verdicts and 94.5% with the automatic rule. Each source misses bursts the other finds (Figure 9.3), so neither alone is complete. Of the 58 contaminated time bins, 37 could be replaced by a clean observation from the same bin; 21

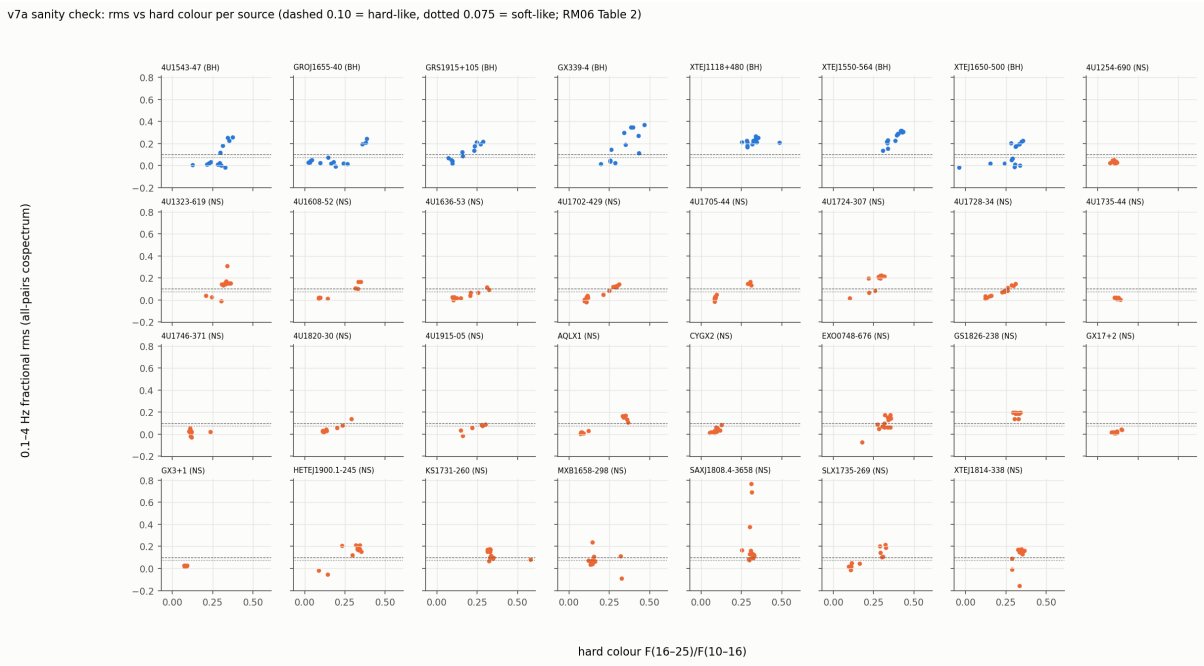


Figure 9.2: The state index (0.1–4 Hz fractional rms, all-pairs cospectrum) against the hard colour for every S1 source (BH blue, NS orange; dashed 0.10 = hard-like threshold, dotted 0.075 = soft-like threshold, RM06 Table 2). XTE J1118+480 and XTE J1550−564 are hard-like throughout; the Z sources Cyg X-2 and GX 17+2 are soft-like; in most sources rms rises with the hard colour. Repo: `figures/v7a/v7a_rms_vs_colour_by_source.png`, `scripts/34_v7a_states.py`.

had none (mostly frequent bursters with few eligible pointings).

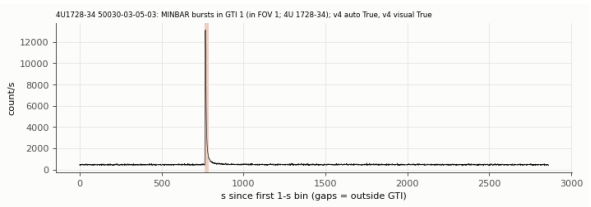
Table 9.3: Observations with a burst according to MINBAR (primary rule) and to the v4 Standard-1 detector (`results/v7b1/minbar_vs_std1_agreement.csv`).

Standard-1 set	both	MINBAR only	Std1 only	neither	agreement
v4 automatic	47	11	14	384	94.5%
v4 visual	56	2	2	396	99.1%

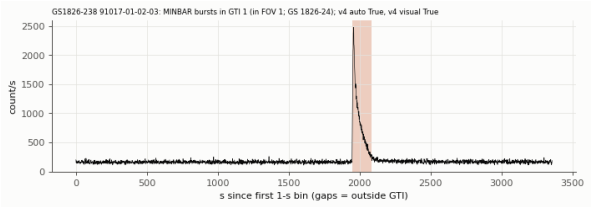
9.3.3 Persistent and additional BHs (v7b2, v8b)

Marcel et al. (2026) (Table 1) list Cyg X-1 (21.2 $M_\odot$, O9.7 Iab companion), LMC X-1 (10.9 $M_\odot$, O8(f)p) and LMC X-3 (7.0 $M_\odot$, B2.5 Ve) as dynamically confirmed. With the v2 rules (epoch 5, exposure $\geq$ 1000 s, `STD1RATE` $\geq$ 5, $\geq$ 10 eligible, 15 time-stratified observations) and pointings within 0.1° of the SIMBAD position (the approved rule), Cyg X-1 has 1,205, LMC X-1 31 and LMC X-3 830 eligible pointings; all 45 sampled observations passed (Table 9.4). The catalogue median pointing of LMC X-1 is 0.263° from the source: 450 of its pointings pass the epoch, exposure and `STD1RATE` cuts, but only 31 lie within 0.1° of the SIMBAD position (the v7 report quotes 450 for a catalogue-median rule). Median deadtime fractions are 2.3% (Cyg X-1, maximum 3.8%), 1.1% (LMC X-1) and 1.3% (LMC X-3); no observation is MINBAR-contaminated. v8b searched the HEASARC TAP `xtemaster` table for the Table 1 BHs not yet used: only GS 1354−64 (1997–98, gain epoch 3; 11 eligible, 8 accepted, 3 without Standard Products) and SS 433 (108 eligible, 15 accepted) were available, with the minimum-eligible rule relaxed to 3.

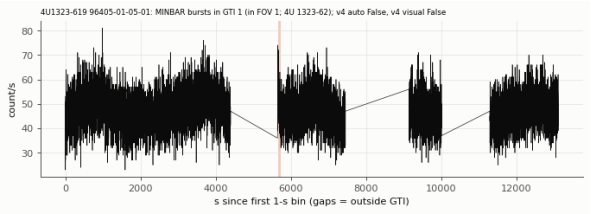
States of the new observations: Cyg X-1 14 hard-like and 1 unknown; LMC X-3 9 soft-like and 6 unknown; LMC X-1 mixed (3 hard-like, 5 intermediate, 4 soft-like, 3 unknown); SS 433 15



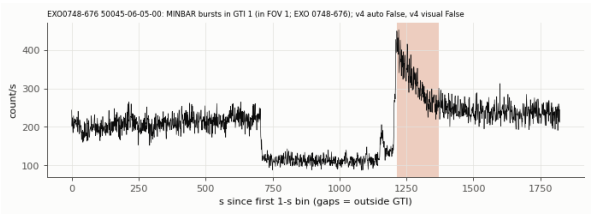
(a) 4U 1728−34: MINBAR and v4 agree.



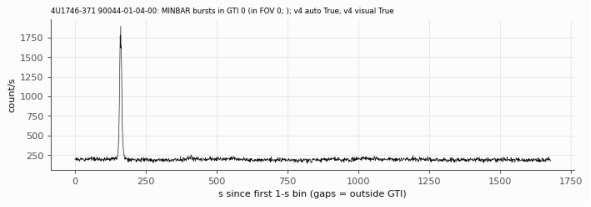
(b) GS 1826−238: MINBAR and v4 agree.



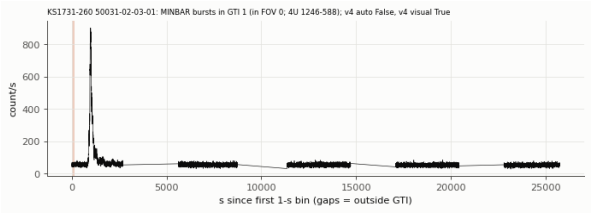
(c) 4U 1323−62: MINBAR only (burst at a GTI break, missed by v4).



(d) EXO 0748−676: MINBAR only (burst after an eclipse, missed by v4).



(e) 4U 1746−37: Standard-1 only (clear burst not in MINBAR).



(f) KS 1731−260: Standard-1 only (the long burst of Figure 7.2).

Figure 9.3: MINBAR versus the v4 Standard-1 burst detector: 1-s Standard-1 light curves (counts s^{-1}, time since the first bin; gaps are outside the GTIs) with MINBAR burst intervals shaded; the title states the MINBAR match and the v4 automatic and visual verdicts. Repo: `figures/v7b1/` (one PNG per agreeing or disagreeing observation), `scripts/31_v7b1_minbar.py`.

Table 9.4: Catalogue-level attrition of the additional BHs (`results/v7b2/attrition_catalog.csv`, `results/v8b/attrition_catalog.csv`, `.../attrition_download.csv`). "sep" = separation of the catalogue median pointing from SIMBAD.

Source	rows	sep [°]	window	exposure	`STD1RATE`	pointing	accepted
Cyg X-1	1,740	0.001	1,370	1,205	1,205	1,205	15/15
LMC X-1	557	0.263	501	450	450	31	15/15
LMC X-3	1,273	0.001	1,102	951	830	830	15/15
GS 1354−64	12	–	12	11	11	11	8/11
SS 433	131	–	131	109	108	108	15/15

soft-like. The external set of v8b, X, is E (23 sources) plus LMC X-1, GS 1354−64 and SS 433: 26 sources (7 BH, 19 NS), 379 observations; its hard-like stratum has 99 observations of 5 BHs (Cyg X-1 14, GS 1354−64 7, V4641 Sgr 5, LMC X-1 3, XTE J1859+226 1) and 15 NSs. The four slow pulsars stay a stress set outside X.

9.3.4 Event-mode data (v8c)

Archival event files (`SE*.evt.gz`, processed in 2012) contain a `PCUID` column, so cross-PCU cospectra can be formed at high frequency. File headers were inventoried by HTTP range requests (`results/v8c/event_survey.csv`, 483 files). A file qualifies if its `DATAMODE` matches `^E_\d+us_\d+M_0_\d+s$` (all channels), `TDDES2` contains all six xenon layers and `TIMEDEL` $\leq 2^{-12}$ s. Of the 311 hard-like observations of S1 and X, 201 had a qualifying file (2.44 GB; modes `E_125us_64M_0_1s` 327 files, `E_125us_64M_0_8s` 5, `E_16us_64M_0_8s` 5, `E_250us_128M_0_1s` 4); the hard-like observations of GRS 1915+105 have only binned modes. The estimator (Section 2.9) uses 2^{-12}-s bins, 16-s segments inside the GTIs and away from bursts, all-pairs cospectra and bands LC 1–4 Hz, HF1 4–64 Hz, HF2 64–512 Hz, HF3 512–1024 Hz and a null band 1536–2048 Hz; quality cuts ≥ 10 segments and $|z_{\rm null}| \leq 5$ (0 of 148 S1 observations removed). S1-HF has 148 observations of 23 sources (6 BH); X-HF 44 observations (2 BH, 11 NS).

Gate IC4 failed. The event-mode 1–4 Hz rms should agree with the Standard-1 T_3^* at Spearman ≥ 0.90; it gave 0.813 (0.818 after the estimator change below). The diagnosis (`results/v8c/ic4_diagnosis.csv`), made before any v8c model, is a noise ceiling rather than a bias: within the hard-like state the 1–4 Hz rms spans a narrow range (IQR 0.11–0.14); T_3^* itself has split-half Spearman 0.563 (Spearman–Brown full-data reliability 0.72), and two Standard-1 estimators agree only at 0.896; in the high-S/N half the event and Standard-1 values agree at 0.978 with unit slope (median ratio 0.94). Following the pre-registration, **all v8c results are exploratory**. The diagnosis also showed that the pre-registered per-observation mean of per-segment ratios is unstable for faint sources (one Aql X-1 segment at 3.6 c s^{-1} PCU^{-1} on 25 c s^{-1} PCU^{-1} background gave −195); it was replaced by the ratio of means (Eq. (2.20)), with the old values kept in `*_mean_of_ratios` columns.

9.3.5 MAXI/GSC (v7c, v8d, v9c)

Reproduction input (v7c1). The processed data of de Beurs et al. (2022) from their GitHub repository (pinned at commit `488b785`, MIT licence): 44 systems, ten 20% subsamples D1–D10 of their MAXI on-demand colours (CC1 = SC, CC2 = HC, CC3 = RelInt), three classes (BH including 7 candidates, non-pulsing NS, pulsar).

Strict pipeline (v7c2). Labels were frozen before the light curves were downloaded: BH = dynamically confirmed systems of Marcel et al. (2026, Table 1); non-pulsing NS (NPNS) = type-I bursters of the MINBAR source table minus those within 0.1° of an accretion-powered pulsar; pulsars (PSR, secondary task) = HMXBs with a pulse period in Liu et al. (2006) plus Patruno and Watts (2021, Table 1). MAXI source names were resolved with Sesame and matched within 0.1° (J-name position within 0.3° as fallback when Sesame fails); blended light curves ("with" in the name) and ambiguous identifications were excluded. Data are the MAXI/GSC standard one-day light curves in 2–4 (L), 4–10 (M) and 10–20 keV (H), with the cuts of de Beurs et al. (2022): every band $\geq \sqrt{3}\,\sigma$ with positive errors, points with total $\geq$ mean + 10 std removed, ≥ 100 points per source. Passing: **13 BHs** (Cyg X-1, GRO J1655−40, GRS 1915+105, GS 1354−64, GX 339−4, LMC X-1, LMC X-3, MAXI J1305−704, MAXI J1820+070, V4641 Sgr, SS 433, Swift J1727.8−1613, XTE J1650−500), **48 NPNS** and **39 PSR**; 93,461 points in the main task. Features (Eq. (2.23)): SC = $(M-L)/(M+L)$, HC = $(H-L)/(H+L)$, RelInt =

total / the source's 99.99th percentile, C4 = $(H-M)/(H+M)$; sets CCI = [SC, HC, RelInt], C2D = [SC, HC], C1D = [C4] (no 2–4 keV information).

Absorption control (v8d). HI4PI (HI4PI Collaboration et al., 2016) Galactic $N_{\rm H}$ from HEASoft `nh` (weighted average within 0.1°); transmissions $t_b(N_{\rm H})$ of the three bands for `tbabs` (Wilms et al., 2000) × a $\Gamma = 2$ power law, tabulated on $N_{\rm H} \in \{0\} \cup$ log-grid 10^{19}–$10^{23.5}$ (200 points) and interpolated in $\log N_{\rm H}$; corrected fluxes $L' = L/t_L$ etc. and recomputed SC′, HC′, C4′, RelInt′. Gate IC5 passed ($t = 0.864/0.979/0.997$ at $10^{22}\,{\rm cm}^{-2}$, monotone); IC6 (uncorrected models reproduce v7c2) passed with maximum difference 1.1×10^{-16}. Median $N_{\rm H}$: BH 0.39, NPNS 0.46 $\times 10^{22}\,{\rm cm}^{-2}$.

Strata (v9c). The main-task points split into tertiles of C4 (cuts −0.671 and −0.445; 31,154 / 31,153 / 31,154 points).

9.4 Data structure in this phase

Most RXTE analyses of this phase do not build new design matrices: they attach a state label or a burst mask to the existing LOSO OOF scores of Phases 1–3 and recompute metrics within strata (Table 9.5). New matrices are the burst-excluded and extended RXTE samples, the event-mode features and the MAXI point tables, where each row is one MAXI day and the group is the source.

Table 9.5: Arrays of Phase 4. Most RXTE analyses re-use existing OOF scores and only add a stratum label or a mask; new design matrices are listed with their shapes.

Analysis	Array	Shape	Content
v7a, v8a	state $\boldsymbol{\sigma}$	456	soft-like / intermediate / hard-like / unknown (`results/v7a/states.csv`)
	OOF scores	$456 \times K$	v2 (H, B; LR, RF), v5 (H+T), v6 (H+T*, H+N(+T), H+T$_k$)
	hard-only $\boldsymbol{X}_H$, $\boldsymbol{X}_{H+T}$	180×2, 180×6	retraining on hard-like observations, 24 groups
v7b1	$\boldsymbol{X}_H$, $\boldsymbol{X}_B$	398×2, 398×45	MINBAR set (389 literal, 396 union, 435 replaced)
v7b2	$\boldsymbol{X}_H$, $\boldsymbol{X}_B$	501×2, 501×45	v2 + 45 observations; 34 groups (10 BH)
v8b	$\boldsymbol{X}_{H_R}$, $\boldsymbol{X}_{H_R+T}$	379×2, 379×6	external X, frozen S1 models; hard-like rows 99
v8c	$\boldsymbol{X}_{H+HF}$	148×5	$h_1, h_2, {\rm HF}_1, {\rm HF}_2, {\rm HF}_3$ (signed $\sqrt{R}$, Eq. (2.20)); also H+T+HF (148×9)
v7c1	per subsample	∼rows ×3	CC1, CC2, CC3 of 44 systems; 3 classes
v7c2, v8d	$\boldsymbol{X}_{\rm CCI}$, $\boldsymbol{X}_{\rm C2D}$, $\boldsymbol{X}_{\rm C1D}$	$93461 \times \{3, 2, 1\}$	one row per MAXI day; training ≤ 200 random points per source (seed 42); 61 groups
	N$_{\rm H}$ variants	same	primed features; $+\log N_{\rm H}$ column
v9c	stratum	93,461	C4 tertile of each point

9.5 Models in this phase

The RXTE analyses use the fixed v2 LR and RF; only the MAXI analyses select hyper-parameters (kNN, SVM), always inside the training sources (Table 9.6). Within-state metrics use the stratified bootstrap of Section 5.5: sources are drawn as usual and the observations of the drawn sources that fall in the stratum are pooled.

Table 9.6: Models of Phase 4.

Model	Where	Settings
LR, RF (§4.3, §4.4)	all RXTE analyses; MAXI	v2 fixed settings
kNN (§4.8)	v7c1	$k = 24$ on raw features, vote fractions (as de Beurs et al., 2022)
	v7c2, v8d	standardised, $k \in \{5, 15, 24, 35\}$ chosen by inner LOSO source AUC, prior correction
SVM-RBF (§4.7)	v7c1	$C = 0.655298$, $\gamma = 0.585144$, standardised, Platt probabilities (seed 42), 3 classes
	v7c2	standardised, balanced, $\sigma(f)$; $C \in \{0.1, 1, 10\} \times \gamma \in \{0.1, 0.585, 3\}$ by inner `GroupKFold(5)` source AUC
stratified bootstrap (§5.5)	v7a, v8a, v8b, v9c	draws over all sources; observations of the drawn sources pooled within a stratum; AUC undefined if a class is absent

9.6 Results

9.6.1 Data quality: burst contamination from MINBAR (v7b1)

PRIMARY Retraining H-LR without the 58 MINBAR-contaminated observations leaves all 31 source decisions unchanged (Table 9.7): source BA difference 0.000 [0, 0], source AUC −0.006 [−0.036, 0.000] – **not detected** under the pre-registered criterion. The other exclusion sets agree. Other models change source BA by −0.021 to 0 and source AUC by −0.036 to 0. The observation BA of H-LR drops slightly (−0.021 [−0.037, −0.007]) because the removed observations are mostly correctly classified NS observations of bursting sources – a change of composition, not a worse model. Within every set the Phase 1 conclusion stands: B − H changes source BA by −0.042 to 0 and source AUC by −0.060 to −0.012; as in v4, the RF observation-level B − H is +0.025 to +0.030 (interval above 0) in the MINBAR, literal and union sets.

Table 9.7: PRIMARY v7b1: H-LR retrained (LOSO) on the observations kept by each burst set, paired by source with the v2 H-LR (456 observations). From `results/v7b1/exclusion_bootstrap.csv` and `results/v7b1/exclusion_counts.csv`.

Set	obs.	Δ source BA	Δ source AUC	Δ obs. BA
★ MINBAR (primary rule)	398	0.000 [0, 0]	−0.006 [−0.036, 0.000]	−0.021 [−0.037, −0.007]
literal rule (any origin)	389	0.000 [0, 0]	0.000 [0, 0]	−0.023 [−0.040, −0.008]
MINBAR ∪ v4 visual	396	0.000 [0, 0]	−0.012 [−0.054, 0.000]	−0.015 [−0.032, +0.002]
same-bin replacement	435	0.000 [0, 0]	0.000 [0, 0]	−0.026 [−0.044, −0.009]

9.6.2 States: soft-like separable, hard-like not (v7a)

PRIMARY Table 9.8 and Figure 9.4 show the central result of the phase. The v2 H-LR has an observation-level AUC of 0.954 [0.801, 1.000] on soft-like observations and 0.646 [0.445, 0.841] on hard-like observations; the difference +0.309 [+0.099, +0.510] is **detected** (95% CI excludes 0) (both strata contain ≥ 3 BH sources: 5 and 7). All four models agree, the observation-BA

differences are +0.29 to +0.32, and with the MINBAR-excluded OOF scores the H-LR difference is +0.389 [+0.160, +0.592]. The intermediate stratum has a single BH source (one observation) and is only described.

Table 9.8: PRIMARY v7a: observation-level AUC of the v2 LOSO OOF scores within variability states, with the soft-like − hard-like difference. Observations (BH/NS): soft-like 38/183, intermediate 1/20, hard-like 59/121, unknown 7/27. From `results/v7a/state_stratified_metrics.csv`.

Model	soft-like	intermediate	hard-like	unknown	soft-like − hard-like
H-LR	0.954 [0.801, 1.000]	0.450	0.646 [0.445, 0.841]	0.672	★ +0.309 [+0.099, +0.510]
H-RF	0.947	0.350	0.693	0.696	+0.254 [+0.113, +0.393]
B-LR	0.946	0.350	0.653	0.656	+0.293 [+0.079, +0.530]
B-RF	0.946	0.600	0.633	0.619	+0.314 [+0.091, +0.545]

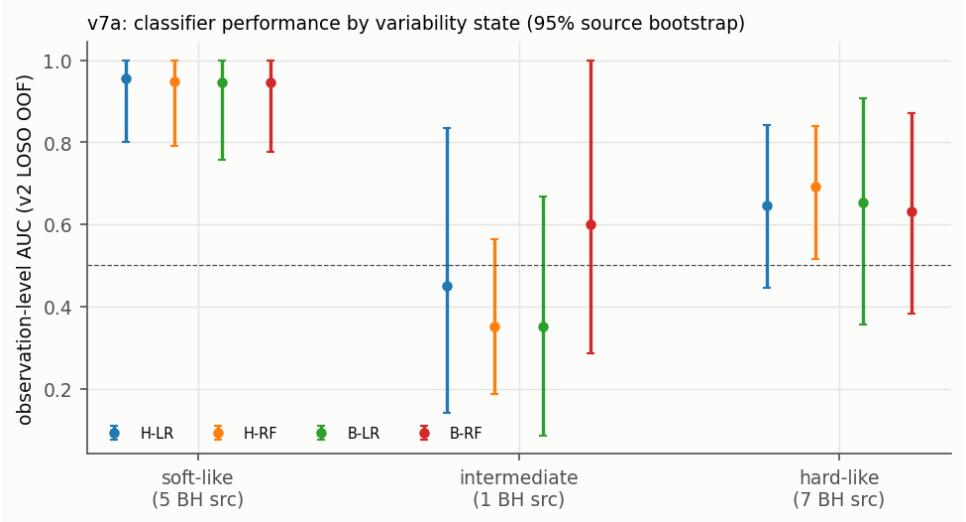


Figure 9.4: Observation-level AUC of the v2 LOSO OOF scores within each variability state (95% source-bootstrap intervals; H-LR, H-RF, B-LR, B-RF). Repo: `figures/v7a/v7a_auc_by_state.png`, `scripts/34_v7a_states.py`.

Descriptively, the colour-overlap box of v4 (defined from the classification features themselves) gives the same picture: H-LR AUC 0.637 inside and 0.954 outside. Retraining only on hard-like observations (exploratory; 180 observations, 24 sources, 7 BH) gives H-LR source AUC 0.681 [0.412, 0.916] and H-RF 0.807 [0.597, 0.958]; the differences from the v2 OOF on the same observations all include 0 (`results/v7a/hard_only_bootstrap.csv`): the hard state has some ranking information in colour, but not enough to measure a gain.

9.6.3 Timing inside the hard state (v8a, v8b)

PRIMARY **v8a.** Within the S1 hard-like stratum, the existing v5 H+T-LR OOF scores reach AUC 0.910 [0.803, 0.975] against 0.646 for H-LR: +0.264 [+0.048, +0.455], **detected** (95% CI excludes 0) (Table 9.9). The gate IC1 (recomputing the v7a value with the v8 code: 0.6456 [0.4446, 0.8414]) passed. The result holds for RF, for the all-pairs features, after controlling the nuisance variables, after excluding bursts and when training only on hard-like observations. The information is in T1 and T2; T3 adds nothing. The state gap shrinks from 0.309 to 0.080 [0.014, 0.181] (difference in differences −0.229 [−0.420, −0.005]; RF −0.207 [−0.354, −0.056]). At source level (hard-like observations only, 24 sources) the AUC difference is +0.185 [+0.008, +0.403] and the BA difference +0.147 [−0.084, +0.391]. This direction was predictable from v5's overlap-box result; it is not a blind test.

Table 9.9: v8a: observation-level AUC within the S1 states (existing LOSO OOF unless stated); differences against the colour-only counterpart. From `results/v8a/v8a_metrics.csv`.

Model	soft-like AUC	hard-like AUC	hard-like difference
H-LR (v2)	0.954	0.646 [0.445, 0.841]	–
H+T-LR (v5)	0.990	0.910 [0.803, 0.975]	★ +0.264 [+0.048, +0.455]
H+T*-LR (v6)	0.993	0.916	+0.270 [+0.059, +0.457]
H-RF → H+T-RF	0.947 → 0.970	0.693 → 0.923	+0.230 [+0.074, +0.408]
H+N-LR → H+N+T-LR	0.995 → 0.999	0.748 → 0.949	+0.201 [+0.019, +0.418]
H+T1-LR	0.989	0.926	+0.280 [+0.059, +0.481]
H+T2-LR	0.982	0.874	+0.228 [+0.042, +0.411]
H+T3-LR	0.961	0.654	+0.008 [−0.029, +0.060]
burst-excluded (395 obs.): H → H+T	–	0.595 → 0.875	+0.280 [+0.055, +0.484]
hard-only retrained: H → H+T	–	0.620 → 0.887	+0.267 [+0.060, +0.460]

Per source (Table 9.10; mean hard-like scores, H-LR → H+T-LR): the BHs XTE J1118+480 0.41 → 0.94, XTE J1550−564 0.55 → 0.87, GX 339−4 0.75 → 0.91 and 4U 1543−47 0.85 → 0.91 rise, while GRO J1655−40 (3 observations) 0.59 → 0.40 and GRS 1915+105 0.49 → 0.43 fall; most NSs fall (4U 1608−52 0.64 → 0.23, 4U 1702−429 0.54 → 0.15, SAX J1808.4−3658 0.71 → 0.42, Aql X-1 0.69 → 0.41) and 4U 1724−307 rises from 0.42 to 0.66.

Table 9.10: v8a: mean LOSO scores of the hard-like S1 observations per source (computed for this report from `results/v2/oof_predictions_loso.csv`, `results/v5/v5_oof_predictions.csv` and `results/v7a/states.csv`; **bold** = wrong side of 0.5).

Source	label	hard-like obs.	H-LR	H+T-LR
4U 1543−47	BH	5	0.85	0.91
GRO J1655−40	BH	3	0.59	**0.40**
GRS 1915+105	BH	6	**0.49**	**0.43**
GX 339−4	BH	8	0.75	0.91
XTE J1118+480	BH	15	**0.41**	0.94
XTE J1550−564	BH	15	0.55	0.87
XTE J1650−500	BH	7	0.68	0.72
4U 1323−62	NS	12	0.34	0.10
4U 1608−52	NS	4	**0.64**	0.23
4U 1636−53	NS	1	**0.52**	0.06
4U 1702−429	NS	6	**0.54**	0.15
4U 1705−44	NS	3	0.44	0.12
4U 1724−307	NS	12	0.42	**0.66**
4U 1728−34	NS	4	0.30	0.05
4U 1820−30	NS	1	0.19	0.02
Aql X-1	NS	5	**0.69**	0.41
EXO 0748−676	NS	6	0.24	0.04
GS 1826−238	NS	15	0.32	0.13
HETE J1900.1−2455	NS	11	**0.66**	0.46
KS 1731−260	NS	10	0.48	0.11
MXB 1659−298	NS	3	0.32	0.26
SAX J1808.4−3658	NS	10	**0.71**	0.42
SLX 1735−269	NS	6	**0.65**	**0.58**
XTE J1814−338	NS	12	0.49	0.11

PRIMARY **v8b.** On the external set X the frozen v6 models do *not* confirm the hard-state gain

(Table 9.11): hard-like AUC 0.817 → 0.882, difference +0.064 [−0.185, +0.367], **not detected** under the pre-registered criterion. Gate IC2 (the frozen models reproduce the v6a scores on 401 observations, maximum difference 1.1×10^{-16}) passed. In the soft-like stratum timing lowers the AUC (−0.118 [−0.337, +0.055]). Table 9.12 lists the hard-like sources: the new BH GS 1354−64 moves from 0.54 to 0.97, LMC X-1 (3 observations, rms 0.101–0.113, at the threshold) from 0.82 to 0.99, and three AMXPs are corrected (XTE J0929−314, XTE J1807−294, IGR J17511−3057), but the AMXP IGR J00291+5934 moves from 0.45 to 0.89 and EXO 1745−248 from 0.32 to 0.73. SS 433 is soft-like in all 15 observations and is called NS by both models (0.31 → 0.02); its X-rays are jet-dominated, so it is a poor test of accretion-state classification but is kept by rule.

Table 9.11: PRIMARY v8b: frozen models on the external set X (26 sources) by state; the hard-like stratum has 99 observations of 5 BH and 15 NS sources. From `results/v8b/v8b_metrics.csv`.

Stratum (X, frozen)	H_R-LR	H_R+T-LR	difference
hard-like (99 obs.)	0.817 [0.595, 0.981]	0.882 [0.710, 0.993]	★ +0.064 [−0.185, +0.367]
soft-like	0.828	0.710	−0.118 [−0.337, +0.055]
all states	0.817	0.799	−0.019 [−0.144, +0.093]
hard-like, source AUC	0.880	0.960	+0.080 [−0.093, +0.307]
hard-like, source BA	0.733	0.867	+0.133 [−0.033, +0.300]

Table 9.12: v8b hard-like observations of the external set X: mean frozen-model scores per source (**bold** = wrong side of 0.5). From `results/v8b/v8b_per_source_state.csv`.

Source	label	origin	hard-like obs.	H_R-LR	H_R+T-LR
Cyg X-1	BH	E_{new}	14	0.74	0.80
GS 1354−64	BH	v8b	7	0.54	0.97
LMC X-1	BH	v7b2	3	0.82	0.99
V4641 Sgr	BH	v4b2	5	0.91	0.95
XTE J1859+226	BH	v4b2	1	0.76	0.60
4U 0513−40	NS	v4b2	3	0.25	0.23
4U 0919−54	NS	v4b2	2	0.38	0.02
4U 2129+12	NS	v4b2	1	0.33	0.12
EXO 1745−248	NS	v4b2	2	0.32	**0.73**
IGR J00291+5934	NS	E_{new}	9	0.45	**0.89**
IGR J17498−2921	NS	E_{new}	6	0.49	0.14
IGR J17511−3057	NS	E_{new}	4	**0.51**	0.06
SAX J1748.9−2021	NS	v4b2	2	**0.69**	**0.59**
SAX J1750.8−2900	NS	v4b2	1	**0.66**	0.20
XTE J0929−314	NS	E_{new}	14	**0.59**	0.27
XTE J1709−267	NS	v4b2	1	**0.68**	0.44
XTE J1710−281	NS	v4b2	4	**0.85**	**0.59**
XTE J1751−305	NS	E_{new}	11	**0.52**	0.35
XTE J1759−220	NS	v4b2	2	0.45	0.03
XTE J1807−294	NS	E_{new}	7	**0.65**	0.31

9.6.4 More BHs: the persistent systems (v7b2)

PRIMARY Training H-LR on 34 sources (adding Cyg X-1, LMC X-1, LMC X-3) and evaluating on the original 31 gives source BA difference 0.000 [−0.214, +0.214] and AUC −0.048 [−0.167, +0.048] – **not detected** under the pre-registered criterion (Table 9.13). The zero BA difference hides two opposite changes: GRS 1915+105 moves from 0.41 to 0.52 (corrected) and XTE

J1550−564 from 0.55 to 0.36 (now wrong). Held out, the new BHs are called BH by all four models (LMC X-1 0.90–1.00, LMC X-3 0.71–0.99, Cyg X-1 0.56–0.73), Cyg X-1 – hard-like in 14 of 15 observations – being closest to the threshold. In the colour plane (Figure 9.5) the LMC BHs sit with the soft BHs on the left, and Cyg X-1 in the overlap region at the upper end of the NS track.

Table 9.13: PRIMARY v7b2: (i) models trained on 34 sources evaluated on the 31 v2 sources, paired with the v2 models; (ii) performance on all 34 sources. From `results/v7b2/v7b2_bootstrap.csv`.

Comparison	source BA	source AUC	obs. BA
★ (i) H-LR − v2	0.000 [−0.214, +0.214]	−0.048 [−0.167, +0.048]	−0.080 [−0.214, +0.032]
(i) H-RF − v2	0.000 [0, 0]	−0.012 [−0.042, 0.000]	+0.013 [−0.011, +0.053]
(i) B-LR − v2	+0.042 [0, +0.104]	+0.012 [0, +0.042]	+0.022 [+0.009, +0.037]
(ii) H-LR, 34 sources (10 BH)	0.838 [0.688, 0.958]	0.908 [0.771, 0.996]	0.725
(ii) H-RF, 34 sources	0.900 [0.750, 1.000]	0.925 [0.796, 1.000]	0.811
+LMC X-3 only, (i) H-LR	+0.021 [−0.193, +0.235]	−0.054 [−0.190, +0.071]	−0.082 [−0.217, +0.032]

POST HOC After adding the persistent BHs the H-LR scores of NSs rose generally (Cyg X-2 0.21 → 0.40). Two faint LMC X-3 observations (92095-03-12-11, 93113-01-54-00) have hard colours of about −0.9 to −1.0 (negative net counts at 16–25 keV, kept by the v2 rule) and therefore high leverage on the LR. Removing them (descriptive) brings the NS scores back close to v2 (Cyg X-2 0.24) and gives a source-AUC difference of +0.006 [−0.036, +0.054]; RF is unaffected.

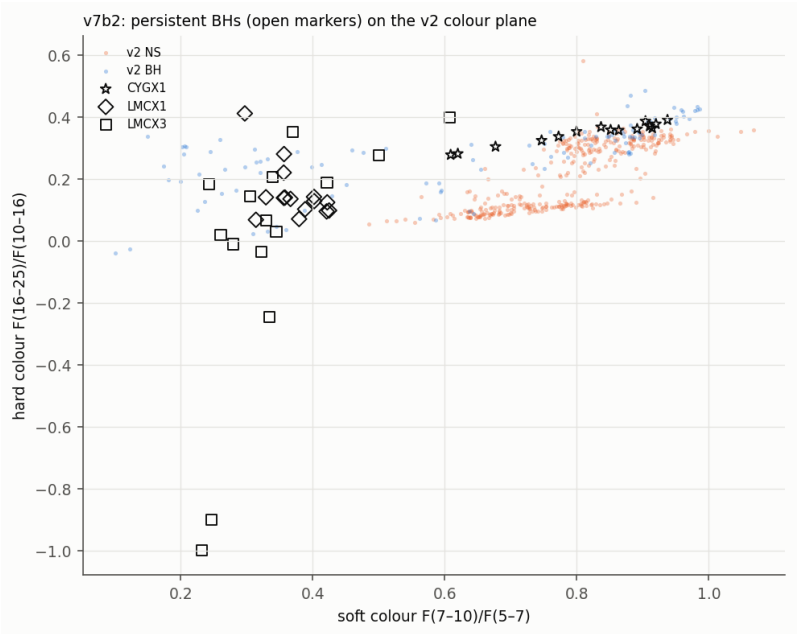


Figure 9.5: The 45 observations of the persistent BHs (open symbols: star Cyg X-1, diamond LMC X-1, square LMC X-3) on the v2 colour plane (dots: v2 BH blue, NS orange). The two LMC X-3 points near hard colour −1 are the high-leverage faint observations. Repo: `figures/v7b2/v7b2_colour_positions.png`, `scripts/36_v7b2_analysis.py`.

9.6.5 Above 4 Hz: event-mode frequencies (v8c, exploratory)

EXPLORATORY Because IC4 failed, none of these results is confirmatory (Table 9.14). The pre-declared comparison H+HF-LR − H-LR is −0.056 [−0.154, +0.010]; adding HF to H+T changes nothing (−0.007 [−0.022, +0.007]); HF alone is worse than chance-level ranking (0.384). For RF,

HF helps on top of H (+0.244 [+0.098, +0.411]) and slightly on top of H+T (+0.034 [+0.005, +0.075]). Without the three AMXPs the LR differences are −0.027 and −0.001. On the external X-HF (44 observations), the frozen H_R+HF-LR reaches 0.976 against 0.997 for H_R-LR.

Table 9.14: EXPLORATORY v8c: observation-level AUC on S1-HF (148 observations, 23 sources, 6 BH; LOSO within S1-HF). From `results/v8c/v8c_metrics.csv`.

Model	obs. AUC	difference	against
H-LR	0.797	–	
H+HF-LR	0.740	−0.056 [−0.154, +0.010]	H-LR (pre-declared)
H+T-LR	0.943	+0.146 [+0.023, +0.331]	H-LR
H+T+HF-LR	0.936	−0.007 [−0.022, +0.007]	H+T-LR
HF-LR	0.384	−0.413 [−0.653, −0.148]	H-LR
H+HF-RF	0.926	+0.244 [+0.098, +0.411]	H-RF (0.682)
H+T+HF-RF	0.954	+0.034 [+0.005, +0.075]	H+T-RF (0.920)
H+HF_c-LR (post hoc)	0.871	+0.074 [−0.023, +0.201]	H-LR
H+HF_c-RF (post hoc)	0.871	+0.189 [+0.019, +0.400]	H-RF

A cross-PCU white component (post hoc). In 4U 1323−619 the null band (1536–2048 Hz) carries more power than 512–1024 Hz (0.096 vs 0.072), and in GS 1826−238 they are comparable, revealing a frequency-independent component that is correlated between PCUs – plausibly particle events hitting several PCUs at once, amplified for faint sources by the source-rate normalisation (not verified). Subtracting "null band × bandwidth ratio" from each band (post hoc) changes the Sunyaev and Revnivtsev (2000)-type rule "512–1024 Hz power above $3\sigma \Rightarrow$ NS" from 6/17 NSs to **1/17** (SAX J1808.4−3658, $z = 3.3$), with all 6 BHs at $|z| < 1.5$ (Table 9.15). At single-observation RXTE sensitivity the kHz criterion is essentially undetectable in this sample.

Table 9.15: SR2000-type rule on S1-HF: per-source 512–1024 Hz fractional rms, its z-value and the call (NS if $z > 3$), as pre-registered (left) and after null-band subtraction (right, post hoc). Negative "rms" values are signed square roots of negative variance estimates. From `results/v8c/v8c_sr2000_rule_S1.csv` and `results/v8c/v8c_sr2000_rule_null_subtracted.csv`.

			pre-registered			null-subtracted (post hoc)		
Source	label	obs.	rms	z	call	rms	z	call
4U 1543−47	BH	4	0.076	+2.0	BH	-0.035	-0.2	BH
GRO J1655−40	BH	1	0.107	+2.0	BH	-0.086	-0.6	BH
GX 339−4	BH	6	-0.013	-0.4	BH	+0.012	+0.2	BH
XTE J1118+480	BH	6	0.057	+0.9	BH	-0.078	-0.9	BH
XTE J1550−564	BH	13	0.021	+0.9	BH	+0.028	+0.8	BH
XTE J1650−500	BH	7	0.013	+1.1	BH	+0.018	+1.0	BH
4U 1323−62	NS	12	0.267	+5.1	NS	-0.165	-1.0	BH
4U 1608−52	NS	4	0.085	+1.7	BH	+0.145	+2.6	BH
4U 1636−53	NS	1	0.133	+1.4	BH	+0.248	+2.2	BH
4U 1702−429	NS	6	0.082	+5.4	NS	+0.076	+2.3	BH
4U 1705−44	NS	3	0.032	+0.5	BH	-0.087	-1.9	BH
4U 1724−307	NS	9	0.089	+2.8	BH	+0.047	+0.4	BH
4U 1728−34	NS	4	0.059	+3.2	NS	+0.064	+1.8	BH
4U 1820−30	NS	1	0.056	+1.0	BH	+0.060	+0.6	BH
Aql X-1	NS	4	0.050	+0.9	BH	-0.086	-1.4	BH
EXO 0748−676	NS	3	0.078	+0.4	BH	-0.178	-1.1	BH
GS 1826−238	NS	15	0.040	+2.9	BH	-0.023	-0.4	BH
HETE J1900.1−2455	NS	11	0.086	+2.9	BH	-0.014	-0.0	BH
KS 1731−260	NS	10	0.038	+3.2	NS	+0.025	+0.7	BH

Source	label	obs.	rms	z	call	rms	z	call
MXB 1659−298	NS	1	0.120	+1.0	BH	+0.128	+0.5	BH
SAX J1808.4−3658	NS	10	0.073	+7.9	NS	+0.068	+3.3	NS
SLX 1735−269	NS	6	0.159	+3.1	NS	+0.152	+1.4	BH
XTE J1814−338	NS	11	0.089	+2.4	BH	-0.030	-0.1	BH

9.6.6 An independent instrument: MAXI/GSC (v7c, v8d, v9c)

Reproduction (v7c1). With the published inputs of de Beurs et al. (2022) the reproduction succeeded exactly (Table 9.16): SVM 36/44 (BH 5/12) as in the paper, kNN 38/44 (BH 7/12; paper 37/44 and 6/12), both within the pre-declared tolerance. Against their published per-source SVM medians: Spearman 0.9989, maximum BH-probability difference 0.011, 100% class agreement on the 44 systems. Of the five systems in their BH class that this project treats as dynamically confirmed, LMC X-3, LMC X-1, GX 339−4 and Cyg X-1 are correct and GRS 1915+105 is called NS by both methods; Cyg X-3 is called a pulsar. Their high overall accuracy comes mostly from the two NS classes; BHs are about half right.

Table 9.16: v7c1 reproduction of de Beurs et al. (2022) (`results/v7c/v7c1_reproduction.csv`, `results/v7c/v7c1_vs_published_svm.csv`).

Method	correct	BH	NPNS	pulsar	paper (total / BH)	within tolerance
SVM	36/44	5/12	19/20	12/12	36 / 5	yes
kNN	38/44	7/12	19/20	12/12	37 / 6	yes

PRIMARY **Strict pipeline (v7c2).** With dynamically confirmed BHs, MAXI standard products and LOSO, LR has almost no discriminating power (source AUC 0.45–0.59), and both pre-declared LR comparisons are **not detected** under the pre-registered criterion (Table 9.17): (i) adding intensity, CCI − C2D, −0.029 [−0.066, +0.002]; (ii) the 2–4 keV colour, C2D − C1D, +0.139 [−0.032, +0.306]. Non-linear models do better with two colours (kNN 0.800, RF 0.774), and descriptively all three non-linear models show a 2–4 keV gain with intervals above 0, while intensity does not help (SVM: −0.119 [−0.212, −0.040]). Per source (Table 9.18), GRS 1915+105 is called NS by every model (0.05–0.42); LR calls only half of the NPNS correctly (24/48), RF calls all 48 NPNS NS but most BHs too. In the secondary task (BH vs pulsars, LR and RF only) LR with two colours reaches source AUC 0.815 [0.679, 0.931], (ii) +0.308 [+0.099, +0.511] and (i) 0.000.

Table 9.18: v7c2 LOSO source scores of the 13 MAXI BHs (n = one-day points; **bold** = below 0.5). From `results/v7c/v7c2_per_source.csv`; SAX J1819.3−2525 = V4641 Sgr.

Source	n	LR CCI	LR C2D	LR C1D	RF C2D	RF C1D	kNN C2D	SVM C2D
Cyg X-1	3,965	0.50	0.50	**0.44**	0.51	**0.22**	0.63	**0.49**
GRO J1655−40	162	**0.49**	**0.49**	0.52	**0.25**	**0.28**	**0.44**	**0.39**
GRS 1915+105	3,272	**0.28**	**0.28**	**0.42**	**0.05**	**0.17**	**0.08**	**0.24**
GS 1354−64	251	**0.50**	**0.50**	**0.50**	**0.30**	**0.30**	0.51	**0.46**
GX 339−4	846	0.51	0.51	**0.49**	**0.40**	**0.25**	0.58	**0.49**
LMC X-1	599	0.57	0.60	0.51	**0.40**	**0.26**	0.60	0.60
LMC X-3	440	0.59	0.59	0.52	**0.36**	**0.29**	0.57	0.53
MAXI J1305−704	100	0.55	0.55	0.52	**0.28**	**0.22**	**0.43**	**0.40**
MAXI J1820+070	146	0.51	0.51	**0.47**	**0.41**	**0.32**	0.56	**0.44**
SAX J1819.3−2525	308	**0.50**	0.51	0.52	**0.27**	**0.28**	**0.42**	**0.45**
SS 433	429	**0.45**	**0.45**	0.52	**0.22**	**0.26**	**0.40**	**0.34**

Source	n	LR CCI	LR C2D	LR C1D	RF C2D	RF C1D	kNN C2D	SVM C2D
Swift J1727.8−1613	205	0.54	0.54	**0.47**	0.50	**0.28**	0.66	0.57
XTE J1650−500	105	**0.50**	0.51	0.53	**0.32**	**0.28**	**0.43**	**0.50**

Same sources, two instruments (v7c3). DESCRIPTIVE Thirty sources (8 BH, 22 NS) are in both the RXTE and the MAXI main samples. Their RXTE H-LR and MAXI LR (CCI) LOSO source scores correlate at Spearman 0.63 and agree in label for 22/30 (73%) (Figure 9.6 and Table 9.19). The MAXI LR scores are compressed into 0.28–0.59; disagreements are mostly NSs pushed just above 0.5 by MAXI (2S 0918−549, 4U 1323−62, SLX 1735−269, GRS 1747−312, XB 1916−053) or BHs just below (XTE J1650−500, GRO J1655−40, V4641 Sgr); GRS 1915+105 (RXTE 0.41, MAXI 0.28) and M15 X-2 / 4U 2129+12 (0.54, 0.55) are wrong on both. The instruments differ in epoch (RXTE 1996–2011, MAXI from 2009), band and burst handling, so this is descriptive only.

Table 9.19: v7c3: RXTE (H-LR, LOSO; sample of origin) and MAXI (LR CCI, LOSO) source scores of the 30 common sources (**bold** = wrong side of 0.5). From `results/v7c/v7c3_rxte_vs_maxi.csv`.

MAXI name	MAXI id	label	RXTE sample	RXTE	MAXI
Cyg X-1	J1958+352	BH	v7b2	0.56	0.50
GRO J1655−40	J1654-398	BH	v2	0.82	**0.49**
GRS 1915+105	J1915+109	BH	v2	**0.41**	**0.28**
GX 339−4	J1702-487	BH	v2	0.81	0.51
LMC X-1	J0539-697	BH	v7b2	0.90	0.57
LMC X-3	J0538-640	BH	v7b2	0.71	0.59
SAX J1819.3−2525	J1819-254	BH	v4b2	0.93	**0.50**
XTE J1650−500	J1650-499	BH	v2	0.85	**0.50**
2S 0918−549	J0920-552	NS	v4b2	0.33	**0.56**
4U 1254−69	J1257-692	NS	v2	0.11	0.45
4U 1323−62	J1326-621	NS	v2	0.32	**0.53**
4U 1608−522	J1612-524	NS	v2	0.35	0.47
4U 1636−536	J1640-537	NS	v2	0.22	0.47
4U 1705−44	J1708-441	NS	v2	0.18	0.39
4U 1722−30	J1727-308	NS	v2	0.37	0.49
4U 1735−444	J1738-444	NS	v2	0.03	0.42
4U 1746−37	J1750-370	NS	v2	0.08	0.48
4U 1820−303	J1823-303	NS	v2	0.04	0.43
Cyg X-2	J2144+383	NS	v2	0.21	0.44
GRS 1747−312	J1750-312	NS	v4b2	0.27	**0.52**
GS 1826−24	J1829-237	NS	v2	0.32	0.46
GX 17+2	J1816-140	NS	v2	0.09	0.38
GX 3+1	J1747-265	NS	v2	0.09	0.40
M15 X-2	J2129+121	NS	v4b2	**0.54**	**0.55**
SAX J1747.0−2853	J1747-288	NS	v4b2	0.28	0.37
SLX 1735−269	J1738-269	NS	v2	0.40	**0.52**
Ser X-1	J1839+050	NS	v4b2	0.08	0.44
XB 1916−053	J1918-052	NS	v2	0.29	**0.51**
XTE J1709−267	J1709-266	NS	v4b2	0.25	0.47
XTE J1710−281	J1710-281	NS	v4b2	**0.52**	**0.52**

PRIMARY **Absorption control (v8d).** If the 2–4 keV gain merely encoded Galactic absorption (tied to sky position and hence to source identity), correcting for $N_{\rm H}$ should remove it. It does not (Table 9.20 and Figure 9.7): after the HI4PI correction the RF two-colour model still beats the ≥ 4 keV colour by +0.377 [+0.160, +0.595] in source AUC – **detected** (95% CI excludes 0) –

Table 9.17: PRIMARY v7c2: MAXI main task (13 BH vs 48 NPNS, LOSO over sources), source BA / source AUC, and the paired differences. From `results/v7c/v7c2_bootstrap.csv`.

Model	CCI	C2D (2 colours)	C1D (≥ 4 keV colour)
LR	0.519 / 0.561	0.596 / 0.590	0.498 / 0.450
RF	0.538 / 0.729	0.577 / 0.774	0.500 / 0.364
kNN	0.668 / 0.736	0.738 / 0.800	0.397 / 0.428
SVM	0.591 / 0.623	0.574 / 0.742	0.397 / 0.446

Comparison	Δ source BA	Δ source AUC
★ (i) LR: CCI − C2D	−0.077 [−0.192, +0.010]	−0.029 [−0.066, +0.002]
★ (ii) LR: C2D − C1D	+0.098 [−0.087, +0.280]	+0.139 [−0.032, +0.306]
(i) RF / kNN / SVM	−0.038 / −0.070 / +0.018	−0.045 / −0.064 / −0.119 [−0.212, −0.040]
(ii) RF	+0.077 [0, +0.192]	+0.410 [+0.229, +0.591]
(ii) kNN	+0.341 [+0.135, +0.550]	+0.372 [+0.184, +0.566]
(ii) SVM	+0.176 [−0.047, +0.410]	+0.296 [+0.096, +0.514]

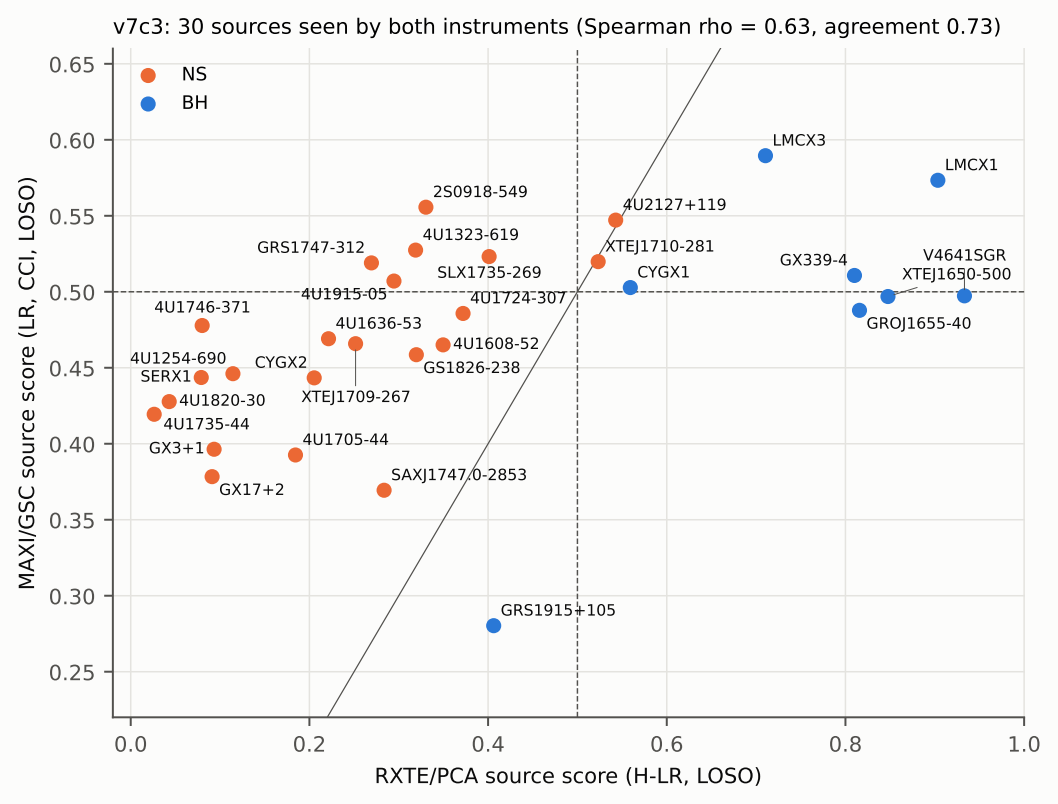


Figure 9.6: RXTE/PCA H-LR source score (horizontal) against MAXI/GSC LR (CCI) source score (vertical) for the 30 common sources (BH blue, NS orange; dashed lines at 0.5; solid line $y = x$). The vertical axis is limited to 0.22–0.66, the range of the MAXI scores. Redrawn for this report from `results/v7c/v7c3_rxte_vs_maxi.csv` (same points as the repository figure `figures/v7c/v7c3_rxte_vs_maxi.png` of `scripts/39_v7c2_analysis.py`, whose labels overlap; Spearman ρ and agreement recomputed from the CSV and identical to the repository values).

against +0.410 uncorrected (difference in differences −0.034 [−0.223, +0.130]). $\log N_{\rm H}$ alone does not separate the classes (AUC 0.410 [0.244, 0.588]), and with $\log N_{\rm H}$ as an extra feature the gain remains (+0.147 [+0.059, +0.240]). The softest 2–4 keV colours belong to BHs (LMC X-1, LMC X-3, Swift J1727.8−1613, Cyg X-1). Two caveats are stated in the repository: choosing RF as the primary was made after seeing v7, and RF on C1D alone has AUC below 0.5, so part of the gain is relative to a worse-than-random baseline.

Table 9.20: PRIMARY v8d: MAXI main task, source AUC before and after the HI4PI $N_{\rm H}$ correction. From `results/v8d/v8d_bootstrap.csv`.

Model / difference	uncorrected	$N_{\rm H}$-corrected
RF, ≥ 4 keV colour (C1D)	0.364 [0.216, 0.532]	0.361 [0.204, 0.534]
RF, two colours (C2D)	0.774 [0.590, 0.934]	0.737 [0.550, 0.907]
RF C2D − C1D	+0.410 [+0.229, +0.591]	★ +0.377 [+0.160, +0.595]
kNN C2D − C1D	+0.372 [+0.184, +0.566]	+0.268 [+0.082, +0.470]
LR C2D − C1D	+0.139 [−0.032, +0.306]	+0.123 [−0.026, +0.269]
$\log N_{\rm H}$ alone	–	0.410 [0.244, 0.588]
RF [C2D + $\log N_{\rm H}$] − RF [C1D + $\log N_{\rm H}$]	–	+0.147 [+0.059, +0.240]
difference in differences, RF	−0.034 [−0.223, +0.130]	
difference in differences, kNN	−0.104 [−0.231, +0.006]	
difference in differences, LR	−0.016 [−0.059, +0.021]	

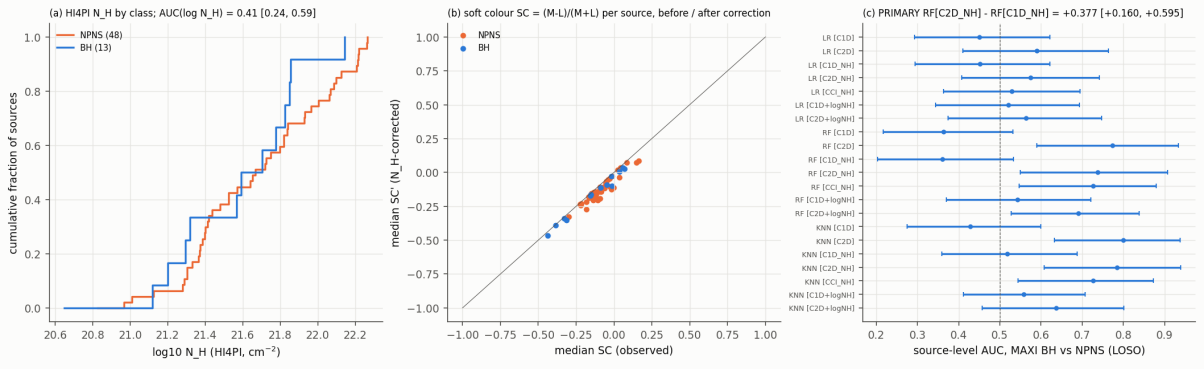


Figure 9.7: v8d: (a) cumulative HI4PI $\log N_{\rm H}$ of the 13 BHs and 48 NPNS (AUC of $\log N_{\rm H}$ alone 0.41); (b) per-source median soft colour SC before and after the correction; (c) source AUC of all model/feature combinations (95% source-bootstrap intervals). Repo: `figures/v8d/v8d_nh_control.png`, `scripts/45_v8d_nh.py`.

DESCRIPTIVE **Strata (v9c).** Is the 2–4 keV gain confined to soft (thermal) states? The pre-declared point-level AUCs are near 0.5 in every C4 tertile (RF C2D 0.49–0.56) because the point-level metric is dominated by sources with thousands of points (Cyg X-1 has 3,965); the C2D − C1D point-level difference is +0.167 [−0.185, +0.580], +0.060 [−0.184, +0.264] and +0.052 [+0.007, +0.103] (softest to hardest). The POST HOC source-level version (Table 9.21) shows a gain of about +0.27 to +0.40 in every tertile, before and after the $N_{\rm H}$ correction: the 2–4 keV information is not confined to thermal states.

Figure 9.8 shows the pre-declared point-level version.

9.6.7 Phase summaries

Figures 9.9 and 9.10 reproduce the repository's summary figures of v7 and v8.

Table 9.21: v9c: RF C2D − C1D within tertiles of the ≥ 4 keV colour C4. Point-level AUC differences are pre-declared (descriptive); source-level ones are post hoc. From `results/v9c/v9c_maxi_c4_strata.csv` and `results/v9c/v9c_maxi_c4_strata_source_level_post_hoc.csv`.

C4 tertile (BH / NS sources)	point AUC Δ	source AUC Δ (post hoc)	source AUC Δ, $N_{\rm H}$-corr.
low, softest (8 / 38)	+0.167 [−0.185, +0.580]	+0.388 [+0.151, +0.628]	+0.257 [+0.003, +0.543]
middle (11 / 44)	+0.060 [−0.184, +0.264]	+0.269 [−0.048, +0.570]	+0.399 [+0.159, +0.647]
high, hardest (13 / 48)	+0.052 [+0.007, +0.103]	+0.388 [+0.160, +0.603]	+0.361 [+0.122, +0.601]

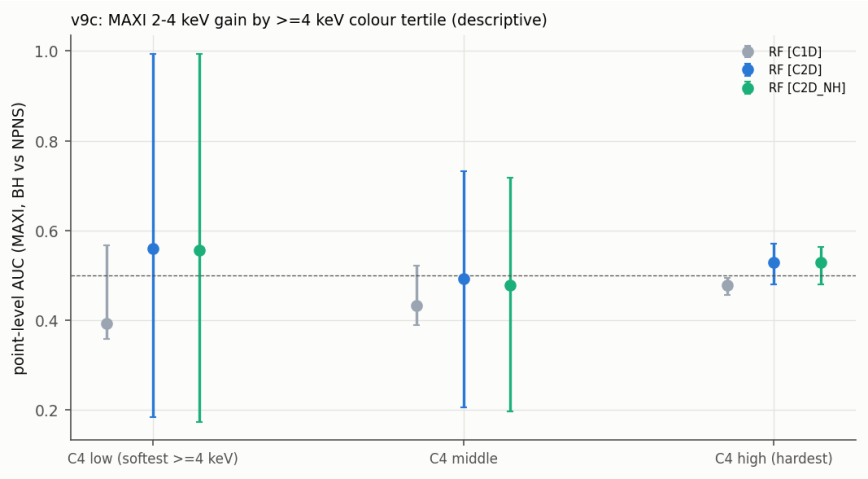


Figure 9.8: v9c (pre-declared, descriptive): point-level AUC of RF on C1D, C2D and $\rm C2D_{NH}$ within tertiles of the ≥ 4 keV colour C4 (95% source-bootstrap intervals). The point-level metric is dominated by the sources with most points, which is why all values are near 0.5; the source-level version is in Table 9.21. Repo: `figures/v9/v9c_maxi_c4_strata.png`, `scripts/50_v9c_maxi_strata.py`.

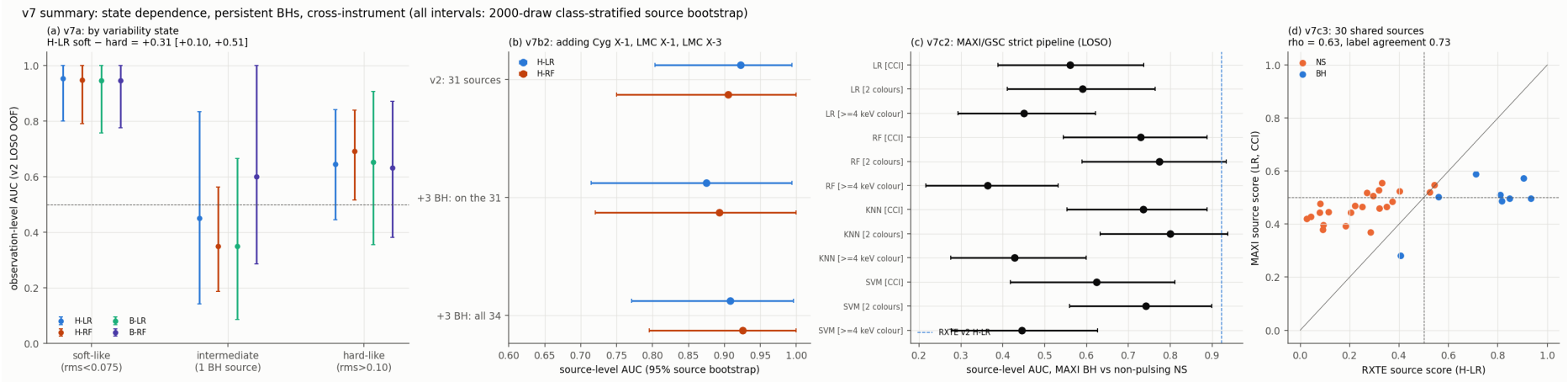


Figure 9.9: v7 summary: (a) AUC by variability state for four models; (b) source AUC of H-LR and H-RF with 31 sources, with 34 sources evaluated on the 31, and on all 34; (c) MAXI source AUC of all models and feature sets (dashed: RXTE v2 H-LR); (d) RXTE vs MAXI source scores of the 30 common sources. Repo: `figures/v7/v7_summary.png`, `scripts/40_v7_summary_figure.py`.

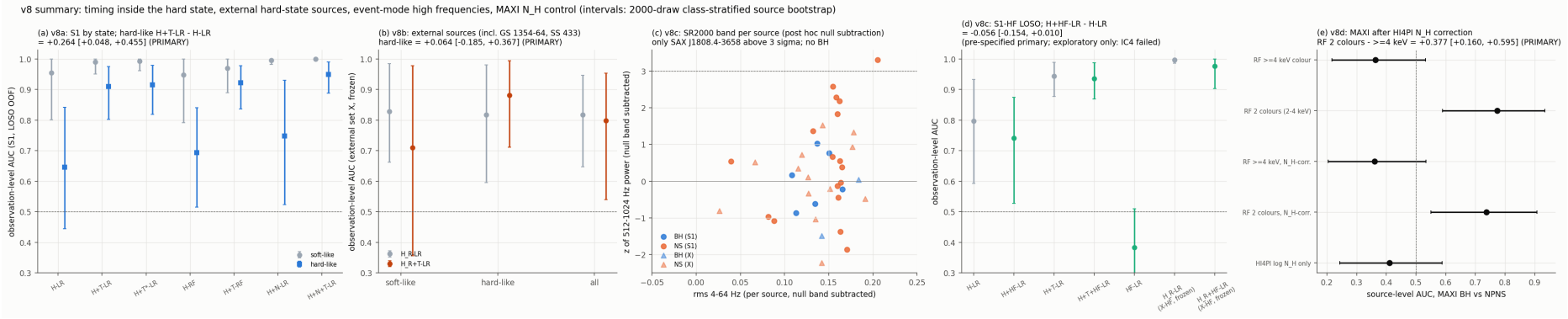


Figure 9.10: v8 summary: (a) S1 AUC by state for colour-only and timing models (hard-like squares); (b) external X with frozen models by state; (c) per-source null-subtracted 512–1024 Hz z against 4–64 Hz rms (post hoc; only SAX J1808.4−3658 above 3σ); (d) S1-HF observation AUCs (exploratory); (e) MAXI RF source AUC before and after the $N_{\rm H}$ correction. Repo: `figures/v8/v8_summary.png`, `scripts/46_v8_summary_figure.py`.

9.7 Interpretation

What the data support directly.

1. The two-colour classifier depends strongly on the state: AUC ≈ 0.95 on soft-like observations and ≈ 0.65 on hard-like ones, difference +0.31 [+0.10, +0.51]; consistent across four models and after burst exclusion. The Phase 1 inference "hard BHs and hard NSs overlap in colour" is now a number with an interval.
2. Type-I burst contamination does not affect the Phase 1 conclusions: with the official MINBAR catalogue the 31 H-LR source decisions are unchanged. MINBAR and the independent Standard-1 detector agree on 99.1% of observations; each misses 2 bursts the other finds.
3. Inside the hard-like state of S1, low-frequency timing raises the AUC from 0.65 to 0.91 (interval excludes 0), robustly across models, estimators, nuisance control, burst exclusion and hard-only training; the information is in 0.016–1 Hz, and the state gap shrinks from 0.31 to 0.08. This is where the v5/v6 gain came from.
4. On external sources the hard-state gain is *not* confirmed: the point estimate is positive (source-level BA +0.13) but the interval is wide and one AMXP is pushed strongly towards BH.
5. Adding three persistent BHs does not change performance on the original 31 sources; all three are recognised as BHs when held out, Cyg X-1 (hard-like) closest to the threshold.
6. At single-observation RXTE sensitivity, after null-band subtraction, broad-band 512–1024 Hz power is detected in none of 6 BHs and in 1 of 17 NSs; high frequencies add essentially nothing beyond the low-frequency features (exploratory).
7. The de Beurs et al. result reproduces exactly from their published inputs. With confirmed labels, standard MAXI products and LOSO, LR has hardly any discriminating power; non-linear models reach ≈ 0.8 with two colours; the 2–4 keV colour adds information that survives a Galactic $N_{\rm H}$ correction and is present in every C4 tertile.

Possible explanations (not verified).

- In soft states the disc dominates and the BH colours lie outside the NS track; in hard states both classes are dominated by Comptonisation with similar 5–25 keV shapes.
- At equal hard colour, BHs show stronger < 1 Hz variability, consistent with lower characteristic frequencies (the ν_c test of v6 was significant on S1 but not externally); amplitude and frequency are not separated here (taken up with Lorentzian fits in Phase 6).
- The external failure may reflect the BH composition (an HMXB, a gain-epoch-3 transient, an LMC source at the state threshold), the AMXP-dominated NS sample (IGR J00291+5934 has strong low-frequency noise), and less room for improvement (external colour-only hard-like AUC already 0.82 vs 0.65 on S1).
- The SR2000 criterion may fail here because the sample is post-2000 (fewer PCUs, gain epoch 5), single observations last 1–3 ks while Sunyaev and Revnivtsev (2000) combined observations with all five PCUs, and a broad 512–1024 Hz band dilutes narrow kHz QPOs.
- The MAXI 2–4 keV information may reflect the disc temperature (cooler BH discs in thermal states, NS boundary-layer emission harder; cf. Done and Gierliński, 2003); since the gain appears in all C4 tertiles it is not limited to thermal states. LR may fail on MAXI because BHs lie on both sides of the NS distribution in the colour plane (soft and hard states), which is not linearly separable.

Hypotheses that need more data. A prospective confirmation of the frozen H+T model on more *hard-state* confirmed BHs – the RXTE era is exhausted, so this needs NICER, Insight-HXMT, AstroSat or XRISM and a recalibrated timing feature (Phase 5); fits of Lorentzian characteristic frequencies (Phase 6); stacked kHz searches with a model of the cross-PCU white component; state-resolved MAXI analyses.

9.8 Deviations and caveats

- **Written after earlier results**: v7 after v1–v6 (user-approved), v8 after v1–v7 (self-approved); the state analysis tests inferences already drawn, and v8a was predictable from v5.
- **v7b1**: the primary burst rule (same ObsID or origin within 1°) replaced the literal rule after seeing the matching and before any classification result; the MINBAR ∪ visual set was added after the cross-check; the comparison "paired with v2 on common sources" was implemented as pairing with the 456-observation v2 OOF by source (same-observation comparisons also reported).
- **v7b2**: post hoc check of the two high-leverage LMC X-3 observations (main analysis unchanged).
- **v7c2 labels** (fixed before any C2 model): Liu+2007 LMXB pulse periods include burst oscillations, so LMXB pulsars come only from Patruno and Watts (2021); HMXB pulsars from Liu et al. (2006) were added (the pre-registration named only Patruno & Watts); MAXI matching changed to Sesame-resolved names within 0.1° (the J-name position within 0.35° had mismatched IGR J16358−4726 and 4U 1630−47); blends excluded; name fixes (V616 Mon, IGR J17480−2446, IGR J16358−4726).
- **v7c2 computation**: secondary task only with LR and RF; the SVM inner selection uses grouped 5-fold instead of LOSO (estimated > 6 h), decided before any SVM result; kNN used inner LOSO as pre-declared. Unspecified details fixed in code: standardisation of kNN/SVM inputs, equal-prior kNN probabilities, SVM score $\sigma(f)$ with balanced weights.
- **v8 gates and estimator**: the IC3 criterion "squared HF $< 0.01^2$ under pure Poisson" was unattainable and replaced by $|z| < 3$, and a hand-computed segment-test expectation corrected (before models); IC4 failed ⇒ v8c exploratory; the estimator changed from mean of ratios to ratio of means after IC4 and before any v8c model; the null-band subtraction is post hoc.
- **v8 tooling**: downloads switched from `urllib` (≈ 0.1 MB/s) to `curl` with the same logging; HTTP retries added after one inventory connection error; a trailing `\r` from a Windows CSV read in bash fixed in v8d before results. Only two of the planned v8 figures were produced; the other results are in CSV files.
- **States**: the index stops at 4 Hz, uses BH thresholds for NSs and is an operational variability class, not a full state assignment (no disc fraction, no photon index); 34 S1 observations (7 BH) are "unknown"; external states use MINBAR-only burst removal; the state of the faint LMC X-1 is uncertain.
- **Small samples**: 7 BH sources in the S1 hard-like stratum (6 in S1-HF), 5 in the external hard-like stratum.
- **MAXI**: standard products (2–4/4–10/10–20 keV photon fluxes) instead of on-demand bands; one-day averages mix states; globular-cluster light curves may blend sources; HI4PI gives only the Galactic atomic column (no molecular or intrinsic absorption, e.g. LMC X-1); RF as the v8d primary was chosen after v7; epochs differ between RXTE and MAXI.

- **SS 433** is jet-dominated and a poor test of accretion-state classification (kept by rule and listed separately); SR2000 sources overlap with this sample, so v8c is not an independent test of their criterion.
- **Bootstrap** intervals keep predictions fixed.

9.9 Takeaway

The failure of the colour classifier is a hard-state failure, and in the hard state of the training sources low-frequency variability repairs most of it – but that repair is not confirmed on external sources, the persistent BHs and higher frequencies add nothing measurable, and bursts never mattered. On MAXI, with strict labels, linear colour models fail while the 2–4 keV colour carries information that is not absorption. The hard-state timing claim therefore needs a test on genuinely new observations with a different instrument: Phase 5 takes it to NICER.

Chapter 10

Phase 5 (v9–v10): a second instrument, and the pooled evidence

Phase at a glance: v9–v10

Question. Do the two timing claims of Phases 3–4 – (1) inside the hard state, low-frequency variability adds BH/NS information to colour; (2) at fixed colour, BHs have a lower characteristic frequency ν_c – reappear on a different instrument (NICER/XTI, 2017 onwards, 2–10 keV), and how strong is the evidence when all usable sources of RXTE and NICER are combined?
Pre-declared primaries. v9a: NICER hard-like observation AUC, H_N+T_N-LR − H_N-LR; v9b: colour-conditional source permutation test of $\log_{10}\nu_c$ in NICER hard-like observations; v10a: equal-weight mean of the hard-state timing gains of three subsamples (RXTE v2, RXTE external, NICER full observations); v10b: pooled colour-conditional $\log_{10}\nu_c$ difference over both instruments with a joint permutation.
Data. NICER: 368 observations of 63 sources (9 BH, 54 NS), first ≈ 6 segments per observation (14.2 GB) in v9, full observations (further 24.2 GB) in v10; the RXTE samples of Phases 3–4. Union of sources: 86 (16 dynamically confirmed BHs).
Verdict. NICER reproduces both effects in direction and size (hard-like AUC 0.68 → 0.80; $\Delta\log_{10}\nu_c = -0.31$), but both v9 primaries are **not detected** by a small margin (v9a +0.112 [−0.004, +0.229], v9b $p = 0.136$). Pooled, both v10 primaries are **detected**: +0.145 [+0.028, +0.257] and $D = -0.368$ dex, $p = 0.0012$, robust to leaving out any single BH. **v10 is the overall strength of the evidence, not an independent confirmation**: the hypotheses and the pooling were specified after v5–v9, and two of the three subsamples had been seen.

10.1 Motivation

At the end of Phase 4 the hard-state timing gain was detected on the RXTE v2 sources but not on external RXTE sources, and the RXTE archive offered no further confirmed BHs. NICER is independent in instrument, band (≈ 0.5–12 keV; 2–10 keV used here), epoch (2017 onwards) and source mix; it covers two dynamically confirmed BHs never used before in this project (MAXI J1820+070, Swift J1727.8−1613) with many hard-state observations, and its time resolution reaches the full 0.1–10 Hz band of Remillard and McClintock (2006) and beyond. The RXTE models are not transferred (bands and responses differ): the same pre-declared design is repeated on NICER. v10 then asks the question that the single-instrument tests could not settle because each was at the edge of significance: what do all usable sources say together?

Figure 10.1 maps the phase.

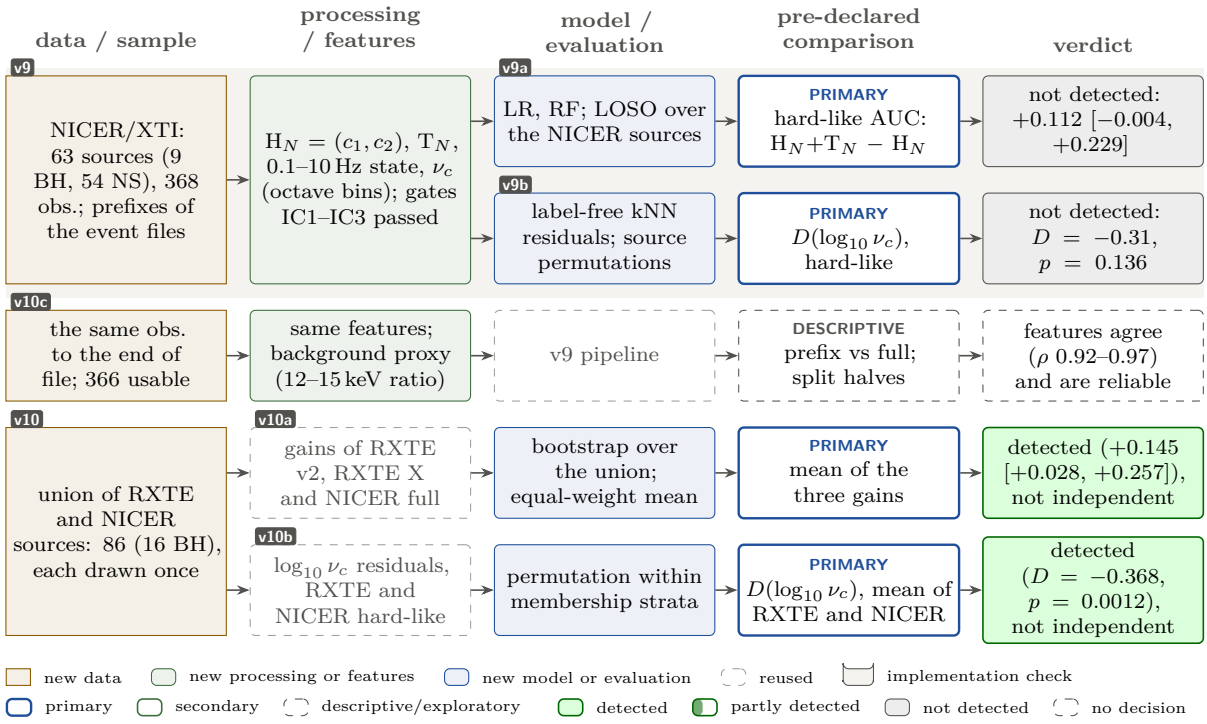


Figure 10.1: Phase 5 analysis map (reading as in Figure 6.1). v9 repeats the hard-state design on NICER (the RXTE models are not transferred); v10c extends the same observations to full length; v10a and v10b pool all usable sources of both instruments, each source drawn or relabelled once. **v10 is the overall strength of the evidence, not an independent confirmation**: its hypotheses and pooling were specified after v5–v9. Values from Tables 10.3, 10.5 and 10.7 to 10.9. Drawn for this report.

10.2 Pre-registration

Table 10.1 summarises the two pre-registrations.

Section 0 of the v10 pre-registration states its position explicitly: “v10’s hypotheses and pooling were proposed after seeing the v5–v9 results: the v10 results represent the overall strength of evidence with all data, not an independent confirmation; the report must say so” (translated). Decision rules as before: bootstrap 2000, seed 42, class-stratified; “difference” iff the interval excludes 0 or $p < 0.05$; one primary per sub-analysis.

10.3 Data

10.3.1 NICER sources and observations (v9)

Sources. BH: the 25 dynamically confirmed systems of Marcel et al. (2026, Table 1); NS: the 115 burster entries of the MINBAR source table, minus 9 that have another entry within 0.05° (unresolvable in the NICER field of view, radius ≈ 0.05°). Observations were found in the `nicermastr` table (target within 0.05°, exposure ≥ 1 ks); sources with ≥ 5 such observations: 11 BH and 55 NS (`results/v9/attrition_catalog.csv`).

Table 10.1: Phase 5 pre-registrations (`reports/preregistration_v9.md`, `reports/preregistration_v10.md`), both written and self-approved by the analyst under the user's instructions ("merge and pull and get on to next step"; for v10, in Chinese, "commit everything to git and start v10").

Ver.	Written	Primary	Descriptive
v9	2026-10-05 (≈23:00), after all v1–v8 results; pushed (`ee4b7db`) before any NICER download	v9a: NICER hard-like observation AUC, H_N+T_N-LR − H_N-LR (LOSO on all states); v9b: permutation test of $\log_{10}\nu_c$ in hard-like observations	RF, rate control, bands, state gap, hard-only training, source level, new BHs only, bright observations; T_k and rms permutation tests; v9c MAXI strata (reported in Chapter 9)
v10	2026-10-06, after all v1–v9 results; pushed (`aeedac7`) before any v10 download or pooled statistic	v10a: equal-weight mean of three hard-state gains, bootstrap over the union of sources; v10b: mean over instruments of the colour-conditional $\log_{10}\nu_c$ difference, permutation within instrument-membership strata	components; BH-weighted mean; RF; background-screened NICER; colour-window version of v10b; leave-one-BH-out; v10c NICER full-observation quality

Sampling. Each source's observations were split into 8 time bins; within a bin candidates were tried in a random order (`default_rng(SEED + position)`) up to 3 times: 522 bins, 792 tries, **368 accepted** (BH 9 sources, NS 54). Rejections: 2–10 keV rate $< 20\,\mathrm{c\,s^{-1}}$ (239) and fewer than 3 valid segments (185) (`data/v9/observations.csv`). SS 433 and GS 2023+338 (V404 Cyg) had no qualifying observation.

Download. Cleaned event files (`ni<obsid>_0mpu7_cl.evt.gz`) were read as a stream through the HEASARC AWS mirror (byte-identical, $\approx 3.5\,\mathrm{MB\,s^{-1}}$ instead of 0.1–0.4 $\mathrm{MB\,s^{-1}}$), with 16 MB HTTP range requests, stopping after six gap-free 128-s segments, at the end of the file or at 300 MB (14.2 GB in total, stored outside the repository).

Features (Section 2.10; all pre-declared): 2–10 keV events (PI 200–999), 128-s segments of 1/256-s bins ($N = 32768$, Nyquist 128 Hz) cut from the good intervals (gap > 1 s ends an interval), burst segments removed with the v4 rules on the 1-s 2–10 keV light curve ([start −20 s, end +200 s]); an MPU is on if all eight 16-s blocks have counts; ≥ 2 MPUs. All-pairs cross-MPU cospectra, ratio of means and signed square root (the v8 lesson); no background subtraction (rate cut $20\,\mathrm{c\,s^{-1}}$). Colours $H_N = [c_1, c_2]$ with $c_1 = C(4\text{–}6)/C(2\text{–}4)$ and $c_2 = C(6\text{–}10)/C(4\text{–}6)$; timing T_N = fractional rms in the v5 bands (0.016–0.094, 0.10–1.0, 1.0–4.0 Hz); state from the 0.1–10 Hz rms with the RM06 thresholds; ν_c from 13 octave bins over 1/128–64 Hz (Eq. (2.18)).

Gates. IC1: the streaming parser equals `astropy` on complete and truncated synthetic files; IC2: synthetic events recover sinusoids at 0.05, 0.5 and 2 Hz as 0.099–0.102 (true 0.1), the 0.1–10 Hz rms as 0.143 (true 0.141), pure Poisson $|z| \leq 1.3$, and Lorentzians with $\nu_0 = 0.5$ and 5 Hz give $\nu_c = 0.55$ and 4.18 Hz; IC3: Cyg X-2 and GX 17+2 are soft-like in 15/16 observations (0.94). All passed.

States. Table 10.2 gives the NICER states. The hard-like stratum has 31 BH observations of 7 sources (MAXI J1820+070 7, Cyg X-1 7, Swift J1727.8−1613 5, GX 339−4 4, GRS 1915+105 4, 4U 1543−47 3, LMC X-3 1) and 111 NS observations of 32 sources.

10.3.2 Full NICER observations (v10c)

For the same 368 observations the files were downloaded beyond the v9 prefix up to 1 GB compressed (24.2 GB; 180 observations to the end of file, 12 truncated at 1 GB); prefix and tail are parsed as one gzip stream (IC1: one-shot and concatenated reads of 6403520101 agree, 16,982,648 bytes, 729,479 events). 366/368 observations remain usable (two at 19.2 and 19.8 $\mathrm{c\,s^{-1}}$, just below 20); the median number of 128-s segments rises from 9 to 16; 9 observations change

Table 10.2: States of the 368 NICER observations (0.1–10 Hz rms; `data/v9/processed/nicer_features.csv`).

State	BH obs.	NS obs.	total
soft-like	30	172	202
intermediate	4	20	24
hard-like	31 (7 sources)	111 (32 sources)	142
total	65 (9 sources)	303 (54 sources)	368

state; IC4 (Z sources soft-like) gives 15/16. Table 10.3 shows that the full-observation features agree with the prefix values and are reliable in split halves. A background proxy, the 12–15 keV to 2–10 keV count ratio, flags 13 observations above the 99th percentile of the bright ($\geq 500\,\mathrm{c\,s^{-1}}$) observations (threshold 0.2687).

Table 10.3: v10c: agreement of full-observation features with the v9 prefix values, and split-half reliability of the full observations (odd vs even segments; Spearman–Brown corrected). From `results/v10c/prefix_vs_full.csv` and `results/v10c/split_half.csv`.

Feature	prefix vs full (n)	split-half ρ (n)	Spearman–Brown
T1	0.915 (366)	0.681 (350)	0.810
T2	0.948 (366)	0.796 (350)	0.886
T3	0.961 (366)	0.674 (350)	0.806
0.1–10 Hz rms	0.968 (366)	0.760 (350)	0.864
ν_c	0.932 (238)	0.805 (195)	0.892
c_1, c_2	0.969, 0.948 (366)	–	–

10.3.3 Source identity across instruments (v10)

RXTE sources (from the v2, v4b2, v6, v7b2 and v8b source lists) and NICER sources within 0.1° are the same object; the NICER name is canonical. 34 pairs matched with no label conflict (`results/v10a/source_identity.csv`); the union used by the pooled statistics has **86 sources, 16 of them BHs** (`logs/v10/54_pooled.log`; the slow-pulsar stress set is excluded). All pooled resampling and permutation is done on this union, so a source observed by both instruments is drawn, or relabelled, once.

10.4 Data structure in this phase

10.5 Models and statistics in this phase

Models are the fixed LR and RF (§4.3, §4.4); there is no selection. The new statistics are the pooled bootstrap and the pooled permutation test (Sections 5.5 and 5.6):

- **Pooled gain** (v10a). For a bootstrap draw b of the source union (BH and NS resampled separately, sorted names, `default_rng(42)`), each component m computes the hard-like AUC difference on the observations of its drawn sources (with multiplicity), $\Delta_m^{(b)}$; the pooled value is $\bar{\Delta}^{(b)} = \frac{1}{3}\sum_m \Delta_m^{(b)}$ (draws where a component is undefined are dropped). The interval is the 2.5–97.5% percentile range.

Table 10.4: Arrays of Phase 5. NICER rows from `data/v9/processed/nicer_features.csv` (v9) and `data/v10/processed/nicer_full_features.csv` (v10); RXTE timing table `data/v10/processed/rxte_timing_table.csv`.

Analysis	Array	Shape	Columns / content
v9a	$\boldsymbol{X}_{H_N}$	368×2	c_1, c_2
	$\boldsymbol{X}_{H_N+T_N}$	368×5	c_1, c_2, T_1, T_2, T_3 (no missing values: ≥ 3 segments required)
	$\boldsymbol{X}_{H_N+R(+T_N)}$	368×3 ($\times 6$)	adds $\log_{10}$ rate (2–10 keV)
	$\boldsymbol{y}, \boldsymbol{g}$	368	65 BH; 63 sources; LOSO over sources, metrics on hard-like rows
v9b	per-source residuals	39 sources	$\log_{10} \nu_c$, T_k, rms after label-free kNN on standardised (c_1, c_2) ($k = 25$), hard-like rows (141–142)
v10a	three gain components	180 / 99 / 144 hard-like obs.	RXTE v2 (v5 vs v2 OOF), RXTE X (frozen v6), NICER full (LOSO, 366×5)
	union labels	86	BH/NS of each canonical source; bootstrap draws over the union
v10b	per-instrument residuals	44 (RXTE), 39 (NICER)	hard-like source-mean residuals of $\log_{10} \nu_c$; strata: RXTE only / NICER only / both

- **Pooled ν_c difference** (v10b). For instrument i, D_i = mean residual of BH sources minus mean residual of NS sources (label-free kNN residualisation on the instrument's colours, leave-one-source-out); $D = \frac{1}{2}(D_{\mathrm{RXTE}} + D_{\mathrm{NICER}})$. The null permutes labels within the three membership strata (RXTE only, NICER only, both) with the number of BHs per stratum fixed and the same label on both instruments for shared sources; 20,000 permutations, seed 42, $p = (1 + \#\{|D_\pi| \geq |D|\})/(1 + 20000)$. Synthetic calibration: null rejection rate 0.065 (200 simulations), power 0.98 for a BH shift of -1.5 SD (50 simulations).

10.6 Results

10.6.1 NICER alone: hard-state timing (v9a)

PRIMARY On NICER, colour alone gives a hard-like AUC of 0.683 – close to RXTE's 0.646 – and a soft-like AUC of 0.894; adding the timing features raises the hard-like AUC to 0.795 (Table 10.5). The difference +0.112 [−0.004, +0.229] is **not detected** under the pre-registered criterion, its lower limit just below 0. Descriptively, the RF gain (+0.236 [+0.104, +0.372]), the hard-only LR (+0.191 [+0.039, +0.329]) and the T2 and T3 bands have intervals above 0, and the state gap shrinks (difference in differences −0.125 [−0.246, −0.001]; RF −0.209 [−0.319, −0.095]). Controlling for count rate reduces the LR gain to +0.069 [−0.013, +0.161]; using only the two never-before-used BHs (MAXI J1820+070, Swift J1727.8−1613) and all NSs gives +0.074 [−0.094, +0.252]; observations brighter than $100\,\mathrm{c\,s^{-1}}$ give +0.115 [−0.010, +0.241]. At source level the hard-like AUC rises from 0.683 to 0.804 (+0.121 [−0.004, +0.254]) while the BA falls slightly (0.741 → 0.710).

Table 10.6 lists the hard-like sources. The BHs MAXI J1820+070 (0.56 → 0.74), Cyg X-1 (0.59 → 0.73), 4U 1543−47 (0.57 → 0.81), GX 339−4 (0.51 → 0.66) and Swift J1727.8−1613 (0.60 → 0.65) rise; GRS 1915+105 stays at 0.01–0.05 (in the NICER era it is in an obscured/low state whose colours and timing resemble an NS). Most NSs fall or stay, but the Rapid Burster MXB 1730−335 (0.16 → 0.92), SAX J1808.4−3658 (0.65 → 0.82) and the dipper 4U 1323−62 (0.04 → 0.45) move towards BH. POST HOC Excluding the very hard NS observations ($c_2 > 0.7$, likely background-dominated or dipping; 11 NS sources) gives +0.135 [−0.001, +0.273] for LR and

Table 10.5: PRIMARY v9a: NICER observation-level AUC by state (LOSO over 63 sources trained on all states; hard-like stratum 7 BH / 32 NS sources). From `results/v9a/v9a_metrics.csv`.

Model	soft-like AUC	hard-like AUC	hard-like difference
H_N-LR	0.894	0.683 [0.432, 0.874]	–
H_N+T_N-LR	0.881	0.795 [0.517, 0.958]	★ +0.112 [−0.004, +0.229]
H_N-RF → H_N+T_N-RF	0.909 → 0.936	0.703 → 0.939	+0.236 [+0.104, +0.372]
H_N+R-LR → H_N+R+T_N-LR	0.780 → 0.790	0.753 → 0.822	+0.069 [−0.013, +0.161]
H_N+T1-LR	0.895	0.731	+0.048 [−0.041, +0.140]
H_N+T2-LR	0.890	0.826	+0.143 [+0.028, +0.271]
H_N+T3-LR	0.827	0.798	+0.115 [+0.030, +0.217]
hard-only training: H_N → H_N+T_N	–	0.640 → 0.831	+0.191 [+0.039, +0.329]

+0.244 [+0.100, +0.385] for RF (`results/v9a/v9a_post_hoc_exclude_c2_gt_0.7.csv`).

Table 10.6: v9a: mean LOSO scores of the NICER hard-like observations per source (**bold** = wrong side of 0.5). From `results/v9a/v9a_per_source_state.csv`.

Source	label	hard-like obs.	H_N-LR	H_N+T_N-LR
4U 1543−475	BH	3	0.57	0.81
Cyg X-1	BH	7	0.59	0.73
GRS 1915+105	BH	4	**0.01**	**0.05**
GX 339−4	BH	4	0.51	0.66
LMC X-3	BH	1	0.69	0.68
MAXI J1820+070	BH	7	0.56	0.74
Swift J1727.8−1613	BH	5	0.60	0.65
1A 1744−361	NS	1	0.45	**0.55**
2S 0918−549	NS	1	**0.63**	**0.66**
4U 0614+09	NS	2	**0.63**	**0.61**
4U 1246−588	NS	4	0.22	0.15
4U 1323−62	NS	2	0.04	0.45
4U 1543−624	NS	7	**0.59**	**0.62**
4U 1608−522	NS	3	**0.55**	**0.51**
4U 1636−536	NS	6	**0.51**	**0.50**
4U 1702−429	NS	6	0.37	0.40
4U 1722−30	NS	6	0.49	0.46
4U 1728−34	NS	3	0.24	0.18
4U 1812−12	NS	8	**0.53**	**0.59**
4U 1820−303	NS	2	0.26	0.21
4U 1850−086	NS	1	**0.66**	**0.69**
Aql X-1	NS	2	**0.61**	**0.56**
Cen X-4	NS	1	0.00	0.00
EXO 0748−676	NS	4	0.08	0.01
GS 1826−24	NS	2	**0.57**	0.48
GX 17+2	NS	1	0.29	0.19
IGR J17498−2921	NS	4	0.28	0.25
IGR J17591−2342	NS	5	0.34	0.36
MAXI J1807+132	NS	4	**0.62**	**0.62**
MXB 1730−335	NS	3	0.16	**0.92**
SAX J1750.8−2900	NS	5	0.27	0.21
SAX J1808.4−3658	NS	3	**0.65**	**0.82**
SAX J1810.8−2609	NS	7	**0.55**	**0.55**
SLX 1732−304	NS	5	0.48	**0.58**
Swift J1922.7−1716	NS	1	0.02	0.00

Source	label	hard-like obs.	H_N-LR	H_N+T_N-LR
XB 2129+47	NS	1	0.04	0.00
XTE J1709−267	NS	3	0.13	0.01
XTE J1739−285	NS	2	0.35	0.27
XTE J1810−189	NS	6	0.25	0.29

10.6.2 NICER alone: the characteristic frequency (v9b)

PRIMARY At fixed colour, NICER BHs have $\log_{10}\nu_c$ lower by $D = -0.310$ (a factor ≈ 2), the same size as on the RXTE v2 sources (−0.31, v6d), but with 7 BH sources the permutation p is 0.136 (Holm over five features 0.544): **not detected** under the pre-registered criterion (Table 10.7). POST HOC Restricting to the colour window occupied by the hard-like BHs ($c_2 \approx 0.11$–0.40) and comparing sources directly, the median ν_c is 0.90 Hz for 7 BHs and 4.25 Hz for 25 NSs (Mann–Whitney two-sided $p = 0.013$, AUC 0.81); exceptions are LMC X-3 (one hard-like observation, 11.7 Hz) and the NSs MXB 1730−335 (0.07 Hz) and XTE J1739−285 (0.8 Hz). The pre-declared label-free residualisation is conservative when BHs cluster in colour (they become each other's neighbours).

Table 10.7: PRIMARY v9b: colour-conditional source permutation test on the NICER hard-like observations (39 sources, 7 BH; 20,000 permutations; Holm over the five features). From `results/v9b/v9b_permutation_test.csv`.

Feature	D (BH − NS)	null SD	p	p_{Holm}	role
$\log_{10}\nu_c$	−0.310	0.207	0.136	0.544	★
T1 (0.016–0.094 Hz)	+0.017	0.136	0.902	0.928	
T2 (0.10–1.0 Hz)	+0.072	0.104	0.425	0.928	
T3 (1.0–4.0 Hz)	+0.068	0.033	0.038	0.188	
0.1–10 Hz rms	+0.102	0.108	0.309	0.928	

Figure 10.2 shows the NICER hard-like observations in the timing and ν_c planes.

Figure 10.3 reproduces the repository's v9 summary figure.

10.6.3 Pooled: the hard-state timing gain (v10a)

PRIMARY Table 10.8 shows the three components and the pooled value. The equal-weight mean is +0.145 [+0.028, +0.257]: **detected** (95% CI excludes 0). Weighting by the number of BH sources (+0.154 [+0.041, +0.260]), using RF (+0.224 [+0.108, +0.333] from the RXTE v2 and NICER components) and removing the 13 background-suspect NICER observations (+0.152 [+0.036, +0.263]) agree. Gate IC2a passed: with only the RXTE v2 component the pooled bootstrap reproduces v8a exactly (+0.2643 [+0.0484, +0.4549]). Leaving out one BH at a time (Figure 10.4) the pooled gain ranges from +0.110 (without XTE J1118+480, which v5 moved from 0.41 to 0.94) and +0.118 (without GS 1354−64) to +0.158 (without Cyg X-1) (`results/v10a/v10a_influence.csv`). The NICER component with full observations (+0.107) is nearly the same as with prefixes in v9 (+0.112) and on its own still touches 0.

10.6.4 Pooled: the characteristic frequency (v10b)

PRIMARY Table 10.9 gives the pooled colour-conditional difference: $D = -0.368$ dex (BH ν_c lower by a factor $10^{0.368} \approx 2.3$), $p = 0.0012$ – **detected** (95% CI excludes 0). Each instrument alone

Table 10.8: PRIMARY v10a: hard-like observation-level AUC gain from timing (H+T-LR − H-LR) per component and pooled over the union of 86 sources (16 BH). From `results/v10a/v10a_pooled_gain.csv`.

Component	hard-like obs.	BH / NS sources	gain
RXTE v2 (v8a)	180	7 / 17	+0.264 [+0.044, +0.469]
RXTE external X (v8b, frozen)	99	5 / 15	+0.064 [−0.171, +0.341]
NICER full observations	144	7 / 32	+0.107 [−0.010, +0.224]
★ pooled, equal weights		16 BH in the union	+0.145 [+0.028, +0.257]
pooled, weighted by BH sources			+0.154 [+0.041, +0.260]
RF, RXTE v2 + NICER			+0.224 [+0.108, +0.333]
NICER component, background-screened (13 obs. removed)			+0.127 [+0.016, +0.242]
pooled with screened NICER			+0.152 [+0.036, +0.263]

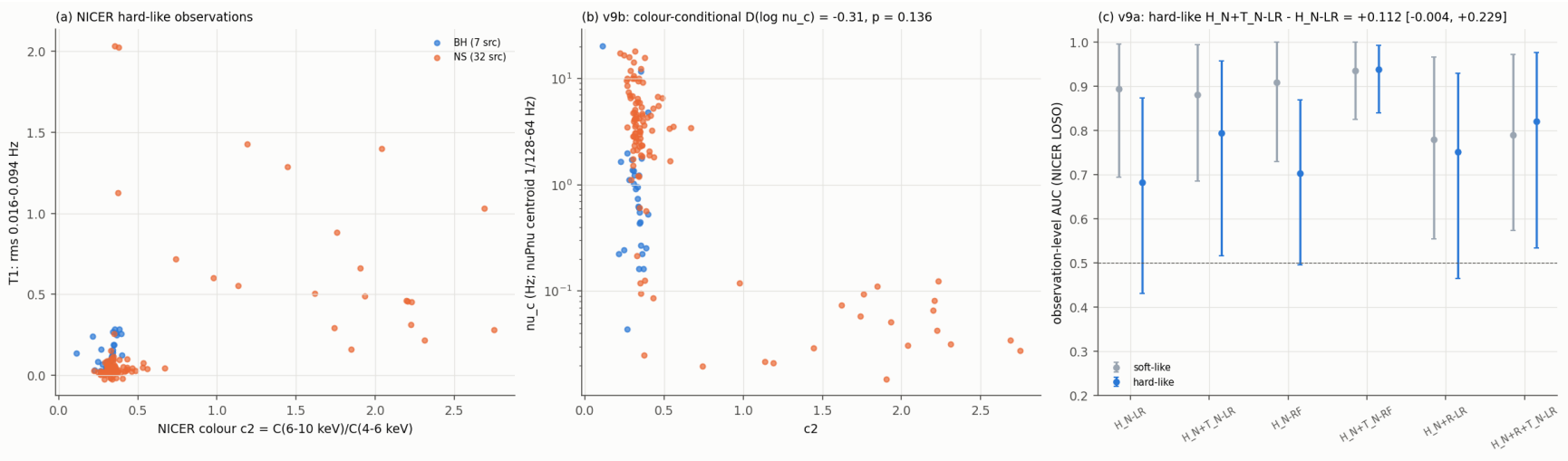


Figure 10.2: NICER hard-like observations (BH blue, 7 sources; NS orange, 32 sources): (a) T1 rms against the colour $c_2 = C(6\text{–}10)/C(4\text{–}6)$; (b) ν_c against c_2 – at $c_2 \approx 0.2$–0.4 the BHs lie at lower ν_c, and the very hard NS observations ($c_2 > 0.7$) have very low ν_c; (c) observation-level AUC by state for the v9a models. Repo: `figures/v9/v9_nicer.png`, `scripts/49_v9_analysis.py`.

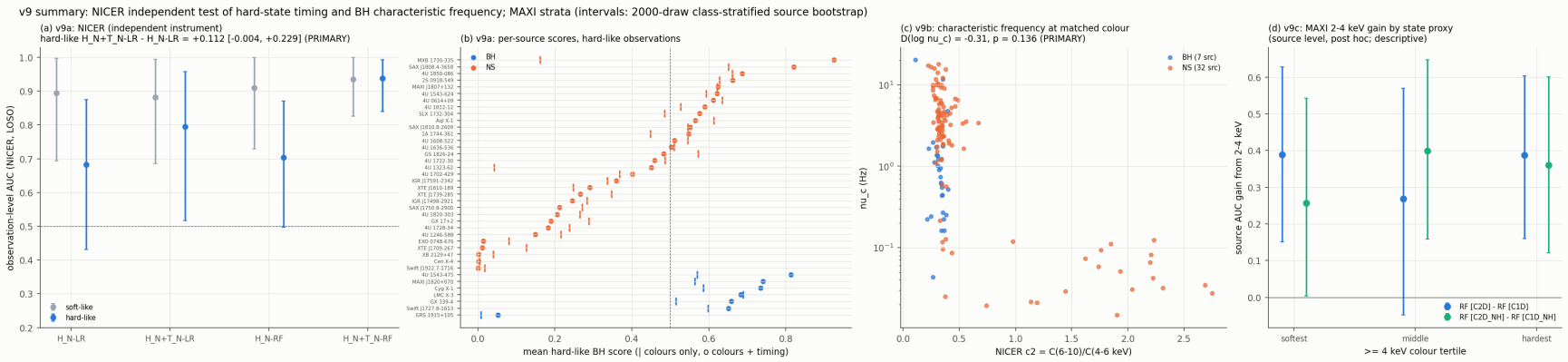


Figure 10.3: v9 summary: (a) NICER AUC by state; (b) per-source mean hard-like scores, colour only (bars) and with timing (circles); (c) ν_c against c_2; (d) MAXI 2–4 keV source-level gain by C4 tertile (v9c, post hoc; Table 9.21). Repo: `figures/v9/v9_summary.png`, `scripts/51_v9_summary_figure.py`.

is negative (RXTE −0.265, $p = 0.0018$; NICER full −0.471, $p = 0.033$; with full observations NICER moved from $p = 0.136$ to 0.033). Gate IC2b passed: the single-instrument D values equal those of `v6lib.perm_test`. Leaving out one BH at a time, D ranges from −0.329 (without GRS 1915+105) to −0.420 (without 4U 1543−47). The pre-declared descriptive colour-window version gives −0.381, $p = 0.015$.

Table 10.9: PRIMARY v10b: colour-conditional $\log_{10}\nu_c$ difference (BH − NS, hard-like observations) per instrument and pooled; permutation within instrument-membership strata. From `results/v10b/v10b_pooled_nuc.csv`.

Analysis	sources (BH)	D [dex]	p
RXTE hard-like (v2 + X; Standard-1 ν_c, 1/128–4 Hz)	44 (12)	−0.265	0.0018
NICER hard-like (full observations, 1/128–64 Hz)	39 (7)	−0.471	0.033
★ pooled (mean of instruments, joint permutation)	64 (15)	−0.368	0.0012
colour window, RXTE (descriptive)	44 (12)	−0.368	0.0001
colour window, NICER (descriptive)	36 (7)	−0.394	0.182
colour window, pooled (descriptive)	61 (15)	−0.381	0.015

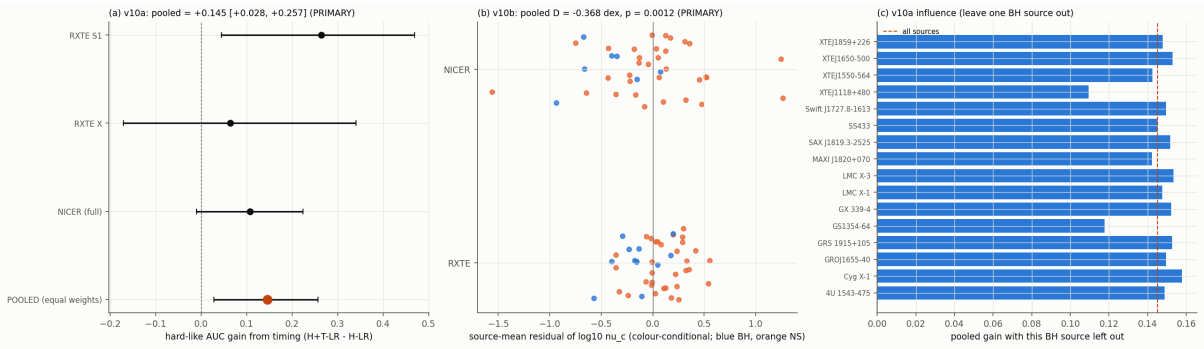


Figure 10.4: v10 pooled tests: (a) hard-like timing gain per component and pooled (red); (b) source-mean residuals of $\log_{10}\nu_c$ at fixed colour per instrument (BH blue, NS orange); (c) pooled gain with each BH source left out (dashed: all sources). Repo: `figures/v10/v10_pooled.png`, `scripts/54_v10_pooled.py`.

10.7 Interpretation

What the data support directly.

1. On an independent instrument and epoch the colour-only hard-like AUC is 0.68 (RXTE 0.65) and the soft/hard gap reappears; timing raises the hard-like AUC by +0.11 (LR) to +0.24 (RF), but the pre-declared LR interval touches 0.
2. At fixed colour the NICER BHs have ν_c lower by ≈ 2 (same size as on RXTE), with $p = 0.136$ in v9 and $p = 0.033$ with full observations.
3. Pooling all usable sources of both instruments (86, 16 BH), both effects are detected – hard-state gain +0.145 [+0.028, +0.257], $\Delta\log_{10}\nu_c = -0.37$ dex, $p = 0.0012$ – and neither depends on any single BH.
4. RF gains more than LR, so timing and colour interact non-linearly.
5. NICER timing features are reliable (split-half 0.8–0.9).

What the data do not support. The pooled results are not an independent confirmation: both hypotheses and the pooling rule were written after v5–v9, and the RXTE v2, RXTE external and NICER prefix results were known; only the NICER full-observation measurement was new (on the same observations). The 16 BHs are the only dynamically confirmed BHs in the analyses.

Possible explanations (not verified). A lower characteristic frequency fits the picture of time scales growing with the compact-object mass or inner disc radius (BHs $\approx$ 5–20 $M_\odot$, NSs $\approx 1.4\,M_\odot$), but the factor ≈ 2.3 is smaller than the mass ratio and may also reflect accretion rate/luminosity, which is not controlled (no Eddington ratios). On NICER the information sits in 0.1–4 Hz (T2, T3), on RXTE in 0.016–1 Hz (T1, T2), plausibly because NICER prefixes are short (T1 noisy) and its band softer. GRS 1915+105's obscured NICER-era state is the main BH error. Some NS "hard-like" NICER observations may be dips, eclipses or background-dominated. The external RXTE component is smallest, perhaps because the external NSs are mostly AMXPs with strong low-frequency noise (IGR J00291+5934).

Hypotheses that need more data. A genuinely independent, prospective test with frozen features and rules on sources and instruments never used (Insight-HXMT, AstroSat, future NICER transients) – this is Phase 6; control of the Eddington ratio; Lorentzian fits instead of the νP_ν centroid; a usable rule such as "hard-like and $\nu_c < 1$ Hz $\Rightarrow$ BH-like".

10.8 Deviations and caveats

- **Written after earlier results, self-approved**; v10 explicitly not an independent confirmation.
- **v9 download** (before any model): HEASARC AWS mirror instead of the main site (byte-identical), curl range requests; raw files outside the repository (14 GB).
- **v9 post hoc**: colour-window ν_c comparison, the source-level v9c version and the $c_2 > 0.7$ exclusion were decided after the results.
- **v10 permutation** (before any v10b statistic): union-wide permutation replaced by permutation within instrument-membership strata, motivated by a synthetic power test; a bug with single-instrument strata (object array of equal-length tuples) was fixed without changing the statistics; scatter jitter seeded by `crc32`.
- **v10 resources**: feature parallelism reduced from 4 to 2 workers (OneDrive used $\approx$ 10 GB RAM); results unaffected.
- **Sample sizes**: 7 BH sources in each instrument's hard-like stratum (GRS 1915+105 anomalous, LMC X-3 one observation); only two BHs had never been used before.
- **Measurement**: NICER prefixes in v9 (T1 noisy); no NICER background subtraction (only a 20 c s^{-1} cut and a background sensitivity analysis); 12 bright observations truncated at 1 GB; RXTE and NICER bands and ν_c ranges differ, so their D values are not strictly comparable; the two ν_c definitions differ in their bin weights (see the CHECK in the data section).
- **Bootstrap and permutation** keep the predictions and features fixed.

10.9 Takeaway

NICER repeats both timing effects with the same size as RXTE, each just short of the pre-declared threshold on its own; pooled over 86 sources and 16 dynamically confirmed BHs, both are detected and robust to dropping any BH. Because the hypotheses and the pooling were chosen after the component results, Phase 5 measures the overall strength of the evidence rather than confirming it. Phase 6 therefore freezes a rule and tests it prospectively on new observations.

Chapter 11

Phase 6 (v11–v13): prospective tests on new observations

Phase at a glance: v11–v13

Question. Do the two timing claims survive a *prospective* test – pipeline, thresholds, models and decision rule frozen and committed before any new data are seen – on NICER observations never downloaded before? And are they an artefact of the (so far uncorrected) NICER background?
Pre-declared primaries. v11a: frozen out-of-source H_N+T_N-LR − H_N-LR on new hard-like observations; v11b: colour-conditional $\log_{10}\nu_c$ difference on the new observations; v12a: the same after 3C50 background correction and screening of the v10+v11 observations; v12b: RF hard-state gain after correction (RF chosen *post hoc*); v13a: frozen RF hard-state gain on a third, disjoint set of new observations (RF chosen *before* download); v13b: ν_c difference on that set.
Data. The same 66 NICER sources; 318 new observations in v11 (39.2 GB streamed), 3C50 for 503 faint observations in v12, 238 new observations in v13 (198 after correction and screening).
Verdict. The **RF hard-state timing gain** is detected after background correction (v12b +0.174 [+0.071, +0.300]) and **prospectively** on new observations with a pre-chosen RF (v13a +0.184 [+0.049, +0.337]); the LR gain is not detected in v11 (+0.066). The ν_c **difference** is detected in v11 (−0.663, $p = 0.023$) but depends on a few very hard faint observations, and is not detected after correction (v12a −0.514, $p = 0.073$) or prospectively (v13b −0.427, $p = 0.21$); its size is stable at −0.4 to −0.6 dex. These are **observation-level** prospective tests on the *same sources*, not new sources; only 16 dynamically confirmed BHs exist in the whole project.

11.1 Motivation

Phase 5 measured the overall strength of the evidence but could not confirm it, because the hypotheses and pooling followed the component results. The cleanest remaining test is prospective: freeze everything, then look at data never seen. New dynamically confirmed BHs were exhausted (16 of the 25 BHs of Marcel et al. (2026) had been used across RXTE and NICER), and batch access to Insight-HXMT or AstroSat data could not be verified, so the maximum independence available was *new observations of the same NICER sources*. v11 did this; its ν_c result raised the suspicion of background-dominated faint observations, so v12 applied

the official 3C50 background model (Remillard et al., 2022); v13 repeated the prospective test with the v12 pipeline and a model (RF) chosen before the download. v14 then stopped the series (Chapter 12).

Figure 11.1 maps the three versions.

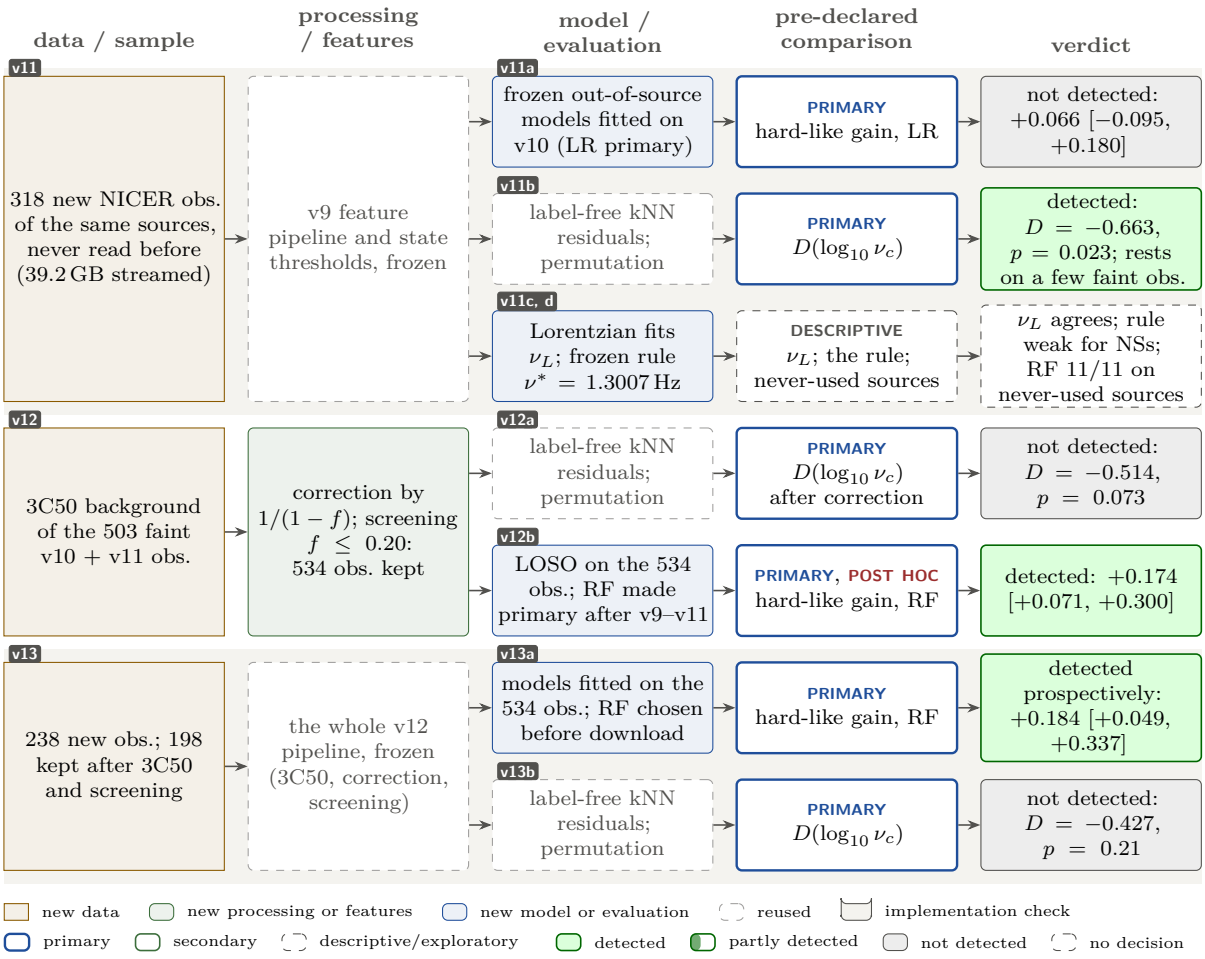


Figure 11.1: Phase 6 analysis map (reading as in Figure 6.1). Every test uses new observations of the *same* sources (observation-level independence, not new sources). v11 freezes the v9 pipeline and the v10 models; v12 adds the 3C50 background correction and screening; v13 freezes the whole v12 pipeline. **RF was made the primary model post hoc in v12 and was pre-specified in v13.** Values from Tables 11.3 to 11.5. Drawn for this report.

11.2 Pre-registration

Table 11.1 summarises the three pre-registrations.

The frozen rule (Section 4.13) is $\nu^* = \sqrt{\tilde{\nu}_{\rm BH}\,\tilde{\nu}_{\rm NS}}$, the geometric mean of the BH and NS median ν_c of the hard-like observations of the reference data: v10 NICER full observations ($\tilde{\nu}_{\rm BH} = 0.5917$ Hz, 32 observations; $\tilde{\nu}_{\rm NS} = 2.8594$ Hz, 111) $\Rightarrow \nu^* = 1.3007$ Hz for v11; v12 corrected and screened data (0.6294 Hz, 62; 2.8362 Hz, 138) $\Rightarrow \nu^* = 1.3361$ Hz for v13. Observation rule: hard-like and $\nu_c < \nu^* \Rightarrow$ BH; source rule: BH if at least half of its new hard-like observations are BH.

Table 11.1: Phase 6 pre-registrations (`reports/preregistration_v11.md`, `_v12.md`, `_v13.md`), all written and self-approved after all earlier results (user instructions: "merge and continue v11 at 1:40 am", "merge and continue", and, for v13 onwards, in Chinese, "when it finishes continue to the next step (v13, 14, ...) until it is worth wrapping up, then wrap up").

Ver.	Written	Primary	Frozen before new data
v11	2026-10-06 (≈01:50), after v1–v10; commit `6fea016`	v11a: hard-like obs. AUC, frozen out-of-source H_N+T_N-LR − H_N-LR; v11b: colour-conditional $D(\log_{10}\nu_c)$	v9 feature pipeline and state thresholds; v10 fold models (each source predicted by the model trained without it); rule $\nu^* = 1.3007$ Hz committed in `58ded89`
v12	2026-10-06 (≈03:20), after v1–v11; commit `2126e9d`	v12a: $D(\log_{10}\nu_c)$ after 3C50 correction and screening ($f \leq 0.20$); v12b: RF hard-like gain after correction (POST HOC choice of RF, after v9–v11)	3C50 settings, correction formulae, screening threshold
v13	2026-10-06 (≈05:00), after v1–v12; commit `42fe1d2`	v13a: frozen out-of-source RF hard-like gain on a third set of new observations; v13b: $D(\log_{10}\nu_c)$	the full v12 pipeline (incl. 3C50 and screening); models fitted on the 534 corrected v12 observations; RF chosen before download; $\nu^* = 1.3361$ Hz

11.3 Data

11.3.1 New NICER observations (v11, v13)

v11. For each of the 66 v9 sources, eligible observations not among the 792 tried in v9 were split into 8 time bins and tried in a random order (`default_rng(SEED + 1000 + position)`, up to 3 per bin); sources never used before with only 1–4 eligible observations contributed all of them (7 NS and XTE J1859+226; 19 observations). 422 source–bin combinations, 588 tries, **318 accepted**; 39.2 GB were streamed (raw bytes not stored; URLs and SHA256 in `logs/downloads.jsonl`). New hard-like observations: 33 of 9 BH sources (Cyg X-1 8, MAXI J1820+070 6, Swift J1727.8−1613 5, GRS 1915+105 4, 4U 1543−47 3, GX 339−4 3, LMC X-1 2, LMC X-3 1, V4641 Sgr 1) and 110 of 33 NS sources. Of the never-used sources, 3 NSs have hard-like observations (7); XTE J1859+226 had none that qualified. Gates: IC1 LOSO refit on the v10 data reproduces the v10 OOF scores (maximum difference $\leq 1.6 \times 10^{-15}$); IC2 Z sources soft-like 15/16; IC3 no overlap with the v9 obsids; IC4 synthetic Lorentzians of 0.3 and 5 Hz recovered as 0.302 and 5.10 Hz (two-component low frequency 0.101); streamed parsing equals the v10 prefix+tail parsing (6403520101, 729,479 events).

v13. Eligible observations of the v9 sources not tried in v9 (792) or v11 (588), same scheme (`default_rng(SEED + 2000 + position)`): 291 groups, **238 accepted**; 3C50 for the 148 faint ones (109 ok, 39 failed: 38 old gain calibration, 1 other; median $f = 0.006$, one with $f > 0.20$); 198 kept after correction and screening. Hard-like: 29 observations of 8 BH sources, 45 of 19 NS sources (74). Gates: Z sources 15/16 soft-like; disjoint from v9 and v11.

11.3.2 Background: 3C50 (v12)

The 3C50 model (Remillard et al., 2022) was run with HEASoft 6.37.1 `nibackgen3C50` (CALDB background library and index, `gainepoch=AUTO`, default `hbgcut`=0.5, `s0cut`=30; CALDB tarball 112,180,364 bytes, SHA256 `572954c8...`). A feasibility run on 3657027601 gave a 3C50 total rate of 58.0 c s^{-1} against 61.3 from the pipeline and a background of 0.67 c s^{-1}. For the 503 observations of v10+v11 with 2–10 keV < 500 c s^{-1} the cleaned and unfiltered event files were downloaded, processed and deleted (spectra kept outside the repository): 362 succeeded, 141

failed – 139 because `nibackgen3C50` rejects data processed with the original gain calibration (`nixtiflightpi20170601v001`; recorded as return code 255, missing total spectrum, in `data/v12/processed/bkg3c50.csv`) and 2 with return code 218. The background fraction f (2–10 keV) has median 0.008, 90th percentile 0.026 and 99th percentile 0.37; 9 observations exceed 0.20 (Figure 11.3a). Gates: IC1 3C50/pipeline total-rate ratio within [0.7, 1.3] for 93.4% (median 0.961); IC2 all $f \in [0, 1]$; IC3 Z sources 15/16 soft-like after correction.

Correction (Section 2.10.5): fractional rms features and the state rms are multiplied by $1/(1 - f_T)$; colours are recomputed from the band counts scaled by $(1 - f_A)$, $(1 - f_B)$, $(1 - f_C)$ (PI 200–400, 400–600, 600–1000); ν_c is not corrected; bright observations ($\geq 500\,\mathrm{c\,s^{-1}}$) get $f = 0$. **Screening**: $f > 0.20$ and 3C50 failures are excluded, leaving **534 of 684** observations; 3 change state.

11.4 Data structure in this phase

Table 11.2 lists the arrays.

Table 11.2: Arrays of Phase 6 (NICER features as in Table 10.4: $H_N = [c_1, c_2]$, $T_N = [T_1, T_2, T_3]$).

Analysis	Array	Shape	Content
v11	training (frozen)	$(366 - n_s) \times \{2, 5\}$	v10 NICER full observations without source s; models LR, RF on H_N, $H_N + T_N$
	test	$318 \times \{2, 5\}$	new observations (`data/v11/processed/nicer_new_features.csv`); never-used sources predicted by the model fitted on all 366
	ν_c, Lorentz fits	142 hard-like	$\log_{10} \nu_c$; ν_L from 13 octave bins (`results/v11/v11c_lorentz_fits.csv`)
v12	corrected features	$534 \times \{2, 5\}$	v10+v11 after 3C50 correction and screening (`data/v12/processed/nicer_corrected_screened.csv`); LOSO
	f_A, f_B, f_C, f_T	503	3C50 background fractions (`data/v12/processed/bkg3c50.csv`)
v13	training (frozen)	$(534 - n_s) \times \{2, 5\}$	v12 corrected data without source s
	test	$198 \times \{2, 5\}$	new, corrected, screened (`data/v13/processed/nicer_new_features.csv`); 74 hard-like

11.5 Models in this phase

LR and RF with the fixed v2 settings (§4.3, §4.4), always *out-of-source*: a new observation of source s is scored by the reference-data model trained without s, so no source contributes to the model that scores it. The permutation test, the Lorentzian fits (Eq. (2.22): one or two zero-centred Lorentzians integrated over each octave bin, weighted least squares on log-parameters with multiple starts, the second component kept if $\mathrm{BIC}_1 - \mathrm{BIC}_2 > 6$, ν_L = lowest width) and the frozen ν^* rule are defined in Chapters 2, 4 and 5.

11.6 Results

11.6.1 Hard-state timing gain on new observations

PRIMARY **v11a.** On the 143 new hard-like observations (9 BH / 33 NS sources) the frozen LR gain is +0.066 [−0.095, +0.180] (0.660 → 0.726): **not detected** under the pre-registered criterion (Table 11.3). The LR gain of v8a, v9a and v10a is therefore *not* prospectively confirmed. Leaving out one BH at a time gives +0.044 to +0.110 (`results/v11/v11a_influence.csv`); at source level the LR AUC falls slightly (0.724 → 0.710). The RF gain, descriptive in v11, is +0.247 [+0.128, +0.384].

PRIMARY **v12b.** After 3C50 correction and screening (LOSO on the 534 observations), the RF gain is +0.174 [+0.071, +0.300] (0.750 → 0.924): **detected** (95% CI excludes 0); RF was chosen as primary after seeing v9–v11 (POST HOC choice). LR: +0.090 [−0.056, +0.243]. Source-level RF AUC 0.853 → 0.926 (+0.074 [−0.033, +0.206]).

PRIMARY **v13a.** With everything frozen before the download – the v12 pipeline including 3C50, the models fitted on the 534 corrected observations, and the choice of RF – the gain on the 74 new hard-like observations (8 BH / 19 NS sources) is +0.184 [+0.049, +0.337] (0.680 → 0.864): **detected** (95% CI excludes 0) *prospectively*. LR is +0.182 [+0.022, +0.347] this time; leaving out one BH gives RF gains of +0.146 to +0.212 (`results/v13/v13a_influence.csv`); recomputing the faint observations' features on the 3C50-filtered time intervals gives +0.190 [+0.065, +0.330]; source-level RF AUC 0.717 → 0.842 (+0.125 [−0.053, +0.303]).

Table 11.3: Hard-like observation-level AUC with and without timing on new or corrected NICER observations (frozen out-of-source models in v11 and v13; LOSO in v12). From `results/v11/v11a_metrics.csv`, `results/v12/v12b_gain.csv`, `results/v13/v13a_gain.csv`.

Set	Model	soft-like AUC	hard-like AUC	hard-like gain
v11 new (143 hard-like)	LR	0.826 → 0.801	0.660 → 0.726	★ +0.066 [−0.095, +0.180]
	RF	0.896 → 0.899	0.665 → 0.912	+0.247 [+0.128, +0.384]
v12 corrected (534 obs.)	RF†	0.949 → 0.945	0.750 → 0.924	★ +0.174 [+0.071, +0.300]
	LR	0.871 → 0.861	0.641 → 0.730	+0.090 [−0.056, +0.243]
v13 new (74 hard-like)	RF	0.938 → 0.943	0.680 → 0.864	★ +0.184 [+0.049, +0.337]
	LR	0.870 → 0.843	0.503 → 0.685	+0.182 [+0.022, +0.347]
	RF, 3C50 GTIs	–	–	+0.190 [+0.065, +0.330]

† RF chosen as the v12 primary after seeing v9–v11 (post hoc choice).

Across the NICER analyses the RF gain is remarkably stable: +0.236 (v9 prefixes), +0.219 (v10 full), +0.247 (v11 new), +0.174 (v12 corrected), +0.184 (v13 new); the LR gain is smaller and less stable (+0.112, +0.107, +0.066, +0.090, +0.182).

11.6.2 Characteristic frequency on new observations

PRIMARY **v11b.** On the new hard-like observations (42 sources, 9 BH) the colour-conditional difference is $D = -0.663$, $p = 0.0225$: **detected** ($p < 0.05$) (Table 11.4). Adding the log count rate as a brightness control gives −0.709 ($p = 0.018$; on the v10 data −0.459, $p = 0.046$); the lowest Lorentzian width ν_L gives the same picture (median BH 0.148 Hz, NS 0.618 Hz; $D = -0.670$, $p = 0.023$; 142 fits, 58% with two components, 12 at the lower bound 1/128 Hz). Leaving out one BH gives −0.51 (without LMC X-1) to −0.74. POST HOC However, the new "hard-like" observations of LMC X-1, LMC X-3 and V4641 Sgr are very hard (c_2 0.5–1.0) with very low ν_c (0.02–0.09 Hz), like background-dominated faint observations; excluding observations above the frozen v10 background-proxy threshold gives −0.604 ($p = 0.058$), and additionally

excluding $c_2 > 0.7$ gives -0.458 ($p = 0.19$, 7 BHs) (`results/v11/v11b_post_hoc_screens.csv`).

PRIMARY **v12a.** After 3C50 correction and screening, $D = -0.514$, $p = 0.073$: **not detected** under the pre-registered criterion. Across every background treatment the effect size is -0.50 to -0.57 dex and only the sample size changes p (Table 11.4); the 141 observations that 3C50 could not process (2017–2019, including early LMC X-1, LMC X-3 and V4641 Sgr data) are the main loss. The very hard observations suspected in v11 are mostly not background-dominated by 3C50: of the 45 hard-like observations with $c_2 > 0.7$, the 15 with a 3C50 result have median $f = 0.125$ (0.008–0.91); the NSs EXO 0748−676, 4U 1246−588 and 4U 1323−62 have $f > 0.2$, while the BH observations (LMC X-3 two, GRS 1915+105 one) have f of 1–2% (`results/v12/v12_very_hard_obs.csv`). POST HOC 3C50 filters high-background intervals for its own spectra, whereas the timing features use all GTIs, so f may underestimate the background in the timing data – the motivation for the GTI-restricted check in v13.

PRIMARY **v13b.** Prospectively, $D = -0.427$, $p = 0.209$ (27 sources, 8 BH): **not detected** under the pre-registered criterion; without screening or correction -0.520 ($p = 0.081$); with features recomputed on the 3C50 intervals -0.300 ($p = 0.41$); leaving out one BH -0.31 to -0.50.

Table 11.4: Colour-conditional $\log_{10}\nu_c$ difference (BH − NS) on hard-like NICER observations. From `results/v11/v11b_nuc.csv`, `.../v11b_post_hoc_screens.csv`, `results/v12/v12a_nuc.csv`, `results/v13/v13b_nuc.csv`.

Analysis	obs.	sources (BH)	D [dex]	p
★ v11b new observations	142	42 (9)	−0.663	0.0225
+ log count rate	142	42 (9)	−0.709	0.018
v11c lowest Lorentzian width ν_L	142	42 (9)	−0.670	0.023
post hoc: background proxy ≤ 0.2687	134	38 (9)	−0.604	0.058
post hoc: + $c_2 \leq 0.7$	122	34 (7)	−0.458	0.19
★ v12a corrected + screened ($f \leq 0.20$), v10+v11	200	42 (8)	−0.514	0.073
$f \leq 0.10$ / $f \leq 0.30$	196 / 203	42 (8)	−0.509 / −0.497	0.079 / 0.086
screened only, no correction	199	42 (8)	−0.523	0.075
no screening, no correction	292	51 (9)	−0.564	0.021
3C50 failures included uncorrected	288	51 (9)	−0.571	0.022
v10 / v11 observations only	99 / 101	32 (7) / 35 (7)	−0.515 / −0.472	0.036 / 0.24
pooled with RXTE ($f \leq 0.20$)	294	55 (15)	−0.310	0.029
★ v13b new observations, frozen v12 pipeline	74	27 (8)	−0.427	0.209
no screening, no correction	91	31 (8)	−0.520	0.081
3C50-GTI-restricted features	73	26 (7)	−0.300	0.41

11.6.3 The frozen rule and never-used sources

DESCRIPTIVE The simple rule "hard-like and $\nu_c < \nu^* \Rightarrow$ BH" (Table 11.5) finds BH sources reliably but NSs only about half to two thirds of the time: it is not usable as a classifier on its own. On the never-used sources of v11 (IGR J17511−3057, SAX J1748.9−2021, SLX 1735−269, SAX J1712.6−3739, XB 1916−053; 11 observations with ν_c), the frozen RF scores every observation below 0.5 (all correct), the frozen LR puts 8 of 11 in 0.51–0.60 (wrong: a threshold drift between epochs), and the rule calls one observation (IGR J17511−3057, 1.16 Hz) BH (`results/v11/v11d_never_used_sources.csv`).

Figure 11.2 summarises v11, Figure 11.3 the background correction of v12 and Figure 11.4 the prospective v13 test.

Table 11.5: Frozen ν^* rule on new observations (`results/v11/v11_frozen_rule.csv`, `results/v13/v13_frozen_rule.csv`).

Set	ν^* [Hz]	obs. sensitivity	obs. specificity	BH sources	NS sources	source specificity
v11	1.3007	0.79 (33 BH obs.)	0.71 (110 NS obs.)	9/9	18/33	0.55
v13	1.3361	0.79	0.76	7/8	13/19	0.68

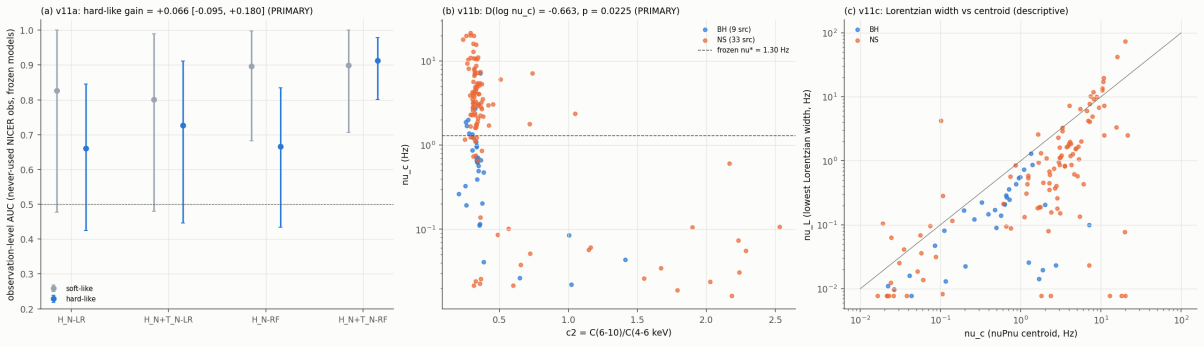


Figure 11.2: v11 on new observations: (a) observation-level AUC by state of the frozen models; (b) ν_c against c_2 for the new hard-like observations (BH blue, 9 sources; NS orange, 33) with the frozen $\nu^* = 1.30$ Hz; note the very hard ($c_2 > 0.5$), very low-ν_c points discussed in the text; (c) lowest Lorentzian width ν_L against ν_c (v11c). Repo: `figures/v11/v11_prospective.png`, `scripts/57_v11_analysis.py`.

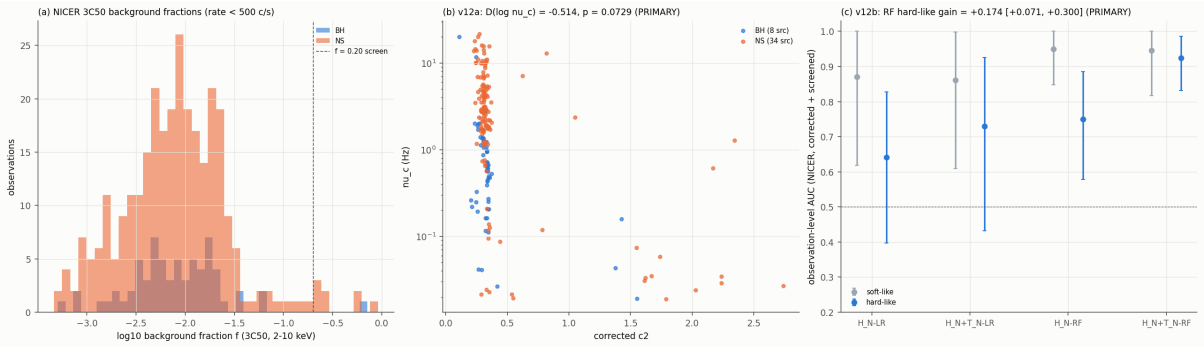


Figure 11.3: v12: (a) 3C50 background fractions of the faint ($< 500\,\mathrm{c\,s^{-1}}$) observations with the $f = 0.20$ screen; (b) ν_c against the corrected c_2 (8 BH, 34 NS sources); (c) AUC by state after correction and screening. Repo: `figures/v12/v12_background.png`, `scripts/59_v12_analysis.py`.

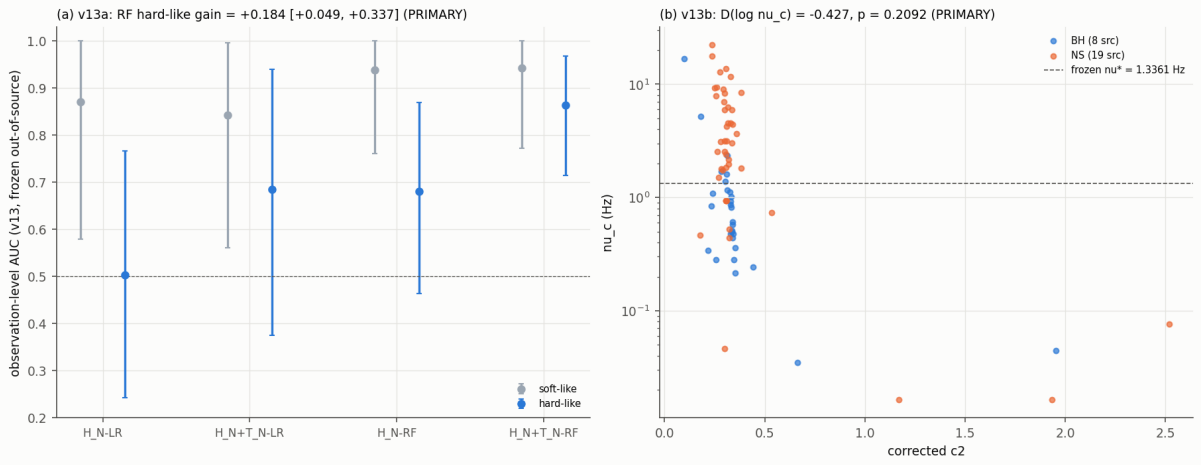


Figure 11.4: v13 on a third set of new observations with the frozen v12 pipeline: (a) AUC by state of the frozen out-of-source models; (b) ν_c against the corrected c_2 with $\nu^* = 1.3361\,\mathrm{Hz}$ – at $c_2 \approx 0.3$ the BHs lie mostly below the NSs. Repo: `figures/v13/v13_prospective.png`, `scripts/63_v13_analysis.py`.

11.7 Interpretation

What the data support directly.

1. With pipeline, background treatment, models and model type frozen and the observations new, timing raises the hard-state BH/NS separation by ≈ 0.18 in observation-level AUC with an interval excluding 0 (v13a). The RF gain has been stable across five NICER analyses (+0.17 to +0.25).
2. The LR gain was not confirmed prospectively in v11; it was detected in v13. LR's calibration drifts between epochs (never-used NSs at 0.51–0.60), RF's does not.
3. The colour-conditional ν_c difference keeps its sign and size (-0.3 to -0.7 dex) in every NICER analysis, but it is detected only once (v11, where it depends on a few very hard faint observations) and not after background correction (v12) or prospectively (v13).
4. NICER backgrounds are mostly small (median $f \approx 0.8\%$); background correction does not change the size of either effect.
5. The frozen ν^* rule identifies BH sources (16/17 over v11 and v13) but has poor NS specificity.

Possible explanations (not verified). The timing information acts non-linearly with colour (e.g. strong low-frequency variability is BH-like only in a certain colour range), which RF captures and LR captures unstably. The label-free colour residualisation is conservative when BHs cluster in colour; with 8 BH sources the ν_c test has little power. The RF gain may use combinations of T_1–T_3 beyond ν_c.

What would be needed. New BH sources – future transients or other missions – tested with the same frozen procedure; that would be a source-level confirmation, which this project could not make.

11.8 Deviations and caveats

- **Written after earlier results, self-approved**; in v12 the choice of RF as primary was made after v9–v11.

- **Same sources**: the prospective tests are independent at the observation level only; new and old observations of a source may belong to the same outburst.
- **v11**: observation-level parallelism reduced from 3 to 2 (memory), six concurrent 64-MB range requests per file; ν^* computed and committed before any new data, as pre-registered; the background/very-hard sensitivity analysis of v11b is post hoc and does not change the verdict; raw bytes not stored (re-downloadable with the logged URLs and SHA256).
- **v12**: 3C50 cannot process data with the original gain calibration; a 2018 gain-epoch fallback did not work and was removed, the failures excluded as pre-registered, and an "include uncorrected" sensitivity added before modelling; 3C50 jobs 3 → 6 (speed only); 3C50 estimates a mean background, does not model its time variation, uses its own GTI filtering, and its library does not cover data after May 2023.
- **v13**: no analytical deviation; after an interruption the analyst wrongly believed the faint observations' event files had been deleted (they were kept; recorded in the decision log).
- **No Eddington-ratio control**: the MINBAR source table has no distances; the count rate is the only brightness proxy.
- **Lorentzian fits**: only 13 octave bins; 12 widths at the lower bound.
- **Bootstrap and permutation** keep predictions and features fixed.

11.9 Takeaway

On genuinely new NICER observations, with everything frozen in advance, low-frequency variability adds BH/NS information inside the hard state (+0.18 in AUC, RF, prospectively detected); this is the most robust positive result of the project, but it is an observation-level confirmation on the same 16 dynamically confirmed BHs. The lower characteristic frequency of BHs at fixed colour is consistent in size but not robustly significant once backgrounds are treated and data are new. The series stopped here because further versions could only re-use the same sources (Chapter 12).

Chapter 12

Synthesis across phases (v14)

Version v14 (2026-10-06) ran no new test. It collected the results of v1–v13 into a final summary, two tables and one figure, and decided to stop the series because any further version could only re-use the same sources. The files are:

- `reports/final_summary_zh-TW.md` (summary text);
- `results/final_timing_gain_across_versions.csv` and `results/final_nuc_difference_across_versions.csv` (tables);
- `figures/final/summary_across_versions.png`, made by `scripts/64_final_summary_figure.py`.

This chapter puts all pre-declared primary comparisons side by side, shows the two timing results across samples and instruments, and states the conclusions in three tiers together with the degree of independence behind each.

12.1 Every primary comparison

Table 12.1 lists each pre-declared primary comparison of v1–v13 with its result and verdict under the rule that was written down before the comparison. It is the shortest honest summary of the project.

Table 12.1: All pre-declared primary comparisons, v1–v13 ("det." = detected: 95% interval excludes 0 or $p < 0.05$; "n.d." = not detected under the pre-registered criterion). Each pre-registration was written after the results of all earlier versions. Sources: the results tables of Chapters 6 to 11.

Ver.	Comparison (sample)	Result	Verdict	Ch.
v1	none formal: LOSO evaluation of A and B (8 sources)	source BA 0.50–0.88 by combination	descriptive	6
v2	B − A, LR (31 sources), source BA	−0.092 [−0.268, +0.021]	n.d.	6
v2	B − A, RF, source BA	+0.235 [+0.042, +0.429]	det. (shape beats raw intensity for RF)	6
v3a	H+B − H, LR, source AUC	−0.048 [−0.179, +0.048]	n.d.	6
v3a	H+B − H, RF, source AUC	−0.012 [−0.077, +0.048]	n.d.	6
v4a	nested automatic selection − H-LR, source BA / AUC	−0.062 [−0.146, +0.021] / −0.173 [−0.399, +0.012]	n.d.	7
v5	H+T-LR − H-LR (S1), source BA / AUC	+0.113 [−0.021, +0.277] / +0.065 [0.000, +0.179]	n.d.	8

Ver.	Comparison (sample)	Result	Verdict	Ch.
v6a	frozen H_R+T-LR − H_R-LR (external E), source BA / AUC	+0.105 [0.000, +0.237] / 0.000 [0, 0]	n.d. (AUC saturated)	8
v7b1	H-LR without MINBAR bursts − v2, source BA / AUC	0.000 [0, 0] / −0.006 [−0.036, 0.000]	n.d. (no burst effect)	9
v7a	H-LR observation AUC, soft-like − hard-like	+0.309 [+0.099, +0.510]	**det.**	9
v7b2	+ 3 persistent BHs, H-LR on the 31, source BA / AUC	0.000 [−0.214, +0.214] / −0.048 [−0.167, +0.048]	n.d.	9
v7c1	reproduction of de Beurs et al. (2022)	SVM 36/44, kNN 38/44	reproduced	9
v7c2	MAXI LR: CCI − C2D / C2D − C1D, source AUC	−0.029 [−0.066, +0.002] / +0.139 [−0.032, +0.306]	n.d. / n.d.	9
v8a	S1 hard-like AUC, H+T-LR − H-LR	+0.264 [+0.048, +0.455]	**det.**	9
v8b	external X hard-like AUC, frozen H_R+T-LR − H_R-LR	+0.064 [−0.185, +0.367]	n.d.	9
v8c	S1-HF AUC, H+HF-LR − H-LR	−0.056 [−0.154, +0.010]	exploratory (IC4 failed)	9
v8d	MAXI RF $C2D_{NH}$ − $C1D_{NH}$, source AUC	+0.377 [+0.160, +0.595]	**det.**	9
v9a	NICER hard-like AUC, H_N+T_N-LR − H_N-LR	+0.112 [−0.004, +0.229]	n.d.	10
v9b	NICER hard-like $D(\log_{10}\nu_c)$	−0.310, $p = 0.136$	n.d.	10
v10a	pooled hard-state gain (equal weights)	+0.145 [+0.028, +0.257]	**det.** (not independent)	10
v10b	pooled $D(\log_{10}\nu_c)$	−0.368, $p = 0.0012$	**det.** (not independent)	10
v11a	new obs., frozen LR hard-like gain	+0.066 [−0.095, +0.180]	n.d.	11
v11b	new obs., $D(\log_{10}\nu_c)$	−0.663, $p = 0.0225$	**det.** (background-sensitive)	11
v12a	3C50-corrected $D(\log_{10}\nu_c)$	−0.514, $p = 0.073$	n.d.	11
v12b	3C50-corrected RF hard-like gain (RF chosen post hoc)	+0.174 [+0.071, +0.300]	**det.**	11
v13a	third new set, frozen RF hard-like gain (RF pre-chosen)	+0.184 [+0.049, +0.337]	**det.** (prospective)	11
v13b	third new set, $D(\log_{10}\nu_c)$	−0.427, $p = 0.209$	n.d.	11

Excluding v1 (no formal primary), the v7c1 reproduction and the exploratory v8c, and counting the two models of v2 and v3a and the two comparisons of v7c2 separately, the table contains 25 primary comparisons, of which 9 were detected: v2-RF (shape beats raw intensity, an early question unrelated to the later hypotheses), v7a (state dependence), v8a, v10a, v12b and v13a (hard-state timing gain), v10b and v11b (characteristic frequency) and v8d (MAXI 2–4 keV colour). Every primary that asked whether *more spectral information* helps was not detected.

12.2 The two timing results across samples and instruments

Tables 12.2 and 12.3 and Figure 12.1 collect the two results that Phases 3–6 tested repeatedly.

Table 12.2: Hard-like observation-level AUC gain from the low-frequency timing features (H+T − H) across samples, instruments and models, with 95% source-bootstrap intervals; ★ = pre-declared primary. From `results/final_timing_gain_across_versions.csv`.

Analysis	Instrument	LR	RF
v8a RXTE v2 sample	RXTE	★ +0.264 [+0.048, +0.455]	+0.230 [+0.074, +0.408]
v8b RXTE external (frozen)	RXTE	★ +0.064 [−0.185, +0.367]	–
v9a NICER prefixes	NICER	★ +0.112 [−0.004, +0.229]	+0.236 [+0.104, +0.372]
v10a pooled RXTE + NICER	both	★ +0.145 [+0.028, +0.257]	–
v10 NICER full observations	NICER	–	+0.219 [+0.100, +0.337]
v11 NICER new observations (frozen)	NICER	★ +0.066 [−0.095, +0.180]	+0.247 [+0.128, +0.384]
v12 NICER 3C50-corrected	NICER	+0.090 [−0.056, +0.243]	★ +0.174 [+0.071, +0.300]
v13 NICER third set (frozen)	NICER	+0.182 [+0.022, +0.347]	★ +0.184 [+0.049, +0.337]

Table 12.3: Colour-conditional difference of $\log_{10}\nu_c$ (BH − NS, dex) and permutation p; ★ = pre-declared primary. From `results/final_nuc_difference_across_versions.csv`.

Analysis	D [dex]	p
v6d RXTE v2 (all states)	−0.313	0.0037
v6d RXTE external (all states)	−0.037	0.783
★ v9b NICER prefixes (hard-like)	−0.310	0.136
v10b RXTE hard-like	−0.265	0.0018
v10b NICER full (hard-like)	−0.471	0.033
★ v10b pooled RXTE + NICER	−0.368	0.0012
★ v11b NICER new observations	−0.663	0.0225
★ v12a NICER 3C50-corrected	−0.514	0.073
★ v13b NICER third set	−0.427	0.209

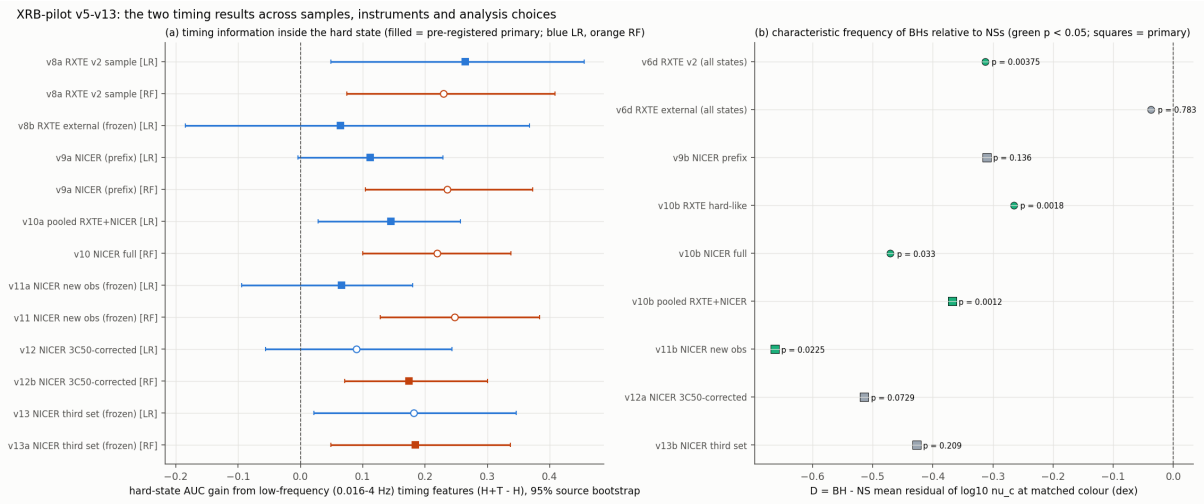


Figure 12.1: The two timing results of v5–v13. (a) Hard-state AUC gain from the 0.016–4 Hz timing features with 95% source-bootstrap intervals (blue LR, orange RF; filled squares = pre-declared primaries). (b) Colour-conditional $\log_{10}\nu_c$ difference BH − NS (green: $p < 0.05$; squares: primaries). Every RF interval lies above 0; the LR intervals of the external and new-observation sets include 0; every D except the four-BH external RXTE set is between −0.27 and −0.66 dex. Repo: `figures/final/summary_across_versions.png`, `scripts/64_final_summary_figure.py`; values in the two `results/final_*.csv` files.

12.3 Conclusions in three tiers

Supported by the data

1. Two count-space hardness ratios capture essentially all the BH/NS information in a single 5–25 keV RXTE spectrum: neither the full shape, nor intensity, nor 3–5 keV, nor 25–60 keV, nor flexible algorithms, nor 17× more observations add anything detectable (Phases 1–2).
2. The colour information works in soft states and fails in hard states: observation-level AUC 0.954 (soft-like) vs 0.646 (hard-like) on RXTE, difference +0.31 [+0.10, +0.51]; colour-only hard-state AUC 0.50–0.75 on NICER (Phase 4–6).
3. Inside the hard state, 0.016–4 Hz variability raises the BH/NS separation, consistently on RXTE, on three disjoint NICER observation sets, in the cross-instrument pool, after background correction, and prospectively with a frozen pipeline and a pre-chosen RF on new observations (v13: +0.18 [+0.05, +0.34]).
4. Type-I burst contamination does not drive any of the spectral conclusions (v4, v7b1).

Consistent in direction, marginal in significance

1. At the same colour, the characteristic frequency of BH variability is lower than that of NSs by 0.3–0.6 dex in every data set except the four-BH external RXTE set; pooled $p = 0.0012$ (v10b), but single NICER data sets give p between 0.02 and 0.21. No Eddington-ratio control was possible, so mass and accretion state cannot be separated.
2. The timing gain is stable for the non-linear RF (+0.17 to +0.25) and unstable for LR (+0.07 to +0.26).
3. On MAXI, the 2–4 keV colour carries BH/NS information beyond the ≥ 4 keV colour that is not explained by Galactic absorption (v8d detected; v9c post hoc in all C4 tertiles).

Not supported

1. A broad-band 512–1024 Hz criterion in the style of Sunyaev and Revnivtsev (2000): after removing a cross-PCU white component it is essentially undetectable at single-observation sensitivity (v8c, exploratory).
2. High accuracy of linear colour models on MAXI with confirmed BHs: LR has source AUC 0.45–0.59 (v7c2); the high accuracy of de Beurs et al. (2022) comes mostly from the two NS classes.
3. More flexible models, nested model selection, wider bands or more observations per source as a route beyond two colours (v4).

12.4 How independent is each piece of evidence?

Table 12.4 places the timing results on an "independence ladder". Every pre-registration was written after the results of all earlier versions, so later hypotheses were increasingly guided by earlier data; the project compensated by freezing pipelines and rules before new data (v6, v8b, v11, v13). No step reaches source-level independence: the BH side always consists of the same 16 dynamically confirmed BHs (7–9 in the hard state per instrument).

Table 12.4: Levels of independence behind the timing claims.

Level	Analyses	What is independent / what is not
Same observations, LOSO	v5, v8a, v9a, v10c	sources held out in turn; features and hypotheses chosen with knowledge of earlier results
Frozen model, other sources	v6a, v8b	new sources (23–26), but only 4–5 BHs; v4b2 sources had been seen in other models
Other instrument, same design	v9, v10 NICER	new instrument and epoch; design repeated, not transferred; 2 never-used BHs
Pooled evidence	v10a, v10b	all data; hypotheses and pooling chosen after the components (not a confirmation)
Prospective, new observations	v11, v13	pipeline, rules and (v13) model type frozen before download; same sources
Prospective, new sources	–	not achieved: no further dynamically confirmed BHs with usable RXTE/NICER timing

12.5 Why the series stopped

The decision log records the v14 decision to stop after v13: 16 of the 25 dynamically confirmed BHs of Marcel et al. (2026) had been used, the remaining ones have no usable RXTE or NICER timing data, and batch access to other missions could not be verified. A further version could only add new observations of the same sources, which would not change the level of independence. The natural next step is a source-level prospective test on new BH transients (Chapter 14).

Chapter 13

Limitations

The phase chapters list the caveats of each version. This chapter collects those that limit the project as a whole, ordered by how much they constrain the conclusions.

13.1 Sample size and labels

- **Sixteen dynamically confirmed BHs.** Across RXTE and NICER only 16 dynamically confirmed BHs have usable data, and 7–9 of them per instrument have hard-state observations. One source changes a source-level balanced accuracy by ≈ 0.07 (e.g. XTE J1118+480), and every bootstrap interval is wide because it resamples these few sources. The remaining BHs of Marcel et al. (2026) have no usable RXTE or NICER timing data.
- **Label definitions.** BHs are dynamically confirmed (BlackCAT, then Marcel et al. 2026); NSs are type-I bursters (Galloway et al. 2008, MINBAR) and pulsars. The NS class mixes atolls, Z sources, AMXPs and globular-cluster sources, whose hard states differ; BH candidates are used only in sensitivity analyses (v3c). The v2 source rule was executed incompletely (13 NSs missed, found in v4b2), so the v1–v3 numbers describe a smaller NS sample than the rule implies.
- **Population differences between samples.** Training BHs are LMXB transients; external sets add HMXBs (Cyg X-1, LMC X-1, LMC X-3) and SS 433, external NSs are mostly AMXPs; this complicates the external comparisons in either direction.

13.2 Independence and pre-registration

- **Every pre-registration was written after the results of all earlier versions**, and from v5 onwards each was designed and approved by the analyst under a standing instruction of the user, without external review. Later hypotheses were therefore guided by earlier data (Table 12.4).
- **The strongest tests are observation-level.** v11 and v13 used observations never downloaded before, with frozen pipelines and rules, but of the same sources. v10 pooled all data but chose its hypotheses and pooling after the components. No test is source-level independent.
- **Post hoc elements** are labelled throughout (e.g. the source-equal-weight scale fix, the MINBAR matching rule, the v8c estimator change and null-band subtraction, the v12

choice of RF), and none of them is used as a primary conclusion except the v12 RF choice, which v13 then tested with RF pre-chosen.

- **Multiplicity.** Descriptive comparisons were many (76 in v4a, 45 in v5, 16+16 in v4b1, dozens in v7–v9) and uncorrected; they are reported but not interpreted individually.

13.3 Measurement

- **Count space.** RXTE spectral features are net count rates on a fixed channel-energy grid, not response-unfolded fluxes; the power-law-normalised R and H_R only partly remove response differences; PCU-combination tilts of $\approx \pm 10\%$ at 25 keV remain.
- **Backgrounds.** RXTE uses the Standard-Products background model without systematic errors; NICER timing is not background-subtracted before v12; 3C50 models only the mean background, uses its own GTI filtering, cannot process data with the original gain calibration (139 + 38 faint observations excluded) and does not cover data after May 2023.
- **Timing.** Standard-1 has no energy resolution and stops at 4 Hz; the RXTE state index therefore omits 4–10 Hz and uses BH thresholds for NSs; event-mode estimates failed their consistency gate (v8c); NICER v9 used only file prefixes. The two ν_c definitions (Standard-1 and NICER) differ in band and bin weighting, so their D values are not strictly comparable.
- **States** are operational variability classes (rms thresholds), not full spectral-state assignments.
- **No physical normalisation.** Eddington ratio, distance, mass and inclination are not controlled; brightness enters only as a count rate. The ν_c result cannot be attributed to mass rather than accretion state.

13.4 Statistics

- **Fixed-prediction resampling.** All bootstrap intervals and permutation tests keep the out-of-fold predictions or features fixed; they do not include the variability of model fitting or of nested model selection. Nested source AUCs are approximations (shifted scores).
- **Observation-level metrics within states** pool the observations of the drawn sources and are dominated by sources with many observations; the source-level versions are reported alongside but have even less power.
- **Saturation.** Headline metrics saturated on the external set of v6 (AUC 1.000), which the decision rule did not anticipate.
- **The colour-conditional permutation test** residualises with a label-free kNN; when BHs cluster in colour they become each other's neighbours, which makes the test conservative.

13.5 Process and reproducibility

- Visual verdicts (burst candidates, figure checks) were made by one person.
- Raw data are not in the repository (by design); they can be re-downloaded with the scripts, and every file's URL, size and SHA256 is in `logs/downloads.jsonl`. Some NICER raw bytes were streamed and not stored (v11).

- Analyses ran on Windows (Python 3.12.10) and, for some re-checks, macOS (Python 3.12.14); feature files reproduced to $\leq 10^{-16}$ and RF source means to ≤ 0.001 across machines. HEASoft steps (XSPEC, `nh`, `nibackgen3C50`) ran in WSL.
- This report re-ran no analysis; all numbers are read from committed files, and a small number of descriptive quantities (per-source hard-like means in Phase 4, the v9b Mann–Whitney check) were recomputed read-only from committed CSVs, as stated where they appear.

Chapter 14

Future work

The repository's final summary (`reports/final_summary_zh-TW.md`, §6) and the "hypotheses that need more data" of the version reports point to a small number of next steps. They are listed in order of how directly they address the main limitation – the lack of a source-level test.

14.1 A source-level prospective test

Freeze the v13 classifier (RF on colours + 0.016–4 Hz rms, the v12 NICER pipeline including 3C50 and screening) and the ν^* rule, commit them, and apply them unchanged to *new* dynamically confirmed BH transients and to NSs not used so far, as they become available (NICER, or missions whose data can be retrieved by script: Insight-HXMT, AstroSat, XRISM). The decision rule, the metrics (including a proper scoring rule, which does not saturate) and the minimum number of new BH sources should be fixed in advance. This would turn the observation-level confirmation of v13 into a source-level one.

14.2 Physical normalisation

Collect distances and masses for the sources (BH masses from Marcel et al. 2026 and Corral-Santana et al. 2016; NS distances, e.g. from photospheric-radius-expansion bursts) to compute Eddington ratios, and test whether the colour-conditional ν_c difference scales with compact-object mass at fixed accretion state. Replace the νP_ν centroid by fitted Lorentzian characteristic frequencies (begun in v11c) as the primary frequency measure.

14.3 Data improvements

- NICER: reprocess the 2017–2018 observations with the current gain calibration (or use the SCORPEON background model) to recover the $\approx 28\%$ of faint observations excluded by 3C50; model the time variation of the background.
- RXTE: response-unfolded colours or fluxes instead of count space; a model of the cross-PCU white component before any kHz search; stacked, narrow-band searches for kHz QPOs as a better test of the Sunyaev and Revnivtsev (2000) criterion.
- MAXI: state-resolved (or hardness-resolved) analyses with dynamically confirmed BHs; source-level rather than point-level metrics as the pre-declared ones.

14.4 Methods

- A pre-registered, matched-design alternative to the label-free kNN residualisation, which is conservative when BHs cluster in colour.
- A formal source-level meta-analysis across instruments, with the pooling rule fixed before any component is seen.
- A reproducible classification tool built on the RF timing gain, with source-level conformal prediction sets (v6e) so that an astronomer obtains {BH}, {NS} or "cannot tell" with a stated coverage.

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Appendix A

Hyper-parameters and fixed settings

Every setting below is read from `scripts/config.py` (names in typewriter; blocks are activated by the environment variable `XRB_VERSION`) or, where stated, from the analysis script. Model hyper-parameters are also given, with their mathematical role, in Chapter 4.

Table A.1 is the master table.

Table A.1: Master table of models, hyper-parameters and fixed settings (`scripts/config.py` unless stated).

Versions	Setting (name)	Value	Used in
Global			
all	random seed `SEED`	42	all models, bootstrap, sampling
v2–	bootstrap resamples `N_BOOT`	2000, class-stratified over sources	§5.5
all	decision threshold	0.5 (fold-specific only in v4a 1d/1e)	Eq. (3.1)
v1–v4	observations per source `N_PER_SOURCE`	15 (time-stratified); v4b1 `'all'`	Algorithm 1
v2–	`MIN_ELIGIBLE`, `MJD_MAX`	10 eligible pointings; MJD 55931 (v8b: 3)	source rule
v2–	`DEADTIME`, `VLE_DT_ALT`	`'std1'` (GOF recipe); 150 µs VLE (sensitivity only)	Eq. (2.4)
Reference models			
v1–	`LR_PARAMS`	$C = 1$, L2, `class_weight='balanced'`, `max_iter=5000`, `lbfgs`	§4.3
v1–	`RF_PARAMS`	500 trees, `max_features='sqrt'`, `min_samples_leaf=2`, balanced, `n_jobs=2`, seed 42	§4.4
v4a fixed settings (`V4_FIXED`)			
v4	ExtraTrees	500 trees, `sqrt`, leaf 2, balanced, seed 42	§4.5
v4	HistGB	lr 0.1, 200 iterations, 15 leaves, leaf 20, $\ell_2 = 0$, no early stopping, balanced, seed 42	§4.6
v4	SVM-RBF	$C = 1$, γ=`'scale'`, balanced; score $\sigma(f)$	§4.7
v4	kNN	$k = 15$, uniform; prior correction	§4.8

Versions	Setting (name)	Value	Used in
v4	QDA	`reg_param=0.1`, priors $(0.5, 0.5)$	§4.9
v4	MLP	hidden (32,), $\alpha = 10^{-3}$, early stopping (20%), 500 iterations; seeds `V4_MLP_SEEDS_OUTER` 0–4 (inner 0–2)	§4.10
v4	source-equal weights `V4_SOURCE_WEIGHTED_MODELS`	LR, RF, HistGB	Eq. (4.3)
v4	nested grids `V4_GRID`	see Table 7.5; `V4_INNER_TREES`=200, `V4_OUTER_TREES`=500	§5.4
v4 data rules			
v4	burst detector `V4_BURST`	1-s bins, 4σ, merge 30 s, pre-window 60 s, peak $\geq 1.5\times$, rise ≤ 10 s, half-decay 3–300 s, ≥ 3 bins above 25%	Eq. (2.14)
v4b3	HEXTE `V4_HEXTE_*`	cluster 1 (B); last MJD 55179.6736; bands 25–40, 40–60 keV; min denominator S/N 3	Eq. (2.13)
v4b3	overlap box `V4_OVERLAP_BOX`	$c_1 \geq 0.70$, $c_2 \geq 0.20$ (post hoc definition)	Table 7.10
v4b2	gain-epoch stops `V4_EPOCH_STOP_MJD`	50163.7729, 50188.9618, 51259.7340, 51677.0000	diagnostic only
v4b2	`V4_R_GAMMA`, `V4_R_NH`; bridge `V4_BRIDGE_MIN_SPEARMAN`	$\Gamma = 2$, $N_{\rm H} = 0$; 0.90	Eq. (2.8)
Standard-1 timing (v5–v13)			
v5–	`V5_N_BINS`, `V5_DT`	1024 bins of 0.125 s (128-s rows)	§2.8
v5–	`V5_BANDS_J`	T1 j=2–12, T2 13–128, T3 129–511	Table 2.6
v5–	`V5_MIN_PCUS`, `V5_BLOCKS`, `V5_MIN_ROWS`	2 PCUs; 8 blocks with counts; 3 valid rows	§2.8.3
v5	cross-check `V5_CHECK_RATE`, `V5_CHECK_MIN_SPEARMAN`	20–500 c s^{-1} PCU^{-1}; 0.90	gate
v6–	`V6_AUTO_MAX_RATE`	single-PCU auto-power rows only below 500 c s^{-1} PCU^{-1}	Eq. (2.17)
v6–	`V6_NLOGBINS`, `V6_NUC_MIN_RMS`	9 log bins 1/128–4 Hz; ν_c missing if $T^*_{\rm total} < 0.05$	Eq. (2.18)
v6	CCTLR `V6_T_DOF`, `V6_SIGMA_FLOOR`	Student-t with 4 d.o.f.; $\sigma \geq 0.01$	Eq. (4.13)
v6–	permutation `V6_KNN`, `V6_N_PERM`	$k = 25$; 20,000 permutations	Algorithm 16
v6	`V6_CONFORMAL_EPS`	0.1, 0.2	Eq. (5.9)
v6	`V6_BUDGET_K`, `V6_BUDGET_DRAWS`	$k \in \{1, 2, 3, 5, 8, 13, 21\}$; 500 draws	§5.8
v6	`V6_PROB_CLIP`	$[0.01, 0.99]$ for proper scores	Eq. (5.5)
v7			
v7–	state index `V7_RMS_J`, `V7_RMS_HARD`, `V7_RMS_SOFT`	j=13–511 (0.102–3.99 Hz); > 0.10 hard-like, < 0.075 soft-like	Eq. (2.19)
v7–	`V7_STATE_MIN_ROWS`, `V7_BURST_PRE_S`	3 rows (≥ 2 PCUs); burst rows from 20 s before start	§9.3.1
v7	sanity `V7_SANITY_FRAC`; stratum test `V7_MIN_BH_SOURCES_PER_STRATUM`	2/3; ≥ 3 BH sources	gates
v7b1	`V7_BURST_DEFAULT_DUR_S`	300 s when MINBAR duration missing	§9.6.1

Versions	Setting (name)	Value	Used in
v7c1	de Beurs kNN / SVM (script `37_v7c1_repro.py`)	$k = 24$ raw features; $C = 0.655298078515272$, $\gamma = 0.585144090708118$, Platt, seed 42 (`V7_DEBEURS_SVM` in `config.py` holds rounded values 0.655, 0.585)	Table 9.16
v7c1	`V7_REPRO_TOL_SOURCES`, `_BH`	2 sources, 1 BH	gate
v7c2	MAXI cuts `V7_MAXI_SIGMA`, `_OUTLIER_SIGMA`, `_MIN_POINTS`	value 3 (the selection script applies flux/$\sigma \geq \sqrt{3}$ in every band, `38_v7c2_select.py`); mean + 10 std; 100 points	§2.11
v7c2	`V7_MAXI_TRAIN_CAP`	200 random points per training source	Table 9.5
v7c2	`V7_C2_KNN_GRID`, `V7_C2_SVM_GRID`	$k \in \{5, 15, 24, 35\}$; $C \in \{0.1, 1, 10\} \times \gamma \in \{0.1, 0.585, 3\}$	Table 9.6
v8			
v8b	`MIN_ELIGIBLE`, `V8_MIN_SOURCES_PER_CLASS`	3 eligible; ≥ 3 BH and NS sources per stratum	§9.6.3
v8c	`V8_HF_MODE_REGEX`, `V8_HF_CHAINS`, `V8_HF_MAX_TIMEDEL`	`^E_\d+us_\d+M_0_\d+s$`; all six xenon layers; $\leq 2^{-12}$ s	event survey
v8c	`V8_HF_DT`, `V8_HF_SEG_S`, `V8_HF_BANDS_J`	2^{-12} s; 16-s segments; LC 16–63, HF1 64–1023, HF2 1024–8191, HF3 8192–16383, NULL 24576–32767	Eq. (2.20)
v8c	QC `V8_HF_MIN_SEG`, `_MAX_NULL_Z`, `_MAX_NULL_FAIL_FRAC`, `_MIN_SPEARMAN_LC`	10 segments; $\lvert z\rvert \leq 5$; 20%; IC4 0.90	gates
v8d	`V8_NH_GAMMA`, `V8_NH_GRID`, `V8_NH_BANDS`	$\Gamma = 2$; $\log N_{\mathrm{H}}$ 19–23.5 (200 points, plus 0); L 2–4, M 4–10, H 10–20 keV	§2.11
NICER (v9–v13)			
v9–	`V9_TOL_DEG`, `V9_MIN_EXPOSURE_S`, `V9_MIN_ELIGIBLE`	0.05°; 1000 s; 5	§10.6.1
v9–	`V9_N_PER_SOURCE`, `V9_MAX_TRIES`	8 time bins; 3 tries per bin	sampling
v9	`V9_CAP_BYTES`, `V9_TARGET_SEG`, `V9_GAP_S`	300 MB; 6 segments; gap > 1 s	prefix download
v9–	`V9_SEG_S`, `V9_DT`, `V9_PI_TIMING`, `V9_PI_BANDS`	128 s; 1/256 s; PI 200–1000 (2–10 keV); A 200–400, B 400–600, C 600–1000	§2.10
v9–	`V9_T_BANDS_J`, `V9_STATE_J`, `V9_WHITE_J`, `V9_NUC_EDGES_HZ`	as v5; 13–1280 (0.1–10 Hz); 12288–16383; 13 octave bins 1/128–64 Hz	Eq. (2.18)
v9–	`V9_MIN_MPUS`, `V9_BLOCKS`, `V9_MIN_RATE`, `V9_MIN_SEG`	2 MPUs; 8 blocks; 20 c s^{-1}; 3 segments	quality
v9–	`V9_BURST_PRE_S`, `V9_BURST_POST_S`	20 s, 200 s	burst removal
v10	`V10_FULL_CAP_BYTES`, `V10_MATCH_DEG`, `V10_N_PERM`	1 GB; 0.1°; 20,000	§10.6.3
v10	`V10_BKG_PI`, `V10_BKG_BRIGHT_RATE`, `V10_BKG_QUANTILE`	PI 1200–1500; 500 c s^{-1}; 0.99 (threshold 0.2687)	background proxy
v11	`V11_N_PER_SOURCE`, `V11_SEED_OFFSET`, `V11_MAX_TRIES`	8; 1000; 3	sampling
v11	`V11_STREAM_CHUNK`, `V11_N_WORKERS`, `V11_RANGE_CONCURRENCY`	64 MB; 2; 6	streaming
v11	`V11_LORENTZ_NU`, `V11_LORENTZ_DBIC`	width 1/128–128 Hz; ΔBIC > 6	Eq. (2.22)

Versions	Setting (name)	Value	Used in
v11, v13	frozen ν^* (results files)	1.3007 Hz; 1.3361 Hz	Eq. (4.16)
v12	`V12_BKG_MAX_RATE`, `V12_FBKG_MAX`, `V12_FBKG_SENS`	$500\,\mathrm{c\,s^{-1}}$; $f \leq 0.20$; 0.10, 0.30	Eq. (2.21)
v12	`V12_PI_BANDS`, `V12_IC_TOT_RATIO`, `V12_N_WORKERS`	A, B, C as v9, T 200–1000; [0.7, 1.3]; 6	gates
v13	`V13_SEED_OFFSET`, `V13_KEEP_CL`	2000; keep cleaned files of faint observations	sampling

Appendix B

Reproduction

B.1 Run scripts

Each version has a PowerShell and a Bash driver in the repository root that runs its unit tests (gates) first and then the analysis scripts in order: `run_all.ps1`/`.sh` (v1–v2), `run_v3` ... `run_v13` (`.ps1` and `.sh`). The version reports end with a "re-run" section listing the exact order, durations and download volumes; for example `./run_v4.sh test a burst b3 b2 b1` runs the v4 parts selectively. Scripts select their configuration block through the environment variable `XRB_VERSION` (`scripts/config.py`). Raw data outside the repository are found through `XRB_EXTERNAL_RAW` (default `~/xrb-pilot-data/raw`).

B.2 Scripts and their outputs

Table B.1 maps every script to its outputs.

Table B.1: Analysis scripts (`scripts/`) with their purpose (first line of the docstring, shortened) and main outputs.

Script	Purpose	Main outputs
`01_inspect_archive.py`	official index products; inspect two spectral triplets	`data/cache/` (not in git)
`02_catalogs.py`	MissionLongData catalogues of the sources	`data/raw/catalogs/` (not in git), `data/catalog_observations.csv`
`03_select_observations.py`	source table and time-stratified candidate list	`data/sources.csv`, `data/observation_candidates.csv`, `results/attrition_catalog.csv`
`04_fetch_process.py`	download Std2 products, quality rules, features	`data/processed/features.npz`, `data/observations.csv`, `results/attrition_download.csv`
`05_two_source_check.py`	data-flow validation on one BH and one NS	`figures/step2_*`
`06_dataset_checks.py`	integrity checks and diagnostics	`results/data_checks.md`, `figures/step3_*`
`07_train_eval.py`	LOSO (Dummy, LR, RF), leaky split	`results/oof_predictions_loso.csv`, `metrics_*.csv`, `folds_loso.csv`
`08_error_analysis.py`	error analysis, A vs B	`results/error_analysis.md`, `per_source_results.csv`
`09_bootstrap_sources.py`	source bootstrap	`results/bootstrap_source_level.csv`

Script	Purpose	Main outputs
`10_xspec_crosscheck.py`	XSPEC check of background subtraction (WSL)	`results/v2/xspec_crosscheck.csv`
`11_v3a_increment.py`, `12_v3b_extended_band.py`	v3a, v3b	`results/v3/`
`13_v3c_resolve_catalogs.py`, `14_v3c_candidates.py`	v3c candidates	`data/v3c/`, `results/v3c/`
`15_colour_colour.py`	colour–colour wrap-up figure	`figures/final/colour_colour_decision_boundary.png`, `results/final/colour_colour_source_medians.csv`
`16_v4a_algorithms.py`	v4a 1a–1g	`results/v4/v4a_*`, `figures/v4/v4a_*`
`17_v4_burst_detect.py`, `18_v4_burst_exclusion.py`	burst detection and exclusion	`results/v4/burst_*`, `figures/v4/burst_candidates/`
`19_v4b3_hexte.py`	HEXTE colours	`data/v4b3/processed/hexte_features.csv`, `results/v4b3/`
`20–22_v4b2_*`	source verification, download, analysis	`data/v4b2/`, `results/v4b2/`, `figures/v4b2/`
`23–25_v4b1_*`	all epoch-5 pointings	`data/v4b1/`, `results/v4b1/`, `figures/v4b1/`
`26_v5_timing_features.py`, `27_v5_analysis.py`	Standard-1 timing features and v5 analysis	`data/v5/processed/timing_features.csv`, `results/v5/`, `figures/v5/`
`28–30_v6_*`	external sources, T^*, ν_c, v6a–h	`data/v6/`, `results/v6/`, `figures/v6/`
`31–33_v7b1_*`	MINBAR matching, replacement, exclusion	`results/v7b1/`, `figures/v7b1/`
`34_v7a_states.py`	variability states	`results/v7a/`, `figures/v7a/`
`35`, `36_v7b2_*`	persistent BHs	`results/v7b2/`, `figures/v7b2/`
`37_v7c1_repro.py`	de Beurs reproduction	`results/v7c/v7c1_*`
`38`, `39_v7c2_*`	MAXI strict pipeline, RXTE vs MAXI	`data/v7c/`, `results/v7c/v7c2_*`, `v7c3_*`, `figures/v7c/`
`40_v7_summary_figure.py`	v7 summary	`figures/v7/v7_summary.png`
`41_v8b_select.py`, `42_v8_states_timing.py`	new BHs; external states and timing	`data/v8b/`, `results/v8b/`
`43_v8c_events.py`	event-mode features (exit 3 if IC4 fails)	`data/v8c/processed/`, `results/v8c/ic4_*`, `results/v8c/event_survey.csv`
`44_v8_analysis.py`	v8a–c analysis	`results/v8a/`, `results/v8b/`, `results/v8c/`
`45_v8d_nh.py`	HI4PI $N_{\rm H}$, `tbabs` transmissions, v8d (WSL)	`data/v8d/`, `results/v8d/`, `figures/v8d/`
`46_v8_summary_figure.py`	v8 summary	`figures/v8/v8_summary.png`
`47`, `48_v9_*`	NICER selection and prefix features	`data/v9/`, `results/v9/attrition_catalog.csv`
`49_v9_analysis.py`	v9a, v9b	`results/v9a/`, `results/v9b/`, `figures/v9/v9_nicer.png`
`50_v9c_maxi_strata.py`, `51_v9_summary_figure.py`	v9c; summary	`results/v9c/`, `figures/v9/`
`52_v10_nicer_full.py`, `53_v10_rxte_nuc.py`, `54_v10_pooled.py`	full NICER observations; RXTE ν_c table; pooled tests	`data/v10/`, `results/v10a/`, `results/v10b/`, `results/v10c/`, `figures/v10/`
`55–57_v11_*`	new observations, frozen tests	`data/v11/`, `results/v11/`, `figures/v11/`

Script	Purpose	Main outputs
`58_v12_bkg.py`, `59_v12_analysis.py`	3C50 (WSL), corrected analysis	`data/v12/`, `results/v12/`, `figures/v12/`
`60–63_v13_*`	third new set, frozen v12 pipeline	`data/v13/`, `results/v13/`, `figures/v13/`
`64_final_summary_figure.py`	v14 wrap-up	`figures/final/summary_across_versions.png`, `results/final_*_across_versions.csv`
`common.py`, `spectra.py`, `config.py`, `plotstyle.py`	shared I/O, Std2 reader without HEASoft, pre-registered settings, plot style	–
`v3lib.py` … `v11lib.py`	version helpers (representations, bootstrap, models, timing estimators, NICER streaming)	–
`test_v4lib.py` … `test_v11lib.py`	pre-declared implementation checks (gates)	`logs/v*/test_*.log`

B.3 Data locations

- **In git**: source tables, observation lists with every exclusion reason (`data/*/observations.csv`), processed feature files (`features.npz`, `*_features.csv`), all result tables (`results/`), figures and logs.
- **Not in git** (`.gitignore`): raw RXTE products in `data/raw/` (v1–v3, v8 events), raw files of v4b1 and NICER prefixes, tails and v13 cleaned files under `~/xrb-pilot-data/raw/`, 3C50 spectra under `~/xrb-pilot-data/nicer_bkg_spectra/`, the NICER CALDB under `~/xrb-pilot-data/caldb/`. All can be re-downloaded with the scripts; v11 streamed ≈ 39 GB without storing the bytes.
- **Download log**: `logs/downloads.jsonl` has one JSON record per downloaded or streamed file (50,777 lines at commit `d9730c9`) with the fields `url`, `path`, `bytes`, `sha256` and `utc`, so every input can be verified byte for byte after re-download.

B.4 Environment

- Python 3.12.10 (Windows, `logs/environment.txt`; re-checks on macOS with Python 3.12.14), with the versions pinned in `requirements-lock.txt`: numpy 2.4.4, pandas 2.3.3, scipy 1.17.1, scikit-learn 1.8.0, astropy 7.2.0, matplotlib 3.10.9, joblib 1.6.0, requests 2.33.1 (plus their dependencies); software references: scikit-learn (Pedregosa et al., 2011), NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020), Astropy (Astropy Collaboration et al., 2022), Matplotlib (Hunter, 2007).
- Second-machine check (decision log, 2026-09-30): on macOS the 2,689 v1–v2 raw files were re-downloaded from the URLs in `logs/downloads.jsonl` with all SHA256 values matching; `features.npz` agreed with the Windows file to $\leq 10^{-16}$, LR scores were identical, single RF scores differed by ≤ 0.006 (source means ≤ 0.001), and all predicted labels and accuracies were identical.
- HEASoft 6.37.1 (NASA High Energy Astrophysics Science Archive Research Center (HEASARC), 2014) in WSL Ubuntu 22.04 for the XSPEC (Arnaud, 1996) cross-check (v2), `nh` and XSPEC `tbabs` (v8d) and `nibackgen3C50` with the NICER CALDB (v12, v13).

B.5 This report

- Sources: `report_en/main.tex` and `report_en/sections/*.tex`; bibliography `report_en/references.bib`.
- Figures: repository figures copied unchanged to `report_en/figures/` with their original sub-paths; new data figures G1–G5 are produced read-only from committed files by `report_en/figures/generated/make_figures.py` (for G1, G4 and G5 it also writes the plotted values as CSV next to the PDF). G6 and G7 are redraws of two repository scatter plots whose source labels overlapped (Figures 8.3 and 9.6) and G8 of the v4a forest plot whose row labels were unreadable (Figure 7.4): same committed CSVs, same points and intervals; the original PNGs remain in `report_en/figures/`. The very wide summary figures of v7–v9 are printed on landscape pages. Method diagrams are TikZ code inside the chapters; the flowcharts (the chain of questions, the six phase analysis maps, the processing pipelines and the evaluation protocols) are TikZ files in `report_en/figures/flowcharts/`, drawn with the shared `fc` styles of `main.tex`.
- Build: `latexmk -pdf main.tex` in `report_en/` (TeX Live 2026; packages listed in the preamble).
- Tables of per-source scores were generated from the committed CSV files by small read-only scripts; no model was refitted and no statistic recomputed except where explicitly stated in the text (hard-like per-source means in Table 9.10; the Mann–Whitney check in Section 10.6.2; the Spearman ρ and label agreement printed in Figure 9.6, identical to the repository values).

Appendix C

Pre-registration summary

Table C.1 summarises the thirteen pre-registrations (`reports/preregistration*.md`). The column "after" states which results were already known when the document was written; "approval" states who approved it. The primary comparisons and their verdicts are listed in Table 12.1.

Table C.1: Pre-registration documents, timing and approval.

Ver.	Written	After	Approval	Content (primary in bold)
v1	2026-09-29, before any model (during download)	–	user-directed project	sources (4 BH, 4 NS), selection and quality rules, A/B representations, Dummy/LR/RF, **LOSO evaluation**, leaky split as a demonstration
v2	2026-09-29, before any v2 download	v1	user ("run the suggested next step")	mechanical source rule (31 sources), Standard-1 deadtime, **source bootstrap; B−A**, H and HI baselines, XSPEC check
v3	2026-09-30, before any v3 model	v1, v2 (+ two descriptive statistics)	not stated in the decision log	**v3a H+B − H**; HI+A; v3b 3–5 keV; v3c BH candidates
v4	2026-09-30, before any v4 model or download	v1–v3	user ("start and finish full if no big problem")	**v4a nested automatic selection**; 1b–1d descriptive; burst check; v4b3 HEXTE; v4b2 earlier epochs (bridge gate); v4b1 all observations
v5	2026-10-01, before any v5 feature	v1–v4	analyst, under the user's instruction to design and run v5	Standard-1 timing T; **S1 H+T-LR − H-LR**; gates IC (synthetic, cross-check, plausibility)
v6	2026-10-01 (≈03:00), before any v6 download	v1–v5	analyst ("deep think . . . innovations")	**v6a frozen external test**; all-pairs estimator, ν_c; CCTLR; permutation test; conformal; budget; proper scores; nuisance control
v7	2026-10-05, before any v7 download	v1–v6	user, 07:23 (three pending items adopted)	one primary each: **v7b1 MINBAR**, **v7a states**, **v7b2 persistent BHs**, **v7c1 reproduction**, **v7c2 MAXI (two LR comparisons)**
v8	2026-10-05 (≈21:30), pushed `a481fa0` before download	v1–v7	analyst ("go directly to the next step")	**v8a S1 hard-like**, **v8b external hard-like**, **v8c event mode (gate IC4)**, **v8d MAXI N_H**
v9	2026-10-05 (≈23:00), pushed `ee4b7db` before NICER download	v1–v8	analyst ("merge and pull and get on to next step")	NICER pipeline; **v9a hard-like gain**; **v9b ν_c permutation**; v9c MAXI strata (descriptive)

Ver.	Written	After	Approval	Content (primary in bold)
v10	2026-10-06, pushed `aeedac7` before download and pooled statistics	v1–v9	analyst ("commit everything and start v10")	declared not an independent confirmation; **v10a pooled gain**, **v10b pooled** ν_c; v10c full NICER observations
v11	2026-10-06 (≈01:50), `6fea016`; ν^* committed in `58ded89` before new data	v1–v10	analyst ("merge and continue v11")	frozen pipeline and fold models; **v11a LR gain**, **v11b** ν_c; Lorentzian widths; never-used sources
v12	2026-10-06 (≈03:20), `2126e9d`	v1–v11	analyst ("merge and continue")	3C50 correction and screening; **v12a** ν_c, **v12b RF gain** (RF chosen after v9–v11)
v13	2026-10-06 (≈05:00), `42fe1d2` with $\nu^* = 1.3361$ Hz	v1–v12	analyst ("continue until it is worth wrapping up")	third disjoint new set; frozen v12 pipeline and models; **v13a RF gain (pre-chosen)**, **v13b** ν_c

Common rules, unchanged from v4 on: LOSO over sources; source score = mean OOF score; threshold 0.5 unless chosen inside the training sources; class-stratified source bootstrap with 2000 resamples and seed 42, paired across models; "improvement/difference" only if the 95% interval lies entirely on one side of 0 (one-sided "improvement" wording in v4–v6); every change recorded in `reports/decision_log.md` with a flag saying whether it was made after seeing results.

Appendix D

Decision-log summary

`reports/decision_log.md` records every change to data, code or analysis with a flag stating whether it was made after seeing results. Table D.1 summarises the entries that affect the interpretation; purely technical entries (paths, logging, parallelism, performance) are omitted unless they touch the numbers. "After results?" copies the flag of the log; "post hoc" entries are labelled as such wherever their results appear in this report.

Table D.1: Decisions and deviations that matter for interpretation (from `reports/decision_log.md` and the "deviations" sections of the version reports).

Ver.	Decision / change	After results?
v1	catalogue rows with NaN coordinates broke the pointing check; fixed with `nanmedian` (before any spectrum)	no
v1	deadtime applied to the net rate as DCOR $\times (r_s - r_b)$ (before download)	no
v2	H colours kept linear instead of $\log_{10}$ (two BH spectra with slightly negative 16–25 keV sums)	no (before v2 training)
v2	paired B−H and A−HI differences added to the bootstrap (same pre-declared question "only hardness?")	yes (comparison added, labelled)
v2	output-path bug in `08_error_analysis.py` overwrote two v1 figures; restored and fixed	no effect on numbers
v3	v3a run in a cloud sandbox (pandas 3.0.2); reproduction of v2 OOF to $\leq 8 \times 10^{-10}$ (LR); vectorised bootstrap identical bit for bit	reproduction only
v3	v3c candidate list drafted from memory, verified against BlackCAT and frozen before download; one catalogue-name fix	no
v3	wrap-up colour–colour figure and integrated v1–v3 report (descriptive only)	yes (documentation)
v4	1e cross-algorithm tie-break completed (algorithm, representation, grid order)	no (before the run)
v4	**source-equal weights scale bug** (classes summed to 1 instead of $n/2$; LR regularisation $\approx n/2$ stronger); fixed, 1a–1c re-run, conclusion unchanged; pre-fix results in commit `9120afb`	yes (found after v4a results)
v4	v4b2: duplicate source (XB 1254−690 = 4U 1254−69) fixed before download; 13 additional NSs from executing the v2 rule correctly	no
v4	v4b2: comparison with the same-sample H-LR added (required by §0, omitted in the script)	yes
v4	v4b1 raw files outside the repository; download threads 4 → 6	no
v5	unit-test case "PCU off within a row" corrected before any real feature (rule unchanged)	no
v6	figure annotation fixed before the first run	no

Ver.	Decision / change	After results?
v7	v7b1 primary burst rule changed from "any origin" to "same ObsID or within 1°"; literal rule kept as sensitivity; MINBAR ∪ visual set added	yes (after matching, before any classification)
v7	v7b2 post hoc check of two high-leverage LMC X-3 observations (descriptive)	yes
v7	v7c2 labels and matching fixed (burst oscillations in Liu+2007 periods; HMXB pulsars added; Sesame names within 0.1°; blends excluded; name fixes)	yes (after first matching, before any model)
v7	v7c2 SVM inner selection by grouped 5-fold instead of LOSO (compute)	no (before any SVM result)
v8	IC3 criterion reformulated ($\lvert z\rvert < 3$); segment-test expectation corrected	no (before models)
v8	IC4 failed ⇒ v8c exploratory; estimator changed to ratio of means	yes (after IC4, before models)
v8	null-band subtraction for the HF bands (post hoc)	yes
v8	downloads switched to curl; one inventory connection error recovered; `\r` from a Windows CSV fixed in v8d	no
v9	HEASARC AWS mirror (byte-identical) and raw files outside the repository	no (before models)
v9	colour-window ν_c comparison, $c_2 > 0.7$ exclusion, source-level v9c (post hoc, descriptive)	yes
v10	permutation changed from union-wide to within instrument-membership strata (synthetic power test)	no (before any v10b statistic)
v10	single-instrument stratum bug fixed; jitter seed via `crc32`; parallelism 4 → 2	no effect on statistics
v11	ν^* computed and committed before new data; parallelism 3 → 2, six concurrent range requests	no
v11	v11b background / very-hard sensitivity (post hoc, descriptive)	yes
v12	3C50 rejects original gain calibration (2018 fallback removed); failures excluded as pre-registered; "include uncorrected" sensitivity added	no (before any v12 result)
v12	RF chosen as the v12b primary (stated in the pre-registration as chosen after v9–v11)	yes
v13	resumed after an interruption; the faint cleaned files had been kept (an earlier note wrongly said otherwise)	no
v14	stop after v13: further versions could only re-use the same sources	yes (decided after v13)
report	English report: no analysis re-run, no raw download; discrepancies marked % CHECK	documentation only

Appendix E

Glossary

The repository reports are written in Traditional Chinese. Table E.1 gives the English terms used in this report for the recurring Chinese ones, with a short definition.

Table E.1: Chinese terms of the repository reports, the English terms of this report, and their meaning.

Report term	This report	Meaning
硬度比, 顏色	hardness ratio, (X-ray) colour	ratio of net counts in two energy bands (count space unless stated)
軟態, 硬態	soft state, hard state	spectral states; in this project defined operationally by variability
軟態類, 硬態類, 中間, 無法判定	soft-like, hard-like, intermediate, unknown	variability classes: rms < 0.075, > 0.10, in between, too few rows
硬態類觀測	hard-like observations	observations whose state index exceeds 0.10
天體, 來源	source	one astrophysical object (all its observations share a label and a fold)
來源層級, 觀測層級	source-level, observation-level	metrics on source mean scores vs on individual observations
預先設定, 預先登記	pre-registration	analysis plan written and committed before the data or models of a version
主要 (★)	primary	the pre-declared comparison that decides the version's verdict
描述性, 探索性	descriptive, exploratory	reported without a confirmatory claim
事後	post hoc	decided after seeing the corresponding results
凍結	frozen	fixed and committed before new data are seen (models, rules, pipelines)
前瞻性檢驗	prospective test	test of frozen procedures on data not seen before
有差異, 有提升	detected	the 95% source-bootstrap interval excludes 0 (or $p < 0.05$)
未偵測到	not detected	not detected under the pre-registered criterion (not "absent")
動力學確認 BH	dynamically confirmed BH	BH with a dynamical mass measurement (BlackCAT; Marcel et al. 2026)
候選者	BH candidate	BH classified by X-ray behaviour only; never used as a training label
type-I 爆發, 爆發源	type-I X-ray burst, burster	thermonuclear burst on an NS surface; proof of an NS
脈衝星	pulsar	NS with coherent pulsations (AMXP or slow pulsar)

Report term	This report	Meaning
持續黑洞, 暫現源	persistent BH, transient	BHs in HMXBs (Cyg X-1, LMC X-1, LMC X-3) vs outbursting LMXBs
特徵頻率 ν_c	characteristic frequency	νP_ν-weighted centroid frequency of the variability
洩漏	leakage	information from the test source reaching the training set (e.g. observation-wise splits)
設計矩陣	design matrix	$\boldsymbol{X}$, one row per observation, one column per feature
out-of-fold 分數	out-of-fold (OOF) score	score from a model that never saw the observation's source
天體等權重	source-equal weights	sample weights giving every training source the same total weight
顏色重疊區	colour-overlap box	$h_1 \geq 0.70$, $h_2 \geq 0.20$ in the count-space colour plane (post hoc definition)
橋接檢查	bridge check	gate showing that a new representation reproduces the old one on the old sample
實作檢查	implementation check (gate)	pre-declared synthetic or consistency test that must pass before modelling
外部天體, 從未用過	external sources, never used	sources or observations not used in any training or analysis before
壓力測試	stress test	slow pulsars used to test whether coherent pulses mimic BH-like noise
敏感度分析	sensitivity analysis	re-analysis under an alternative rule
置換檢定	permutation test	test with labels permuted among sources
差中差	difference in differences	change of a state gap between two models
背景比例, 篩選, 修正	background fraction, screening, correction	3C50 f; exclusion of $f > 0.20$; scaling by $1/(1-f)$
增益	gain	paired metric difference (model with − without a feature block)
門檻, 校準	threshold, calibration	decision cut (0.5); agreement of scores with frequencies
信度	reliability	split-half agreement of a feature (Spearman–Brown corrected)